

ngspice-46 — Full Change Report

Every modification and its reason (original → current)

Ngspice + OpenVAF Enhancements project

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ngspice-46 — Full Change Report

Every modification applied to ngspice-46 in this project, with its reason. The baseline is the pristine ngspice-46 source tree (as vendored in the project's `original/` snapshot, which already contains OpenVAF-Reloaded's stock OSDI support); the current state is the tree committed in this repository under `ngspice-46/`. Each entry links the enhancement write-up in [enhancements_doc/](#) that carries the full engineering detail and the verifying example suite. The companion document [openvaf_changes_full-report.md](#) covers the compiler side.

Maintenance note: this report is updated whenever an enhancement touches ngspice sources. If you are reading it alongside a newer tree, the per-enhancement index at the end tells you the last change it covers.

Scope summary. Two new source files and 51 modified ones carry all the functional changes (~2,100 substantive diff lines). A further ~100 `Makefile.in` files and `config.h.in` differ only because the auto-tools were regenerated with a current libtool ([E-77](#)); they contain no hand-written changes and are not listed individually. Everything below is behavior, interface, or diagnostics.

1. The OSDI ABI, version 0.4 → 0.7

`src/osdi/osdi.h` is the authoritative C contract between ngspice and the `.osdi` objects `openvaf-r` produces. The project extended it six times — three times additively (new exported symbols old loaders simply never look up), three times with genuine version bumps (layout changes):

Change	Kind	Enhancement	Why
OSDI_ABSDELAY_COUNTS / OSDI_ABSDELAY_INFOS exports	additive symbols	E-1	<code>absdelay()</code> needs simulator-side waveform history and synthetic delay-equation rows; the descriptors tell ngspice which nodes/offsets each delay slot uses
OSDI_LAST_CROSSING_COUNTS / OSDI_LAST_CROSSING_INFOS exports	additive symbols	E-6	<code>last_crossing()</code> needs waveform history and crossing interpolation — same additive pattern as <code>absdelay</code>
OSDI_TERM_SHORT_COUNTS / OSDI_TERM_SHORT_INFOS exports	additive symbols	E-401	$V(a,b) <+ 0$ between two TERMINALS cannot be served by node collapsing (terminals are simulator-allocated), so the compiler emits a real 0 V source and lists the branch; the simulator drops its current when the netlist already ties those terminals to one node, which is what stops the redundant equation being a singular row
<code>OsdiNode.nodeset</code> field	ABI 0.5 (node stride change)	E-45	Verilog-A net initializers (<code>electrical a = 5.0;</code>) become solver initial-guess nodesets; every node record carries the hint
<code>num_ac_stim_src / ac_stim_sources / load_ac_stim</code> descriptor tail	ABI 0.6 (descriptor grows)	E-51	full <code>ac_stim()</code> support: a Verilog-A module can <i>be</i> the AC stimulus, so the descriptor must enumerate its AC right-hand-side sources

Change	Kind	Enhancement	Why
load_noise destination stride 1 → 2 (signed (flat, react) power pairs)	ABI 0.7	E-54	correct noise physics: operating-point-dependent factors and <code>ddt()</code> -shaped ($j\omega$) noise need a complex amplitude per source, $re + j\omega \cdot im$, so coherent same-named sources (LRM 4.6.4) sum with correct sign and frequency shape

Two additive flag conventions ride on the existing `eval()` interface (not version bumps):

- `EVAL_FLAG_IS_FINAL_STEP` (bit $1 \ll 21$) in the `eval input` flags — `@(final_step)` firing ([E-53](#));
- `eval return` flags for `$finish/$stop` requests and `$discontinuity`-driven step rejection ([E-55](#)).

Pairing consequence: this ngspice requires $OSDI \geq 0.7$ and rejects older objects with a clear version message; `.osdi` files from older compilers must be recompiled ([E-54](#)).

2. The OSDI runtime — `src/osdi/`

`osdi.h` — `PARAM_KIND_*` unsigned shifts

The parameter-kind flag macros were `PARAM_KIND_MASK` ($3 \ll 30$) and `PARAM_KIND_OPVAR` ($2 \ll 30$) — both shift into the sign bit of a signed int, which is undefined behavior (a UBSan build flags it on every OSDI load, at `osdiinit.c:41`). The shifts are now unsigned ($3u \ll 30$, $2u \ll 30$, and the two zero/one companions for consistency). The bit patterns are unchanged (`0xC0000000` / `0x80000000`), so the OSDI ABI is byte-identical, and they now match the u32 constants `openvaf-r` emits in `metadata/osdi_0_4.rs`. Found by the 2026-07 ASan/UBSan pass.

`osdiaccept.c` — new file ([E-55](#), [E-1](#), [E-6](#))

The accepted-timepoint hook (`DEVaccept`) OSDI previously lacked. It advances the `absdelay/last_crossing` waveform histories only at *accepted* points (so rejected timesteps don't corrupt the delay lines) and reads the per-attempt latched `$finish/$stop` request flags at the one boundary where honoring them is safe. Why: acting on `$stop` mid-Newton-iteration was treated as a step failure and ground the timestep into a rejection loop; `$finish` was ignored outright.

`osdidefs.h` (104 diff lines)

- `OsdiExtraInstData` grows the per-instance fields behind every runtime feature: `dt/temp` offsets with given-flags (instance temperature), `eval_flags`, `absdelay` waveform-history rows and pre-allocated KLU/SPARSE matrix-pointer sets for the delay-equation rows (re-pointed on every DC↔AC transition, mirroring the regular Jacobian handling), and the `last_crossing` history ([E-1](#), [E-6](#), [E-55](#)).
- `ALIGN` macro renamed `OSDI_ALIGN`: the macOS SDK's `<sys/param.h>` defines an unrelated `ALIGN`, producing a redefinition warning in every OSDI translation unit ([E-77](#)).

`osdiitf.h` / `osdiext.h`

The registry-entry structure gains the `absdelay/last_crossing` descriptor pointers, and `OSDIpendingRequests()` is exported so the analyses can ask "did any device request `$finish/$stop` at this accepted point?" ([E-1](#), [E-6](#), [E-55](#)). It also gains the terminal-short pointers and the `OsdiTermShortInfo` record ([E-401](#)).

osdiregistry.c (68 diff lines)

- Loads the `absdelay/last_crossing` descriptor arrays from each `.osdi` and threads them into the registry entries (E-1, E-6).
- Loads the terminal-short arrays the same way (E-401).
- Requires $\text{OSDI} \geq 0.7$ (E-54).
- Descriptor-count hardening (E-228): `load_object_file` strided/indexed the descriptor arrays by the counts the `.osdi` itself declares (`OSDI_NUM_DESCRIPTOR`s, and per-descriptor `num_params/num_opvars/num_terminals/...`) with no bounds, so a corrupt, truncated, or ABI-mismatched object that still `dlopens` but declares an out-of-range count read past the array (the inner loop dereferences each `param_opvar` entry's `name[]` pointers) $\rightarrow$ `SIGSEGV` during `pre_osdi`. Two generous ceilings (`OSDI_MAX_DESCRIPTOR`s 4096, `OSDI_MAX_DESC_COUNT` $1 < < 20$, far above any real model) now reject such a file with a clean diagnostic — the outer count before it sizes the registry / strides the array, and each descriptor's own count fields in the loop before they index anything (a single load-time choke point protecting all downstream consumers). Targets the egregious/out-of-range count (the truncated / byte-corrupted / ABI-drift signature — an ABI mismatch reads a pointer's bits as a count $\rightarrow$ huge); a count that lies within the plausible range is an internally inconsistent binary indistinguishable from a valid one and is out of scope, as is a hostile object (a `.osdi` is native code whose loading executes it).

osdiinit.c (8 diff lines)

- Registers the `DEVaccept` hook (E-55).
- Publishes the synthetic instance-temperature parameter under both spellings, `dt` and the conventional `dtemp` every built-in device uses; `n1 a b mod dtemp=10` used to fail with *"unknown parameter"* while the underlying plumbing was exact (E-80).

osdiload.c (441 diff lines — the largest OSDI change)

- Stamps the `absdelay` synthetic rows (`y_synth`, `z`) for DC, AC and transient, driving the delay from the recorded history (E-1).
- Evaluates `last_crossing` from the recorded waveform with linear interpolation between the bracketing timepoints (E-6).
- Passes the final-step flag through to `eval()` (E-53).
- Latches `eval`-return flags per timepoint attempt (`point_eval_flags`, OR-ed across Newton iterations, reset per attempt) so `$finish/$stop` under solution-dependent conditions survive to the accepted-point boundary (E-55).
- Folds the stride-2 signed noise-power pairs (`fac · | fac | semantics`) when transferring `load_noise` results (E-54, E-42).

osdisetup.c (178 diff lines)

- Creates the synthetic delay-equation unknowns and matrix entries for `absdelay` (E-1).
- Applies the OSDI nodesets (`Osdinode.nodeset`) as solver initial guesses (E-45).
- Terminal shorts (E-401): `collapse_nodes` documents that terminals can never be collapsed (`ngspice` allocates them, not OSDI) and skips such a pair — which left $V(a,b) <+ 0$ between two terminals connecting NOTHING, a silent open circuit where the model wrote a short. The compiler now emits a real 0 V source there and lists the branch in `OSDI_TERM_SHORT_INFO`s; setup drops that branch's current unknown (`drop_node`, the same renumbering an ordinary collapse-to-ground does) whenever the two terminals do not resolve to two DISTINCT CONNECTED circuit nodes. Without that half the equation is redundant wherever the netlist already ties the terminals together — the normal wiring of a self-heating model — and a redundant 0 V source is a singular row.

- A `$fatal/$finish` raised during setup (models rejecting their configuration by design — HiSIM's `$port_connected` guards) used to surface as ngspice's baffling *"impossible error — can't occur"*; it now reads *"a Verilog-A device rejected its configuration during setup"* (E-56).
- An internal OSDI node that appears in no Jacobian entry — a net structurally decoupled from the matrix, e.g. an explicit ground `gnd` reference whose branch contribution $V(p, gnd) \leftarrow \dots$ drops the `gnd` column — used to be allocated its own solver row. That all-zero row/column is benign under Sparse (it decouples to $V=0$) but makes the KLU matrix structurally singular, so KLU returned a wrong DC answer (groundcontrib: $v(p)=0$ not 1.5; hierbranch: branch currents 0). Such nodes are now tied to ground (node 0) at setup, matching how OpenVAF already treats them and fixing both solvers (E-116).

osdiacld.c (89 diff lines)

- AC loading gains the `ac_stim` right-hand-side injection: the partitioned AC-stimulus source array is loaded through `load_ac_stim` and added to the AC RHS with exact magnitude and phase, `$mfactor` applied linearly (deterministic signals scale with multiplicity) (E-51, E-26).
- Complex ($j\omega$ -shaped) noise coupling entries participate in the AC matrix (E-54).

osdinoise.c (93 diff lines)

- Per-instance grouping of same-named noise sources: coherent summation of complex amplitudes $(a + j\omega \cdot b) \cdot T$ before squaring, per LRM 4.6.4 — same-named sources correlate (including exact anti-phase cancellation), differently-named ones stay independent (E-42).
- Reads the stride-2 signed pairs of ABI 0.7 (E-54).

osditrunc.c (34 diff lines)

- Honors `$discontinuity(n)`'s next-step clamp: a negative `bound_step` sentinel from the model limits the step after the event (E-24).
- Honors the step-rejection return flag: when an event fired inside the step, `OSDItrunc` requests `delta/8` (with a $20 \cdot \text{CKTdelmin}$ termination floor) so the integrator bisection onto the discontinuity instead of extrapolating across it (E-55).

osdiparam.c, osdisetup.c, osdiregistry.c, osdiinit.c, osdidefs.h — the collapse-owner map

A terminal collapsed onto an internal node ($V(d, di) \leftarrow 0$, the `rd = 0` default path of essentially every compact model) reported a terminal current of exactly 0.0: the model writes its current into `di`'s residual and terminal `d`'s slot stays zero, while `OSDIask` read only `descr->nodes[t].resist_residual_off`. With `rd` and `rs` both zero every terminal read 0.0 and the reported currents summed to zero, so a Kirchhoff check passed on $0 = 0$. The solution was never affected — `osdiloadd.c` stamps the whole group into one matrix row, which is why nothing a user plots ever moved.

`OSDIask` now sums the terminal's whole collapse group, resistive residual plus (in a transient) the integrated reactive residual, striding the state cursor over non-members exactly as `osdiloadd.c` does. The grouping cannot be recovered after setup — `write_node_mapping` replaces the local mapping with global node numbers, and two terminals on one net share a global number while ground additionally collects grounded terminals, nodes collapsed to ground, E-116's decoupled internal nodes and E-401's dropped term-short flow nodes. It is therefore snapshotted the moment `collapse_nodes` returns, into one `uint32_t` per descriptor node trailing `OsdiExtraInstData(osdi_collapse_owner)`, holding `t + 1` for "owned by terminal `t`" so that the clobbered zero means "setup has not run" (E-416).

osdiparam.c, osdisetup.c, osdiinit.c, osdiregistry.c, osddefs.h, cktsens.c, breakp2.c — the collapse decided once, re-decided later

OSDItemp re-runs `setup_instance` on every temperature update, so a model can re-decide its node collapse — but nothing derived from the first decision can be rebuilt there (the node mapping, the matrix element pointers, the state layout, E-416's owner map). The file's own `// TODO` check that there are no changes in node collapse? is now answered: OSDItemp clears `collapsed[]`, re-decides, compares against a snapshot taken in `OSDIsetup`, and restores the snapshot unconditionally before the error check, because the matrix implements the snapshot. Clearing first is the whole fix — OpenVAF's collapse callback only ever STORES TRUE, so without it a collapse that stopped happening is invisible, which is the direction the defect takes. The snapshot is one `bool` per collapsible pair trailing E-416's owner map in the same instance block.

Three consumers acted on the mismatch. `cktsens.c` perturbed a collapse-selecting parameter and differenced two loads that shared one matrix element: the branch's `+g` and `-g` cancelled on the same diagonal and what survived was $\text{eps} * E / (Y * \text{delta}^2)$ — roundoff whose magnitude tracks $1/\text{rd}$ and whose sign moves with unrelated parameters ($2.5289\text{e-}02$ printed where the truth is $2.2676\text{e-}04$). It is now reported as 0 with a warning naming instance and parameter; it cannot be computed correctly, since `DEVsetup` runs only at the base value and re-running it would allocate nodes and trip `cktsens`'s own `CKTlastNode` guard. `.dc temp` ran a whole sweep on one topology — exactly 0 A sweeping into a collapse, a plausible 2x error sweeping out of one — and now warns once per instance; a hard error would mean propagating five bare `CKTtemp(ckt)`; returns in `dctrcurv.c` and changing error handling for every device model in the tree. `breakp2.c` stopped skipping the per-terminal expansion at exactly two terminals, where `@dev[i_p]` had been a length-1 scalar while `@dev[i_d]` was a waveform at three; the bare `@dev[i]` is kept beside it and synthesized names are de-duplicated against explicit `.save` entries (E-417).

osdisetup.c, outitf.c, com_measure2.c — three things nobody checked

`absdelay()` and `last_crossing()` are the only analog operators whose OUTPUT row the compiler deliberately leaves empty: `ngspice` fills it from its own history buffer, through matrix elements `OSDIsetup` allocates later. E-116's decoupled-node scan can only see coupling the COMPILER declared, so it found the output node in no Jacobian entry, tied it to ground, and `eval()` read `CKTrhs0ld[0]` -- exactly 0.0, forever. The value was right only when the model also CONTRIBUTED it, because that is what put the node into a Jacobian entry; `transition()` shares the same lowering and worked only because its output feeds the slew residual. Those nodes are now marked used before the scan decides. E-415 had looked for this in the compiler and reverted when the descriptor came out byte-identical -- correctly, since nothing was wrong there. No OpenVAF change and no ABI change: the descriptor already carries the infos and `osdiregistry.c` already parses them.

`.save` validated nothing. Nothing between `settrace` and the per-point read looked a name up, and `getSpecial`'s caller discards `INPaName`'s `E_NODEV/E_BADPARM`, so a misspelled device, a bogus parameter or an unexpanded wildcard produced a registered vector that stayed 0 long in silence. The check now calls `INPaName` itself -- the same routine the per-point read uses -- and a hierarchical spelling is REWRITTEN rather than rejected, so E-410's `@x1.r1[i]` resolves while the display name stays the user's. Unresolvable entries are warned about and still added, because dropping them would change what the plot contains.

`meas ...` when interpolated a crossing-free interval. The initialisation counted a crossing in the FIRST interval -- which runs from the operating point to the first timepoint and so straddles thresholds for reasons unrelated to the waveform -- and the evaluation, which deliberately starts one sample in, applied that leftover count to a later interval, dividing by a difference that was exactly zero or one ULP: $-inf$, $1.15292\text{e+}05$ s in a 3 us run, negative times. The count is no longer taken, every interpolation is gated on the interval actually bracketing the target (mirroring `measure_at`), and a `.dc` sweep restart resets to sample 0 so the previous sweep's last point is never a bracketing endpoint (E-418).

pz filled the simulator-stamped row nowhere. Because those rows are the simulator's to fill, every load path has to fill them -- `osdiload.c` does for `dc/tran` and `osdiacld.c` for `ac`, but `osdipzld.c` did for neither, so the row was identically zero, the matrix was singular at every trial `s`, every trial looked like a root, and pz blamed the netlist: "the input signal is shorted on the way to the output". This predates the fix above for a model that CONTRIBUTES the value (its row was already live), and marking the nodes used would have widened it to every observed-only model. The AC stamp cannot be reused: `e^-s*t` overflows across pz's own search range and a transport delay has infinitely many roots, so there is no finite pole-zero set to find. `absdelay` is therefore stamped as the zero-delay wire -- the same linearization the operating point uses -- and warned about once per instance; `last_crossing` is stamped exactly, with no warning, its small-signal sensitivity being zero. All three cases now report the delay-free pole to the digit, including the two that aborted before E-418 (E-418).

osdi/Makefile.am

Adds `osdiaccept.c` to the build (E-55).

3. Analyses — **src/spicelib/analysis/**

dctran.c (46 diff lines)

- Advances OSDI delay/crossing histories via the accept path and skips history corruption on rejected steps (E-1, E-6).
- Issues the dedicated `@(final_step)` evaluation at the converged end of a successful transient (E-53).
- Honors accepted-point `$finish` (ends the analysis cleanly, firing `final_step` at the requesting point), `$stop` (pauses resumably), and aborts with a clear error on `$fatal`'s `E_PANIC` instead of retrying it as nonconvergence (E-55).

dcop.c (9) and **acan.c** (10)

- Fire `@(final_step)` at their successful end (an operating point fires both step events — a single point is first *and* last) (E-53).
- An AC job's operating-point phase carries the analysis name bit (only the name — adding the reactive `CALC_*` bits would wrongly enable integration at an op), so `analysis("ac")` holds through the whole AC run per LRM 4.6.1 and `@(initial_step("ac"))` fires in AC (E-53).

dctrcurv.c (150) + **include/ngspice/trcvdefs.h** (13)

- Generic `.dc @inst[param]` sweeps (a new sweep code): the DC sweep previously hardcoded `Vsource/Isource/Resistor/temperature`, so sweeping any other device parameter was a fatal error. The generic sweep resolves `@inst[param]` through the device's own parameter tables, refreshes per point via `DEVtemperature` (for OSDI exactly the `alter` path), nests with other sweep variables, and restores the original value afterwards (E-62).
- Fires `final_step` at the last sweep point; honors a `$finish` raised mid-sweep (E-53, E-55).

noisean.c (37)

- A singular AC matrix during noise analysis crashed ngspice with a SIGABRT: the code ignored `NIacIter`'s return and the adjoint solve asserted on the unfactored matrix. It now aborts cleanly ("AC solution failed at ... Hz") (E-56).
- Honors deferred `$finish`/`$stop` raised at the noise operating point and fires `final_step` on success (E-55, E-53).

span.c (31) + **cktspdum.c** (14)

- 1-port S-parameter analyses enabled: the "we need at least two ports" error was over-strict (a 1-port is a plain reflection measurement; the matrix machinery is N-general) — and it was hiding the cadjoint crash fixed in **dense.c** (E-64).
- **Rbase** published into the **sp** plot from port 1's **z0**, making the Touchstone writers work with no manual `let Rbase = 50` incantation (E-64).
- Stock NaN fixed in **donoise** noise-parameter extraction: the uncorrelated noise conductance G_u is analytically zero for fully-correlated single-source topologies, and floating-point rounding could land $\sqrt{Y_{cor}^2 + G_u/R_n}$'s argument at $-10^{-18} \rightarrow \text{NaN}$ for the noise figure. Clamped to the physical range ≥ 0 — found by parity testing against an OSDI resistor that produced the exact textbook NF where the built-in returned NaN (E-63).

cktdisto.c (16)

- **.disto** silently reported zero distortion for OSDI devices — the distortion kernel needs higher-order Taylor coefficients the OSDI ABI cannot provide (first derivatives only), and such devices were skipped without a word. A prominent warning now names each affected OSDI device type (E-62).

4. Frontend — **src/frontend/**

plotting/pyplot.c (new) + **com_pyplot.c** (new) + **plotit.c**, **commands.c** (E-94)

A new interactive command, **pyplot**, plots simulated vectors with matplotlib — a Python counterpart to gnuplot. **ft_pyplot()** (**plotting/pyplot.c**) mirrors **ft_gnuplot()**: it writes a `<file>.data` table and a `<file>.py` matplotlib script and **system()**s **python3** (synchronously for a PNG, backgrounded for an interactive window). A new "pyplot" device arm in **plotit()** routes to it, so it accepts the same plot expressions as **plot/gnuplot**; the **com_pyplot()** wrapper mirrors **com_gnuplot()**, and **pyplot** is registered in both command tables. Two set variables tune it: **pyplot_terminal=png** renders headless (Agg) to a PNG, and **pyplot_python** picks the interpreter (default **python3**). Purely additive (E-94). The output file base name is optional (E-95): **com_pyplot()** treats the first word as a file name only when it is not a plot expression (no `(`, and not a vector by **vec_get()**), otherwise it defaults to **pyplot** — so **pyplot v(out)** plots directly (the command's minimum argument count was lowered from 2 to 1). **ft_pyplot()** also grew stacked subplots (**set pyplot_subplots=N** $\rightarrow$ `plt.subplots(nrows, 1, sharex=True)`, **N** traces per panel; **vs** stays the x-axis, so it is not a panel separator) and matplotlib style sheets (**set pyplot_style=<name>**, **dark** $\rightarrow$ **dark_background**, guarded) (E-98). The headless **pyplot_terminal** was extended from PNG-only to the vector export formats **svg** and **pdf** (**set pyplot_terminal=svg/pdf** $\rightarrow$ `fig.savefig(<file>.<fmt>)`, matplotlib picking the writer from the extension), and a figure size control was added (**set pyplot_figsize="W,H"** $\rightarrow$ `plt.subplots(..., figsize=(W, H))`; quote the value so ngspice keeps the comma) (E-99). A **-eye** flag turns **pyplot** into a one-line eye-diagram renderer over the E-207 eye analysis: **com_pyplot()** detects the **-eye** marker, runs **com_eye()** on the trailing tokens (which folds the waveform and leaves **eye_wave/eye_t** + the scalar metrics in a fresh current eye plot), then a new **ft_pyplot_eye()** reads those back and writes a **hist2d**-based persistence eye script (**LogNorm**, **turbo**, empty cells masked; **threshold** + eye-centre lines, eye-height/width double-arrows, a metric title), launched by the same **system(python ...)** mechanism as **ft_pyplot()** and honouring the same **pyplot_terminal/pyplot_python/pyplot_backend/pyplot_figsize/pyplot_style** settings. **ft_pyplot()** and the **gnuplot/asciiplot** back-ends are untouched (E-208).

Four further groups of additions. Appearance controls — seven set variables (**pyplot_grid**, **pyplot_legend**, **pyplot_markers**, **pyplot_axhline**, **pyplot_axvline**, **pyplot_dpi**, **pyplot_transparent**) threaded into **ft_pyplot()**, all additive so a control-free plot is byte-identical to before (E-296). A **-fft** mode — **ft_pyplot()**'s **bool hist** became an **int mode** (**LINE/HIST/FFT**); **-fft** reuses the time-

domain data table and the emitted script resamples onto a uniform grid (`np.interp`, because transient data is adaptively sampled) before `np.fft.rfft`, scaled by `2/sum(window)` so a tone reads back its amplitude (`pyplot_fft_window/_db/_points/_logf`) (E-297). Complex-aware AC views — a new `ft_pyplot_ac()` renderer for `-bode` (mag dB / phase deg vs log-f, stacked, `np.unwrap`), `-nyquist` (imag vs real) and `-polar` (mag at phase), emitting a `<vec-index> <freq> <re> <im>` table that KEEPS the imaginary part an ordinary `pyplot` discards (E-298). And finishing touches — cross-run overlay (`pyplot_tran1.v(out) tran2.v(out)`) now sizes the data table by the longest scale so a finer second run is not truncated, plus a `pyplot_cursor` hover crosshair (matplotlib's built-in `Cursor`, window-only) (E-299). A `pyplot_mplcursors` variable then selects the `mplcursors` package (data cursors that snap to a trace) as the cursor backend, turning the cursor on by itself, with a try/except fallback to the built-in `Cursor` if the package is absent where the emitted script runs (E-300). The cursor gating was then consolidated so `pyplot_cursor` is the single master switch (off by default, the only enable); `pyplot_mplcursors` only selects the backend when the cursor is on (E-301).

postcoms.c (414 diff lines) + **postcoms.h** + **commands.c** (16)

The Touchstone I/O subsystem (E-64, E-72):

- **wrsnp** (with `wrs2p` dispatching to the same handler): Touchstone v1 for any port count — the classic 2-port S11 S21 S12 S22 column order preserved byte-identically, $N \geq 3$ in the spec's row-major layout with at most four complex pairs per line;
- writer options `wrsnp <file> [ri|ma|db] [s|y|z] [hz|khz|mhz|ghz]`, any combination in any order: MA/DB formats, Y/Z parameters (normalized to Rbase, $Y \cdot R / Z/R$, per the v1 spec), frequency units — the option line reflects every choice;
- **rdnsnp**: reads any Touchstone v1 file into a new plot with a Hz frequency scale and complex vectors matching the `.sp` plot's conventions (MA/DB converted back, Y/Z de-normalized, the 2-port column order handled, Rbase published so imports round-trip) — measured data diffs against simulation in one `let` expression.

Why: E-63 showed the S-parameter *analysis* was exact but its results could not leave ngspice in the industry-standard format (`wrs2p` demanded a never-created Rbase vector), and nothing could bring VNA data in.

outitf.c (16)

- Integer operating-point variables recorded per point: `getSpecial` masked with `IF_REAL` only, so an integer opvar saved with `.save @n1[flag]` produced garbage in transient vectors (E-32).
- The plot-memory guard's message printed `%Id` — a Windows-only `printf` length modifier that is undefined behavior elsewhere and rendered the message as the numberless "memory required (Id Bytes)". Now `%zu`: real byte counts on every platform (E-77).

resource.c (27)

`ft_ckptspace()` — the "approaching max data size" warning (`RSS > 95%` of `RSS + free RAM`) — now honors `set no_mem_check` (the same opt-out the fatal output-size estimate in `outitf.c` respects; one variable governs both memory guards) and warns once per excursion above the threshold, re-arming below 90%, instead of repeating on every check through a long analysis (E-81).

display.c (1)

`#undef NODEV` before the local macro definition — the macOS SDK's `<sys/param.h>` defines an unrelated `NODEV` (E-77).

inpcom.c, parse.c, vectors.c, device.c + analysis/dctrcurv.c — bracketed names (E-408)

Three defects on the path from a name written in a deck to the object it denotes, none of which reported anything.

A leading zero split a node in two. `n[1]`, `n[01]` and `n[001]` were three distinct nodes — `print all` listed all three — while the vector lookup canonicalised the index before resolving it, so all three probes returned the `n[1]` value. With a single node the same mismatch is worse: a netlist that built `m[02]` was unreachable by every spelling, its own included, because the lookup only ever asked for `m[2]`. Enhancement-221's own header settles the question (*"the scalar names produced use the same bracket form, so a bus `a[0:1]` and an explicit `a[0]` denote the same node"*), so `inp_canon_bus_index_token()` canonicalises the index in the netlist, negative indices included; it is a byte-identical no-op when the spelling is already canonical.

`@dev[param]` could not name a bracketed parameter. `show` lists a bus device's terminal currents `i_a[0]..i_a[3]` (E-394) and its array parameter elements `ap[0]..ap[2]` with correct values, and the instance line can set them, but four separate splitters each cut the name at the inner `]` and reported *"no such parameter `i_a[0]`".* All four had to change because each serves a different command: `Pplex` (`parse.c`) lexes the token at all, `vec_get` (`vectors.c`) serves `print` and `let`, `com_alter` (`device.c`) serves `alter`, and `DCTfindInstParam` (`dctrcurv.c`) serves E-62's `.dc @inst[param] sweep`, which had been aborting with a fatal error. The E-269 wildcard alias `@*[param]` works *because* of the old first-`]` split, so a name beginning with `[` keeps it deliberately.

A bus range expanded everywhere except where the answer is read. `[lo:hi]` expanded on instance lines, R/C node lists, subcircuit calls and port lists, but not on `.save/.print/.plot` (the card produced nothing) or `.ic/.nodeset` (warned and ignored). Those cards were skipped by design: `inp_expand_buses` admits lines carrying bare node fields, and an output card writes `v(a[0:3])`. `inp_bus_rewrite_wrapped()` emits one copy of the whole token per index, so the wrapper and any `=value` ride along; a spaced `.ic v(a[0:1]) = 0.25` absorbs its value into the token first. A token naming two ranges has no unambiguous expansion and stays literal.

com_sweep.c + spiceif.c — a wildcard sweep knob is put back (E-409)

`sweep @*[wavelength_nm] ...` ran correctly and then printed

```
Error: no such device or model name
PError: syntax error in line segment
    @*[wavelength_nm]
near
    wavelength_nm]
```

The name looks mangled, which is what made the message alarming, but nothing was wrong with the parameter: the expression lexer's specials set contains `*`, so `@*[p]` cannot lex as one token and the parser reports the leftover text.

The message was the harmless part. `sw_read_knob()` is how E-385 captures the nominal it puts back at the end of a sweep, and it parses the knob name as an expression — the call that cannot succeed for a wildcard. The read reported failure, and E-385's deliberate all-or-nothing rule then skipped restoring everything: a wildcard sweep left every target at its last swept value, and with `-vs` it took a concrete co-knob down with it (`@dev1[wavelength]` left at 3.0 although it restores perfectly on its own).

One number could not fix it. A wildcard sets every matching model or instance to one value, but the values it overwrites can all differ, so undoing it needs one reading per target. `if_saveparam_wildcard()` / `if_restoreparam_wildcard()` walk exactly the same targets in exactly the same order as the existing `if_setparam_wildcard{,_instance}` — same device-type loop, same model and instance chains, same `paramlookup(..., inout=set)` predicate — so index *i* of the saved array names the same target on the way out as on the way in, reading through the parameter's askable twin and the same `doask/doset`

pair a plain `alter/altermod` uses. Saving stays all-or-nothing (a set-only or vector-valued parameter refuses the whole capture), and the replay is pinned to the circuit the readings came from, so a `.param` co-knob that re-sources the deck cannot have them replayed onto a different one.

`spiceif.c`, `dctrcurv.c`, `com_sweep.c`, `gens.c` — a subcircuit device without its type letter ([E-410](#))

A resistor inside a subcircuit is `@r.x1.r1[resistance]`, while the node beside it is plain `x1.m`. The obvious spelling `@x1.r1[resistance]` named nothing.

The letter is not an accessor convention — `r.x1.r1` is the device's real name. ngspice flattens a subcircuit by rewriting its cards back into the deck and re-parsing them as ordinary element lines, and the parser takes the device type from the FIRST CHARACTER of the card (`inppas2.c`), so the flattened refdes has to keep a type letter in front: the card `x1.r1 a m 1k` would otherwise be re-read as another subcircuit call. Two details confirm it: `translate_inst_name()` (`subckt.c`) exempts `x` devices, whose names already start with the right letter, so `x2` inside `x1` is plain `x1.x2`; and `NODES`, having no type, keep plain hierarchical paths.

The reconstruction needs no search. The letter prepended is literally the leaf name's own first character, and a device's name must already begin with its type letter, so `x1.r1` can only mean `r.x1.r1` — at any depth, and for OSDI devices alike (`x1.x2.nd1 -> n.x1.x2.nd1`). `if_find_instance_hier()` is consulted only after the exact lookups fail, so every name that resolves today is unaffected.

Following [E-408](#)'s lesson, every consumer is routed through it rather than just `print: finddev()` (`print/let/alter/altermod`), `finddev_special()`, `DCTfindInstParam()` for [E-62](#)'s `.dc @inst[param]` sweeps, and `sw_fp_bind()` for the sweep knob. For sweep the old behaviour was not even an error — the knob silently failed to bind and the three points came out identical.

`show` needed a different answer: its grammar takes the query's first character as the device-TYPE selector and uses `:/#`, not `.`, as the subcircuit delimiter, so `x1.r1` parsed as *type x* and could never match a resistor. `dgen_hier_match()` is consulted alongside that grammar as a whole-word alternative and can only add a match. A first attempt widened the `strcmp` inside the decomposition instead; it was measured as DEAD CODE (the query bails at the type check long before) and reverted.

`inpcom.c` — a descending bus range binds terminals in reverse ([E-411](#))

`nd1 d[3:0] 0 mm` against a model declaring `inout [3:0] a` compiles clean, simulates clean, and connects the terminals backwards. Both sides say `[3:0]`, which is exactly why it reads as correct.

Two conventions disagree. The COMPILER declares a bus port's terminals in ASCENDING bit order whatever direction the declaration uses -- `hir_def`'s `item_tree` lowering sorts the endpoints and loops upward, *"declare from lsb to msb (ascending), matching natural bit order; direction of the original [msb:lsb] only affects range checks"* -- so `inout [3:0] a` and `inout [0:3] a` give the same positional terminal list. The NETLIST does honour direction: [E-221](#) expands `d[3:0]` to `d[3] d[2] d[1] d[0]`. Since the instance line's Nth node binds to the model's Nth terminal, a descending node list reverses the mapping.

Measured with one conductance per bit (`a[k] -> (k+1) mS`, so the current at a node identifies the bit it landed on): wired `d[0:3]` the nodes read 1, 2, 3, 4 mS; wired `d[3:0]` they read 4, 3, 2, 1 -- and the model's own declaration direction changes nothing, the `[3:0]` and `[0:3]` rows being identical.

`inp_expand_bus_token()` now reports whether the range ran msb-first, and `inp_expand_buses()` warns on element instance lines only. The scope came from scanning what the shipped suite actually does with descending buses rather than from taste: a `.subckt` descending PORT list is a deliberate interface choice that [E-221](#) documents and `busnodes_examples` tests; output and IC cards bind nothing; and a range too wide to expand never binds at all ([E-338](#)'s guard), so warning there would be false. The opt-out is set `nobusdirwarn` in `.spiceinit` -- a set inside `.control` is too late, because bus ranges expand while the netlist is read, and the message says so.

osdi/osdiloadd.c — an operating point that changed with frequency (E-412)

An OSDI operating-point variable read after `.ac` returned the SMALL-SIGNAL solution rather than the bias, and returned a different value at every frequency: an opvar defined as nothing but `vbias = V(a, c)` gave 0.4939733 after `op` (the true bias, from solving the circuit's KCL cubic by hand) but 0.4823410 / 0.4819028 / 0.0473587 after `.ac` at 1 k / 10 k / 1 meg.

An operating point cannot depend on frequency, which is what separates this from a tolerance argument -- the value is identical at `reltol` 1e-3, 1e-6 and 1e-10, and a multi-point sweep leaves the LAST point's.

CAUSE: E-53 fires `@(final_step)` by issuing one dedicated evaluation per instance when an analysis completes, at `CKTrhsOld` -- which after a frequency sweep holds the small-signal solution. The model is genuinely re-evaluated, not mis-read: opvars chosen to tell those apart confirmed it (3V+1 and V^2 both matched the ac solution to nine digits, while a constant opvar stayed 42). Built-ins are unaffected, having no such post-analysis evaluation.

NOT WRONG, and established before fixing anything: the analyses themselves. A bias-dependent `white_noise(kf*V^2)` source integrated to `onoise_spectrum = 1.342573180366e-01`, matching the dc-bias prediction exactly and not the ac-bias one (7.21e-02). This is a READBACK defect; it is also not sticky, any following `op/dc/tran` restoring the correct value.

The evaluation cannot simply be skipped -- it is the only thing that fires `@(final_step)`, and `@(final_step("ac"))` and the noise variant are supported and tested by `finalstep_examples`. So `OSDIfinalStep` snapshots the instance data around the call and restores it, for `MODEAC` and `MODEACNOISE` only: the event bodies still run and their `$strobe` output stands, while what the evaluation wrote is discarded -- correct, since those results are by design never loaded into the matrix or RHS. DC, DC-sweep and transient are deliberately not snapshotted, their final solution being a real operating point.

WHY IT SURVIVED: with a purely resistive device the dc and ac solutions coincide, so the natural small test case cannot see it at all.

breakp2.c + osdi/osdiparam.c — `savecurrents` for a multi-terminal OSDI device (E-413)

`.options savecurrents` on a four-terminal compact model produced `@nd1[i] : current, real, 0 long` -- a vector that exists and holds nothing, with no warning -- while a built-in BJT in the same deck produced four filled waveforms (`@q1[ic]`, `@q1[ie]`, `@q1[ib]`, `@q1[is]`).

CAUSE: `inp_savecurrents()` is a TEXTUAL pre-pass over the deck. It runs long before any `.osdi` is loaded, so it cannot know a compact model's terminal names and emits the same bare `@dev[i]` that R/C/L use -- its own comment (E-394) says exactly that. E-394 defines the bare `i` alias only for TWO-terminal devices, where it is unambiguous; for anything wider the save named a parameter that does not exist, so the vector was registered and never written.

The values were never wrong, only uncapturable: read as scalars after `op` the terminal currents were exact (1.000e-3 / 4.000e-3 / 1.200e-2 / -1.700e-2, summing to zero by KCL) and an explicit `.save @nd1[id]` always worked. What was missing was the waveform, which is what `savecurrents` exists to provide.

Since the names cannot be known during a textual pre-pass, the expansion moves to where they can: `ft_getSaves()` runs at analysis start with the circuit set up, so a bare `@dev[i]` belonging to an OSDI instance that does not define it is expanded there into one entry per terminal, via a new `OSDIterminalNames()`. The two-terminal case is deliberately untouched -- `@dev[i]` is real there and is byte-identical to the pre-413 binary -- and built-ins never enter the path.

5. Parser and device registry — `src/spicelib/`

`parser/ingtok.c` (10)

A `.model` card override silently dropped its first parameter when the type name and the opening parenthesis had no space between them (`modname devtype(p1=v1 ...)`): `INPgetNetTok`'s scan did not treat `(` as a token boundary, so the type token swallowed the paren and the first parameter name. Found while verifying event-function counters whose model-card overrides mysteriously kept defaults (E-8).

`parser/ingmod.c` (5)

`find_model_parameter()` dereferenced `*(device->numModelParms)`, which is `NULL` for built-ins that take no model cards (`VCVS`, `CCCS`, ...) — any *referenced* `.model m vcvs()` card segfaulted, with no OSDI involved at all (an ordinary MOS instance sufficed). This was the true root of the long-documented “module named like a built-in crashes” gotcha. One `NULL` guard; both shapes now produce clean, located errors (E-76).

`parser/numparse.c` — out-of-range int test

`ft_numparse` decides whether a parsed number is integer-valued with the round-trip `(double)(int)val == val`. Casting a double that lies outside `int` range to `int` is undefined behavior, reached by any control-language literal `>= 2^31` (e.g. `let x = 3e9`) — a UBSan build flags it at `numparse.c:166`. The value is now range-checked (`val >= -2147483648.0 && val < 2147483648.0`) before the cast; a value outside `int` range is, by definition, not `int`-representable, so the answer is unchanged for every in-range input (boundary-tested at `INT_MIN`, `INT_MAX`, and $\pm(2^{31})$). Found by the 2026-07 ASan/UBSan pass.

`snp2va.c` — `lstsq_real` heap overflow on a tiny Touchstone file

`lstsq_real` (the Householder-QR least-squares primitive behind `pre_snp`'s vector fit) documented itself as requiring an overdetermined system ($m \geq n$), but its back-substitution read `b[i]` and `A[i*n+i]` for every unknown `i` up to `n-1`, past the `m`-row buffers when $m < n$. `pre_snp` on a Touchstone file with very few frequency points (≤ 3) drives the fit underdetermined — the stacked system has fewer rows than poles, and the buffers shrink to a `malloc(0)` — so the back-substitution read off the end: a heap-buffer-overflow (ASan: READ of size 8 ... 7 bytes after a 1-byte region), reachable from an ordinary coarse `.snp` measurement. The back-substitution now leaves any unknown with no constraining row ($i \geq m$) at zero instead of reading past the matrix; for the normal $m \geq n$ path the guard never fires and the fit is bit-identical (verified: a 21-point file fits to 2 poles, rms 1.77e-05, before and after). Found by the 2026-07 ASan/UBSan pass (heavy-deck sweep, via a degenerate `pre_snp` input).

`snp2va.c` — unbounded filename-derived port count → heap corruption (E-227)

`parse_touchstone` infers the port count `N` from the file extension `.snp` (`N = atoi(dot+2)`) with no upper bound. A filename such as `.s2147483647p` (`N = INT_MAX`) slips past the parse — the record stride `rec = 1 + 2*N*N` and the per-record inner-loop bound `N*N` share the same wrapped `int`, so `N*N` overflows to a small/zero value and the token layout stays self-consistent — but `N` is stored in `out->N` and then used to size the downstream vector-fit allocations (`malloc(nf*N*N*sizeof(cplx))`, `malloc(N*N*sizeof(cplx))`). The overflowing allocation and the writes that follow corrupt the heap (the program typically aborts later on an unrelated `malloc`, so it presents as flaky). The brute-force fallback already caps port inference at 512; the filename path now honors the same bound — `if (N > 512) N = 0`; falls back to inferring `N` from the data. Valid files (`.s2p`, ...) are far below the cap and unaffected. Found by the 2026-07 Touchstone-parser fuzzing pass (`rdsnp` was clean over 4,500 iters; `pre_snp` hit this on 103/3,600). Verify: `examples/snpfuzz_examples/verify_snpfuzz.py`.

parser/inp2n.c — an OSDI instance line with too FEW nodes ([E-402](#))

The node count was bounded on one side only: `numnodes > *dev->terms` errors with *“too many nodes connected to instance”*, while too FEW was never asked about. The binding loop then wrote `GENnode(fast)[i] = -1` for each unsupplied terminal and said nothing, so a typo silently produced a different circuit — a 4-terminal device given 3 nodes still simulates, with the omitted pin dangling and every branch touching it dead (`v(g)` 1.333 fully connected vs 2.000 mistyped, `rc=0`, empty diagnostic stream). Every built-in rejects the same mistake.

Not fixed by rejecting it. That `-1` is the SAME sentinel `osdisetup.c` reads as *“this terminal is deliberately unconnected”* (`terminals[i] == -1` `connected_terminals = i`) — the LRM optional-terminal feature that 32 corpus models query through `\$port_connected` (BSIMSOI, BSIM-CMG/IMG/BULK, BSIM6, PSP-HV). A dangling node is not a substitute either: electrically it matches an omitted pin, but it reports `\$port_connected == 1` where the omission reports 0.

So the lower bound WARNS, naming each absent terminal from `dev->termNames`, and states the consequence — including that an omitted terminal is not grounded, since that is the natural assumption and it is wrong (grounding the pin moves the measured node by 67%). Semantics unchanged; regression 322/322 with the warning firing zero times across the suite.

device noise routines — a nominal temperature added to a difference ([E-403](#))

`res/resnoise.c`, `bjt/bjtnoise.c`, `dio/dionoise.c`, `vbic/vbicnoise.c`, `hicum2/hicum2noise.c` all computed the thermal-noise temperature delta as

```
dtemp = inst->REStemp - ckt->CKTtemp + (model->REStnom - CONSTCtoK);
```

`NevalSrcInstanceTemp` evaluates $4 * k * (CKTtemp + param2) * G$, so `param2` is a DELTA. The third term adds the NOMINAL temperature expressed in CELSIUS (27 by default) to that difference, so an instance sitting at ambient got +27 K: `rd x 0 1k temp=27` read 4.4% more noise voltage (9% power) than the identical `rd x 0 1k`. Instance temperatures are stored in Kelvin, so the first two terms were already the whole answer; the third is dropped. `temp=100` and `dtemp=73` -- the same temperature written two ways -- now agree exactly, at the analytic ratio.

Surfaced as `sens` poisoning every later noise in a session: `sens` restores parameter VALUES but leaves the `...tempGiven` flag set, which flips the noise routine into the branch above. Correcting the arithmetic fixes both, since an instance at ambient then contributes a zero delta whether or not the flag is set.

devices/dev.c (51)

- Duplicate device-type registration warned and skipped: `osdi_add_device()` appended every descriptor unconditionally, and the model-card lookup scans front-to-back — so a module name duplicated across two loaded libraries silently resolved to the *first* registration (loading an updated model library gave the stale device with no hint), and a module named like a built-in corrupted model creation. Registration now checks the table case-insensitively and skips duplicates loudly ([E-76](#)).
- **pre_osdi** path dedup: loading an already-loaded file notes it and skips ([E-76](#), [E-81](#)).
- **pre_osdi -f** force reload ([E-229](#)): the three dedup gates (`path`, `dlopen handle`, `device-name`) plus `dlopen`'s own path cache meant a recompiled `.osdi` never took effect without restarting ngspice. A `-f/-force` flag (parsed in `com_osdi`, `frontend/com_dl.c`) now force-reloads: `load_osdi(path, force)` stages a byte-for-byte copy of the recompiled file under a fresh unique path and loads *that* (a new path is always re-read, sidestepping the cache; no `dlopen`, which on macOS does not reliably unmap and where overwriting a mapped file faults the process), removes the staged copy once mapped, and `osdi_add_device(..., replace)` swaps the registered device to the new descriptor in place at its existing table index. The old SPICEdev + mapping are left resident

so a circuit still built against the prior model stays valid. Without `-f` the re-load is still skipped (now with a hint pointing at `-f`).

osdi/osdiparam.c (E-93)

Setting a fixed (**localparam**) OSDI parameter from the netlist used to be swallowed silently: `openvaf` exports every `localparam` as a parameter, its “given” flag is forced false, and the written value is overwritten by the parameter-init default — so `.model ... N=8` on a frozen structural width parameter changed nothing, with no feedback. `OSDIparam/OSDImpParam` now test the new `PARAM_FLAG_FIXED` descriptor flag (set by `openvaf`, a free bit of the parameter `flags` field) and emit *“parameter ‘N’ is a fixed (localparam) value and cannot be set from the netlist; ignored”* instead of silently dropping it (E-93).

6. Maths — **src/maths/**

dense/dense.c (11 substantive lines; the file also carries a whitespace/line-ending normalization from editing)

cadjoint() had no 1×1 base case: its cofactor loop allocated 0×0 and then negative-sized minors, killing `ngspice` with `malloc: can't allocate -8 bytes` — the crash that had been hiding behind the “at least two ports” guard in `span.c`. The adjugate of `[a]` is `[1]` (the empty minor’s determinant is 1 by convention), making `cinverse` of a 1×1 equal $1/a$; a $100\ \Omega$ load on a $50\ \Omega$ port now writes a proper `.s1p` with $S_{11} = 1/3$ exactly (E-64).

sparse/spfactor.c (6)

The Markowitz-product overflow guards compare a `double` against `LARGEST_LONG_INTEGER (= LONG_MAX)`, which is not exactly representable as a `double`; the conversion is now explicit — identical semantics, intentional and warning-free (E-77).

KLU/klu_multiply.c (16)

`SMPmultiply`’s KLU path passed `NULL` internal↔external ordering maps to `KLU_matrix_vector_multiply`, which dereferenced them unconditionally (`&NULL[n] = 0x14` for `n = 5`) and segfaulted — so `.option linesearch` crashed under `.option klu`. The line search is the only caller of `SMPmultiply` under KLU, so this path had never been exercised. A `NULL` map now means the identity ordering (`ext = i + 1`, since the KLU CSR is 0-based and the RHS/Solution vectors are 1-based), making the KLU matrix-vector product — and hence the E-111 line-search residual merit — correct under KLU, with a merit sequence numerically identical to Sparse 1.3 (E-112).

KLU/klusmp.c (SMPcaSolve)

`SMPcaSolve` is the complex adjoint (transposed) solve used by noise and S-parameters. Its Sparse branch calls `spSolveTransposed`, but the KLU branch called the *non-transposed* `klu_z_solve` — silently wrong for any asymmetric matrix (every circuit with a transistor or controlled source), which is why `.noise` was disabled under KLU. It now calls `klu_z_tsolve` (transposed), so KLU noise matches Sparse exactly (including OSDI device models). This is the core of E-113, which also removes the KLU refusal guards in `noisean.c / pzan.c` (single-ended pole-zero runs under KLU). Balanced-output pole-zero was Sparse-only at first but E-172 closed that too (a KLU `SMPcAddCol` branch plus a union-pattern reservation in `CKTpzSetup`), so no analysis is Sparse-only under KLU any more.

spicelib/analysis/cktsens.c

Sensitivity builds an auxiliary perturbation matrix `delta_Y` (holding $\partial Y / \partial p$) via `SMPnewMatrix`, which allocates a plain Sparse 1.3 matrix (`delta_Y->CKTkluMODE = 0`, no `SMPkluMatrix`). It is only *multiplied* against the solution (`SMPmultiply` $\rightarrow$ `spMultiply`), never factored, and `DEVbindCSC` always binds into `ckt->CKTmatrix`, not `delta_Y` — so it is correctly Sparse in every case. But two KLU-only setup blocks were gated on the main matrix's `CKTkluMODE`, so under `.option klu` they ran and dereferenced the NULL `delta_Y->SMPkluMatrix` — a segfault on every DC/AC `.sens` deck. The blocks now gate on `delta_Y->CKTkluMODE` (its own flag, 0), keeping `delta_Y` Sparse under KLU exactly as it already is under the default solver, while the main `Y` matrix stays KLU. DC and AC sensitivity now match Sparse exactly (E-114).

spicelib/analysis/distoan.c

Distortion is a complex analysis (Volterra series solved at the harmonic / intermodulation frequencies via `NIdIter` $\rightarrow$ `SMPcSolve`), but `distoan.c` had no KLU code at all: under `.option klu` the KLU matrix stayed in *real* mode, the complex solves ran against an unconverted matrix, and every harmonic came back zero — a silent wrong answer. The fix mirrors `acan.c`: before the solve loop it converts the matrix to complex (`DEVbindCSCComplex` per device + `KLUmatrixIsComplex`), and on exit converts back to real (`DEVbindCSCComplexToReal`) so a later real analysis in the same session is unaffected. KLU distortion now matches Sparse bit-for-bit (single-tone, two-tone intermodulation, and OSDI models) (E-115).

misc/equality.c (AlmostEqualUlp)

`AlmostEqualUlp` is ngspice's ULP-distance float comparison — the primitive behind breakpoint matching, `.measure` triggers, transient truncation, PSS, and source timing (18 files). It reinterprets each double as an `int64_t`, maps the sign, and compared them with `intDiff = labs(aInt - bInt)`. That signed subtraction overflows `int64_t` when the two keys straddle zero, and `labs(INT64_MIN)` is itself undefined — surfaced by a UBSan build. It is not merely theoretical UB: `AlmostEqualUlp(-2.0, +2.0)` returned TRUE (declaring `-2` and `+2` "almost equal"), because those two keys differ by exactly `INT64_MIN`, whose `labs` stays negative and slips under `maxUlp`. The difference is now taken in `uint64_t` as larger-minus-smaller, where the wraparound is well defined and the true magnitude of any two-`int64_t` difference always fits; the comparison against `maxUlp` is unsigned. Behavior is identical wherever the old code did not overflow, and correct (FALSE) where it did. Found by the 2026-07 ASan/UBSan pass.

7. Build system

xspice/icm/makedefs.in (4)

The XSPICE codemodel (`.cm`) link rule hardcoded the pre-macOS-10.3 idiom `-bundle -flat_namespace -undefined suppress`, deprecated loudly by the modern linker on all 14 codemodel links (CI macOS builds included). Now `-bundle -undefined dynamic_lookup` — the modern spelling for plugins whose host symbols resolve at load time (E-77).

Autotools regeneration (the ~100 **Makefile.in** files + **config.h.in**)

Regenerated by `./autogen.sh` with libtool 2.5.4 during the zero-warning work — a build tree whose configure predates libtool 2.4.7 selects the deprecated `darwin -undefined suppress` flag whenever `MACOSX_DEPLOYMENT_TARGET` is unset. No hand-written content (E-77).

8. Per-enhancement index

Every enhancement that touched ngspice, oldest first:

E-1 *osdi.h, osdidefs.h, osdiitf.h, osdiregistry.c, osdiload.c, osdisetup.c, dctrans.c (+accept path)*

`absdelay()`: synthetic delay rows + waveform history

E-6 *osdi.h, osdidefs.h, osdiitf.h, osdiregistry.c, osdiload.c*

`last_crossing()`: history + interpolated crossings

E-8 *parser/ingtok.c*

`.model` card first-parameter drop fix

E-24 *osditrunc.c*

`\$discontinuity` next-step clamp

E-25 *osdi callbacks (get_simparams)*

expose `analysis_name` + simulator `simparams`

E-32 *outitf.c*

integer `opvars` recorded per point

E-42 *osdinoise.c*

coherent same-named noise grouping (LRM 4.6.4)

E-45 *osdi.h (ABI 0.5), osdisetup.c*

Verilog-A nodesets applied to the solver

E-51 *osdi.h (ABI 0.6), osdiacld.c*

full `ac_stim` AC-RHS injection

E-53 *dctrans.c, dcop.c, dctrans.c, acan.c, noisean.c, osdiload.c*

`@(final_step)` + analysis-name phase bits

E-54 *osdi.h (ABI 0.7), osdinoise.c, osdiload.c, osdiacld.c, osdiregistry.c*

node-free complex noise factors, stride-2 pairs

E-55 *osdiaccept.c (new), osdiload.c, osditrunc.c, osdiinit.c, dctrans.c, dctrans.c, osdiitf.h, Makefile.am*

`\$finish/\$stop/\$fatal` honored; `\$discontinuity` step rejection

E-56 *noisean.c, osdisetup.c*

noise SIGABRT fix; setup-rejection diagnostics

E-62 *dctrans.c, trcvdefs.h, cktdisto.c*

generic `.dc @inst[param]` sweeps; `.disto` warning

E-63 *span.c*

`donoise` NaN clamp (stock defect, found by parity)

E-64 *span.c, cktspdum.c, dense.c, postcoms.c/.h, commands.c*

Touchstone export, auto-Rbase, 1-port `.sp`, `cadjoint 1x1`

E-72 *postcoms.c/.h, commands.c*

Touchstone round 2: MA/DB/Y/Z/units + rdsnp reader

E-76 *dev.c, inpgmod.c*

duplicate-registration warning, load dedup, stock `.model` segfault fix

E-77 *osdidefs.h, display.c, outitf.c, spfactor.c, makedefs.in* (+autotools regen)

zero-warning build, 33 → 0

E-80 *osdiinit.c*

dtemp instance-parameter alias

E-81 *resource.c, dev.c*

memory-warning opt-out + once-per-excursion; reload-note hint

E-93 *osdi.h, osdiparam.c*

warn (not silently ignore) when a netlist sets a `PARA_FLAG_FIXED` (localparam) OSDI parameter

E-94 *plotting/pyplot.c (new), com_pyplot.c (new), plotit.c, commands.c*

new `pyplot` command — plot vectors with matplotlib (a Python `gnuplot`)

E-95 *com_pyplot.c, commands.c*

make the `pyplot` output file name optional (defaults to `pyplot`)

E-98 *plotting/pyplot.c*

`pyplot` stacked subplots (`pyplot_subplots=N`) + matplotlib style sheets (`pyplot_style`)

E-99 *plotting/pyplot.c*

`pyplot` vector export formats (`pyplot_terminal=svg/pdf`) + figure size (`pyplot_figsize`)

E-110 *optdefs.h, tsdefs.h, cktsopt.c, cktntask.c*

`.option errpreset=conservative|moderate|liberal` — one knob for a coordinated tolerance/robustness set; explicit options override regardless of order (`moderate` = historical defaults)

E-111 *optdefs.h, tsdefs.h, cktddefs.h, cktsopt.c, cktdjob.c, cktntask.c, cktdtest.c, niiter.c*

`.option linesearch` — globalized (damped) Newton via Armijo backtracking on a new KCL-residual merit $\|F\| = \|G \cdot x - b\|$; result-neutral, off by default

E-112 *maths/KLU/klu_multiply.c*

KLU support for `.option linesearch` — `SMPmultiply`'s KLU path passed NULL ordering maps that `klu_matrix_vector_multiply` dereferenced (SIGSEGV); NULL now means identity ordering. Line search verified merit-identical KLU vs Sparse

E-113 *maths/KLU/klusmp.c, spicelib/analysis/noisean.c, pzan.c*

KLU support for noise + single-ended pole-zero — `SMPcaSolve`'s adjoint KLU branch used the non-transposed solve (silently wrong noise on asymmetric matrices); now `klu_z_solve`. Guards removed; balanced-output `pz` keeps a targeted guard

E-114 *spicelib/analysis/cktsens.c*

KLU support for sensitivity — the auxiliary perturbation matrix `delta_Y` is Sparse, but two KLU setup blocks gated on the *main* matrix's flag dereferenced its NULL `SMPkluMatrix` (segfault on every DC/AC `.sens`); now gated on `delta_Y`'s own flag. DC/AC `.sens` match Sparse exactly

E-115 *spicelib/analysis/distoan.c*

KLU support for distortion — `.disto` is a complex analysis but `distoan.c` had no KLU code, so under KLU the matrix stayed real and every harmonic came back zero (silent wrong answer); now converts $\text{real} \leftrightarrow \text{complex}$ around the solve loop like `acan.c`. Matches Sparse bit-for-bit

E-116 *osdi/osdisetup.c*

KLU wrong-DC fix — an OSDI internal node with no Jacobian entry (a decoupled ground reference) was given an all-zero solver row that made the KLU matrix structurally singular; now tied to ground. Fixes the `groundcontrib` and `hierbranch` KLU_XFAILs

E-117 *configure.ac (+configure)*, *spicelib/analysis/dcpss.c*

Periodic steady state (PSS) productionized — built by default (was experimental `--enable-pss`, so `.pss` was unimplemented in shipped builds); ~230 lines of shooting-loop trace routed through `set ngdebug` (232 $\rightarrow$ 31 lines by default); fail-fast KLU guard (PSS hangs under KLU) directing `.pss` to `.option sparse`

E-118 *spicelib/analysis/dcpss.c*

PSS runs under KLU — the shooting loop hung under KLU (`klu_refactor`'s reused pivots inflated the truncation error into a ~20M-step timestep explosion); forcing a full re-factor (`NISHOULDREORDER`) each PSS step makes KLU converge to the same result as Sparse. E-117's Sparse-only guard removed

E-119 *pssdefs.h*, *spicelib/analysis/dcpss.c*

Retain the PSS periodic operating point — PSS sampled the node voltages per period for its DFT then freed them, and never captured device states; now the voltages and `CKTstate0` states are captured per sample and retained on the PSSan job (with frequency + dims) as the substrate for the periodic small-signal suite (PAC/pnoise/PXF). Self-checks that the retained samples reproduce the fundamental

E-120 *spicelib/analysis/dcpss.c*

Periodic small-signal Jacobian harmonics — walk the retained operating point, re-linearize each sample (`CKTload MODEINITSMSIG`) and stamp $G+jC$ (`CKTacLoad  $\omega=1$`), read the osc-node diagonal $\rightarrow g(t), c(t)$, DFT to harmonics. The blocks the PAC conversion matrix is built from. (Fixed a complex-mode-not-set bug where $C(t)$ accumulated across samples.) Verified: RC gives $G=1/R1, C=C1$, no harmonics

E-121 *spicelib/analysis/dcpss.c*

PAC conversion-matrix engine — extend E-120 from the osc diagonal to every matrix nonzero, complex-DFT to harmonics G_k, C_k , assemble the $(2M+1)N$ harmonic conversion matrix $H_{\{nm\}}=G_{\{n-m\}}+j\omega_m \cdot C_{\{n-m\}}$, inject a unit current at the osc node in sideband 0 and solve by dense complex LU (`pss_csolve`), report the sideband conversion gains. The numerical heart of PAC/pnoise/PXF. Verified: linear RC gives sideband-0 = AC driving-point $|Z|(f0/2) = 303.3 \Omega$ with the ± 1 sidebands at floating-point zero (no spurious conversion)

E-122 *include/ngspice/pssdefs.h*, *spicelib/analysis/psssetp.c*, *spicelib/parser/inp2dot.c*, *spicelib/analysis/dcpss.c*

`.pac` command (periodic AC) — the user-facing analysis on the E-121 engine. `.pac Fguess StabTime OscNode Points Harmonics SC_iter Steady_coeff <dec|oct|lin> Npts Fstart Fstop` runs PSS then sweeps the input frequency, solving the conversion matrix at each point and emitting the 0-th-sideband node responses as a complex PAC Analysis plot (`print/plot/wrdata`). Reuses the PSS analysis via a `PSSdoPAC` flag; engine factored into `pac_extract_harmonics` (once)

+ pac_solve_at (per freq). Verified: swept sideband-0 $|b(f)| == \text{analytic AC driving-point } |Z(f)|$ to $1.6e-7$ across 10 kHz–1 MHz

E-123 *include/ngspice/pssdefs.h, spicelib/analysis/psssetp.c, spicelib/parser/inp2dot.c, spicelib/analysis/dcpss.c*

finish .pac — (1) source-referenced stimulus: capture the netlist AC-source RHS from CKTacLoad as the sideband-0 stimulus B_0 (a periodic-AC transfer/conversion gain), unit-current-at-osc-node fallback; (2) multi-sideband output: optional trailing maxsideband Ksb emits every conversion sideband $f_{in} + k \cdot f_0$ ($k = -Ksb \dots Ksb$) as its own vector — sideband 0 keeps the node name, others $\langle \text{node} \rangle_usb\langle k \rangle / \langle \text{node} \rangle_lsb\langle k \rangle$ via IFnewUId. Verified: RC with V1 AC 1 gives sideband-0 = low-pass transfer $1/\sqrt{1 + (2\pi fRC)^2}$ ($0.998 \rightarrow 0.157$) with b_usb1/b_lsb1 ~0 (no conversion)

E-124 *include/ngspice/pssdefs.h, spicelib/analysis/psssetp.c, spicelib/parser/inp2dot.c, spicelib/analysis/dcpss.c*

.pnoise command (periodic noise) — fold each device's noise through the conversion matrix. pac_solve_adjoint solves $H^T \Psi = e_{\{out, 0\}}$; pnoise_sweep loads the sideband-k adjoint into CKTrhs/CKTirhs and calls the existing device noise routines (NevalSrc, OSDI load_noise) once per sideband, accumulating $\Sigma_k S \cdot |\Delta \Psi_k|^2$ — reuses every device noise model via a minimal local NOISEAN context. .pnoise <pss> OutNode InSrc <dec|oct|lin> Np Fstart Fstop; emits onoise_spectrum/inoise_spectrum. Verified: linear RC reduces exactly to .noise ($4kTR/(1 + (2\pi fRC)^2)$), matches the .noise reference to every digit)

E-125 *include/ngspice/pssdefs.h, spicelib/analysis/psssetp.c, spicelib/parser/inp2dot.c, spicelib/analysis/dcpss.c*

.pxf command (periodic transfer function) — the adjoint of PAC, completing the PSS $\rightarrow$ PAC $\rightarrow$ Pnoise $\rightarrow$ PXF suite. pxf_sweep solves $H^T \Psi = e_{\{out, 0\}}$ per frequency and dots each sideband block with the AC-source pattern $B_0 \rightarrow x_{f_k} = \Sigma_j \Psi_k(j) \cdot B_0(j)$, the input $\rightarrow$ output transfer at each sideband. .pxf <pss> OutNode <dec|oct|lin> Np Fstart Fstop [maxsb]; emits xf + xf_usb<k>/xf_lsb<k>. Verified: by the identity $(H^{-1}B)_{out} = (H^{-T}e_{out})^T B$ the sideband-0 transfer is bit-identical to the PAC response ($0.998 \rightarrow 0.157$ low-pass), conversion sidebands $\sim 2e-16$

E-126 *include/ngspice/pssdefs.h, spicelib/analysis/psssetp.c, spicelib/parser/inp2dot.c, spicelib/analysis/dcpss.c*

cyclostationary noise — .pnoise ... cyclo evaluates each device's noise PSD $S(t)$ at every PSS sample's bias (CKTload per sample) and folds it through the time-domain adjoint transfer $A_s(j) = \Sigma_k \Psi_k(j) e^{j2\pi k s/P}$, averaging: $onoise(f) = (1/P) \Sigma_s S(t_s) \cdot |\Delta A_s|^2$. Reuses the device noise routines. Verified: (1) reduces exactly to .noise on the linear RC (bias-independent thermal $\rightarrow$ Parseval); (2) a flicker resistor carrying the RC current gives $onoise \cdot f = R I^2 \cdot KF \cdot \langle I^2 \rangle = 4.88e-10$ (uses the period-average $\langle I^2 \rangle$, matched to 5 digits)

E-127 *include/ngspice/{optdefs,tskdefs,cktdefs}.h, spicelib/analysis/{cktsopt,cktnntask,cktdojob,cktop,cktdest}.c, maths/ni/niiter.c*

pseudo-transient continuation — .option ptcont embeds $f(x) = 0$ in a backward-Euler pseudo-transient $f(x) + Gps \cdot (x - x_{prev}) = 0$ and marches $d\tau$ small $\rightarrow$ large ($Gps \rightarrow 0$). Gps diagonal via the gmin path; the $Gps \cdot x_{prev}$ RHS coupling (added in NIiter) makes each step a stable-trajectory move, not a static gmin step. New pseudo_transient in the CKTop cascade, off by default. Verified under KLU+Sparse: result-neutral on normal circuits; on a stiff no-limiting exp, reaches the correct DC 0.837922 V where plain Newton lands on a spurious 70.5 V (21/21)

E-128 *include/ngspice/{optdefs,tskdefs,cktdefs}.h, spicelib/analysis/{cktsopt,cktnntask,cktdojob}.c, spicelib/analysis/dctran.c*

LTE-based dynamic integration-order control — .option dynorder selects the Gear order per step from the local-truncation-error limit instead of the stock $1 \leftrightarrow 2$ toggle, so orders 3–6 (already

coded in NIcomCof but never used) are actually exercised. The selector compares the raw LTE-limited step (uncapped CKTtrunc) at the current order and its ± 1 neighbours and moves with hysteresis ($1.2\times$), a settling hold after each change (let the BDF divided differences rebuild), and an order-dependent step-growth cap ($2\times$ at order $\leq 3 \rightarrow 1.3\times$ at 6). Off by default; bounded by maxord; inert at the default maxord=2. Stiff-transient guards (post-breakpoint order hold + leaky-bucket rejection-rate order drop) keep it from collapsing the step on a violent slew. Verified under KLU+Sparse: RC decay $3\text{--}5\times$ fewer steps at matched accuracy with monotone error; smooth RLC ringdown $8.9\times$ fewer steps AND more accurate (0.13 % vs 0.34 %); nonlinear rectifier matches stock to 5 sig figs; the transistor-level $\mu\text{A}741 \pm 5\text{ V}$ slew completes under dynorder and matches fixed Gear-2

E-129 *frontend/oututf.c*

sweep progress bar — the throttled Reference value status line (redrawn in place every 0.25 s during a sweep) now carries a live progress bar + percentage, e.g. Reference value : 5.91926e-04 [=====] 74%. outp_progress_frac() computes the 0-1 fraction per analysis: transient from $(\text{CKTtime}-\text{TSTART})/(\text{TSTOP}-\text{TSTART})$, AC/noise from the frequency's linear/log position in the band, DC from the accepted-point count over the nested step-count product; analyses with no span (op, ...) keep the plain line. Fixed-width (no stale chars), std-out status-line only (never in the rawfile/wrdata), same !ft_norefprint && !cp_background gating. Verified 22/22: printed % matches the analytic sweep fraction to 0.5 % across tran/AC/DC/noise; op shows no bar

E-130 *frontend/com_optimize.{c,h}, commands.c, options.c, include/ngspice/fteext.h, spicelib/analysis/cktdobj.c, frontend/oututf.c*

built-in Nelder-Mead optimizer — optimize -param <name> <init> <lo> <hi> [-param ...] -analysis <cmd> -minimize <expr> [-maxiter N] [-tol T] [-verbose] varies device/alter parameters, re-runs an analysis, and minimizes a scalar objective expression via a derivative-free downhill simplex in normalized [0,1] space (scale-invariant across orders-of-magnitude params). Sub-commands dispatched synchronously through cp_coms (not the deferring re-entrant cp_evloop); the hundreds of inner analyses are silenced via a new ft_optimizing flag gating the banner/row-count/E-129 progress bar at source (-verbose to show). Verified 9/9 against analytic optima: DC divider $R1 \rightarrow 2333.3\ \Omega$, AC low-pass $R1 \rightarrow 2756.6\ \Omega$, 2-D compound objective $R1=3k/R2=2k$ exactly

E-131 *frontend/com_checkpoint.{c,h}, commands.c, com_commands.h, Makefile.am, spicelib/analysis/dctran.c, include/ngspice/cktdefs.h*

transient checkpoint / restart — new savestate <file> / loadstate <file> commands serialize the full transient integration state (solution vector CKTrhsOld, device state history CKTstates[], time/step/order/mode, pending breakpoints) to a binary file and resume it, including in a fresh process — so a long run survives a crash, splits across sessions, or moves between machines (stock ngspice could only resume in memory). Dctran() gains a CKTcheckpoint-gated branch that opens a fresh output plot (no live plot to 666-relink across a reload), initializes the XSPICE breakpoint markers, and fixes up the breakpoint list before jumping into the stepping loop; the rhs length is keyed off SMPmatSize+1 (not CKTmaxEqNum). A stored signature rejects a mismatched circuit; Sparse-solver only (KLU rejected with a clear message). Verified 19/19: resumed waveform bit-identical to an uninterrupted run for RC-step/pulse/diode built-ins and $\sim 2e-7$ for a compiled OSDI diode, across a separate process

E-132 *spicelib/analysis/dcpss.c, spicelib/parser/inp2dot.c, spicelib/analysis/psssetp.c, include/ngspice/pssdefs.h*

periodic S-parameters (.psp) — small-signal scattering parameters around a PSS operating point, including conversion between the input frequency and its sidebands $f_{in}+k \cdot f_0$ (mixers / switched circuits, where a static-DC .sp cannot see the conversion). Sits on the PSS $\rightarrow$ conversion-

matrix suite (E-117–126): after PSS, psp_sweep excites each RF port (portnum/z0, the .sp framework) by driving its branch source (V=1, like .sp's VSRCspupdate) through the shared (2M+1)N conversion matrix, reads per-sideband port waves in the same Kurosawa power-wave convention, and forms $S^{(k)} = B^{(k)} \cdot A^{-1}$ (dense-complex cinverse/cmultiply). pac_solve_at's matrix assembly factored into a reusable pac_build_matrix; PSSdoPSP flag + psp PSS param + dot_psp card. Because $S = B \cdot A^{-1}$ is excitation-basis-invariant, sideband 0 reduces exactly to .sp for a time-invariant network. Runs under both linear solvers (the conversion matrix is a standalone dense LU; PSS runs under both since E-118). Verified 8/8: sideband-0 matches .sp to $\sim 1e-16$ for 1/2/3-port resistive + reactive networks (magnitude and phase) incl. OSDI Verilog-A devices (G + reactive ddt stamps), conversion sidebands correctly ~ 0

E-133 *frontend/com_qpss.{c,h}, commands.c, com_commands.h, Makefile.am*

quasi-periodic (two-tone) steady state (qpss) — qpss <expr> <f1> <f2> [periods] [max-order] computes the two-tone steady-state spectrum: every mixing product $k_1 \cdot f_1 + k_2 \cdot f_2$ including third-order intermodulation (IM3 at $2f_1 - f_2$ / $2f_2 - f_1$) that single-tone AC/PSS cannot show. For commensurate tones (common beat $fb = \gcd(f_1, f_2)$) it runs an ordinary transient over a few beat periods to reach steady state, then evaluates the Fourier coefficient directly at each exact intermod frequency (a direct DFT, exact — no FFT-bin rounding) and labels it by the 2-D index (k1, k2). Deliberately not a slow beat-frequency shooting PSS; a front-end command driving a transient, so solver-independent and works with built-in and OSDI devices. Verified 11/11: analytic fundamentals/IM3/3f of a cubic, no even-order products, the 3:1 IP3 slope law (fund $\times 2$, IM3 $\times 8$ per $2\times$ drive), beat-frequency derivation, and an OSDI cubic matching the built-in

E-134 *spicelib/analysis/dcpss.c, frontend/com_hb.{c,h}, commands.c, com_commands.h, Makefile.am, include/ngspice/cktdefs.h*

Harmonic Balance (hb <f0> <K>) — solves the periodic steady state in the frequency domain by Newton (each node a truncated Fourier series), instead of integrating in time; the real analysis behind ngspice's unimplemented WITH_HB stub. Residual $F_k = I_{R,k}(V) + [dq/dt]_k - I_{s,k} = 0$ with the E-121 (2K+1)N conversion matrix as the exact Jacobian. Per iteration: inverse-DFT $V \rightarrow v(t_s)$; drive DC+AC device loads at those voltages for the resistive current $I_R = G \cdot v - b$ (companion b saved before acLoad, so it's the ACTUAL current not the tangent $G \cdot v$), and $G(t)/C(t)$; DFT; dense complex Newton (pss_solve). Nonlinear reactive with NO charge extraction: $dq/dt = C(v) \cdot v'$, so the reactive current is the conversion matrix's jwC term applied to V. Solver-independent (KLU + Sparse): the dense complex Newton is HB's own; the sparse solver only reads $G(t)/C(t)$ off the device matrix, so hb_extract carries the same #ifdef KLU complex-CSC binding (DEVbindCSCComplex) as the PAC extraction, and com_hb copies the task's KLU mode before building so a bare hb honours .option klu — verified bit-identical under both. Built-in + OSDI. Verified 8/8 vs transient/fourier, quadratic convergence: nonlinear R (analytic 3rd harmonic), R+C, a real diode rectifier (junction limiting), OSDI varactor Q(v) 2nd harmonic, KLU==Sparse parity

E-135 *spicelib/analysis/dcpss.c, examples/hb_examples/verify_hb.py*

HB source-stepping continuation — makes E-134 Harmonic Balance robust on strongly-driven circuits where a cold full-strength Newton diverges ($|F| \rightarrow 1e69$). HBanalyze's Newton loop is wrapped in an adaptive homotopy: every independent source scaled by λ : $0 \rightarrow 1$, each level warm-started from the last converged V. The residual becomes $F = I_R + I_C - \lambda \cdot I_s$; on level success V is checkpointed and $d\lambda$ grows ($\times 1.7$), on failure (Newton exhausted, residual non-finite, or singular Jacobian) V is restored and $d\lambda$ halves; $d\lambda < 1e-5 \rightarrow$ reported as no reachable steady state. First level is full strength ($d\lambda=1$) so easy circuits converge at $\lambda=1$ immediately, bit-identical to the plain solve. Automatic (no new syntax); set hb_verbose shows the λ ramp, summary reports iterations + continuation steps. Verified 9/9: a strongly-driven diode rectifier (5 V into 20 Ω , sharp $I_S=1e-14$) diverges cold but converges in 3 continuation steps, matching transient fourier to $<0.1\%$ for DC/f0/2f0; the 8 existing HB checks stay bit-identical

E-136 *spicelib/analysis/dcpss.c, frontend/com_qpss.c, frontend/commands.c, include/ngspice/cktdefs.h, examples/qpss_examples/verify_qpss_hb.py*

two-tone Harmonic Balance (qpss <expr> <f1> <f2> hb [K1] [K2]) — the TRUE, incommensurate-capable quasi-periodic steady state, a frequency-domain HB engine alongside the E-133 transient qpss (which is unchanged, commensurate-only). Each node is a 2-D Fourier series $v(t) = \sum \sum V_{\{k1,k2\}} e^{j(k1\omega_1 + k2\omega_2)t}$; QPSShb samples devices on a 2-D phase grid (θ_1, θ_2) — time never appears, so incommensurate tones (no beat period) just work — and 2-D DFTs to the conversion matrix $H_{\{n\},\{m\}} = G_{\{n-m\}} + j\omega_m \cdot C_{\{n-m\}}$ (qp_build_matrix), Newton-solved by pss_csolve with the E-135 source stepping. Sources captured by an oversampled least-squares APFT ($\Gamma^H \Gamma$) $I_s = \Gamma^H b$ (a square Vandermonde is catastrophically ill-conditioned past a few harmonics). Reactive needs NO charge extraction ($dq/dt = C(v)v'$); the converged operating point is retained (qpss_hb_saved) for qpac (E-137). Solver-independent (KLU + Sparse) — same `#ifdef KLU` complex-CSC binding as PAC/HB. Built-in + OSDI. Verified 7/7: analytic cubic $|IM3(2, -1)| / |3rd(3, 0)| = 3$, even products ~ 0 , 3:1 IP3 slope, incommensurate $\sqrt{2}$ tones (E-133 cannot), HB==transient (commensurate), KLU==Sparse; E-133 verify_qpss.py stays 11/11

E-137 *spicelib/analysis/dcpss.c, frontend/com_qpac.{c,h}, commands.c, com_commands.h, Makefile.am, include/ngspice/cktdefs.h, examples/qpss_examples/verify_qpac.py*

two-tone small-signal QPAC (qpac <f_in>) — completes the quasi-periodic gap: the two-tone analogue of PAC. Run after qpss ... hb, QPACanalyze injects a small signal at f_in around the retained QPSS operating point (qpss_hb_saved) and reports the response at every sideband $f_{in} + k_1 \cdot f_1 + k_2 \cdot f_2$, mixing it through the SAME 2-D conversion matrix $H_{\{n\},\{m\}} = G_{\{n-m\}} + j\omega_m \cdot C_{\{n-m\}}$ (qp_build_matrix at f_in) that the QPSS Newton used as its Jacobian. Stimulus placed in the $(0, 0)$ sideband — a netlist AC-source RHS B0 (captured bias-independent at the op-point) or a unit-current fallback — one dense pss_csolve. Adds no device evaluation → solver-independent (KLU + Sparse) by construction. Verified 7/7: reduce-to-AC (pump→0 $\sqrt{2}$ direct $(0, 0) = \text{plain .ac response} = R$, sidebands vanish), v^2 -pump conversion ratio $| (1, 1) | / | (2, 0) | = 2$, equal-tone symmetry, clean no-op-point error, KLU==Sparse; E-136 (7/7) + E-133 (11/11) unaffected

E-138 *spicelib/analysis/dcpss.c, frontend/com_qpnoise.{c,h}, commands.c, com_commands.h, Makefile.am, include/ngspice/cktdefs.h, examples/qpss_examples/verify_qpnoise.py*

two-tone small-signal QPnoise (qpnoise <output_node> <f_in>) — quasi-periodic noise, the two-tone analogue of pnoise (E-124), on the retained qpss ... hb operating point. Reports output + input-referred noise density at f_in, folding every device's noise over all sidebands $f_{in} + k_1 \cdot f_1 + k_2 \cdot f_2$ (mixer/PA noise conversion invisible to a static .noise). QPnoiseAnalyze biases the devices at the QPSS operating point, then qp_solve_adjoint transposes the 2-D conversion matrix (qp_build_matrix) and solves $H^T \Psi = e_{\{out, (0, 0)\}}$ — one adjoint gives the transimpedance from every (node, sideband) to the output; each device's DEVnoise computes $S \cdot |\Psi|^2$ (reading the transfer from CKTrhs/CKTirhs) and the sum over all N_h harmonics is the folded onoise; input-referred via the QPAC gain. Reads only retained data → solver-independent (KLU + Sparse) by construction. With no pump the conversion matrix is block-diagonal so only $(0, 0)$ survives $\sqrt{2}$ reduces to .noise. Verified 6/6: reduce-to-noise (pump→0 $\sqrt{2}$ onoise == plain .noise == $4kTR$ exactly), conversion active under pump, inoise=onoise/gain², clean no-op-point error, KLU==Sparse; E-133 (11/11) + E-136 (7/7) + E-137 (7/7) unaffected

E-139 *spicelib/analysis/dcpss.c, frontend/com_qpnoise.c, frontend/commands.c, include/ngspice/cktdefs.h, examples/qpss_examples/verify_qpnoise.py*

cyclostationary QPnoise (qpnoise <output_node> <f_in> cyclo) — upgrades E-138 to a time-varying device PSD. Under a two-tone pump a device's bias swings over the period (a diode's shot noise $2qI_D(t)$ spikes when it conducts), so QPnoiseAnalyze gains a cyclo branch using the

identity $\text{onoise} = (1/P) \cdot \sum_s S(t_s) \cdot |A_s|^2$, where $A_s(j) = \sum_{\{k1,k2\}} \Psi_{\{k1,k2\}}(j) \cdot e^{j2\pi(k1 \cdot s1/P1 + k2 \cdot s2/P2)}$ is the inverse 2-D DFT of the adjoint transfers (the time-domain transimpedance at phase sample s). Each sample re-biases the devices at the retained op-point (qp_synth , $P1/P2$ now stored in qp_harm) with per-sample junction settling (the E-134 `MODEINITFLOAT` walk — else a limited diode reports a stale bias and the "cyclostationary" result collapses onto the stationary one), evaluates $S(t_s)$, folds through A_s , and averages. By Parseval reduces to the stationary sum (and .noise) when S is constant. Verified 10/10 (6 E-138 + 4 cyclo): cyclo reduce-to-noise, Parseval (bias-independent thermal PSD $\square$ cyclo==stationary even under pump), a hard-pumped diode where cyclo differs from stationary by $\sim 8\times$ (switching-mixer cyclostationary enhancement, visible only with the settling), cyclo $\text{KLU}==\text{Sparse}$; E-133/136/137 unaffected

E-140 *spicelib/analysis/dcpss.c, frontend/com_hbosc.{c,h}, commands.c, com_commands.h, Makefile.am, include/ngspice/cktdefs.h, examples/phasenoise_examples/verify_phasenoise.py*

oscillator phase noise (`hbosc <oscnode> <K> [fguess] [tstab] + phasenoise <fstart> <fstop> [pts]`) — closes the periodic/phase-noise gap with the phase-noise piece. `HBOSC-analyze` is an autonomous harmonic balance: an oscillator has no source, so $F(V)=I_R+[dq/dt]=0$ is solved for the harmonics AND the unknown oscillation frequency ω_0 . The conversion matrix $dF/dV=H$ is singular (right null space = phase mode $u_k=j_k \quad v_k$, since the oscillator's phase is free), so Newton runs on the bordered system $[H, \partial F/\partial \omega_0; u^T, 0]$ (nonsingular by bordering, $\partial F/\partial \omega_0=I_C/\omega_0$, gauge row u^*), transient-seeded (`hbosc` runs a short transient from the deck's `.ic`, reads amplitude + zero-crossing frequency), converging quadratically (LC osc: 4 iters to $|F|=3e-12$; inductors fit the $G+j\omega C$ matrix via branch-current MNA). `PhaseNoiseAnalyze` builds the adjoint of H at `OFFSET df` with the unit at the carrier sideband ($m=1$) and folds device noise (`DEVnoise`); as $df \rightarrow 0$ the limit-cycle matrix goes singular through the phase mode so the transimpedance diverges as $1/df \rightarrow$ folded noise $1/df^2$, normalized to carrier power $2|V1|^2 \rightarrow L(df)$ in dBc/Hz . Verified 8/8 on an LC oscillator: autonomous HB converges to f_0 +describing-function amplitude, $L(df)$ has the -20 dB/dec skirt near carrier flattening into the noise floor at a physical level ($\approx -147 \text{ dBc/Hz}$ @ 1kHz), thermal scaling $L \propto T$ ($2 \times T \rightarrow +3 \text{ dB}$ exactly, pinning the absolute), clean no-op-point error, $\text{KLU}==\text{Sparse}$; `pnoise` (9/9) + `qpnoise` (10/10) unaffected

E-141 *spicelib/analysis/dcpss.c, frontend/com_qpxf.{c,h}, commands.c, com_commands.h, Makefile.am, include/ngspice/cktdefs.h, examples/qps_examples/verify_qpxf.py*

two-tone small-signal QPXF (`qpxf <output_node> <f_in>`) — the quasi-periodic transfer function, the ADJOINT of QPAC (E-137), completing the two-tone small-signal suite ($\text{QPSS} \rightarrow \text{QPAC} \rightarrow \text{QPnoise} \rightarrow \text{QPXF} \equiv \text{PSS} \rightarrow \text{PAC} \rightarrow \text{pnoise} \rightarrow \text{PXF}$). `QPXFanalyze` runs one adjoint solve $H^T \Psi = e_{\{\text{out}, (0,0)\}}$ (`qp_solve_adjoint`, reused from `QPnoise` E-138) and dots each sideband block of Ψ with the netlist AC-source pattern `B0` (the QPAC stimulus, captured at the op-point) $\rightarrow$ the transfer $H_{\{(k1,k2)\}} = \sum_j \Psi_{\{(k1,k2),j\}} \cdot B0_j$ from an input at every sideband $f_{\text{in}}+k1 \cdot f1+k2 \cdot f2$ to the output, all from ONE adjoint. By the reciprocity identity ($H^{-1}B_{\text{out}}=(H^T e_{\text{out}})^T B$) the sideband- $(0,0)$ transfer is bit-identical to the QPAC response (the $\text{PXF} \leftrightarrow \text{PAC}$ cross-check, E-125). Reads only retained data $\rightarrow$ solver-independent ($\text{KLU} + \text{Sparse}$). Verified 6/6: reciprocity ($\text{QPXF}(0,0) == \text{QPAC}(0,0)$ bit-identical), conversion-sideband reciprocity, reduce-to-XF (pump $\rightarrow 0 \square (0,0) = \text{plain transfer} = R$, sidebands vanish), clean no-op-point error, $\text{KLU}==\text{Sparse}$; `QPSS` 11/11 + `QPSS-HB` 7/7 + `QPAC` 7/7 + `QPnoise` 10/10 unaffected

E-142 *spicelib/analysis/dcpss.c, frontend/com_qpac.{c,h}, com_qpnoise.c, com_qpxf.c, commands.c, include/ngspice/cktdefs.h, examples/qps_examples/verify_qps_sweep.py*

input-frequency sweep for the two-tone small-signal analyses. `qpac/qpnoise/qpxf` gain a `<dec|oct|lin> N fstart fstop` sweep of f_{in} that emits a plottable ngspice plot (conversion gain / noise figure / image-rejection curves vs frequency), matching how `.ac/.pnoise/.pxf` sweep. `QPACsweep/QPnoiseSweep/QPXFswep` step f_{in} and reuse the exact single-frequency solve per point: `qpac` records per-node $|(0,0)$ response|, `qpnoise` records `onoise_`

spectrum/inoise_spectrum, qpxf records xf (in-band (0,0)) + xf_conv (total conversion $\sqrt{\Sigma |H_{sb \neq 0}|^2}$). The analysis fills plain arrays; the front-end builds the plot with a frequency scale via the nutmeg vector API (plot_alloc/dvec_alloc/vec_new) — the correct layer for a front-end command (the analysis OUTp framework dereferences a live analysis job's JOBtype, which a front-end command lacks → was crashing). Shared helpers qp_steptype/qp_sweep_maxpts/qp_emit_plot. Verified 5/5: swept value at 0.3G == single-frequency qpac/qpxf/qpnoise (machine precision), reactive roll-off vs f_in, correct point count; single-frequency QPAC 7/7 + QPnoise 10/10 + QPXF 6/6 unaffected

E-143 *frontend/com_optimize.c, frontend/commands.c, examples/optimize_examples/verify_optimize.py, examples/optimize_examples/optimize_lsq_demo.cir*

gradient least-squares curve fitting for **optimize**. The built-in optimizer (E-130, derivative-free Nelder-Mead on a scalar -minimize) gains a least-squares mode: one or more weighted -target <expr> <value> [<weight>] measurements, optionally spread over several -analysis stages (each -analysis opens a stage; following -targets attach to it, so a single fit can combine e.g. a DC operating point AND an AC response), are fitted with Levenberg-Marquardt — a forward/backward finite-difference Jacobian J, normal equations $A=J^T J$, $g=J^T r$, damped solve $(A+\lambda \cdot \text{diag}(A)) \delta = -g$ via a small partial-pivot Gauss solver, standard λ up/down loop. Exploiting the sum-of-squares structure it converges in far fewer analysis runs than the simplex (27 vs 67 evals on a two-target RC fit). -method nm|lm overrides the auto choice (LS→lm, scalar→nm); the scalar -minimize/Nelder-Mead path is unchanged; -minimize and -target are mutually exclusive. Parameters are still applied in place with alter (no re-source) and the search runs in normalized [0,1] space. All changes in com_optimize.c (+ one help string); no new files, no ABI change. Verified 23/23 (E-130 checks [1]–[5] + E-143 [6]–[10]): 1-param/3-target/3-stage RC magnitude fit, 2-param DC+AC multi-analysis fit ($R1=3k, R2=2k$), LM-fewer-evals-than-NM, OSDI/Verilog-A diode extraction (recover both $i_s \rightarrow 1e-14$ and $n \rightarrow 1.2$ from two measured I-V points by weighted least squares), and input validation (lm-without-target, minimize+target, multi-analysis+scalar all rejected)

E-144 *frontend/com_optimize.c, frontend/inp.c, frontend/commands.c, examples/optimize_examples/verify_optimize.py, examples/optimize_examples/optimize_dparam_demo.cir*

optimize tunes symbolic .param values via a new knob kind -dparam. -param remains an alter target (device/instance, changed in place); -dparam names a netlist .param symbol (e.g. .param w=1u, $R1=\{500*k\}$) which — being expanded at parse time — is changed with alterparam <name>=<value> + a reset that re-sources the deck (re-evaluating .param expressions and re-stamping device values). Per parameter a kind (OPT_ALTER / OPT_DECKPARAM) is tracked; when any -dparam is present opt_eval applies ALL deck params first, re-sources ONCE, THEN applies the in-place alter params (reset rebuilds from the deck and would otherwise wipe an earlier alter — this ordering is what lets the two kinds mix; verified on a joint .param+device fit). No -dparam the E-130/-143 fast path is untouched (no reset, no perf hit). The two re-source banners (Reset re-loads circuit ... and Circuit: ... in inp.c) are gated by the existing ft_optimizing flag so hundreds of inner re-sources are silent (only the final leave-at-optimum run shows); ft_optimizing re-asserted after each reset. Also fixed 5 pre-existing -Wsign-conversion warnings in inp.c's *ng_script_with_params handler (argc/size unsigned). Closes the last E-143 follow-up. No new files, no ABI change. Verified 31/31 (E-130/-143 [1]–[10] + E-144 [11]–[15]): scalar .param fit ($r_{top} \rightarrow 2333.3$), mixed -dparam+-param joint fit ($r_{top}=3k, R2=2k$), .param in expression ($k \rightarrow 6$), least-squares -dparam (LM), quiet re-source (≤ 1 banner); AC + OSDI-device-value .param verified manually

E-145 *frontend/com_optimize.c, frontend/commands.c, examples/optimize_examples/verify_optimize.py, examples/optimize_examples/optresm.va, examples/optimize_examples/optimize_mparam_demo.cir*

optimize tunes .model-card parameters via a third knob kind -mparam. -param remains an alter

ter instance target and `-dparam` a symbolic `.param` (re-source); `-mparam <@model[param]>` names a `.model-card` parameter (e.g. `@dmod[is]`, `@rmod[r]`) — which is NOT alter-reachable and NOT `.dc`-sweepable (only sources/resistors/instance params are) — and is changed in place with `altermod <name>=<value>`, no re-source (as cheap per eval as `-param`, unlike `-dparam`). A new kind value `OPT_MODELPARAM`; in `opt_eval` the deck params are re-sourced first (E-144), then the in-place knobs: `alter` for instances, `altermod` for models. `@model[param]` is passed straight to `altermod` (which accepts that accessor form), mirroring how `-param @m1[w]` emits `alter @m1[w]=...`. Circuits with only `-param/-mparam` keep the E-130/-143 fast path (no reset). All three knob kinds mix in one run. Change confined to `com_optimize.c` (+1 help line); no new source files, no `inp.c` change, no ABI change. Verified 39/39 (E-130/-143/-144 [1]–[15] + E-145 [16]–[20]): OSDI model-param fit (`@rmod[r]` → 3k), built-in diode model-param fit (`@dmod[is]` → 1.22e-14), determined model+instance joint fit (`r=3k, R2=2k`), `-mparam` does 0 re-sources (in-place fast path), and all three kinds (`-dparam+-mparam+-param`) coexist and converge

E-146 *frontend/com_sweep.c (new), frontend/com_sweep.h (new), frontend/commands.c, frontend/com_commands.h, frontend/inp.c, frontend/Makefile.am, examples/sweep_examples/**

universal **sweep** command + **.sweep** card. Sweeps ANY circuit knob, generalizing `.dc` (which steps only sources/resistors/instance params). `sw_kind` auto-detects the knob: `@X[y]` with `ft_sim->findModel(ckt,X)` non-NULL `?` model param (`altermod`), else instance (`alter`); a bare name with `nupa_get_param(name,&found)` found `?` `.param` (`alterparam+reset`), else device (`alter`) — so model params and **.params**, impossible with **.dc**, are now sweepable. `sweep <knob> (<start> <stop> <step> | lin|dec|oct N a b | list ...)` [`-analysis <cmd>`] [`-output <name>=<expr> ...`]: sets the knob, runs the inner analysis (default `op`) at each point, evaluates each output (last value; default = all node voltages like `.dc`), and emits a summary plot named `sweep` with the knob values as the scale (nutmeg vector API, front-end layer per E-142). The per-point analysis plots are retained (`tran1,tran2,...`). **.sweep** card: `inp.c` collects a top-level **.sweep** line into the post-parse control list as a sweep command; a static `sweep_active` re-entrancy guard stops a `.param` re-source (reset re-runs the **.sweep** card) from recursing. New source files registered in `commands.c/com_commands.h/Makefile.am`. Verified 11/11 (`verify_sweep.py`): `sweep R1` reproduces built-in `.dc R1` bit-for-bit, model-param (`@rmod[r]`) + `.param (rtop)` sweeps vs analytic divider, auto-detection routing, **.sweep** card == command form (no recursion), AC named-output vs analytic `|H(1kHz)|`, transient settled value, `lin/list/step` point counts, multi-output

E-149 *maths/misc/randnumb.c, include/ngspice/randnumb.h, frontend/numparam/xpressn.c, frontend/numparam/spicenum.c, frontend/commands.c, examples/lhs_examples/**

Latin-Hypercube Monte Carlo sampling (`mcsample`), closing the low-discrepancy-sampling gap vs commercial tools. A stratified sampler in `randnumb.c` holds the `mode/N/sample-index/dimension-counter/seed`; each random dimension's stratum permutation (Fisher-Yates over `0..N-1`) + per-sample jitter are generated lazily from a self-contained `splitmix64` seeded by (`seed, dimension`) — reproducible and order-independent. `mc_sample_uniform()` = (`perm[d][il+jitter[d][il]]/N`); `mc_sample_gauss()` maps it through `inv_normal_cdf` (Acklam probit, `rel err < 1.15e-9`) so Gaussian tails stratify too. Sample boundary: `spicenum.c`'s `nupa_signal(NUPADECKCOPY)` edge (once per reset re-source, guarded by the existing `firstsignals` latch) calls `mc_sample_advance()` (step sample, rewind dimension) before that pass's `.param` draws. Draw hook (`xpressn.c`): `agauss/gauss` take one `mc_sample_gauss()` and `aunif/unif/limit` one `2*mc_sample_uniform()-1` when LHS is active, else the unchanged `gauss1()/drand()` (LHS-off is byte-identical to before). Command `com_mcsample` (`randnumb.c`, registered in `commands.c` both tables) parses `lhs <N> [seed <s>] / random / off`. Front-end only, so solver-independent. Verified 5/5 under both solvers (`verify_lhs.py`): stratification (each of 48 strata hit once vs random's ~32/48), two-param multi-dimension, ~130x lower `Var(sample-mean)` at `N=32` (both converge to the same mean), seed reproducibility, analytic mean/sigma correctness

E-150 *frontend/com_sweep.c, frontend/commands.c, maths/misc/randnumb.c, include/ngspice/randnumb.h, examples/_setup.py, examples/highsigma_examples/**

high-sigma rare-event estimation (highsigma), the last statistical ROW versus a commercial simulator -- 4-6 sigma failure probabilities (SRAM / standard-cell yield) that plain MC cannot reach (1e-7..1e-9 needs 1e7..1e9 runs). Scaled-sigma importance sampling: a new MC_MODE_SSS in the E-149 sampler draws each Gaussian .param from the lambda-inflated normal ($z = \text{lambda} * \text{gauss1}()$) and accumulates that draw's log likelihood-ratio $\log(\text{lambda}) - (z^2/2)(1 - 1/\text{lambda}^2)$ into `sss_logw`; `mc_sample_weight()` = $\exp(\text{sss_logw})$ reweights the sample so the estimator is unbiased for the nominal probability. Uniform .params are bounded so SSS leaves them uninflated (weight 1). Direction-free -- no gradient / sensitivity / most-probable-failure-point. `mc_sss_config(N, lambda, seed)` seeds the global PRNG for reproducibility; `mc_sss_off()` reverts. Command `com_highsigma` lives in `com_sweep.c` (reuses its `sw_run_cmd` reset/analysis runner and `sw_eval_expr`), registered in `commands.c`: `highsigma <N> [-scale <lambda>] [-seed <s>] [-analysis <cmd>] -metric <expr> [-max <hi>] [-min <lo>]` loops N samples (reset -> analysis -> eval metric -> spec test -> read weight), reports P(fail), relative error, equivalent one-sided sigma-to-fail ($-\Phi^{-1}(P)$) and failure count, and leaves them in the `highsigma_*` vectors/vars. The spec is -max/-min values (not a >/< inside the expr, which a control command reads as an I/O redirect). Front-end only, solver-independent; the verify is heavy (thousands of re-sources) so `examples/_setup.py` marks highsigma SPARSE_ONLY. Verified vs analytic Phi(-beta) (verify_highsigma.py): beta=4 -> sigma 4.000, P 3.16e-5 (true 3.17e-5); deep tail beta=5 (P 2.87e-7) recovered from 6000 runs where plain MC expects ~0.0017 failures; two-sided spec ~doubles P; seed reproducibility exact; two independent Gaussians combine as $N(\sqrt{s_1^2 + s_2^2})$

E-151 *maths/misc/randnumb.c, include/ngspice/randnumb.h, frontend/numparam/xpressn.c, frontend/com_sweep.c, frontend/commands.c, examples/_setup.py, examples/yield_examples/**

process/mismatch correlations + packaged Monte Carlo yield -- closes the last two partial (warn) statistical rows vs a commercial simulator. (1) CORRELATIONS: `mc_corr_config(k, matrix)` Cholesky-factors a k x k correlation matrix (row-major, unit diagonal; rejects a non-positive-definite one) into lower-triangular `corr_L`; `mc_corr_component(i)` lazily draws the k underlying z's once per sample THROUGH `mc_sample_gauss()` (so LHS stratification / SSS weighting apply), computes $y = L * z$, caches it, and returns `y[i]`. The cache is reset in `mc_sample_advance()` which now runs its reset in EVERY mode (so correlation works under plain MC, not just LHS/SSS). New .param function `mvnorm(i)` (frontend/numparam/xpressn.c: added to `fmathS`, `XFU_MVNORM` enum in `fmathS` order, and the dispatch) returns the i-th correlated standard normal; with no matrix registered it degrades to an independent draw. `com_mccorr(mccorr <k> <m11> . . . <mkk> | off)` in `randnumb.c`. A reusable `mc_lhs_config(N, seed)` was factored out of `com_mcsample`. (2) YIELD: `com_montecarlo` in `com_sweep.c` (reuses `sw_run_cmd` / `sw_eval_expr`), registered in `commands.c`: `montecarlo <N> [-lhs] [-seed <s>] [-analysis <cmd>] (-spec <metric> [-max <hi>] [-min <lo>])`... runs N samples (reset -> analysis -> eval each spec), counts a sample PASS only if every spec is within limits, and reports the yield with a Wilson 95% score interval and per-spec violation counts, leaving `montecarlo_yield/montecarlo_npass/montecarlo_n`. -lhs uses `mc_lhs_config` for a lower-variance estimate; corners are the ordinary .lib selection. Front-end only, solver-independent; heavy deck so `examples/_setup.py` marks yield SPARSE_ONLY. Verified (verify_yield.py, Sparse): `mccorr rho=+0.7/-0.6 -> empirical corr 0.711/-0.614` with correct means/sigmas; `mvnorm` without `mccorr -> corr ~0`; non-PD matrix rejected; single two-sided spec yield 0.8660 (true $P(|Z| < 1.5) = 0.8664$); two independent specs multiply (0.7508 vs 0.7506); -lhs unbiased and ~800x lower yield-estimate variance; positive correlation raises the joint yield (0.75 -> 0.82). Demo: a matched divider yields ~100% process-correlated (rho=0.9) vs ~74% independent

E-152 *include/ngspice/optdefs.h, include/ngspice/tskdefs.h, include/ngspice/cktdefs.h, include/ngspice/smpdefs.h, spicelib/analysis/cktopt.c, spicelib/analysis/ckntask.c, spicelib/analysis/cktdojob.c, maths/ni/niinit.c, maths/KLU/klusmp.c, examples/klu_tuning_examples/**

KLU matrix reordering + scaling controls. The KLU direct solver ran on the compiled-in `klu_defaults` (AMD ordering, max row scaling, BTF on) with no way to change them, and its one exposed knob `klu_memgrow_factor` was broken (`task->TSKkluMemGrowFactor = (val->rValue == 1.2)` stored a boolean, not the value). E-152 exposes three new `.options` `--klu_ordering=amd|colamd`, `klu_scale=none|sum|max`, `klu_btf=on|off` -- as IF_STRING options (friendly names, mapped to KLU's integer codes 0/1, 0/1/2, 1/0 in the `cktsort.c` handlers) and fixes the memgrow assignment. Plumbing mirrors the existing memgrow path exactly: new `OPT_KLU_ORDERING/SCALE/BTF` enums (`optdefs.h`); `TSKklu*/CKTklu*` int fields on the task (`tskdefs.h`), circuit (`cktdefs.h`) and SMPmatrix (`smpdefs.h`) structs; defaults 0/2/1 that MATCH `klu_defaults` set in `ckntask.c` (so behaviour is unchanged unless the user overrides) plus the task copy; `task->ckt` in `cktdojob.c`; `ckt->CKTmatrix` in `niinit.c`; and in `klusmp.c` `SMPnewMatrix`, right after `klu_defaults()`, `Common->ordering/scale/btf = Matrix->CKTkluOrdering/Scale/BTF`. KLU-only; changes only HOW the matrix factors, never the physical solution; the numerics are untouched. Verified (`verify_klu_tuning.py`, KLU-only) on a resistor grid: every ordering/scaling/BTF setting gives the physically identical solution (max relative spread 2.8e-14); AMD vs COLAMD and `scale=max` vs `scale=none` change the factorization arithmetic so the full-precision result differs in its last digits (rel diff ~1.2e-14 / 2.8e-14 -- a deterministic, nonzero proof the knobs reach KLU); invalid values warn; a plain `.option klu` equals `amd/max/btf-on` bit-for-bit; a wide-dynamic-range network solves correctly under `none/sum/max`

E-153 *maths/ni/niiter.c, maths/KLU/klusmp.c, include/ngspice/smpdefs.h, include/ngspice/optdefs.h, include/ngspice/tskdefs.h, include/ngspice/cktdefs.h, spicelib/analysis/cktsort.c, spicelib/analysis/ckntask.c, spicelib/analysis/cktdojob.c, examples/trustregion_examples/**

Levenberg-Marquardt trust-region Newton (`.option trustregion`, off by default), completing the damped/trust-region-Newton row alongside the E-111 line search. In the Newton loop (`niiter.c`): the E-111 KCL-residual merit is reused (gated on `linesearch||trustregion`); when a dimensionless damping `lambda>0` is in effect, $\mu = \lambda \cdot ||\text{diag}(J)||$ is added to the effective diagonal `gmin` (`trGmin`, passed to `SMPPreorder/SMPluFac`) AND `mux_k` to the RHS (the E-127 pseudo-transient coupling with $x_{\text{prev}}=x_k$), so the solve yields the exact damped step $x_{k+1}=x_k-(J+\mu I)^{-1} F(x_k)$; a post-solve accept/reject block re-loads at the trial point, computes $||F(x_{\text{new}})||$, and either accepts (`relax lambda=0.25`) or rejects (`restore x_k`, grow `lambda`, force another iteration), reusing the E-111 state-save/limiting-reset machinery; a convergence guard forbids converging while `lambda>0`. The $||\text{diag}(J)||$ scale (new `SMPdiagNorm` in `klusmp.c`, for both the KLU and Sparse matrices) makes `lambda` dimensionless (Marquardt / scale-invariant). Unlike the line search (which only shortens a fixed Newton direction) large `mu` re-aims the step toward steepest descent, regularizing an ill-conditioned Jacobian. Result-neutral: the fixed point is `F=0` for any `mu` and `lambda>0` on success, so it converges to the same operating point -- verified BIT-IDENTICAL to plain Newton (diode, BJT, resistor divider, plus transient) under both solvers. Option plumbing mirrors `linesearch` (`OPT_TRUSTREGION`, `IF_FLAG`, `TSK/CKT` fields + `CKTtrLambda`). Solver-independent. HONEST FINDING: instrumented step-rejection counting showed ZERO rejections on every circuit tried (diode strings, behavioral exponentials, cubics, negative-resistance oscillators) -- ngspice globalizes Newton at the DEVICE level (per-device junction limiting: `limexp/pnjlim/fetlim`, 30 families) which damps the controlling voltages before the residual is computed, so a residual-increasing overshoot never reaches the solver-level merit; the trust-region is thus a correct, safe solver-level regularization that stays inert on typical circuits

E-154 *frontend/com_envelope.c (new), frontend/com_envelope.h (new), frontend/commands.c, frontend/com_commands.h, frontend/Makefile.am, spicelib/analysis/envelope.c (new), spicelib/analysis/Makefile.am, include/ngspice/cktdefs.h*

Envelope Following (`envelope <node> <fc> <tstop> [nppp] [m] [maxm] [reltol] [settle]`), the last missing analysis in the RF/periodic-steady-state suite. For a carrier-driven circuit whose amplitude/phase envelope varies slowly over many carrier periods, it samples the state once per carrier period $T=1/f_c$ and integrates the slow drift, jumping `M` periods at a time. The exact per-

period map is $X_{n+1} = \phi(X_n)$, ϕ = one-period DAE integration; the envelope obeys $dX/dn \sim \phi(X) - X$. The naive forward-Euler jump $X_{n+M} = X_n + M(\phi(X_n) - X_n)$ is UNSTABLE on high-Q circuits (the one-period monodromy has unit-circle eigenvalues, so $I + M(\Phi - I)$ amplifies $\rightarrow$ blow-up; this is what shelved an earlier attempt). The new engine (spicelib/analysis/envelope.c, EFanalysis) uses the IMPLICIT backward-Euler jump $X_{n+M} = X_n + M(\phi(X_{n+M}) - X_{n+M})$, solved by Newton $[(1+M)I - M*\Phi] dY = -G$ with $\Phi = d\phi/dY$ the one-period monodromy (finite-differenced, one extra period-integration per state, dense NxN solve capped for modest circuits). A-stable, so it tracks a resonator's envelope without diverging; its fixed point $\phi(X) = X$ is the true steady state. Step size M is chosen by step-doubling local-truncation-error control. The one-period map reuses the transient primitives (NicomCof + Nlter per fixed sub-step, the dctrans state rotation) on a fixed grid of nppp points in TRAPEZOIDAL mode (backward-Euler damps a high-Q resonance); ϕ is SELF-STARTING (a frozen-history variant plateaued at a false low fixed point) with a sub-divided backward-Euler first sub-step to bound the restart damping. The com_envelope.c front-end command runs a short settling transient, calls EFanalysis, and emits an envelope plot ($_amp = 2|V_1|$, $_dc$, $_re$, $_im$ vs time; nutmeg vector API). Solver-independent. Verified against a full .tran under both linear solvers: EF amplitude tracks the transient across a Q~3160 tank ring-up to <3% and to the steady state (<0.5%), stays bounded over 3000 carrier periods (26 envelope samples), and tracks a Q~316 tank to ~1.6%. Completes the RF suite

E-155 *frontend/com_reduce.c (new), frontend/com_reduce.h (new), frontend/commands.c, frontend/com_commands.h, frontend/Makefile.am, spicelib/analysis/rcreduce.c (new), spicelib/analysis/Makefile.am, include/ngspice/cktdefs.h, examples/reduce_examples/**

RC network reduction (reduce <fmax> [factor f] [file fname] [name subckt] [keep node ...]), moving the post-layout "RC reduction / model-order reduction" gap to done. Post-layout extraction produces enormous linear parasitic R/C networks (thousands-millions of interior nodes); reduce collapses the R/C network to a small ELECTRICALLY EQUIVALENT one preserving the port behaviour over DC..fmax, written as an ordinary .subckt of R's and C's. Method (spicelib/analysis/rcreduce.c, CKTreduceRC): TICER (Time-Constant Equilibration Reduction) -- Schur-complement (Gaussian) elimination of interior nodes of $Y(s) = G + sC$, kept first order in s so the reduced network is REALIZABLE as R's and C's (no model-order-reduction black box, no passive-synthesis step). Per elimination of node n, for each neighbour pair (a,b): $G[a,b] := G[a,n]G[n,b]/G[n,n]$; $C[a,b] := (G[a,n]C[n,b] + C[a,n]G[n,b])/G[n,n] - G[a,n]G[n,b]C[n,n]/G[n,n]^2$. The conductance update is the exact Schur complement so DC is preserved EXACTLY; the capacitance is matched to first order. A node is eliminated only when its self time-constant frequency $f_n = G_n/(2\pi C_n) > factor*fmax$ (quasi-static in the band of interest); factor trades reduction against in-band accuracy, monotonically. PORTS (kept nodes) are auto-detected as every node touched by a device that is NOT a resistor/capacitor (sources, transistors, OSDI Verilog-A devices), plus ground and user keep nodes, via the generic GENnode/terminal-count device interface -- so built-in and OSDI devices alike. The com_reduce.c front-end command enumerates the built-in resistor/capacitor instances (CKTtypelook), builds dense G/C over the RC nodes, runs TICER, and emits the reduced subckt (CKTnodName for node names). Solver-independent (reads devices + own dense solve). First cut is dense (node cap ~2500); sparse TICER is the follow-up. Verified under both linear solvers: a huge factor eliminates nothing and the emitted subckt reproduces the full AC response BIT-for-BIT; a moderate factor cuts nodes (e.g. 25->6, 4x) with <0.25 dB in-band error and DC exact; the accuracy/reduction tradeoff is monotone in factor; an OSDI device on a node auto-marks it a kept port

E-156 *spicelib/analysis/rcreduce.c, frontend/com_reduce.c, include/ngspice/cktdefs.h, examples/reduce_examples/**

Sparse RC reduction -- makes the E-155 reduce command scale to real post-layout size. The dense NN TICER engine (capped ~2500 nodes) is replaced by a SPARSE one: the network is stored as per-node adjacency lists (RCadj: a growable edge list of {neighbour, g, c}), and interior nodes are elimi-

nated in a MINIMUM-DEGREE order (a lazy binary heap keyed on current degree, like sparse LU) so fill-in stays tiny -- a degree-2 chain node merges two series elements with ZERO fill. A FILL GUARD (maxdeg, default 12) declines to eliminate a node once its degree grows past the threshold, so a dense mesh core (whose boundary Schur complement is inherently dense) is left intact instead of densifying the whole network (measured: unguarded, one mesh elimination created 3308 fill edges; guarded, 9). The TICER Schur-complement update, DC-exact conductance, frequency criterion (eliminate only when $f_n = G_n / (2\pi C_n) > \text{factorfmax}$), and automatic port detection are unchanged from E-155; in the edge representation the diagonal is implicit so eliminating a node just adds Schur edges among its neighbours. Node cap raised 2500 -> 5,000,000; a 65,017-node network reduces in ~4 s. New maxdeg command option (com_reduce.c) and CKTreduceRC parameter (cktdefs.h). Also fixes a terminal-ordering pitfall: the reduced .subckt's terminals are emitted in internal-index order (not the order keep nodes were typed), so instantiating with the wrong order silently swaps ports -- the command now prints the correct x1 <ports...> <name> instantiation line. Verified under both linear solvers: identity reduction bit-exact (0.00 dB), a moderate factor gives ~4x fewer nodes at <0.25 dB in-band with DC exact, monotone accuracy/reduction tradeoff, OSDI auto-port, and an 8001-node network (past the old dense cap) reduces successfully

E-157 *frontend/com_aging.c (new), frontend/com_aging.h (new), frontend/commands.c, frontend/com_commands.h, frontend/Makefile.am, examples/aging_examples/**

Device aging (aging <t_target> [rate <opvar>] [param <ageparam>] [dynamic <tstop> [tstep]] [verbose]), adding the reliability "stress -> degrade -> re-simulate (fresh/aged)" flow (HCI/NBTI/TDDDB) and moving those reliability gap rows to done. The command finds every aging-capable device in the loaded circuit, computes how much it has degraded after t_target seconds of operation, writes that back into the device, and RE-STAMPS the circuit so any analysis run afterwards (op/dc/tran/ac) sees the aged devices; it runs the fresh stress simulation first, leaving it as the current plot (a "fresh" baseline). The design is MODEL-AGNOSTIC: a device opts in purely by exposing, in its Verilog-A/OSDI model, a degradation-RATE operating-point variable (default agerate, in dose-units/second) and a per-instance AGE parameter (default age, (*type="instance"*)). The engine's only job is to integrate the rate into a dose and feed it back; the model owns the physics that maps age to a parameter shift (a sublinear power law, an Arrhenius factor, ...). Participants are detected by scanning each device TYPE's instance-parameter table (DEVICES[t]->DEVpublic.instanceParms) for the two keywords, so ordinary resistors/sources are silently skipped and probing never errors. Two modes: STATIC (default) reads the rate at the DC operating point and scales by the lifetime, $\text{age} = \text{agerate}(\text{op})t_{\text{target}}$; DYNAMIC (dynamic <tstop>) runs a transient over one representative window, trapezoidally (time-weighted) integrates the rate, and extrapolates, $\text{age} = (\text{INT agerate } dt / t_{\text{stop}})t_{\text{target}}$ -- capturing duty cycle. Because age is per-instance, devices sharing a .model but at different bias age independently. Implementation (frontend/com_aging.c) reuses the com_optimize synchronous command-dispatch pattern to run op/tran, the nutmeg expression engine to read @inst[agerate], and alter @inst[age]=... to write back; console chatter suppressed via ft_optimizing unless verbose. Solver-independent. Verified under both linear solvers with a demo square-law NMOS carrying an NBTI hook (agemos.va): enumeration picks exactly the ageable devices, the dose is exactly $\text{ratet}_{\text{target}}$, the threshold shift matches the analytic $\Delta V_{th} = dv_{th_ref}(\text{age}/\text{age_ref})^{0.25}$ law to 5 sig figs, aged drain current drops, degradation is monotone in lifetime, a near-threshold device loses a larger fraction of its current than a hard-driven one for the same shift, a 30%-duty pulsed gate ages at 0.30x the DC rate (dynamic), and a sub-threshold device accrues zero dose

E-158 *frontend/com_emir.c (new), frontend/com_emir.h (new), frontend/commands.c, frontend/com_commands.h, frontend/Makefile.am, examples/emir_examples/**

Power-grid EMIR (emir [rail V] [thresh frac] [thick m] [jmax A/m²] [n exp] [tref s] [top k] [verbose]), adding electromigration + IR-drop reliability sign-off and moving the last reliability gap row to done. After a DC solve of the power-distribution network the command reports two metrics. IR-DROP: for every node, the drop below the ideal rail (rail - V(node)); the rail

defaults to the highest node voltage (the supply pad) unless given; reports the worst node and every node past a threshold (default 10% of the rail). ELECTROMIGRATION: for every wire-segment resistor, the current DENSITY $J = |I|/(w \cdot thickness)$, ranked; the worst-density segment, every segment past the current-density limit J_{max} , and a Black's-equation relative lifetime $MTTF/ref = (J_{max}/J)^n$ (Black: $MTTF \sim J^{-n}$; a segment exactly at J_{max} has $MTTF =$ the reference lifetime t_{ref} , needing no process-specific prefactor). The physics the analysis exists to expose: EM is set by current DENSITY, not current -- a fat trunk carrying huge current can be safe while a thin wire at a fraction of that current voids first, which is why real grids taper wire width with carried current to hold J roughly constant. Implementation (frontend/com_emir.c) runs a fresh op (com_optimize synchronous-dispatch pattern), walks the current plot's node-voltage vectors (plot_cur->pl_dvecs filtered to SV_VOLTAGE) for IR-drop, and enumerates the resistor instances (the device type named "Resistor") reading each segment's current and width via the nutmeg expression engine (@Rk[i], @Rk[w]) -- so it needs no device-struct access and works for any resistor; segments without a width are skipped and counted. Both tables are qsort-ranked (IR by drop, EM by density) and truncated to top. Solver-independent (reads a DC solution + per-resistor currents). Verified under both linear solvers on a 3-segment tapered ladder off a 1 V rail: worst IR drop 0.30 V (30%) at the far tap matching the exact ladder solve and linear in load current; $J = I/(w \cdot thick)$ exact; the worst-density segment is the narrow low-current wire, not the high-current wide trunk; the Black MTTF ratio between two segments equals $(J_2/J_1)^n$; with J_{max} between the trunk and leaf density exactly the under-sized segments fail; rail auto-detect equals an explicit rail; and an OSDI (Verilog-A) current load at a tap is handled identically to a current source

E-159 (no ngspice source change) examples/compactmodels_examples/*

Real production compact-model bring-up -- a validation/example enhancement that exercises the EXISTING toolchain on actual CMC (Compact Model Coalition) Verilog-A models, so no ngspice or openvaf-r source change. BSIM4 (the ~12,600-line industry-standard bulk MOSFET) is compiled through openvaf-r to OSDI (~6 s) and validated against ngspice's BUILT-IN BSIM4 -- the same model, so a rigorous self-check: the OSDI BSIM4.8 and native BSIM4.8.3 agree to ~2% on the Id-Vds output family and ~4.5% on the Id-Vgs transfer curve (the residual is the point-release version gap). EKV (EPFL compact MOSFET, which ngspice has NO built-in for) is brought up purely through OSDI, giving textbook saturation. A general finding surfaced: OSDI keeps every internal node STATIC (no dynamic node collapsing), so BSIM4's internal drain/source nodes (di/si) -- which the default rdsmod=0 expects the simulator to collapse onto the external d/s -- are left floating and the device conducts ZERO current; enabling the external series-resistance nodes with rdsmod=1 (the model near-shorts them, 1e3 mho, when no S/D geometry is given) connects them and the device works. Models are compiled in place from the bundled OpenVAF integration-test sources (licenses stay with them). Verified under both linear solvers (6 checks)

E-160 (no ngspice source change) examples/cmcsweep_examples/*, examples/_setup.py

CMC compact-model coverage sweep -- the comprehensive follow-up to E-159, again a validation/example enhancement with no ngspice/openvaf-r source change (only _setup.py gains cmcsweep in its SPARSE_ONLY and REGRESSION_EXCLUDE sets so the slow 19-model compile runs single-solver and off the fast sweep). It drives EVERY real CMC (Compact Model Coalition) Verilog-A model bundled with OpenVAF through openvaf-r -> OSDI -> ngspice and reports a coverage matrix, the same idea as the E-84 LRM sweep but for production compact models. Result: 19 of 19 models COMPILE and 19 of 19 LOAD, spanning every device class -- MOSFET (BSIM3, BSIM4, BSIM6, BSIMBULK, EKV, HiSIM2, HiSIMSOTB, MVSG_CMC), FinFET (BSIMCMG, BSIMIMG), SOI (BSIMSOI, HiSIMHV), GaN HEMT (ASMHEMT), surface-potential (PSP102, PSP103), bipolar/SiGe-HBT (HICUML2, MEXTRAM), and diode (DIODE, DIODE_CMC). Deeper conduction+physics validation for a representative model per class: OSDI BSIM4 and BSIM3 match ngspice's BUILT-IN BSIM4/BSIM3 to ~2%/~7%; EKV gives monotonic saturating I-V; HICUML2 shows bipolar current gain beta ~ 100 on a Gummel plot; DIODE_CMC turns on exponentially (x16000 over 0.2-1.0 V). Two findings: (1) the E-159 internal-node/rdsmod=1

issue is BSIM4-SPECIFIC, not universal -- BSIM3 handles the zero-resistance case fine and conducts out of the box (its apparent zero at $V_{gs}=1$ V is just a high ~ 1.7 V default threshold); (2) HiSIMHV (6 terminals d,g,s,b,sub,temp) needs `cosubnode=1` to enable the substrate node, and with the default `cosubnode=0` its `$fatal` correctly guards the node-count mismatch (confirming `$fatal` works through OSDI). Models compiled in place from the bundled integration-test sources (licenses stay put). Verified (7 checks); the compile/load tiers are solver-independent

E-161 *(no ngspice source change) examples/dynmodels_examples/**

Dynamic (AC/RF) compact-model validation -- extends the E-159/160 DC validation into the dynamic domain, again a validation/example enhancement with no ngspice/openvaf-r source change. Where E-159/160 exercised DC I-V, this exercises the DYNAMIC behavior, which flows through a different code path: OSDI's REACTIVE (charge) Jacobian stamping and ngspice's `.ac` analysis, on the real production models (compiled in place from the bundled CMC sources). BSIM4 C-V: the gate capacitance $C_{gg}(V_{gs})$ is extracted from `.ac` as $\text{Im}(I_{\text{gate}})/\omega$ and swept over gate bias; it rises from ~ 40 fF in subthreshold (overlap+fringe) to ~ 129 fF in inversion (approaching $C_{ox}WL$) -- the textbook MOSFET C-V -- and the OSDI-compiled BSIM4 matches ngspice's BUILT-IN BSIM4 to under 1% at every bias (a tighter check than the $\sim 2\%$ DC I-V match, since C_{gg} is dominated by the version-independent oxide term). Cutoff frequency f_T (the frequency where AC current gain $|h_{21}|=|I_{\text{out}}/I_{\text{in}}|$ falls to 1): OSDI BSIM4 $f_T \sim 3.5$ GHz matches the built-in to $\sim 1\%$; HICUML2 (SiGe HBT, which ngspice has no built-in for) has zero default transit time ($t_0=0$) giving infinite f_T , so a realistic dynamic parameter set ($t_0=10$ ps, 1 fF junction caps) is supplied and the resulting f_T lands right at the transit-time limit $1/(2\pi t_0) \sim 15.9$ GHz and rises with collector current -- textbook charge-control behavior. Verified under BOTH the Sparse and KLU solvers (`.ac` is supported by both), 4 checks

E-162 *frontend/inp.c, examples/hb_examples/verify_hb.py*

`.hb` dot-card for harmonic balance -- gives the E-134 `hb` command netlist dot-card parity with the PSS family (`.pss/.pac/.pnoise/.pxf/.psp/.sp` are dot-cards, while the HB family `hb/qpss/hbosc` were control-block commands only). A top-level `.hb <f0> <K> [points] [maxiter]` card now runs single-tone harmonic balance straight from the deck (no `.control` block needed). Rather than duplicating the HB machinery as a separate analysis JOB, `.hb` reuses the deck->control mechanism Enhancement-146 introduced for `.sweep`: during deck loading (frontend/inp.c) a top-level `.hb ...` line is stripped of its leading `.` and appended to the post-parse control list, so it executes as `hb ...` (the `com_hb` command) once the circuit is built -- inheriting all of the E-134 engine (argument parsing, the E-121 conversion-matrix Jacobian, E-135 source-stepping continuation, solver independence). A boundary check (next char after `.hb` is whitespace or end-of-line) keeps it from swallowing any future `.hb*` card. Before this, `.hb` produced unimplemented dot command `'hb'` -- the only `.hb` handler in ngspice was a disabled `#ifdef WITH_HB` upstream stub (`dot_hb`) that merely redirected to the PSS analysis, not the E-134 engine. Verified (examples/hb_examples/verify_hb.py, +2 checks): the `.hb` dot-card run in plain batch mode (no `.control`) converges and produces a harmonic-balance spectrum bit-for-bit identical to the `hb` command form on a junction-limited diode rectifier; it threads its optional `[points]/[maxiter]` arguments and coexists with a following `.control` block (deck order preserved); both Sparse and KLU give identical results. (Like `.sweep`, a bare command-style dot-card with no other analysis card prints a benign "no simulations run" batch notice even though HB ran.)

E-163 *frontend/inp.c, examples/qpss_examples/verify_qpss.py, examples/phasenoise_examples/verify_phasenoise.py*

`.qpss` / `.hbosc` / `.phasenoise` dot-cards -- completes the harmonic-balance family's netlist dot-card parity begun by E-162's `.hb`. Three more top-level command-style cards now run their analyses straight from the deck: `.qpss <expr> <f1> <f2> [periods] [maxorder]` (two-tone quasi-periodic steady state, E-133/136; add the `hb` keyword for the frequency-domain form), `.hbosc <oscnod> <K> [fguess] [tstab]` (autonomous harmonic balance for an

oscillator, E-140; the deck needs a `.ic` to start the oscillation), and `.phasenoise <fstart> <fstop> [points]` (oscillator phase noise, E-140; run after `.hbosc`). Each reuses the same one-branch deck->control mechanism as `.hb` (E-162) / `.sweep` (E-146): in frontend/inp.c a top-level `.qpss/.hbosc/.phasenoise` line is stripped of its leading `.` and appended to the post-parse control list, executing as the corresponding command once the circuit is built -- inheriting the full E-133/136/140 engines and their solver independence. Because the bridge preserves deck order, `.hbosc` (which finds the oscillator PSS + frequency) and `.phasenoise` (which reads it) compose, so a complete oscillator phase-noise run needs no `.control` block. A per-card boundary check (the char after the name is whitespace or end-of-line) keeps the cards distinct -- in particular `.hbosc` is NOT matched by the `.hb` branch, whose own boundary check rejects the trailing `osc`. Verified: the `.qpss` dot-card produces a two-tone spectrum bit-for-bit identical to the `qpss` command (examples/qpss_examples/verify_qpss.py); `.hbosc + .phasenoise` in a deck reproduce the command form's oscillation frequency and phase-noise spectrum exactly with order preserved (examples/phasenoise_examples/verify_phasenoise.py); both under Sparse and KLU. No regression to `.hb/.sweep`.

E-164 (no ngspice source change) examples/rfpa_examples/*

Large-signal RF characterization of a real transistor -- a validation/example enhancement (no ngspice/openvaf-r source change) that closes the production-model validation loop (DC E-159, coverage E-160, small-signal E-161, now large-signal). It drives a common-emitter amplifier built from the bundled HICUM/L2 SiGe HBT (compiled in place, with the E-161 dynamic params $t_0=10\text{ps} + 1\text{fF caps}$) hard and extracts the large-signal RF figures of merit via TRANSIENT + Fourier/FFT: AM-AM gain compression (the power gain expands 17.8->20.7 dB then compresses -- the exponential-transconductance signature of a bipolar -- defining P1dB), harmonic distortion (the third harmonic follows the textbook 3:1 slope, $\text{HD}_3 \sim A^3$), and two-tone third-order intermodulation (the IM3 products $2f_1-f_2 / 2f_2-f_1$ at $\sim 30\text{ dBc}$). IIP3, extracted from the single tone as $A/\sqrt{3*\text{HD}_3}$ (the two-tone IM3 is exactly 3x the single-tone HD3 for a third-order nonlinearity), is constant at $\sim 0.134\text{ V}$ across drive level, confirming the third-order model. A real finding: the frequency-domain harmonic-balance engines (`hb/qpss`, E-134/136, which just gained netlist dot-cards in E-162/163) do NOT converge on this stiff, many-internal-node production model in an amplifier configuration -- `hb` returns error 103 at any drive level (even 1 mV) and with extra iterations/source-stepping, and the two-tone `qpss` is prohibitively slow (>2 min per point). Transient+FFT integrates reliably and is the robust route for large-signal RF on a full production compact model; HB/QPSS remain the tool of choice for lighter, well-conditioned circuits. Verified under both linear solvers (4 checks): transient small-signal gain matches `.ac` (7.744 vs 7.742), HD3 3:1 slope (62.9 vs 64), IIP3 constant (0.134 V, spread <2%), and gain compresses at high drive (7.74->5.97).

E-165 (no ngspice source change) examples/modelnoise_examples/*

Production compact-model noise validation -- a validation/example enhancement (no ngspice/openvaf-r source change) exercising the last untested small-signal path, OSDI's `.noise` stamping of the models' own `white_noise` / `flicker_noise` sources, on production models compiled in place from the OpenVAF integration-test sources. BSIM4 (MOSFET): the output-noise spectral density $S_v(f)$ of a common-source amplifier is compared to ngspice's BUILT-IN BSIM4 over the full band and matches to $\sim 1.5\%$ everywhere, covering both the low-frequency $1/f$ FLICKER region (amplitude density $\sqrt{S_v} \sim 1/\sqrt{f}$) and the flat high-frequency THERMAL floor -- a stringent check since the flicker coefficients, thermal-noise model, and their bias dependence all have to line up. HICUM/L2 (SiGe HBT): no ngspice built-in, so validated against physics -- its default noise is WHITE (shot noise on the junction currents + thermal noise on the parasitic resistances, no flicker by default), a flat spectrum in contrast to the MOSFET's $1/f$ rise; with a small source resistance so the intrinsic device noise dominates, the output-noise floor tracks the collector SHOT-noise line $\sqrt{2qI_c R C^2}$ across two decades of bias current (within 6% at high bias where shot dominates) and scales as $\sqrt{I_c}$. Verified under BOTH the Sparse and KLU solvers (`.noise` works under both since E-113 fixed the KLU adjoint solve), 5 checks: BSIM4 spectrum matches built-in <4%,

BSIM4 1/f flicker slope ($S_v(1\text{Hz})/S_v(10\text{Hz}) \sim \sqrt{10}$), BSIM4 flat thermal floor, HICUM white noise, HICUM shot-noise tracking + $\sqrt{I_c}$ scaling. Completes the production-model validation loop (DC E-159, coverage E-160, small-signal E-161, large-signal E-164, noise E-165).

E-169 *frontend/syntaxhl.c (new), include/ngspice/syntaxhl.h (new), frontend/commands.c, frontend/Makefile.am, main.c, examples/syntaxhl_examples/**

Interactive command-line syntax highlighting -- the interactive prompt now colors the command line AS IT IS TYPED. The command word is shown GREEN the moment it is a recognized command, RED when it can no longer become one, and left the terminal default while it is still a valid PREFIX being typed (so `plo` is neutral, `plot` green, `plt` red); numbers are yellow, "quoted strings" magenta, and -option flags cyan. The command word is classified exactly as the interpreter resolves it -- the same case-insensitive walk of the live command table `cp_coms` that `do_command()` uses, plus the control keywords (`if/while/foreach/begin/end/...`) -- so the color always agrees with whether the word will actually run. Implementation: a pure engine `cp_highlight_line()` (`frontend/syntaxhl.c`) tokenizes a line and wraps each token in ANSI SGR color escapes; live coloring overrides GNU readline's `rl_redisplay_function` (installed by `cp_synhl_init()` from `main.c` right after `rl_outstream` is set). The replacement `redisplay` redraws the colorized line only when the prompt plus line fit on one terminal row and otherwise defers to readline's own `rl_redisplay()`, so a wrapped line is never corrupted; the cursor is repositioned with a plain CR + CUF (cursor-forward) so mid-line editing is unaffected. Because that manual cursor positioning bypasses readline's internal position tracking, readline's own accept-line stops emitting the closing newline, so the Enter key is bound to a wrapper (`cp_synhl_accept`) that emits the newline itself when coloring is active before running the normal accept-line -- otherwise a command's output would begin on the input line (when coloring is off the wrapper is a no-op and readline handles the newline as usual). Gated three ways: it is on by default only at an interactive TTY (`isatty(rl_outstream)`), is disabled by `set no_syntax_highlight` and by the NO_COLOR environment convention, and never fires in batch (`ngspice -b`) or any piped/redirected session -- so simulation output is byte-for-byte unchanged (confirmed by the full example regression). Because as-you-type coloring needs the terminal in raw mode it requires a readline-enabled build, which the shipped binaries are (CI builds with `--with-readline=yes` and the committed binaries link `libreadline`). A new `synhl <command line>` command prints the colorized form of a line non-interactively -- useful for previewing and, since it needs no terminal, what makes the coloring engine regression-testable in batch. Verified (`examples/syntaxhl_examples/verify_syntaxhl.py`, 11 checks): the engine via `synhl` in batch (valid command green, unknown red, valid prefix neutral, numbers/strings/options colored, case-insensitive) and the live as-you-type behavior driven through a pseudo-terminal (green/red/neutral on the fly, NO_COLOR and a non-tty suppress all color); the live layer auto-skips on a build without readline. Solver-independent (front-end only).

E-170 *frontend/syntaxhl.c, include/ngspice/syntaxhl.h, examples/syntaxhl_examples/**

Semantic syntax highlighting -- extends E-169 from lexical to SEMANTIC coloring of the interactive line: it now checks whether the SIGNALS and EXPRESSIONS typed actually exist and parse. (1) A signal reference `v(node)/i(source)/@device[param]` is looked up read-only in the current plot (`vec_fromplot`, no evaluation, no vector creation); it keeps the default color if it exists and is drawn RED if it does not. (2) Inside an otherwise-valid expression only the invalid signal reddens -- `print v(a)+v(zzz)` colors just `v(zzz)`, including signals nested in functions (`sqrt(v(a))-v(bad)`). (3) A genuinely malformed expression argument of an expression command (`plot/print/gnuplot/asciiplot`) reddens as a whole, tested with the real parser `ft_getpnames_from_string(FALSE)` with its tree freed. (4) Error output is drawn RED at an interactive terminal: `cp_err` is wrapped in a coloring stream (`funopen` on BSD/macOS, `fopencookie` on Linux) that prepends the red SGR to each write; `cp_curerr` is pointed at the wrapper too so `ngspice`'s per-command stream reset `cp_ioreset()` keeps it in place instead of restoring and closing it. A half-typed expression is NOT treated as an error and stays neutral -- a signal whose parenthesis has not closed (`v(bP)`), an expression with unbalanced parens (`v(a)+v(b)` or one ending in an operator (`v(a)+`). KEY FIX: the expression parser's

error handler PError writes straight to stderr, bypassing the cp_err muting, so an early version spewed syntax error in line segment ... onto the prompt while typing; expr_pares() now also mutes fd 2 (dup2 /dev/null) across the parse and restores it. Signal-validity is scoped to the unambiguous v()/i()/@ forms (bare words could be plot keywords like vs, functions like sqrt, or options -> false reds). Same three-way gating as E-169 (interactive TTY, set no_syntax_highlight, NO_COLOR); the parser + vector lookups run per keystroke but are read-only and do NOT corrupt command execution (a subsequent let z=v(a)+v(b) still computes correctly). Verified (verify_syntaxhl.py, 19 checks total): the E-169 lexical checks plus a semantic layer driven through a pseudo-terminal after simulating a small circuit -- valid signal not red, invalid signal red, invalid-in-expression reddens only the signal, malformed expression red as a whole, half-typed stays neutral with no parser-error spam, error output red, NO_COLOR suppresses it. Solver-independent (front-end only).

E-171 *maths/KLU/klusmp.c, spicelib/analysis/pzan.c, examples/klupz_examples/**

KLU/Sparse solver audit -- KLU pole-zero fixed (complex determinant + pivot tolerance) -- a deep audit of the KLU<->SMP bridge found that pole-zero analysis under 'option klu' SILENTLY PRODUCED GARBAGE for any circuit with complex poles or zeros: a series RLC reported four bogus real poles (-100, -10, -0.89, 0) instead of its conjugate pair -5000 +- j999987.5. Real-axis roots came out right (the mixed formula is accidentally correct for real pivots), which is why the existing single-real-pole RC regression never caught it. DEFECT 1 (spDeterminant_KLU): the complex determinant built each pivot as the mixed quantity (1/(UxRs), UzRs) and took its complex reciprocal -- KLU's Udiag stores the ACTUAL pivots (its solve divides by them) while Sparse's Diag stores RECIPROCAL pivots (its solve multiplies), and the adaptation kept Sparse's reciprocal dance for the real part only. The real branch was worse: its product loop NEVER RAN (the loop index was left at N by the preceding permutation scan, so det was always +/-1.0), it divided instead of multiplying, and it never wrote *piDeterminant (SMPcDProd consumed an uninitialized stack value). Both branches computed the permutation sign as #non-fixed-points/2, wrong for any cycle longer than 2 (a 3-cycle is EVEN but counted odd). Rewritten: det(A) = sign(P)*sign(Q)*prod(Udiag[k]Rs[k]) with parity computed exactly by cycle decomposition (new PermutationParity helper); the KLU determinant now matches Sparse to ~14 digits at every Muller trial point. DEFECT 2 (SMPcReorder/SMPReorder): pole-zero calls SMPcReorder with PivRel = 0.0; Sparse's spOrderAndFactor SANITIZES a non-positive threshold to its stored default, but the KLU branch assigned it straight to Common->tol, making KLU's partial pivoting accept an EXACTLY-ZERO diagonal as pivot (|diag| >= 0max always holds). At PZ's s=0 trial an inductor branch has a 0.0 diagonal (-sL), so the factorization returned KLU_SINGULAR and PZ recorded a spurious root at the origin -- and the poisoned deflated Muller search never expanded past |s| ~ 10, yielding the garbage above. Both reorder entry points now mirror Sparse's sanitization (in-range PivRel still takes effect; DC/AC/tran pass the 1e-3 default and are byte-identical). With both root causes fixed, the E-113-era guard in pzan.c ('pole-zero finite-zero computation is not supported with option KLU' -- added when the zero search was seen to go spuriously singular, i.e. defect 2) is REMOVED; the finite-zero search now works under KLU and a genuine E_SHORT reports the same 'input shorted' diagnostic as Sparse. The balanced/differential-output PZ guard remains (SMPcAddCol genuinely has no KLU branch). Verified (examples/klupz_examples/verify_klupz.py, 6 checks): full pole/zero root-set parity between the two solvers plus analytic anchors on a series RLC (conjugate pair), RC lowpass (old-good case, no regression), lead network (finite real zero), RC highpass (origin zero), RLC bandstop (PURELY IMAGINARY zeros +-j1e6, the hardest case), and RLC bandpass (the finite-zero search formerly guarded off) -- all six identical across solvers. Full 134-example regression clean. Residual scope (solver-independent, ~1990 Muller-on-determinant fragility): a twin-T's deflated conjugate zero pair can still stall under KLU (4 of 6 roots found) while on the bandpass it is SPARSE that hits its iteration limit (KLU converges cleanly).

E-172 *maths/KLU/klusmp.c, spicelib/analysis/cktpzset.c, spicelib/analysis/pzan.c, examples/_setup.py, examples/klupz_examples/**

KLU balanced pole-zero + full partial pivoting -- closes E-171's two scope items; no analysis card remains Sparse-only under KLU. (1) BALANCED / DIFFERENTIAL-OUTPUT pole-zero ('pz in 0 a b vol', non-ground output reference) was guarded off as unsupported: the algorithm's change of variables needs the per-trial column fold $\text{col}(b) += \text{col}(a)$ (SMPcAddCol), which had no KLU branch -- and KLU's CSC sparsity pattern is FIXED at COO->CSC conversion time, so the fold's fill-in cannot be created on the fly the way Sparse's spcCreateElement does. Fixed in two halves: CKTpzSetup reserves the UNION pattern before the conversion (every row of the solution column is also registered in the balance column via SMPmakeElt; duplicate (row,col) COO entries fold into a single CSC slot by the conversion's group labeling, so the reservation is safe), and SMPcAddCol gains a KLU branch that merge-walks the two columns' sorted row lists adding the complex values in place (with a defensive error if an addend row is missing, impossible after the reservation). The two pzan.c balanced-output guards are removed. (2) FULL PARTIAL PIVOTING for PZ factorizations: validating a fully-differential configuration exposed spurious far-field roots at $|s| \sim 1e19 \cdot 1e21$ (including a positive-real pole). An exact-rational (fraction-arithmetic) determinant of the very matrix KLU factored proved the factorization itself was numerically rotten at extreme $|s|$: at $s=4.52e19$ the exact determinant was $+4.36e11$ while the pivot product gave $-2.05e12$ -- wrong sign AND magnitude. Root cause is architectural: KLU's ordering is fixed at klu_analyze time (pattern-only AMD/BTF) while Sparse re-runs value-aware Markowitz ordering every PZ trial; PZ sweeps $|s|$ across ~ 20 decades, and with the relaxed default $\text{tol}=0.001$ the fixed ordering accepted catastrophically-cancelling pivots. KLU's only value-adaptive lever is the within-block partial pivoting, so the E-171 sanitization fallback is upgraded from 'keep current tolerance' to FULL partial pivoting ($\text{tol}=1.0$) whenever the caller passes an out-of-range PivRel -- only pole-zero does (0.0); explicit in-range tolerances (DC/AC/tran $1e-3$) are honored unchanged. This single change eliminated the far-field junk AND cured the twin-T conjugate-zero-pair stall recorded in E-171's scope (what looked like vintage-Muller noise-floor fragility under KLU was largely this factorization in-accuracy; KLU now finds all 6 twin-T roots with notch-zero real parts $\sim 1e-12$, an order CLOSER to the exact 0 than Sparse's $\sim 1e-11$). Verified (verify_klupz.py, 6 -> 9 checks): the E-171 six plus the differential RC bridge (balanced output, formerly 'not supported'), a balanced output with a complex pole pair, and the twin-T notch -- all root sets identical across solvers. The dual-solver harness's balanced-pz KLU-skip detection is removed from examples/_setup.py (nothing remains Sparse-only). Full example regression 134/134.

E-173 *spicelib/analysis/cktpzeig.c (new), maths/dense/eig.c (new), maths/KLU/klusmp.c, spicelib/analysis/pzan.c, cktsopt.c, cktntask.c, cktdojob.c, include/ngspice/{smpdefs,optdefs,tskdefs,cktdefs}.h, two Makefile.am, examples/pzeig_examples/**

Eigenvalue-based pole-zero ('.options pzeig') -- a modern alternative root finder for .pz. The vintage driver hunts roots of $\det(G+sC)$ with a Muller iteration on determinant values and is famously fragile (iteration limits -- 'giving up after 231 trials' on a plain RLC bandpass -- noise-floor stalls, chaotic search paths; E-171/172 fixed the KLU-side defects but the ~ 1990 algorithm itself stays delicate under BOTH solvers). The new opt-in method computes every pole and zero at once as a dense eigenvalue problem, reusing the existing CKTpzSetup/CKTpzLoad machinery so it applies unchanged to the poles and zeros configurations, to balanced/differential outputs, and to both linear solvers. Steps: (1) the PZ-configured MNA matrix is AFFINE in s , $A(s) = G + sC$, so two loads ($s=0, s=1$) extract the dense pencil via a new solver-agnostic SMPdenseExtractReal (KLU: walk $A_p/A_i/A_x\text{Complex}$; Sparse: walk *FirstInCol* through the *Int->Ext* translation maps); a third load at an arbitrary point VERIFIES affinity so a non-polynomial device falls back with a clear message instead of wrong roots. (2) C is structurally singular, so the pencil is solved by SHIFT-INVERT linearization: factor $(G + \text{sigma}C)$ ONCE at a non-root shift (tried from a small ladder of shifts, with the circuit's own sparse solver), form the dense $M = (G + \text{sigma}C)^{-1}C$ by n sparse solves; from $(G+sC)v=0 \Leftrightarrow Mv = \text{mu}v$ with $\text{mu} = 1/(\text{sigma} - s)$, every finite root is $s = \text{sigma} - 1/\text{mu}$ and the pencil's infinite eigenvalues land harmlessly at $\text{mu} = 0$ (dropped by a noise-floor threshold $64\text{neps} \cdot \|M\|$). (3) eigenvalues of the real dense M from a new self-contained eigensolver maths/dense/eig.c -- the classical balance / Hessenberg / Francis double-shift QR chain (EISPACK bal-

anc/elmhes/hqr lineage, written fresh, no LAPACK dependency), unit-tested standalone against companion matrices, complex conjugate pairs, a 50x50 tridiagonal with known spectrum, and a badly-scaled balance case. Roots are emitted as the same PZtrial list the Muller driver produces, so PZpost and all downstream output are unchanged. Option plumbing follows the standard task-option path (OPT_PZEIG -> TSKpzEig -> CKTpzEig flag -> dispatch in pzan.c). RESULTS: RLC bandpass -- same roots as Muller with NO iteration-limit warning; 10-section RC ladder -- all ten poles matching the analytic tridiagonal-eigenvalue formula $s_k = -(2-2\cos((2k-1)\pi/21))/RC$ and Muller root-for-root; twin-T -- all 6 roots under both solvers; RLC bandstop -- purely imaginary zeros +j1e6 EXACT; balanced/differential output works; purely resistive circuits yield no roots and no crash. DEFAULT REMAINS MULLER (pzeig is opt-in). Dense $O(n^2)$ memory / $O(n^3)$ QR, capped at 2000 unknowns with a clear error above. Accuracy is dense-QR class (absolute error $\sim \text{eps} \times \text{dominant root magnitude}$): an exact origin-zero can print as $\sim 1e-7$ when poles sit at 1e6 rad/s; Muller refines locally and can resolve wider per-root dynamic range, while pzeig never misses or invents a root -- the two methods agree to ≥ 6 digits on every battery circuit. Verified (verify_pzeig.py, 13 checks: both solvers x both methods + analytic anchors + default-method check). Full example regression 136/136.

E-174 *src/frontend/commands.c, examples/helpcmd_examples/**

help all crash fix (help string used as printf format) -- running `help all` (or `help montecarlo`) in the interactive shell crashed the shipped ngspice with 'Error: tvprintf failed / ERROR: fatal error in ngspice, exit(-1)'. Root cause: ngspice's help printer uses each command's help TEXT as a printf FORMAT string -- `out_printf(ccc[i]->co_help, cp_program)` in `com_help.c` (and the parallel `com_ahelp.c`) -- passing only one argument (`cp_program`, the program name, intended for a single `%s` as in the legitimate 'Report a `%s` bug.'). Any other '%' in a help string is an invalid/incomplete conversion specifier, so `tvprintf` fails and ngspice calls a fatal `exit(-1)`. The `montecarlo` command's help (added in E-151) read '... reports the yield with a Wilson 95% CI ...'; the '%' (percent-space) is the invalid specifier. `help all` walks the command table alphabetically and died at `montecarlo`; `help montecarlo` crashed directly too. Present in BOTH command tables (`spcp_coms` for ngspice, `nutcp_coms` for nutmeg). Not X11/terminal/solver specific -- plain format-string handling, so it reproduced on every build; it was invisible in headless CI only because `help all` is an interactive command not exercised by the batch example decks. Fix: escape the literal percent as `%%` (printf convention) in both tables, so the line renders 'Wilson 95% CI' and `help all` completes. New regression guard `examples/helpcmd_examples/verify_helpcmd.py` with two layers: RUNTIME -- drives an interactive ngspice on a pseudo-terminal and asserts `help`, `help all`, and `help montecarlo` each run to completion (no crash, no 'tvprintf failed') and that the `montecarlo` line renders a literal '95% CI'; and a STATIC class-guard that scans `commands.c` and fails if any command help string carries a format hazard (a '%' that is not '%s' or '%%', or more than one '%s' since only one argument is passed), catching a future unescaped '%' even when that command is never exercised at runtime. Full example regression 137/137.

E-175 *spicelib/analysis/dcpss.c, examples/rfconv_examples/**

RF-suite audit -- the dropped parametric term in every periodic small-signal analysis -- a full audit of the RF suite (.sp/Touchstone, PSS/PAC/pnoise/pxf, PSP, HB, QPSS + the two-tone family, hbosc/phasenoise) driven by analytic anchors and cross-analysis invariants. CLEAN: .sp + Touchstone (17 checks: series-R/shunt-C/3-port star S-parameters exact to 10 digits, the asymmetric-2-port S21/S12 column-order trap, reciprocity, RI/MA/DB + Y/Z (v1 normalization) + GHz round-trips through `wrsnp/rdsnp`, N-port 4-pairs-per-line layout); HB (analytic cubic: fundamental $a1A+(3/4)a3A^3$ and $H3=(1/4)a3A^3$ exact to 7 digits, even harmonics at machine zero, Sparse/KLU identical); QPSS-HB/QPSS-tran (two-tone fundamentals, $IM3 (3/4)a3A^3$ exact, commensurate aliasing guard fires correctly). THE BUG: every LPTV small-signal conversion matrix scaled its reactive coupling by the COLUMN (input-sideband) frequency, $H_{nm} = G_{\{n-m\}} + j\omega_{\text{mega}} mC_{\{n-m\}}$. For a small signal riding a FROZEN periodic capacitance $C(t)=dQ/dv|_{\text{orbit}}$ the exact linearization is $di = d/dt[C(t)dv] = C\dot{d}v + Cdv_{\dot{}}$, and the product rule collapses

to the ROW frequency: $H_{nm} = G_{\{n-m\}} + j\omega_n C_{\{n-m\}}$. Using ω_m silently DROPS THE PARAMETRIC-PUMPING TERM $C\dot{v}$ -- the very term that makes a pumped varactor convert. Every conversion sideband through a time-varying capacitance came out scaled by exactly ω_{in}/ω_{out} : on the audit circuit (1k -> OSDI varactor $C(v)=c_0(1+\alpha v)$ pumped 1V@1MHz, 1mV probe at 250kHz) the +1/+2 sidebands were 3x/5x/7x/9x too small, with measured pre-fix/truth ratios matching the omega ratios to FIVE DIGITS (0.3333/0.2000/0.14286/0.11111) -- diagnosis confirmed beyond doubt. AFFECTED: pac, psp, pnoise, pxf, qpac, qpnoise, qpxf, and the phasenoise adjoint, for any circuit with pumped/nonlinear capacitance (varactors, junction and MOS capacitances -- every real mixer and oscillator). UNAFFECTED: LTI circuits (no off-diagonal C harmonics; $\omega_n == \omega_m$ on the diagonal) -- exactly what every prior regression check used; the E-171 accidental-correctness pattern again. THE SUBTLETY: the harmonic-balance residual uses the SAME builders and must KEEP the column frequency -- its reactive current is the exact chain rule $d/dt Q(v(t)) = C(v(t))\dot{v}(t)$ whose n -th harmonic is $\sum_m C_{\{n-m\}}(j\omega_m V_m)$. That is why HB and QPSS-HB always matched transient ground truth while the small-signal analyses did not, and why a blanket fix would have broken HB. FIX: a smallsig mode flag on pac_build_matrix and qp_build_matrix plus the two inline copies (the PAC adjoint and the phasenoise adjoint): smallsig=1 -> row frequency (restores the parametric term); smallsig=0 (HB/HBOSC/QPSS-HB residual + Jacobian) -> chain-rule column frequency, byte-for-byte unchanged. Regression safety proven directly: the linear-circuit PAC control produces IDENTICAL output pre/post fix, and HB/QPSS-HB outputs are byte-identical. Ground truth = plain transient + one-beat Fourier projection (independent of all conversion-matrix machinery); the fixed qpac matches all five sidebands within 2% (integrator error class). Also noted: cktspnoise.c is a dead fossil (#ifdef RFSPICE_ never compiled). Verified (examples/rfconv_examples/verify_rfconv.py, 5 checks x both solvers: truth vs QPSS-HB unchanged, truth vs fixed qpac, pre-fix signature absent, HB + QPSS-HB analytic anchors). Full example regression 138/138.

E-176 [spicelib/analysis/dcpss.c](#), [examples/_setup.py](#), [examples/pssdriven_examples/*](#)

Driven-mode PSS shooting -- no frequency hunt, no breakpoint flood -- diagnosis of the E-175 audit's flagged residual ('PSS shooting robustness: the varcap PSS ran 17+ minutes without converging'). Instrumentation showed 9,627,161 accepted timepoints by $t=2\mu s$ (~0.2 ps per step, zero LTE rejections): ngspice's PSS (the Lannutti shooting) is OSCILLATOR-oriented -- it hunts the fundamental frequency through mid-period forced breakpoints whose spacing is proportional to steady_coeff. On driven circuits (everything the E-117..126 periodic small-signal stack runs on) this is doubly pathological: (1) at the standard decks' steady_coeff=5e-6 the grid spacing is ~0.5 ps -- millions of steps per shooting cycle, and the convergence-tolerance knob doubles as grid density so tighter tolerance quadratically explodes runtime (even the default 1e-3 forces thousands of points/period -- why a linear RC .pss took ~4 minutes); (2) the frequency estimate can never settle EXACTLY on the source frequency, the cycle endpoint phase-drifts against the source, and the period residual FLOORS (~1e-4) -- shooting never converges and burns the full sc_iter budget at flood speed. FIX: a DRIVEN mode, auto-detected when the circuit contains a time-varying independent source (SIN/PULSE/... V or I source, funcTGiven): the period is pinned to the exact source period (no estimation, no estimator grid; each cycle is one plain-transient period; varactor residual 2e-1 -> 8e-9 in 17 cycles / 0.3 s); the shooting-phase max step is clamped to $T/psspoints$ so the orbit is integrated on the SAME discretization as the retained samples (validation caught this: without the clamp LTE grows steps toward $T/2$ and shooting converges -- to 7e-9! -- onto the fixed point of that coarse discretization, several percent off the true orbit: retained swing 0.1651 vs analytic 0.15718; the old flood had been masking the effect by brute force); the post-FFT 'relaunch at the strongest spectral line' is disabled when driven (a rectifier-like circuit with a dominant 2nd harmonic must not retain the wrong period); the AUTONOMOUS oscillator path is untouched (byte-identical behavior verified on oscillator decks; DC-only-supply circuits never trigger detection). IMPACT: rc_pss ~4 min -> 0.05 s with the retained fundamental EXACTLY 1000000 Hz (the old hunt returned 999999.8976); psp example 406 s -> 0.9 s; the rfpss battery (5 verifies, 61 checks)

minutes-each -> <0.5 s each; rfanalyses 'several minutes even Sparse-only' -> 0.56 s; the KLU PSS pass 'prohibitively slow' -> 0.14 s. HARNESS: rfpss and rfanalyses removed from both REGRESSION_EXCLUDE and SPARSE_ONLY -- the whole RF periodic small-signal suite now runs in EVERY regression sweep under BOTH solvers; the direct .pac-on-pumped-varactor check became affordable, closing the E-175 loop on the direct PAC path (sidebands match transient ground truth within 1%). Verified (examples/pssdriven_examples/verify_pssdriven.py, 6 checks x both solvers: driven detection + exact retained frequency, retained swing == |H| within 0.1%, retained-period self-consistency, direct .pac vs transient truth, autonomous non-trigger). Full example regression 145/145.

E-177 *spicelib/analysis/dcpss.c, examples/pnoise_fold_examples/**

pnoise noise-folding referee + the folded-flicker frequency fix -- closes the E-175 audit's last honesty-ledger gap: periodic noise analysis (pnoise/qnoise/phasenoise) was 'correct by construction, not correct by measurement' -- the folding of device noise through LPTV conversion had never been checked against anything independent. THE REFEREE (three independent layers, on a 1-node LPTV-conductance circuit $G(t) = g_0 + g_1 \sin(\omega_0 t)$ chosen so conversion transfers are $O(1)$): (1) a from-scratch Python conversion matrix, $\text{onoise}(f) = \sum_k |Z_k(f)|^2 * S(|f + kf_0|)$ -- same physics, zero shared code; (2) a TRNOISE transient Monte-Carlo (no shared code OR theory; 198-segment Welch PSD, with the pumped/unpumped RATIO cancelling the TRNOISE amplitude convention); (3) the LTI (pnoise == .noise) and white limits. VERDICT ON THE WHITE PATH: measured-correct -- pnoise matched the referee to 6 digits at every frequency and the MC arbiter confirmed the ~1.86x folding ratio within statistical error; the adjoint, sideband sum and thermal densities are right. THE BUG THE REFEREE CAUGHT: the stationary sideband loops set the noise-evaluation frequency ONCE, to the output frequency (data.freq = freq outside the k loop). But the sideband-k adjoint carries noise ORIGINATING at $|f + kf_0|$ -- a frequency-dependent source PSD (flicker $1/f$, noise_table E-109) must be evaluated at the source-side frequency. The proof was digit-exact in both directions: pre-fix pnoise reproduced the referee's deliberately-wrong evaluate-at-f model DIGIT-FOR-DIGIT ($1.114323e-14$ vs $1.114323e-14$ at 1 kHz -- a 21% overestimate on this circuit, unbounded as $f \ll f_0$); post-fix it reproduces the correct model digit-for-digit (9.175472e-15). Why three generations of checks missed it: white noise is frequency-flat and LTI circuits have no $k \neq 0$ transfer -- the THIRD occurrence of the accidental-correctness pattern (E-171 determinant, E-175 parametric term). FIXED in all three stationary loops: pnoise (data.freq = |freq + kf₀| per sideband), qnoise (|f_{in} + k₁f₁ + k₂f₂| per harmonic), and phasenoise -- where it matters most: folded far-sideband blocks near a carrier were evaluated at the tiny OFFSET frequency, inflating the flicker ($1/f^3$) region. CYCLO SCOPE: the cyclo mode's time-domain identity $\text{onoise} = (1/P) \sum_s S(t_s) |A_s|^2$ treats the source PSD as frequency-flat across the folding span -- exact for modulated-white noise and for flicker on conversion-free circuits (the shipped E-125/126 use cases), not for flicker folded through $k \neq 0$ conversion (cannot be collapsed into that product with the aggregate device-noise API); documented at both cyclo sites, with the (now exact) stationary mode as the right tool when folded flicker matters. Verified (examples/pnoise_fold_examples/verify_pnoise_fold.py, 5 checks x both solvers, ~2 s thanks to E-176): white folding vs referee (<=0.1%), flicker folding vs referee (<=0.5%, sample-0 bias read from the retained orbit), pre-fix signature absent, pump/no-pump ratio == referee ratio, LTI limit == .noise. The rfpss/qpss/rcyclo suites pass unchanged (conversion-free or white -- provably unaffected). Full example regression 146/146.

E-178 *spicelib/analysis/dcpss.c, spicelib/devices/dio/dionoise.c, examples/cyclofold_examples/, examples/rfpss_examples/, examples/qpss_examples/**

exact separable cyclostationary folding + the HB DC-source fix -- removes the cyclo mode's frequency-flat limitation E-177 documented. For the separable model $S(t, f) = m(t)^2 g(f)$, $\text{onoise}(f) = \sum_g \sum_q |B_q(g)|^2 * g_g(|f + qf_0|)$ with $B_q(g) = (1/P) \sum_s m_g(t_s) dA_s(g) e^{j(q\theta_s)}$ and $A_s = \sum_k \Psi_k e^{-jk\theta_s}$. Per-generator amplitudes recovered with NO device-API change: the noise-summary machinery (prtSummary/outpVector, one density per generator;

family totals excluded by the name-prefix rule) plus load POLARIZATION against a fixed reference load $R = \Psi_0$ -- five DEVnoise sweeps per orbit sample ($A+-R$, $A+-jR$, R) give $c_{g,s} = \sqrt{S_g(t_s)} dA_s(g)$ up to a constant phase that cancels in $|B_q|^2$. Spectral shapes are MEASURED pointwise at the folded frequencies (no $1/f^{EF}$ assumption -- noise_table works) at several probe biases; flat generators keep the original time-domain identity (exact for ANY quadratic form incl. NevalSrc2-correlated pairs); stationary limit == the E-177 sum, flat limit == the old identity (Parseval). Ported 2-D to qpnoise ... cyclo ($B_{\{q1,q2\}}$ at $|f + q1 f1 + q2 f2|$). THE PHYSICS: 'flicker sees $\wedge^2$, white sees $\langle m^2 \rangle$ ' -- only the envelope's DC feeds the $1/f$ band; the old identity was 23% high ($8/\pi^2$) on $|\sin|$ -modulated flicker and 34% high on the conversion circuit; the new sweep lands on a from-scratch referee to $\leq 3e-4$ (the old binary reproduced the deliberately-flat model to $6e-5$ -- the error was the model, not the code). BONUS BUG (4th accidental-correctness occurrence): the cross-machinery check (same circuit through the QPSS-HB orbit) came out exactly 4x -- BOTH HB Newtons (hb E-134, qpss ... hb E-136) double-subtracted the DC sources (the settle-mode rhs folded into I_R already carries them; $-\lambda$ subtracts them again): every DC bias voltage converged to exactly 2x, silently corrupting all bias-dependent noise and conversion (flicker $\sim I^{AF}$ off by 2^{AF} , shot 2x) while AC content and every AC-driven validation stayed exact. Fixed by adding the unscaled DC source block back (net DC drive exactly $-\lambda$); qpnoise cyclo now agrees with pnoise cyclo DIGIT-FOR-DIGIT across the two orbit machineries. rccyclo's flicker law updated to the exact $\langle |I| \rangle^2 = (8/\pi^2) \langle I^2 \rangle$ (pinned to $1e-4$). 6 checks x both solvers; regression 147/147. FOLLOW-UP (same enhancement): the E-139 hard-pumped-diode deck exploded the cyclo path ($\sim 6.5e1 \text{ V}^2/\text{Hz}$) -- dionoise.c computes its three sidewall generators only when DIOresistSWGIVEN but the prtSummary block copies ALL DIONSRCs slots, so without a sidewall model the summary vector carried uninitialized stack garbage (latent in stock ngspice for decades; E-178 is the first consumer of per-generator densities). Fixed at the device (unwritten slots zeroed, dio/dionoise.c) and hardened in both cyclo consumers (finite non-negative PSD guards; garbage probe ratios fall back to the flat path). verify_qpnoise's 'cyclo > 2x stationary' diode expectation was an artifact of the garbage slots + doubled HB DC bias; replaced with a closed-form torus-average referee (resistive network => memoryless LPTV, instantaneous adjoints) matched to $\sim 1\%$.

E-179 *spicelib/analysis/tfanal.c*, *spicelib/analysis/cktsens.c*, *frontend/com_measure2.c*, *examples/stdaudit_examples/**

standard-analyses audit: referee battery + three fixes. The 'Standard analyses (analog)' gap-table rows are 1990s SPICE3 code whose VALUES had never been physics-checked. FIX 1 (tfanal.c, inherited from Berkeley SPICE3 1988): `.tf i(vm) vin` output impedance was ALWAYS $1e20$ -- the output-impedance solve forces 1V on the sense branch and inverts the branch current as $1/\text{MAX}(1e-20, \text{rhs})$, but that current is NEGATIVE for every passive network (the input-impedance path right above divides by -rhs); fixed with the same sign + fabs guard; feedback-amp referee exact ($1000.990 == RL + \text{node Thevenin}$). FIX 2 (cktsens.c): KLU AC sensitivity silently truncated to ONE frequency point -- the E-114 KLU complex-conversion block inside the frequency loop reused `i` (the OUTER loop variable) for its DEVmaxnum walk, so after the first point `i=DEVmaxnum` ended the sweep; the surviving point was correct, which is why E-62's single-point check passed. Fixed with a local index (dead second block hardened too); full sweep == analytic $dV/dC = -j\omega R/(1+j\omega RC)^2$ to 6 digits, both solvers. FIX 3 (com_measure2.c): `.meas DERIV{ATIVE}` was parsed but evaluation fell into an explicit 'currently not supported' stub (empty `#if 0 measure_deriv()` placeholder) -- implemented: 3-point Lagrange-quadratic derivative on the nonuniform time grid, AT= and WHEN forms sharing FIND's parse/locate flow, verified vs the analytic sine slope to 5 digits; DERIVATIVE/INTEGRAL long spellings accepted (only DERIV/INTEG matched); pre-existing -Wmissing-prototypes warning silenced (`str_has_arith_char2` static). MEASURED-CORRECT: `.disto HD2/HD3` == analytic diode Volterra kernels $\leq 0.06\%$ incl. a frequency-dependent load (harmonics at $Z(2\omega)/Z(3\omega)$ -- no E-177-style frequency bug), SIM2 intermods ($f1+f2$, $f1-f2$, $2f1-f2$) incl. cascades, exact DISTOF1/DISTOF2 scaling, junction-cap harmonics == Harmonic Balance to ~ 6 digits (two independent engines);

.noise onoise_total^2 == band analytic (kT/C) 0.06% + flicker log-integral 6 digits; .sens DC == central finite differences at a nonlinear OP (5-6 digits, model params incl.); nonlinear-OP .pz works via the driving-point form (the E-62 'quirk' was a bias source shorting the injection node -- ngspice's refusal is correct); tran/op/meas battery exact. 8 checks x both solvers; regression 148/148.

E-180 *frontend/com_checkpoint.c, examples/checkpoint_examples/**

checkpoint/restart under KLU (the last Sparse-only feature falls). E-131 guarded savestate/loadstate off under KLU, recording the reason as 'KLU factorization objects absent on the restore path' -- a mis-diagnosis. REAL cause: .option klu lives in the task (TSKkluMODE) and reaches the circuit only in CKTdoJob at analysis dispatch, but loadstate calls CKTsetup DIRECTLY first, so the matrix was built Sparse (SMPkluMatrix NULL); the resume dispatch then set ckt->CKTkluMODE=1 without re-setup (by design -- it must preserve the restored state), and the first NIiter dereferenced the NULL SMPkluMatrix (KLUloadDiagGmin store; reproduced EXC_BAD_ACCESS +0x90). savestate was never broken (the checkpoint file holds only solver-agnostic vectors). FIX: com_loadstate copies the task's solver selection and the E-152 KLU knobs (ordering/scale/BTF/memgrow) into the circuit BEFORE CKTsetup, exactly CKTdoJob's block; both guards removed. BONUS: cross-solver restore -- all four save x load combinations (fresh processes) continue exactly on the uninterrupted trajectory ($\leq 1e-9$ at $t=3.5ms$ of a 4ms diode+RC run). verify_checkpoint's 'KLU rejected' check replaced by the 4-combo matrix (22/22); solver-notes and gap-analysis tables updated. Nothing in the simulator is Sparse-only any more (E-172 closed the last analysis, E-180 the last feature). Regression 148/148.

E-181 *spicelib/analysis/dctran.c, spicelib/analysis/cktsort.c, spicelib/analysis/ckntask.c, spicelib/analysis/cktjob.c, include/ngspice/cktddefs.h, include/ngspice/tskdefs.h, include/ngspice/optdefs.h, examples/corenum_examples/**

core-numeric audit: the integrator certified exactly + **.options ordfix**. Nobody had ever verified that the Gear (BDF) corrector coefficients deliver their nominal order (orders 3-6 were dead code for ~30 years until E-128 selected them; E-128 verified order SELECTION, not coefficient CORRECTNESS). NEW OPTION ordfix=K (off by default): a fixed-order verification mode -- every converged step is accepted (the step is ruled by tmax, not the LTE), the first step is shrunk 1000x (an order-1 starter proportional to the pinned step imposes an $O(h^2)$ floor), the order ramps to K while the step is tiny, then the step grows gently ($\times 1.15$; large ratios destabilize high-order variable-step BDF). Each guard defeats a measurement trap that initially presented as a bug but is real numerics (incl. the LTE step preference $h_k = (c_k \cdot tol)^{1/(k+1)}$ INVERTING at loose tolerance -- the E-128 controller's refusal to raise there is correct). CERTIFIED: accepted trajectories satisfy the exact variable-step BDF-k formula (Lagrange differentiation on the actual nodes) to $\leq 1.3e-13$ at every order 1-6 (~90 uniform stencils each); measured global slopes confirm orders 1-3; trap-vs-Gear dissipation on a lossless LC matches theory ($|R(iy)|=1$ vs order-dependent damping). Also referee'd with NO defects: solver precision == conditioning theory under both solvers (1e18-conductance-spread asymmetric mesh == exact-rational MNA to 1e-15); all four DC convergence-aid paths (default/noopiter/gminsteps=0/srcsteps=0) agree on a 40-diode hard-DC chain (spread 2.2e-6, KLU==Sparse 7e-13); xmu=0.5 == trap bit-identical, xmu=0.45 damps; lvltim=1 works. Drive-by: DCtran_step_quit prototype added to cktddefs.h (pre-existing -Wmissing-prototypes). 8 checks x both solvers; regression 149/149.

E-182 *frontend/plotting/plotit.c, frontend/plotting/pyplot.c, examples/pyplot_examples/**

pyplot autoscale by default -- the matplotlib bridge (E-94/95/98/99) pinned every generated axis with set_xlim/set_ylim taken from ngspice's grid-rounded internal plot ranges. User request: rely on matplotlib's autoscaling + fig.tight_layout() instead. plotit() always fills xlims[]/ylim[] (user args OR data-derived), and the statics that distinguish the user case (xlim/ylim from getlims) are freed BEFORE the device dispatch -- plotit now captures user_xlim/user_ylim flags at the free point and passes NULL limits to ft_pyplot unless the user gave xlimit/ylim explicitly (ft_pyplot

already handled NULL; gnuplot/asciiplot back-ends untouched). `verify_pyplot` +2 checks (no pins + `tight_layout` on auto plots; exact pins for explicit limits). Regression 149/149.

E-183 *frontend/com_pyplot.c, frontend/plotting/pyplot.c, examples/pyplot_examples/, features/*

pyplot usability: distinct default names, deck-folder output, linewidth & backend. (1) REAL BUG: in window (interactive) mode pyplot launches the Python viewer in the BACKGROUND (`python3 pyplot.py &`); two plots that both omit the file name shared `pyplot.py/pyplot.data`, and the 2nd call overwrote them before the 1st viewer read them -- both windows showed the 2nd plot (its title and data). Invisible in PNG mode (synchronous). Fix (`com_pyplot.c`): a per-session counter names successive no-name plots `pyplot`, `pyplot-2`, ... (first unchanged). (2) The `.py/.data/.png` are written next to the CIRCUIT FILE via `ngdirname(ci_filename)` rather than `ngspice's` `cwd`; the script's `loadtxt/savefig` carry the deck-dir path so Python finds the data from any `cwd`; a bare in-folder deck name still lands in the `cwd` (unchanged). (3) `set pyplot_linewidth=<w> -> linewidth` in the `plot()/step()` calls. (4) `set pyplot_backend=<name> -> matplotlib.use()` ahead of `import pyplot`, overriding the auto backend (incl. the Agg forced by the `png/svg/pdf` terminals). All four opt-in; default output byte-unchanged. `filename` buffers in `pyplot.c` grown 128->1024 for full paths. `verify_pyplot` grows to 13 checks x both solvers; packaged as a standalone feature bundle under `features/`. Regression 149/149.

E-184 *frontend/outitf.c, examples/progressbar_examples/**

sweep progress bar reaches 100%. Fix to the E-129 progress bar: it sometimes froze below 100% at end of run (e.g. `'[==== ] 82%'` then `'No. of Data Rows'`). CAUSE: the 'Reference value' status line is throttled -- redrawn only when `>0.25 s` has elapsed since the last update -- so the FINAL sweep point's print is almost always skipped (it lands within 0.25 s of the previous tick), freezing the bar at the last throttled value; reaching 100% was never reliable. FIX: `outp_print_reference` records the latest `refval` + a bar-shown flag every call; a new `outp_finish_reference`, called from `fileEnd/plotEnd` just before the `'No. of Data Rows'` line, reprints the bar full at 100% in place using the sweep's TRUE endpoint (`TRAN CKTfinalTime`, `AC ACstopFreq`, `NOISE NstopFreq`, `DC` last source value). One-shot (reset at analysis start and after firing), silent for non-swept analyses (`op/tf`), same `ft_optimizing/ft_norefprint/cp_background` guards -- quiet/background/optimizer runs and the rawfile unaffected. `verify_progressbar's` end-of-run check tightened from `>=90%` to `==100%` (all four swept analyses); 22/22. Regression 149/149.

E-188 *spicelib/analysis/dcop.c, include/ngspice/cktdefs.h, frontend/com_sweep.c, examples/warmstart_examples/**

Monte Carlo warm-start (`montecarlo ... -warm`). Each MC sample (E-151) re-sources the deck and COLD-solves its DC bias point, running the full `gmin/source-stepping` homotopy from scratch even though consecutive samples move the operating point only slightly. Profiling: the solve, not the deck re-source, dominates for non-trivial circuits, and every op cold-starts -- two IDENTICAL ops back-to-back each burn the full homotopy (~52 Newton iters on a diode ladder; a warm .nodeset at the solution cuts it to ~5). CKTop tries a direct Newton from the caller's `firstmode` before any homotopy, but DCop always passes `MODEINITJCT` (cold junction voltages), discarding the previous solution. FIX: `montecarlo -warm` wraps the loop in `CKTsetWarmStart(1/0)`; DCop keeps a buffer (outside the `CKTcircuit` that reset recreates; indexed by equation number, stable across resets of an identical-topology deck) of the last converged `CKTrhsOld`, preloads it and calls CKTop with `MODEINITFLOAT` (use existing node voltages) when a valid same-size guess exists, and snapshots each converged solution. A poor guess just fails the first NIiter and falls through to the normal cold homotopy, so the converged point is identical -- only a cheap failed attempt is wasted. First sample cold, rest warm; when `-warm` is absent `dcop_warm_enable=0` and the path is byte-identical (one extra `SMPmatSize` read). CORRECTNESS: warm changes only Newton's starting point, so it converges to the same OP to within convergence tolerance -- `warm==cold` yield EXACTLY at `reltol=1e-6`, agreeing to within a couple boundary samples at default `reltol` (`v(3)~3.8V`, where `reltol*|v|~` the narrow spec band; a tolerance effect, removed by tightening `rel-`

tol). Iteration count ~52->~4 (~13x); wall-clock benefit scales with per-iteration cost (large/hard-converging designs win most; small circuits are dominated by deck re-source + command dispatch). Shared DC-op code, so Sparse+KLU; composes with -lhs. 5 checks (exact at tight reltol, close at default, iteration cut, +lhs, +klu). Regression 152/152.

E-189 *frontend/com_sweep.c, examples/sweepwave_examples/**

Sweep waveform overlay (**sweep ... -overlay**). A usability follow-up to the E-146 sweep command: sweep steps any knob, runs an inner analysis per point, and records each output's LAST value into a summary sweep transfer curve -- but overlaying the full per-point WAVEFORMS to compare shapes meant setplot-ing each retained run by hand. -overlay (alias -ov) collects every point's whole output waveform, resamples them onto a common independent-variable grid, and builds one sweepwave plot with a <output>_<value> vector per (output, knob value), left current so plot <out>_<val> ... works at once -- the HSPICE .step overlay in one step. Three static helpers in com_sweep.c: sw_eval_vec (evaluate the output expr and copy its FULL waveform + scale into caller buffers; complex/AC stored as magnitude), sw_interp (binary-search piecewise-linear resample, flat outside range -- each run lands on its own adaptive time/freq grid, so a shared grid is required before a common scale vector), sw_wavename (clean nutmeg name, non-alnum -> _). The sweep loop captures each waveform alongside the unchanged last-value recording; after the loop a common grid over [min,max] of all runs at the finest resolution is built, every waveform resampled onto it. GRACEFUL: a scalar inner analysis (op, no waveform) prints "-overlay ignored (analysis '...' has no waveform to overlay)" and yields the ordinary summary -- never an error. Without the flag the path is byte-identical to E-146. Front-end only, solver-independent. Verify: overlaid RC step responses match $1 - \exp(-t/RC)$ ($RC=1/2/4\mu s$) to worst $<1e-4$, distinct per-value vectors, graceful op, summary curve intact. 5 checks. Regression 153/153.

E-190 *frontend/com_sweep.c, examples/nested_sweep_examples/**

Nested multi-knob sweep (**sweep ... -vs ...**). Second usability follow-up to the E-146 sweep command (after E-189 -overlay). sweep stepped ONE knob; -vs <knob> <spec> (alias -family) adds OUTER knobs whose CARTESIAN PRODUCT forms a curve family: the inner (positional) knob stays the x-axis of the summary sweep plot, and each combination of the outer knobs yields one curve per output, named <output>_<outerknob>_<value>... -- the parametric family of .dc ... SWEEP ... / HSPICE nested .dc. Up to SW_MAXKNOB=4 knobs, cartesian points capped at SW_MAXPTS. The three-form spec parser (lin/dec/oct, list, start/stop/step) was factored into sw_parse_spec, now shared by the inner knob and each -vs knob so all knobs accept the same forms. SET ORDERING (the subtle part): each knob is applied with its E-146 mechanism -- a .param needs alterparam+reset (re-source), an instance/model knob is in-place alter/altermod -- and a reset drops every in-place alter, so the loop iterates inner-fastest and re-resources ONCE only when a .param knob's value actually changed (outer-loop boundaries; an inner .param still resets every point, a pure in-place sweep never resets), staging all .params via alterparam before the single reset, then RE-APPLYING every in-place knob after the reset (so an inner alter survives an outer .param re-source). By construction a single knob reduces EXACTLY to E-146: one outer combination -> curve named <output>, E-146 messages, identical data indexing (existing sweep/sweepwave examples pass untouched). -overlay composes: each cartesian point's full waveform becomes a sweepwave vector named with EVERY knob value (inner first), so a single knob is <output>_<value> exactly as E-189. New static helpers: sw_parse_spec, sw_set_deck/sw_set_inplace (split from the old sw_set), sw_familyname (summary curve name over outer knobs), sw_pointname (overlay name over all knobs; replaces sw_wavename). Verify: RC low-pass step, R inner x C outer, each family curve $== 1 - \exp(-T/(R*C))$ vs R to worst $<5e-6$, incl. a .param-OUTER knob (reset/alter ordering); cartesian run/curve counts; -overlay family names carry both knob values; single knob still names its curve vo. 5 checks. Regression 154/154.

E-191 *spicelib/analysis/acan.c, spicelib/analysis/span.c, examples/aclin2_examples/**

.ac lin 2 / .sp lin 2 off-by-one fix. Found while auditing the data-output path (wldata/wrsnp/save/rawfile write): a ragged-length wldata test wrote an AC vector one point short, which traced not to the writer but to the AC LINEAR frequency sweep. The AC (acan.c) and S-parameter (span.c) analyses computed the linear step behind if (numberSteps - 1 > 1) -- true only for N>=3 -- so exactly **lin 2** fell through to the single-point patch (freqDelta = 0). With a zero step the sweep loop (freq += freqDelta; if (freqDelta == 0) goto endsweep;) stops after ONE point, silently dropping the fstop point: a two-point linear AC/SP sweep produced one row. dec/oct (different step formula) and **lin 1/3/4/...** were all correct; only N=2. The single-point patch (Richard McRoberts) was meant for **lin 1**; - 1 > 1 over-reached by one. FIX: guard on numberSteps > 1 in both files (the form noisean.c already used correctly), so N=1 -> one point at fstart (patch preserved), N=2 -> freqDelta = fstop-fstart -> two endpoints, N>=3 unchanged. The distortion analysis uses a different N+1-point convention (unaffected); noise was already correct. Verify: RC low-pass, both **lin 2** endpoints are GENUINE solved points matching $1/(1+j\omega RC)$ to <1e-6 (not a duplicate); **lin 1** stays one point; **lin N** gives N linearly-spaced points (N=3,5,10); **.sp lin 2** gives two points with analytic series-R S-params (S11=1/3, S21=2/3). Both solvers. The wider output-path audit found the writers themselves CLEAN: Touchstone wrs2p/wrsnp/rdsnp round-trip across RI/MA/DB, S/Y/Z (Rbase normalization Y*R / Z/R), Hz-GHz incl. complex angles; 2-port special S11 S21 S12 S22 order + N-port row-major both correct for round-trip AND spec interop; wldata complex (re,im)/singlescale/ragged padding; save filtering restricts the vector set; rawfile write/load bit-exact at 12 digits. 5 checks. Regression 155/155.

E-192 *frontend/com_checkpoint.c, frontend/com_checkpoint.h, frontend/runcoms.c, examples/autosave_examples/**

Auto-checkpoint on interrupt (**set autosave=<file>**). Follow-up to E-131 checkpoint/restart, prompted by "does savestate fire on Ctrl-C?" -- it did not. E-131 gave savestate/loadstate but a checkpoint had to be taken by hand, so an interrupted long transient lost its progress. E-192: set autosave=<file> makes an interrupted transient auto-write an E-131 checkpoint. SAFE HOOK POINT (the crux): writing a file from a signal handler is undefined behavior, so E-192 does NOT checkpoint in ft_sigintr -- that handler only sets ft_intrpt and returns (during a run ft_setflag is TRUE, so no longjmp). At the next ACCEPTED timepoint, dctran's IFpauseTest (which is OUTstopnow) sees the flag and returns 1, so dctran returns E_PAUSE; if_run maps E_PAUSE to return code 1; dosim (runcoms.c) reports "interrupted" (err equals 1). E-192 hooks exactly that interrupted branch -- on the MAIN thread, at a consistent timestep boundary, with ordinary buffered I/O. CHANGE: com_savestate's body factored into ckt_write_checkpoint(ckt, fname) (the command is now a thin wrapper); the hook calls the SAME function, so an autosave file is byte-for-byte what savestate writes. In the interrupted branch: if cp_getvar("autosave", ...) yields a filename AND the run was a transient (CKTmode and MODETRAN set, CKTtime above 0), call ckt_write_checkpoint. OPT-IN (no var means byte-identical old behavior, nothing written), TRANSIENT-ONLY (the MODETRAN / CKTtime guard blocks stale state after, e.g., an interrupted AC). Interactive-mode only: the SIGINT handler is installed only under non-batch mode, so a batch-mode Ctrl-C still terminates (unchanged). A real Ctrl-C (SIGINT), a stop when ... breakpoint, and a Verilog-A stop task all funnel through the same interrupted branch (they differ only in how the flag is set, upstream of E-192). CORRECTNESS: autosave file byte-identical to a manual savestate at the same pause; loadstate resumes it exactly (E-131 round-trip). Verify (examples/autosave_examples): stop-paused transient writes a checkpoint; equals a manual savestate (byte-identical); loads/resumes; opt-in gate (no var means no file); and a REAL Ctrl-C (SIGINT) via a PTY (interactive) fires it (lenient: skip if the run finishes before the signal, the hook being covered deterministically). E-131 checkpoint example still passes (savestate refactor). 5 checks. Regression 156/156.

E-193 *spicelib/analysis/dcpss.c, examples/pnoiseunits_examples/*, examples/rfpss_examples/verify_rcpnoise.py, examples/rfpss_examples/verify_rcyclo.py, examples/cyclofold_examples/verify_*

cyclofold.py, examples/pnoise/pnoise_examples/verify_pnoise/pnoise_examples/verify_pnoise.py, examples/rfpss_examples/rc_pac.cir

.pnoise honors **sqrnoise** (noise-density units fix). Found auditing the RF/PSS suite. The ngspice manual (sec. **.noise**, pp. 326-327) defines **onoise_spectrum/inoise_spectrum** as a noise spectral DENSITY in V/sqrt(Hz) (or A/sqrt(Hz)) by default, switchable to the squared V^2/Hz form by the **sqrnoise** control variable. **.noise** obeys this (noisean.c reads **cp_getvar("sqrnoise")** and **sqr**'s **onoise/inoise** when unset). But **.pnoise** (periodic noise) ALWAYS emitted the squared V^2/Hz density and IGNORED **sqrnoise** -- so the same-named vectors carried DIFFERENT units across the two analyses (differing by a square), and the documented **sqrnoise** control silently did nothing for **pnoise**. Confirmed by the toggle: **.noise** flips 4.063e-9 (V/sqrt Hz) <-> 1.651e-17 (V^2/Hz) with **set sqrnoise**; **.pnoise** returned 1.651e-17 both ways. FIX: **pnoise_sweep** (dcpss.c) now reads **sqrnoise** via **cp_getvar** (added **#include cpextern.h**) and, when unset (the default), applies **sqr** to the emitted **onoise/inoise** densities -> V/sqrt(Hz); **set sqrnoise -> V^2/Hz**. BOTH emit paths get the conditional **sqr**: the E-178 cyclostationary folding branch and the standard adjoint-folding branch. Now **.pnoise** and **.noise** agree by default (match to 0.00). SCOPE: **.pnoise** only. The two-tone **qpnoise** command's single-point diagnostic already prints BOTH forms explicitly (V^2/Hz and V/sqrt(Hz)) so there is no ambiguity, and its swept + single-point paths share the V^2/Hz convention; an initial **QpnoiseSweep** change was REVERTED to avoid an internal swept-vs-single-point split (**verify_qpss_sweep** [3] compares them). EXAMPLES: **rc_pnoise** now checks the default V/sqrt(Hz) form (**sqr** of the analytic) and cross-checks vs **.noise** without forcing **sqrnoise** (match to 0.00); **rc_cyclo/cyclofold/pnoise/pnoise_examples** are power-density relations (**onoise == ..., referee sums |Z|^2S**) so they set **sqrnoise** to keep V^2/Hz, and two LTI-limit checks that compared **pnoise(V^2)** to **.noise(V/sqrt Hz)^2** became direct equalities. WIDER AUDIT (all correct): PAC driving-point + the +-1 CONVERSION SIDEBANDS (via the **.pac** card's trailing **maxsideband** arg) match a clean transient DFT to ~0.1%; Harmonic Balance matches a transient DFT to 4-5 digits (an apparent 12x **.disto** HD3 and 2.4x PAC-sideband discrepancy both dissolved into **fourier-command** interpolation error at high harmonics, resolved by a proper DFT). Also documented the previously-undocumented **.pac** trailing **maxsideband** arg (dcpss.c comment + **rc_pac.cir**). Verify (**examples/pnoiseunits**): default **pnoise == sqrt(4kTR/(1+(wRC)^2))** (V/sqrt Hz); **set sqrnoise == the density (V^2/Hz)**; default^2 == squared (exact **sqr** relation); default **pnoise == default .noise (0.00)**. 4 checks. Regression 157/157.

E-194 *frontend/com_optimize.c, examples/pssopt_examples/**

Particle swarm optimization (**optimize -method pso**). A new search method for the built-in optimizer. E-130/143 gave two LOCAL methods -- Nelder-Mead simplex (**-method nm, scalar -minimize**) and Levenberg-Marquardt (**-method lm, gradient least-squares over -targets**) -- both of which descend into whichever basin the start point sits in, so on a MULTIMODAL objective they return the local minimum nearest the start. E-194 adds a GLOBAL, population-based, derivative-free method: PSO. N particles fly through the normalized [0,1]^np box; each keeps its own best-seen point (**pbest**) and the swarm shares a global best (**gbest**); each iteration updates every velocity $v = \chi * (v + \phi_1 * r_1 * (pbest - x) + \phi_2 * r_2 * (gbest - x))$ and moves $x = \text{clamp}_{01}(x + v)$ with the standard Clerc-Kennedy constriction ($\chi = 0.72984$, $\phi = 2.05$ so the swarm converges without exploding), velocities clamped to half the box, positions to [0,1]. Particle 0 starts at the user init point, the rest uniform-random, so the whole box is explored. Reuses the existing **opt_eval** (returns the scalar cost for both objective kinds), so PSO works for scalar **-minimize** AND **-target** least-squares -- unlike LM, which needs targets. New options: **-swarmsize <N>** (aliases **-swarm/-npart**; default **auto = 10+4np capped at 60**), **-seed <s>** (PSO draws from a small self-contained **splitmix64** PRNG independent of ngspice's global RNG, so a seed gives byte-identical results), **shared -maxiter/-tol** (**tol** is the **gbest-stagnation tolerance** held over several iterations). The **nm/lm dispatch** is unchanged. Verify (**examples/pssopt**): on the classic multimodal $f(p) = \sin(p) + \sin(10 * p/3)$ over [2.7, 7.5] (global $p = 5.1457$ $f^* = -1.8996$, several higher local minima) started at the $p = 2.7$ corner,

PSO finds the global (-1.8996, p=5.145) while Nelder-Mead from the same start is trapped at a local minimum (-1.19992, p=3.387); a fixed seed is exactly reproducible; independent seeds all reach the global basin; PSO also solves a -target least-squares objective (recovers p to a 1e-12 residual) and a 2-D separable multimodal min (f*=-3.799). Existing optimize example (nm+lm, 20 checks) unchanged. 6 checks. Regression 158/158.

E-195 *frontend/com_optimize.c, examples/deopt_examples/**

Differential evolution (**optimize -method de**). A second GLOBAL method for the built-in optimizer, after PSO (E-194), from the same population-based derivative-free family. DE keeps a population of NP vectors in the normalized $[0,1]^{np}$ box; where PSO pulls trials toward remembered bests, DE (classic DE/rand/1/bin) builds each trial from a scaled DIFFERENCE of random members and crosses it with the target: $v = a + F*(b-c)$ ($F=0.8$; a,b,c distinct and $\neq i$), $u[j] = v[j]$ if $\text{rand} < CR$ or $j == j_{\text{rand}}$ else $x[i][j]$ ($CR=0.9$ binomial crossover, one dimension always taken), then greedy selection (keep u in place of x[i] iff $\text{cost}(u) \leq \text{cost}(x[i])$). The difference vector b-c self-scales to the population's own spread -- large while members are far apart (broad exploration), shrinking as they converge (fine refinement) -- which makes DE robust on rugged / discontinuous / poorly-scaled landscapes with no per-problem step tuning. Reuses the existing opt_eval (scalar cost for both objective kinds), so it works for scalar -minimize AND -target least-squares like PSO. Shares PSO's infrastructure: the same self-contained splitmix64 PRNG (so -seed is byte-reproducible), the -swarmsize population option (default auto $10+4np$ capped at 60; clamped to ≥ 5 because DE needs 4 distinct members to form a mutant), and shared -maxiter/-tol (gbest-stagnation). The nm/lm/psa dispatch is unchanged; DE is only selected by -method de (aliases diffevol, differentialevolution). Verify (examples/deopt): on the multimodal $f(p)=\sin(p)+\sin(10*p/3)$ over $[2.7,7.5]$ (global $p=5.1457$ $f^*=-1.8996$) started at the trapping $p=2.7$ corner, DE finds the global (-1.8996, $p=5.14574$ -- matches p^* to 5 digits) while Nelder-Mead from the same start is trapped at a local minimum (-1.19992, $p=3.387$); a fixed seed is exactly reproducible; independent seeds all reach the global basin; DE also solves a -target least-squares objective (recovers p to a 4e-13 residual) and a 2-D separable multimodal min ($f^*=-3.799$); and DE and PSO both reach the global while the local NM does not. Existing optimize (nm+lm, 20 checks) and pssopt (6 checks) examples unchanged. 7 checks. Regression 159/159.

E-196 *frontend/com_optimize.c, examples/saopt_examples/**

Simulated annealing (**optimize -method sa**). A third GLOBAL method for the built-in optimizer, completing the set: after PSO (E-194) and DE (E-195), both population-based, SA is the classic SINGLE-WALKER global optimizer. It holds one current point in the normalized $[0,1]^{np}$ box; each step it proposes a random neighbour $x_n = \text{clamp}_{01}(x + \text{step} * U(-1,1))$ and applies the Metropolis rule -- accept if $\text{cost}(x_n) \leq \text{cost}(x)$ (downhill), else accept anyway with probability $\exp(-(\text{cost}(x_n)-\text{cost}(x))/T)$ (uphill). While the temperature T is high uphill moves are readily accepted so the walker climbs OUT of local minima; T is then cooled geometrically ($T *= \alpha$, alpha set so T drops ~4 decades over the -maxiter cooling levels, with $L=8+4np$ moves per level), so uphill moves become rare and it settles into a minimum. The best point ever visited is returned. ONE candidate per step (no population) -> the lightest-weight global method, the natural choice when each analysis is expensive. Everything auto-scales to the problem: $T_0 = \text{mean } |D\text{cost}|$ of a dozen random probes (so an uphill move of that size is accepted about half the time when hot), and the step size $= 0.30 * \sqrt{T/T_0} + 0.02$ shrinks as T cools (broad exploration hot, fine refinement cold) -- nothing to tune. Reuses opt_eval (scalar cost both kinds) -> works for scalar -minimize AND -target least-squares. Shares the self-contained splitmix64 PRNG (-seed byte-reproducible); -maxiter is the number of cooling levels. The nm/lm/psa/de dispatch is unchanged; SA is only -method sa (aliases anneal, simulatedannealing). A single walker refines more loosely than a population, so SA reliably reaches the global BASIN rather than machine precision (a short nm/lm polish from its result sharpens it when that matters). Verify (examples/saopt): on the multimodal $f(p)=\sin(p)+\sin(10*p/3)$ over $[2.7,7.5]$ (global $p=5.1457$ $f^*=-1.8996$) from the trapping $p=2.7$ corner, SA finds the global basin ($f < -1.85$, $|p-p^*| < 0.05$) while Nelder-Mead is trapped at a

local minimum (-1.19992, p=3.387); a fixed seed is reproducible; independent seeds all reach the global basin; SA solves -target least-squares and a 2-D multimodal min; and all THREE global methods (sa/psa/de) reach the global while the local nm does not. Existing optimize (20) + pssopt (6) + deopt (7) examples unchanged. 7 checks. Regression 160/160.

E-197 *frontend/com_optimize.c, examples/opt100_examples/**

A 100-parameter circuit optimization (raised **optimize** limits). The built-in optimizer sized its per-run arrays from two compile-time macros; E-197 raises them from OPT_MAXP 16 to 128 (parameters) and OPT_MAXT 64 to 128 (least-squares targets), and lifts the auto-population cap for the PSO/DE population methods from 60 to 256, so a genuinely large problem fits in a single call. Exceeding a cap is reported cleanly (optimize: too many -param (max 128)), not crashed. Testbed: 100 independent 1 mA sources, each into an unknown resistor R_i to ground (so $v(n_i) = 1e-3 * R_i$), fitted to 100 targets $t_i = 1e-3 * (100 + 9800*i/99)$ ohm -- a linear voltage ramp 0.1 .. 9.9 V, a well-posed separable 100-D least-squares. Levenberg-Marquardt is the right tool for a well-posed high-D fit and solves it to machine precision (sum-sq residual 2.5e-29, rms 5e-16, 619 evaluations, ~2 s) with all 100 parameters and 100 targets honored. Two fixes make the derivative-free GLOBAL methods FUNCTION at that scale (they are intrinsically far slower than LM in high dimension, but were failing outright): (1) adaptive DE crossover -- classic DE/rand/1/bin uses CR=0.9, mutating almost every coordinate of a trial; in high dimension a trial that perturbs ~n coordinates at once is essentially always worse than its parent and rejected by greedy selection, so DE freezes at its starting guess. E-197 sets $CR = (n \leq 16) ? 0.9 : 15.0/n$, capping the expected mutated-coordinate count at ~15 for large n (exactly the classic 0.9 for $n \leq 16$, so low-D DE is unchanged), and DE descends again -- from ~1240 to ~882 over 35 generations at 100-D. (2) Dimension-scaled convergence patience -- high-D population runs plateau for several generations between improvements, so the gbest-stagnation counter that stops PSO/DE would cut a still-descending run short; the threshold now grows as $8 + n/4$ (exactly 8 for $n \leq 3$, so small-n tests are unchanged), giving 33 generations of patience at n=100. DE at 100-D remains a genuinely hard global search that does not fully converge in a short budget -- intrinsic to global optimization in high dimension, not a defect; the message is the division of labor between LM (well-posed high-D fits) and the global methods (multimodal, now usable as dimension grows). Verify (examples/opt100): LM fits all 100 params to 100 targets at machine precision (both raised caps proven at once); DE descends substantially at 100-D (self-calibrated against its own starting cost, $\geq 15\%$ reduction); a fixed -seed is byte-reproducible even at 100 dimensions; and a 129-parameter over-cap run is reported cleanly. Existing optimize (39) + pssopt (6) + deopt (7) + saopt (7) examples unchanged. 4 checks. Regression 161/161.

E-198 *frontend/com_stb.c, frontend/commands.c, frontend/com_commands.h, frontend/Makefile.am, examples/stb_examples/**

stb stability / loop-gain analysis. A new front-end command that measures a feedback loop's small-signal loop gain $T(f)$ and reports its phase and gain margin -- the everyday stability check for op-amps, regulators, PLLs -- which ngspice lacked (you had to hand-roll a .control script). It is the one-command equivalent of Spectre's stb (Middlebrook-Tian double injection). Usage `stb <Vprobe> <Iprobe> <ac-sweep>`: the user marks the loop break with a bias-transparent probe pair between the driving node A and the loaded node B -- a series `Vstb A B dc 0 ac 0` (+node = driver A, -node = load B) and a shunt `Istb 0 B dc 0 ac 0` (ground to load B). Breaking a loop and injecting a single test signal mis-reads the loop gain because of the LOADING at the break (a series voltage injection is exact only from a zero-impedance source into an infinite-impedance load); the double injection cancels that error. stb runs two AC sweeps by altering the probe AC magnitudes: voltage injection (`Vstb ac=1`) gives $T_v = -v(A)/v(B)$; current injection (`Istb ac=1`) gives the current loop gain from the probe branch current $a = i(Vstb)$ as $T_i = -a/(a+1)$; combined $T = (T_v T_i - 1)/(T_v + T_i + 2)$. *The combined T is INDEPENDENT of where the probe sits in the loop -- the property a single injection lacks -- verified directly (a loaded break gives the same margins as a clean break in the same loop). Numerically, the common clean-break case has a high*

load impedance so $a \rightarrow -1$ (all injected current returns through the shorted voltage probe) and $T_i \rightarrow \text{inf}$, which makes $T_i = -a/(a+1)$ divide by zero; *stb* evaluates the algebraically identical singularity-free form $T = -(T_v a + a + 1)/(T_v a + T_v + a + 2)$, exact at $a = -1$ where it reduces to $T = T_v$. The complex loop gain is stored as vector *loopgain* vs frequency in a new *stb* plot (`plot db(loopgain) / plot 180/PI*cph(loopgain)`); phase margin = $180 + \text{angle}(T)$ at the first $\text{abs}(T)=1$ crossing, gain margin = $-\text{abs}(T)_{\text{dB}}$ at the first $\text{angle}(T)=-180$ deg crossing (both log-interpolated on an unwrapped phase). Implementation (*com_stb.c*, registered in *commands.c*, declared in *com_commands.h*): recovers the probe's two node names via *CKTinst2Node*, dispatches *alter/ac* through the command table (as *com_optimize/com_sweep* do), and reads the complex vectors $v(A)/v(B)/V_{\text{stb}}\# \text{branch}/\text{frequency}$ with *ft_evaluate*. Reuses the AC analysis, so both the Sparse and KLU solvers. Verify (*examples/stb*): PM/GM match the closed-form loop gain of an ideal-output op-amp loop; a loaded break (finite op-amp output impedance) gives the SAME margins as a clean high-impedance break (proving the current injection corrects the loading -- a single voltage injection is ~10% off and diverges past crossover); a 10x-gain design's reduced phase margin is tracked and matches analytics; the complex *loopgain* is stored; and an unknown probe is reported cleanly. 5 checks. Regression 162/162.

E-199 *examples/nport_examples/snp2va.py, examples/nport_examples/**

N-port Touchstone device (*snp2va.py*). Instantiate a measured or simulated S-parameter block (a filter, cable, connector, package, or amplifier stored in a Touchstone .sNp file) as a circuit element that works in AC AND transient. ngspice had no native n-port element: its .sp "ports" are tagged voltage sources used to MEASURE a circuit's S-parameters, and *rdsnp/wrsnp* (E-64/E-72) are reader/writer commands, not a device. Rather than a native convolution device (with its own history buffers and stability headaches), *snp2va.py* converts the file into a Verilog-A n-port realized with *laplace_nd*, so OpenVAF's OSDI laplace machinery (E-31 complex poles) supplies BOTH the AC response and the transient (recursive-state) response with no convolution code. Pipeline: parse Touchstone (any port count; S/Y/Z data; MA/DB/RI formats; Hz..GHz; arbitrary reference impedance) $\rightarrow S(f) \rightarrow Y(f) = (1/z_0)(I-S)(I+S)^{-1} \rightarrow$ common-pole vector fit (Gustavsen) $\rightarrow$ emit $I(p_i) \leftarrow \sum_j [\text{laplace_nd}(V(p_j), \text{num}_{ij}, \text{den}) + e_{ij} \text{ddt}(V(p_j))]$. Two numerical details are essential. (1) The vector fit MUST be done in NORMALIZED frequency -- without it the *se* column is $\sim 1e9$ while the $1/(s-p)$ columns are $\sim 1e-7$, so the least-squares is hopelessly ill-conditioned and the poles never relocate (the pole relocation is done as polynomial roots of the sigma numerator via Durand-Kerner, not a matrix eigensolve). (2) *Y* is IMPROPER whenever a network has shunt capacitance ($Y \sim sC$ at high frequency), which *laplace_nd* -- a ratio of polynomials -- cannot represent (it gave a 63 percent AC error at the band edge); the fix is to give *laplace_nd* only the strictly-proper (pole) part and emit the s^e term as an explicit ddt (a capacitance), dropping the AC error to $\sim 1e-6$. HARDENING: automatic order selection CLIMBS the pole count to capture high-order and delay-dominated responses (a pure delay is uniformly poor at low orders, so the selection must not quit early on a small between-order improvement), but stops at the KNEE once the error has already come down near a floor -- past that knee extra poles just fit measurement noise into an over-fitted, ill-conditioned, unstable model, so stopping there fits clean data to machine precision and noisy data at its noise floor. It breaks when a fit turns unstable (polynomial-root finding degrades at high degree) and returns the best STABLE fit. Stability is enforced (right-half plane poles reflected into the left half plane) so the emitted model is always BIBO-stable even for a noisy non-passive fit; passivity is checked and reported (enforcement by residue perturbation is a documented non-goal for now). Pure Python standard library, NO numpy: least-squares via Householder QR, complex matrix inverse via Gauss-Jordan, and polynomial roots via Durand-Kerner are all reimplemented (each validated against numpy to machine precision during development). Verify (*examples/nport*): each check generates a Touchstone file from a network whose response is known exactly, runs *snp2va.py* + OpenVAF, and confirms the device matches the ORIGINAL network in ngspice -- an R-L-C resonator to $5e-6$ in AC (including the transmission peak) and $5e-3$ in transient (one model, both analyses); a 5-pole LC ladder (order selection scales past 2 poles); a 3-port star network to $4e-9$ on both coupled outputs; and a noisy measurement

fit at its noise floor, reported non-passive, staying bounded in transient. Stress-tested to near machine precision on resonators, notches, high-Q blocks, and multi-pole ladders; a delay-dominated transmission line degrades gracefully (AC accurate over the fitted band, transient pulse edges ring -- inherent to rational fitting of a pure delay, for which a T/LTRA line is the right model). 6 checks. Regression 163/163.

E-200 `src/frontend/snp2va.c, src/frontend/snp2va.h, src/frontend/com_presnp.c, src/frontend/commands.c, src/frontend/inp.c, src/frontend/inpcom.c, src/frontend/Makefile.am, examples/presnp_examples/*`

Built-in `pre_snp` command. Folds the E-199 Touchstone-to-Verilog-A converter into ngspice itself as a C command, so the workflow no longer needs an external Python script: inside a `.control` block, `pre_snp <file.sNp> [module]` parses the `.sNp`, common-pole vector-fits it, writes the `.va`, and invokes `openvaf-r` to compile the `.osdi` -- a one-line analog of `pre_osdi`. `snp2va.c` is a faithful C port of `snp2va.py` with identical numerics: it uses C99 double *Complex* (`typedef'd cplx`, so it does not collide with ngspice's own `complex.h` macros) and reimplements the three vector-fitting primitives -- least-squares via Householder QR, complex matrix inverse via Gauss-Jordan, polynomial roots via Durand-Kerner -- plus the same parse $\rightarrow S(f) \rightarrow Y(f) = (1/z_0)(I-S)(I+S)^{-1} \rightarrow$ vector fit $\rightarrow$ emit $I(p_i) \leftarrow + \sum_j [\text{laplace_nd}(V(p_j), \text{num}, \text{den}) + e_{ij} \text{ddt}(V(p_j))]$ pipeline, the same normalized-frequency fit, the improper *es* (*shunt-C*) term split out as an explicit ddt, and the same automatic order selection returning the best STABLE fit (RHP poles reflected $\rightarrow$ BIBO-stable). It was checked numerically identical to the Python reference (resonator RMS 4e-6, ladder 7e-7, 3-port 3e-8). The public entry point `snp2va_convert(snpfile, vafile, module, msg, msglen)` is called by the command wrapper `com_pre_snp` (`com_presnp.c`), which then shells out to `openvaf-r`. ORDERING: the command is registered under the name `snp`, so writing `pre_snp` in a deck triggers ngspice's existing `pre` mechanism (`inp.c` strips the `pre_` prefix and runs the command before the circuit is parsed). That alone is not enough -- `pre_snp` must run before the `pre_osdi` that loads its output, and control lines otherwise execute in deck order -- so the collected pre-commands are now executed in TWO PASSES: pass 0 runs every `snp` command (in deck order), pass 1 runs the rest. The `.osdi` a `pre_snp` generates is therefore guaranteed to exist by the time any `pre_osdi` loads it, EVEN IF the deck lists the `pre_osdi` line first; the ordering is a guarantee, not a convention the user must remember. `snp/pre_snp` are also added to `inpcom.c`'s case-preservation list (beside `osdi/pre_osdi`) so the file-path argument keeps its original case. The compiler is located by `find_openvaf()`: the `openvaf` ngspice variable (set `openvaf=...` in `spinit` or on the command line -- a set inside `.control` runs too late, after the pre-commands), then the `OPENVAF` environment variable, then `$SPICE_LIB_DIR/openvaf-r`, then `PATH`; if none resolve, `pre_snp` reports the failing `openvaf-r` invocation and lists all four options. Verify (`examples/presnp`): each check builds a Touchstone file from a network whose response is known exactly, lets `pre_snp` do the whole `convert+compile+load` inside ngspice, and confirms the device matches the ORIGINAL network -- `pre_snp` writes the `.va` and compiles the `.osdi`; the device matches an R-L-C resonator to 5e-6 in AC (including the transmission peak) and to 5e-3 in transient (one compiled block, both analyses); a 3-port star network compiles and matches on both coupled outputs to 4e-9; and with `pre_osdi` written BEFORE `pre_snp` the model still loads, proving the ordering guarantee. 5 checks. Regression 164/164. HARDENING (from an N-port stress sweep -- generate an N-port RLC network, extract S-params with `.sp`, round-trip through `pre_snp`, compare DC/AC/transient vs the original subcircuit): two defects the 2-pole checks never reached, both fixed in `snp2va.c`. (1) Order-selection double-free: the climb that keeps the best STABLE fit let its best and prev buffers alias and then freed the same allocation twice, so `pre_snp` aborted (SIGABRT) on any network whose fit needed ≥ 3 pole orders (the 2-pole resonator/star escaped by breaking at tolerance first); fixed by not freeing a buffer the other pointer still holds. (2) Transient divergence from a non-passive realization: each Y_{ij} is fit independently, so the model could blow up in transient ($v \rightarrow \sim 1e284$) even though every pole is in the LHP and AC is exact -- a single degree-Np rational per element has coefficients spanning $\sim \text{abs}(p)^{Np}$ ($\sim 1e79$ for 8 poles at $1e10$ rad/s, an ill-conditioned companion form), and the improper e^s terms form a capacitance matrix with no passivity constraint (an indefinite "negative-capacitance" matrix $\rightarrow$ unstable DAE);

fixed by realizing each element as a PARALLEL bank of first/second-order laplace_nd sections (every coefficient $\leq O(\text{abs}(p)^2)$) and projecting the e-matrix onto the symmetric PSD cone (symmetrize $\rightarrow$ Jacobi eigendecompose $\rightarrow$ clamp negative eigenvalues to 0). The presnp example gained a 4-port coupled-ladder check (fit climbs past two orders; transient must stay bounded and match) that fails on the pre-fix converter. Regression 164/164.

E-201 *src/frontend/snp2va.c, examples/presnp_examples/**

pre_snp scalability (fast vector fitting + shared-pole realization). The E-200 converter struggled from about 8 ports up (an 8-port took ~190s and larger port counts were infeasible). Two things scaled badly, both in snp2va.c: the pole identification stacked all NN elements into one dense least-squares of size $(2N_sNN) \times (NN(N_p+2)+N_p) \rightarrow O(N^4)$ memory, $O(N^6)$ compute, re-solved dozens of times over the order climb (~8TB of RAM at $N=100$); and the emitted model gave every one of the NN elements its own laplace_nd bank $\rightarrow O(N^2N_p)$ filter sections and OSDI state. Four fixes rebuild the scalability. (1) FAST VECTOR FITTING (Deschrijver/Gustavsen): the stacked pole-ID matrix has block-arrow structure -- each element's residue columns are independent and couple only through the shared common-pole (sigma) columns -- so instead of forming that matrix, each element block is Householder-reduced on its own (a small $2N_s \times (N_p+2)$ QR) and only its residual rows (orthogonal to that element's own columns) are stacked into one tall $((2N_s-N_p-2)N_e) \times N_p$ system solved for sigma; memory $O(N^4) \rightarrow O(N^2)$, compute $O(N^6) \rightarrow O(N^2N_sN_p^2)$. (2) RECIPROCITY: a passive network has symmetric Y, so when detected only the upper triangle $(N(N+1)/2)$ elements is fit and residues are mirrored (~2x fewer solves). (3) ITERATION CAP: pole relocation stops once the poles settle (relative move $< 1e-4$) instead of a fixed 12 sweeps (typically 3-5). (4) SHARED-POLE REALIZATION: all NN elements share the same poles, so the pole-filters are computed ONCE per input port -- a real pole gives one filter $V(p_j)/(s-p)$, a conjugate pair two real basis filters $(s-\sigma)/D$ and ω/D , with the residue entering each output as a real scalar weight -- and each output current is a cheap weighted sum; this is $O(NN_p)$ laplace_nd and OSDI state instead of $O(N^2N_p)$, the difference between a model that compiles/simulates at large N and one that does not. Three latent bugs the stress test also surfaced, all fixed: (a) an order-selection double-free -- at higher pole counts a relocated pole set could lose its adjacent-conjugate-pair structure and build_basis/ctil_to_cres (which walk $i+=2$ on a complex pole) wrote one slot past the residue array; fixed by a canon_poles step that snaps near-real poles to real and rebuilds exact adjacent conjugate pairs. (b) a 256-entry stack array -- the Touchstone parser assembled each frequency's matrix in a fixed cplx pv[1616], hard-capping the port count at 16 ($N=32$ smashed the stack); moved to the heap, auto-detect limit lifted to 512. (c) an OpenVAF crash -- a laplace_nd coefficient array assigned to a variable (as the shared realization does) crashes OpenVAF when a coefficient is an integer literal like '{1}' (which %.12g prints for 1.0); the emitter now forces a real literal '{1.0}'. Results: fit at $N=8$ 220s $\rightarrow$ 3.6s; $N=64$ fits in 47s (was ~860GB of RAM); $N=100$ fits in 2.75min at $9.6e-4$ error; model laplace_nd count now NN_p (65 at $N=8$, 449 at $N=32$) not $\sim N^2N_p/2$ (256, 7168). Correctness preserved: DC/AC/transient still overlay the original subcircuit ($N=8$ unchanged; $N=16$, previously infeasible, matches to 1.3%). The remaining bottleneck is OpenVAF's compile time for the (now far smaller) shared model (~26s at $N=16$, ~43s at $N=20$), so the practical end-to-end ceiling is $\sim N=20-24$ while the fit itself scales to $N=100$. Verify (examples/presnp): an 8-port coupled ladder converts+compiles in a few seconds with the compact shared model (65 laplace_nd for 8 ports) and matches the original network in AC; the E-200 order/realization guards and resonator/3-port checks still pass. 9 checks. Regression 164/164.

E-202 *src/math/dense/dense.c, examples/spscale_examples/**

.sp S-parameter matrix inverse is $O(N^3)$, not $O(N!)$. The pre_snp stress test (E-201) noted a separate wall: producing the input Touchstone file with ngspice's own .sp analysis was itself pathologically slow -- an 8-port took ~13s and a 10-port ~18 minutes, growing by roughly a factor of N per added port. That is in the RFSPICE analysis, not the converter. Profiling put ~99 percent of an 8-port .sp run inside CKTspCalcSMatrix (cktspdum.c), which builds the port S-matrix at every frequency by inverting $N \times N$ complex matrices -- several per frequency, for the S, Y and

Z matrices. The complex inverse (maths/dense/dense.c) was computed by the adjugate/determinant method (Cramer's rule): `cinverse = cadjoint(A) * (1/cdet(A))`. `cdet` is a RECURSIVE COFACTOR EXPANSION -- $O(N!)$ -- and `cadjoint` computes N^2 cofactors (each an $(N-1) \times (N-1)$ determinant), so `cinverse` was $O(N^3)$; with several inverses per frequency across a whole sweep the cost explodes by a factor of N per port ($6! \rightarrow 7! \rightarrow 8!$ is exactly the measured $\times 7, \times 8, \times 9$ per added port). FIX: `cinverse` and `cinversedest` now invert by GAUSS-JORDAN ELIMINATION WITH PARTIAL PIVOTING on the augmented $[A \ I]$, in $O(N^3)$ -- a drop-in replacement with the same signature and the same result (the true inverse). Behavior preserved: the old `cinverse` never returned NULL (the adjugate always produced a matrix, garbage for a singular input), so the new one returns NULL only on a true allocation failure and zero-fills a singular matrix; the singular case does not arise for a well-posed .sp network, and the two `cktspdum.c` call sites that do not null-check are safe. `cdet/cadjoint` remain for any other callers. RESULTS: .sp at $N=8$ 13s $\rightarrow$ 0.3s, $N=10$ ~18min $\rightarrow$ 0.02s (~50000x), $N=16$ 0.03s, $N=32$ 0.09s; the extracted S-parameters are unchanged, verified against the closed-form network to $\sim 2e-7$. The same inverse is used by the periodic S-parameter path (.psp/PSS in `dcps.c`), which benefits too; full example regression (touchstone, rfanalyses, psp, ...) unchanged. Together with E-201 the whole extract-or-import $\rightarrow$ convert $\rightarrow$ simulate workflow now scales to many ports. Verify (examples/spscale): a 12-port RLC ladder -- a port count that took minutes before -- runs through .sp + `wrsnp`, completes in a fraction of a second, and every entry of the extracted 12x12 S-matrix matches the closed-form network across the sweep to $\sim 2e-7$. 2 checks. Regression 165/165.

E-203 *src/frontend/com_measure2.c, src/misc/string.c, examples/acmargin_examples/**

.meas ac gain/phase margin (+ batch vdb/vp auto-save fix). An audit against a 'richer TRIG/TARG, FIND...WHEN, integral/RMS/derivative, AC margins' wish-list found that ngspice's .measure already covers TRIG/TARG (with VAL, RISE, FALL, CROSS, LAST, TD, AT), FIND...WHEN, AVG/RMS/INTEG/DERIV, MIN/MAX/MIN_AT/MAX_AT/PP, ERR, and param='...', across tran/ac/dc/sp -- all present and working. The one genuine gap was AC margins. The textbook way to get them from the existing primitives, .meas ac gm FIND vdb(out) WHEN vp(out)=-180, CANNOT work: `vp()` returns the phase WRAPPED to $(-180, 180]$, so on any loop whose true phase sweeps past -180 the wrapped phase jumps from about -179 straight to about +179 and never equals -180, and the crossover is never found. FIX: two first-class .meas ac functions computed on the UNWRAPPED phase -- `phase_margin` ($PM = 180 +$ the unwrapped phase at the 0 dB gain crossover) and `gain_margin` ($GM = -\text{gain(dB)}$ at the -180 deg phase crossover). The phase is unwrapped by undoing the ± 360 deg `atan2` jumps, the 0 dB and -180 deg crossings are found by linear interpolation, and the crossover frequency is interpolated in log space. A loop with no finite -180 crossover (a one- or two-pole loop, infinite gain margin) is reported as such rather than as a bogus number. Implemented as `measure_margin()` dispatched from `get_measure2()` in `com_measure2.c`. One subtlety: ngspice's phase (`vp`, and the measure 'p' vector type) honors a runtime `cx_degrees` flag that defaults to RADIANS, so `radtodeg()` is a no-op by default; the margin code computes phase in degrees explicitly. TWO BUGS the work surfaced, both fixed: (1) `gettok_iv` (`misc/string.c`), used only by the batch measure auto-save pass, copied the leading v/i then broke after the FIRST following character on `n_paren==0` -- so `vdb(out)` became `vd` (and `vm/vp/...` were all truncated); the stray `vd` then failed to save ('can't parse vd') and left the node unsaved, so a batch .meas ac ... `vdb(out)` died with 'no data saved for AC'. Fixed to stop only once the parenthesised part has actually closed. (2) a companion `strip_ac_vec` reduces `vdb(out)` to `v(out)` so the base node is what gets .save'd. Verify (examples/acmargin): 5 checks against closed-form loop gains (buffered one-pole sections, so poles sit exactly where placed) -- a stable 2-pole loop's phase margin matches analytic to under 1 deg (84.89 vs 84.89) and its gain margin is correctly reported infinite; a 3-pole loop's phase margin (-42.75 deg) and gain margin (-15.92 dB) both match the closed form exactly; and a batch .meas ac ... `vdb(out)` dot-card auto-saves the node and returns the right value. 5 checks. Regression 166/166.

E-204 *src/include/ngspice/optdefs.h, src/include/ngspice/tskdefs.h, src/include/ngspice/cktdefs.h, src/*

*spicelib/analysis/cktsopt.c, src/spicelib/analysis/cktntask.c, src/spicelib/analysis/cktdojob.c, src/spicelib/analysis/cktop.c, examples/convhelp_examples/**

auto-triggering DC convergence aids (.option convhelp). ngspice's CKTop operating-point cascade already auto-escalated through gmin stepping and source stepping, but the two strongest aids -- the globalized damped-Newton line search (E-111) and pseudo-transient continuation (E-127) -- only fired when the user hand-set .option linesearch or .option ptcont, and nothing reported which aid rescued a hard operating point. The ngspice-vs-Spectre gap analysis flagged exactly this: what remained was auto-triggering heuristics (reaching for these aids without the user asking) and folding them into the robustness presets. .option convhelp makes the whole cascade one automatic ladder -- Newton (with line search), then gmin step, then source step, then pseudo-transient, then optran -- and prints a one-line note naming the aid that produced the point. Under convhelp the op-point Newton, and its fallback homotopies (which call NIiter internally), run with line search enabled; E-111's line search reduces to a plain full Newton step whenever the full step already reduces the weighted residual norm, so points that converge easily take the same iterates and are unaffected. The E-127 pseudo-transient gate was widened so it also fires when convhelp is set, with no explicit .option ptcont needed. CKTop was refactored to a single done: exit that saves the caller's line-search flag on entry and restores it on every return (so the setting never leaks into the transient analysis that follows) and emits the aid note there. OFF BY DEFAULT: the default (convhelp unset) path is byte-for-byte the historical behavior with no new output. errpreset=conservative enables it through the E-110 TSKtolGiven given-flag mechanism (a new ERRP_CONVHELP bit), so an explicit .option convhelp=0 still overrides the preset regardless of .options line order. Plumbing mirrors E-127: OPT_CONVHELP enum and ERRP_CONVHELP bit (optdefs.h), TSKconvhelp bitfield (tskdefs.h), CKTconvhelp bitfield (cktdefs.h), the OPT_CONVHELP setter plus convhelp in the errpreset value table plus the convhelp OPTtbl keyword (cktsopt.c), copy and default-off (cktntask.c), CKTconvhelp = TSKconvhelp (cktdojob.c), and the auto-escalation plus reporting in CKTop (cktop.c). Verify (examples/convhelp): the option is accepted; on a battery of normal circuits (diode, BJT, two-diode, resistor network) convhelp is result-neutral (node voltages identical to a plain run) and silent (no aid note, since plain Newton already converges); on a deliberately stiff behavioral-exponential deck with no junction limiting -- root of $1e-14 * (\exp(V/0.026) - 1) = (100 - V)/100$, i.e. $V(1) = 0.837922$ V -- where plain Newton overshoots to a spurious 70.5 V root (and even the transient-op fallback lands there), convhelp reaches the correct 0.837922 V via auto-fired pseudo-transient continuation, reports the aid, and changes the outcome versus plain Newton; errpreset=conservative enables the same ladder; and an explicit convhelp=0 overrides the preset. Both linear solvers (KLU and Sparse 1.3). 35 checks. Regression 167/167.

E-205 *src/frontend/snp2va.c, examples/lowrank_examples/**

pre_snp low-rank residue factorization. The E-200/201 shared-pole realization emits an $O(N^2)$ -term, $O(Nnp)$ -filter Verilog-A model, so OpenVAF's compile time grows super-linearly with the port count and caps the practical N at ~ 20 -24 (an $N=32$ block compiles in ~ 80 s -- the vector fit itself scales to $N=100$, but the emitted model's compile does not). When an N -port block's ports couple through only a few shared modes -- the common case for multi-port filters, cavities, and packages with a shared plane -- each pole's $N \times N$ residue matrix has rank r much less than N . E-205 exploits that: per channel (the conductance d , the capacitance e , and each pole section) the emitter builds the $N \times N$ real weight matrix and, via a new in-C one-sided Jacobi SVD, picks the cheaper of a dense emit (a term per significant entry, one laplace_nd per input port) or a low-rank emit $W = UV^T$ (rank kept above $\text{tol_rank} = 1e-7$, so a clean singular-value gap compresses fully while a slowly-decaying channel stays dense). The key is INPUT COMBINING: because laplace is linear, $\sum_j V[j][m] \text{laplace}(V(p_j)) = \text{laplace}(\sum_j V[j][m] V(p_j))$, so the low-rank form filters the r COMBINED inputs $u_m = \sum_j V[j][m] V(p_j)$ ONCE (r filters, not N) and distributes them to the outputs via U -- collapsing filters from $O(Nnp)$ to $O(rnp)$ (the piece that dominates the compile) and coupling terms from $O(N^2)$ to $O(Nr)$. A full-rank block has no such structure, so every channel keeps the dense form and is emitted bit-identically (the auto-detect leaves it alone); the E-201 PSD projection of the capacitance

e is preserved, so the low-rank transient stays stable. The env var PRE_SNP_DENSE forces the old dense realization (escape hatch / A-B test). Measured on a 3-shared-mode block (openvaf-r compile of the emitted .va): N=16 16.9s to 1.5s (96 to 8 filters), N=24 42s to 5.7s (144 to 10), N=32 81s to 11.6s (192 to 8) -- super-linear to roughly linear, about 7-14x, with the response exact (AC $\sim 4e-7$, transient $\sim 3e-5$ versus a forced-dense build of the same fit). New code: a jacobi_svd and an emit_filter helper plus the per-channel chooser and input-combined emit, all in snp2va.c. Verify (examples/lowrank): a 12-port/3-mode block emits far fewer filters low-rank than dense (7 vs 73); its AC ($3.9e-8$) and transient ($1.2e-4$, bounded) equal the forced-dense build of the same fit; a full-rank 8-port ladder gets no compression with bit-identical AC. 5 checks. Regression 168/168.

E-206 *src/frontend/com_optimize.c, examples/dcenter_examples/**

design centering / yield optimization (optimize -center). The capstone that fuses two whole sub-systems: the optimizer (Nelder-Mead / LM / PSO / DE / SA, E-130-197) and the Monte-Carlo yield suite (agauss/mccorr sampling + montecarlo specs, E-149-151). Design centering optimizes the NOMINAL design point to maximize parametric yield / Cpk under process variation. The outer optimizer searches the design knobs (-dparam/-param/-mparam, which feed the deck's agauss centres); the objective at each candidate is an INNER Monte Carlo of -samples N draws -- each reset re-samples the deck's process variation (agauss/.param, and any mccorr correlations) around the current centre -- scored against -spec [-max hi] [-min lo] limits (the same spec syntax as montecarlo) and reduced to the worst-case Cpk and the pass-fraction yield. The fusion is small and clean in opt_eval: the in-place-knob application was factored into opt_apply_inplace so the MC loop can re-apply the centre after each reset re-sources the deck, and a new opt_eval_center runs the inner MC and returns -min(Cpk); the method drivers (nm/ps/de/sa) are unchanged -- they call opt_eval, which now returns the centering cost. Cpk is the objective (yield is reported alongside) because raw yield is a granular step function that stalls the simplex, whereas $Cpk = \min(\text{gap to each active limit}) / (3 \sigma)$, worst-cased across specs, is continuous and monotone in yield -- so maximizing it centres the distribution in its spec window. A fixed inner seed gives every candidate the SAME process draws (common random numbers), so the objective is deterministic and smooth; with -lhs the stratified sample-mean is about zero, so the optimum lands on the analytic centre. Reuses the E-149/151 sampling (mc_lhs_config, mc_sss_off, setseed); lm is rejected for -center (Cpk is not a least-squares residual) and the default method is Nelder-Mead; results are published as dcenter_yield and dcenter_cpk vectors. com_optimize.c only. Verify (examples/dcenter): a synthetic problem output $\sim N(xc, 0.5)$ with spec [4,6] (agauss(m,v,3) makes the actual $\sigma = v/3 = 0.5$, so the analytic centre is the midpoint 5 and the analytic Cpk is $1/(3*0.5) = 0.667$) is centred from an off-centre start $xc=4.0$ to $xc = 5.00$ with the expected Cpk 0.667 and a high yield; a montecarlo baseline at $xc=4.0$ sits near 50% while the centred design reaches $\sim 96\%$; and a two-knob problem with a lower and an upper spec on two outputs centres both params independently ($xa \rightarrow 5$ on [4,6], $xb \rightarrow 10$ on [9,11]). 5 checks. Regression 169/169.

E-207 *src/frontend/com_eye.c (new), src/frontend/com_eye.h (new), src/frontend/commands.c, src/frontend/com_commands.h, src/frontend/Makefile.am, examples/eye_examples/**

eye diagram / jitter analysis (eye command). A new application domain: the core signal-integrity measurement for serial links / SerDes / high-speed I/O, which ngspice lacked. eye -ui [-tstart t0] [-threshold vth] [-window frac] is a self-contained front-end post-processing command (in the fft / linearize / meas family, registered in commands.c/com_commands.h/Makefile.am), solver-independent. It reads the transient waveform and its time scale via ft_evaluate; auto-detects the two logic rails from the 20th/80th percentiles of the samples (eye_level0, eye_level1, eye_amplitude) and the decision threshold (default the midpoint, or -threshold); finds every threshold crossing by linear interpolation; estimates the UI phase as the circular mean of the crossings modulo UI; computes each crossing's time-interval error (TIE, relative to the ideal UI grid) and reduces it to eye_jitter_rms (standard deviation of TIE) and eye_jitter_pp (peak-to-peak); measures eye_height (the vertical opening at the sampling instant, one half-UI from the transitions -- min of the samples above threshold minus max of the samples below, taken in a +/- windowUI slice)

and `eye_width` (UI minus the peak-to-peak jitter, the horizontal opening at the threshold), plus `eye_width_ber12 = UI - 14.069eye_jitter_rms` (the eye width at BER 1e-12 from the +/-7.035 sigma Gaussian random-jitter tail); and folds the waveform modulo $2 \cdot \text{UI}$ into the `eye_wave` vs `eye_t` vectors (built with `dvec_alloc/vec_new`, as `stb` and `sweep` build their result plots), whose scatter plot IS the eye diagram (plot `eye_wave` vs `eye_t`). No numerical-core changes. Publishes 11 permanent result vectors. Verify (examples/eye): an ideal 0/1 clock (perfectly periodic edges) recovers rails [0,1] and amplitude 1 with jitter about 1e-24 s (machine zero) and a full-UI, fully open eye; a clock whose edges carry a KNOWN injected Gaussian timing jitter ($\sigma = 20$ ps) is recovered to within a few percent (measured about 18.6 ps rms, 99 ps pp) and `eye_width` equals UI minus the measured peak-to-peak jitter; and the folded eye vectors are published and `wrdata eye_wave` writes the folded eye without crashing (the `eye_wave/eye_t` vectors live in their own fresh eye plot -- as `stb` does -- so `wrdata/plot` pair equal-length vectors; dumping the length-m eye vectors into the transient plot, whose scale is the full much-longer time vector, segfaults). 7 checks. Regression 170/170.

E-208 `src/frontend/plotting/pyplot.c, src/frontend/plotting/pyplot.h, src/frontend/com_pyplot.c, examples/pyplot_examples/*`

`pyplot-eye`: one-line matplotlib eye diagrams. A convenience bridge between the E-207 eye analysis and the E-94 matplotlib back-end: `pyplot [name] -eye -ui [-tstart t0] [-threshold vth] [-window frac]` runs the eye analysis AND renders the folded eye in one command. `plot eye_wave vs eye_t` draws a single connected line through the folded samples in file order (a scribble, not an eye); `pyplot-eye` instead draws the samples as a 2-D histogram (`hist2d`, log-scaled `LogNorm`, `turbo colormap`, empty cells masked with `cmin=1`) so overlapping traces accumulate into a persistence eye -- exactly how a sampling scope paints one -- annotated with the decision threshold (dashed), the eye-centre sampling instant at 1 UI (dotted), a vertical eye-height double-arrow, a horizontal eye-width double-arrow along the threshold, and a title carrying UI / eye height / eye width (% UI) / RMS jitter. Implementation: `com_pyplot()` detects the `-eye` marker (a single bare token before it is taken as the output base name; default `eye`), runs `com_eye()` on the trailing tokens -- which folds the waveform and leaves `eye_wave/eye_t` plus the scalar metrics in a fresh current eye plot -- then calls the new `ft_pyplot_eye()` in `plotting/pyplot.c`, which reads those back from the current plot, writes `.data` (the folded sample pairs) and a `.py` matplotlib script, and launches Python with the SAME system(`python ...`) mechanism as `ft_pyplot()`, honouring every existing `pyplot_*` setting: `pyplot_terminal=png/svg/pdf` renders headless (Agg) and runs synchronously, otherwise an interactive window is launched in the background; `pyplot_python` picks the interpreter, `pyplot_backend` forces a matplotlib backend, `pyplot_figsize` sets the figure size, and `pyplot_style` applies a style sheet (a dark sheet flips the annotation colours to stay legible on a dark ground). No numerical-core changes; the eye math is entirely E-207's, and `ft_pyplot()` and the `gnuplot/asciiplot` back-ends are untouched. Verify (examples/pyplot): a self-contained data eye -- a pseudo-random NRZ bit stream (PWL) through a bandwidth-limiting RC channel ($\tau \sim 0.5$ UI) -- is rendered with `pyplot eyefig -eye v(rx) -ui 0.5n`: the eye analysis runs and reports its metrics, `eyefig.png` is a valid PNG, the generated script is a `hist2d` persistence eye referencing the metrics, and the no-name form defaults the base to `eye.png`. 4 checks (17 total in the `pyplot` suite). Regression 170/170 (extends the `pyplot` suite, no new folder).

E-209 `src/spicelib/analysis/dcpss.c, src/frontend/com_hb.c, src/include/ngspice/cktdefs.h, examples/hb_examples/verify_hb.py`

`hb` publishes its spectrum as `nutmeg` vectors. The E-134 harmonic-balance command (`hb`) used to only PRINT a spectrum table: `HBanalyze()` freed the converged two-sided Fourier solution `Vr/Vi` as soon as the table was written, so the result could not be `plot/print/wrdata`-ed without scraping `stdout` (the `amp2n2222` HB study had to parse the table in Python for every figure). Now, after a converged `hb`, a fresh current '`hb`' plot holds a real scale `hbfrequency` (0, `f0`, `2f0`, ..., `Kf0`; `K+1` points) and one COMPLEX vector per node (named by the node, e.g. `out`, `vcc`, `q1#branch`) carrying the single-sided amplitude, so `mag(out)/vp(out)` reproduce the printed

E-210 *src/frontend/inp.c, src/spicelib/analysis/dcpss.c, examples/rfpss_examples/rc_pss.cir, examples/rfpss_examples/verify_rcpss.py*

PSS polish: .pss dot-card runs in batch + complex spectrum vectors. Two usability fixes to the E-117 shooting-method periodic steady state (pss), in the spirit of E-209 (hb). (1) The pss COMMAND (inside .control) worked, but a top-level .pss card by itself silently did nothing in batch mode -- ngspice -b deck.cir with just .pss ... reported "no simulations run", unlike .tran/.ac/.hb which auto-run; a deck had to wrap it in .control ... runendc. .pss is now dispatched to the pss command the same way E-146 (.sweep) and E-162 (.hb) handle their cards: during deck loading (frontend/inp.c) a top-level .pss ... line is stripped of its leading '.' and appended to the post-parse control list, so it executes as pss ... once the circuit is built. A boundary check (next char is whitespace or end-of-line) keeps it from matching a longer name, and it never matches .psp (periodic S-parameters, a distinct card). Decks that already wrap .pss in .control ... run keep working (one PSS run, unchanged output). (2) The pss analysis already published two plots -- pss1 (time-domain periodic waveform) and pss2 (frequency-domain spectrum) -- but pss2 stored MAGNITUDE ONLY (IF_REAL): the DFT computed each harmonic's phase (pssphases) and then discarded it, so vp(out) was unavailable and print out gave a bare magnitude, inconsistent with AC analysis (whose node vectors are complex) and with the hb command's E-209 vectors. pss2 now publishes COMPLEX node vectors (IF_COMPLEX): each harmonic is stored as mag * e^{j*phase} (the DFT returns the phase in degrees; the DC term is kept real), built into IFcomplex rows and emitted with OUTpData (as PAC/PXF/PSP already do) instead of the real-only CKTdumpp. mag(out) reproduces the old magnitude spectrum exactly and vp(out) now recovers the phase; the max-frequency-verification logic still reads the retained magnitude array (pssResults), so it is unchanged, and the time-domain pss1 plot is unchanged (a real waveform). BACKWARD COMPATIBILITY: because pss2 node vectors are now complex, print out / wrdata out emit re,im columns (exactly as AC analysis does) rather than a bare magnitude -- decks that parsed the magnitude directly should use mag(out); the bundled rc_pss.cir was migrated from print b to print mag(b). This is the same magnitude-vs-complex convention the rest of ngspice's frequency-domain analyses use. The shooting-PSS numerical core, the retained periodic operating point, and every analysis built on it (PAC/Pnoise/PXF/PSP) are untouched. Verify (examples/rfpss): a top-level .pss card (no .control) auto-runs in batch and reaches convergence; the frequency-domain spectrum resolves as complex -- both vm(out) (magnitude) and vp(out) (phase) return, with a driven diode rectifier's harmonics showing the expected non-trivial phase; the existing PSS/PAC/PXF/solver-parity checks pass unchanged. 2 checks (27 total in the rc_pss suite). Regression 170/170 (extends the rfpss suite, no new folder).

E-211 *src/spicelib/analysis/dcop.c, src/frontend/inpcom.c, examples/codeanalysis_examples/**

Code-analysis bug fixes: XSPICE DC-op else + LTSPICE table free. A static-analysis pass over the ngspice tree (clang's analyzer run on every project-modified file, each finding then confirmed by reading the code, false positives -- e.g. symbolic-size array indexing in static math helpers -- discarded) surfaced two real defects. BUG 1 (dcop.c): DCop() chooses between the event-driven XSPICE solver (EVTop) and the analog solver (CKTop) with a BRACELESS else whose body was originally converged = CKTop(...), so CKTop ran only when there were no event-driven instances. Enhancement-188 (DC warm-start) inserted its preload code between the else and CKTop, so the else then covered only its first statement (wsize = SMPmatSize(...)) and CKTop ran UNCONDITIONALLY: for a deck with event-driven code-model instances (A-devices) BOTH EVTop and CKTop ran, CKTop re-solving the analog part and overwriting the event-driven DC result, and wsize was read UNINITIALISED at the warm-start guard (line ~110) and snapshot (line ~129) on the EVTop path (undefined behaviour, masked today only because DC warm-start is off by default). FIX: hoist wsize = SMPmatSize(ckt->CKTmatrix) above the branch (always initialised; the warm-start snapshot reads it on both paths) and add braces so the else covers the warm-start preload + CKTop, restoring the original exactly-one-solve semantics. The #ifdef XSPICE braces vanish together when XSPICE is not compiled, so the non-XSPICE build is byte-unchanged. BUG

2 (inpcom.c): `inp_compat()` converts an LTSPICE `E ... table=(...)` controlled source using an UNINITIALISED `char *ckt_array[100]` stack array; the LTSPICE branch sets only `ckt_array[1]`, and a single-pair table (`table=(x0,y0)`, `ipairs==1`) then `tfree(ckt_array[2])` -- a free of a garbage stack pointer (`ckt_array[2]` is written only on the other, non-LTSPICE, branch). FIX: zero-initialise the array (`char *ckt_array[100] = { NULL };`); ngspice's `tfree(NULL)` is a safe no-op (`txfree` guards `if (ptr) free(...)`), so a `tfree` of any slot not written on a given branch does nothing instead of freeing garbage. No functional change to correct analog decks -- the fixes remove undefined behaviour and a redundant analog solve on the XSPICE and LTSPICE-import paths respectively. Verify (examples/codeanalysis, 5 checks x both solvers): a mixed-signal DC op (an analog divider + an `adc_bridge`, an event-driven instance) solves the analog node to `v(a)=0.5` without error; a single-pair LTSPICE `E ... table=(0.5, 3)` imports and yields the constant 3 V; a plain analog DC op is unchanged (a would-fail before/after test is impractical -- the redundant CKTop leaves a simple circuit's DC point unchanged, and freeing a garbage pointer is UB that may or may not crash). 5 checks. Regression 171/171 (new example folder).

E-212 *src/frontend/breakp.c, src/frontend/device.c, src/frontend/vectors.c, src/frontend/dotcards.c, src/frontend/inpcom.c, src/spicelib/analysis/cktpzset.c, examples/crashfix_examples/**

Crash hardening: seven user-triggerable crashes. A static-analysis pass (clang analyzer) plus argument fuzzing of every frontend command and of malformed/degenerate netlists surfaced seven reproducible, user-triggerable crashes (SIGSEGV/SIGABRT) in STOCK ngspice, all in user-input handling (command parsers, the `.op` output path, netlist recursion) and none in the numerical core (device models, OSDI loader, Sparse 1.3, KLU, and the analysis drivers were audited and found clean). Each is a missing edge-guard. (1) `iplot` (`breakp.c`, `com_iplot`): `iplot -w/-d` with the flag as the trailing token consumes its value word (`wl=wl->wl_next`), guards it with `if(wl)` (proving `wl` can be NULL), then the loop-bottom `wl=w1->wl_next` dereferences NULL. FIX: guard the advance. (2) `altermod` (`device.c`, `com_altermod->com_alter_common`): `com_alter` guards `if(!wl)` but `com_altermod` does not, so a bare `altermod` reaches `com_alter_common` with an empty list `-> wlin=w1_head=NULL -> wl_nthelem(100,NULL)=NULL -> eq(wlin->wl_word, "J")` derefs NULL. FIX: guard the empty `w1_head` with a clean error. (3) `measure` (`vectors.c`, `vec_get`): a malformed measure statement (`meas tran x FIND, ... WHEN, FIND v(1)` with no WHEN/AT, across `tran/dc/ac/sp`) leaves `meas->m_vec` NULL, which flows into `com_measure_when -> vec_get(NULL) -> copy(NULL)=NULL -> strchr(NULL,')`. FIX: make `vec_get` NULL-safe (`if(!vec_name) return NULL`) -- the common sink; every caller already handles a NULL return, so each trigger reports "no such vector" instead of crashing. (4) empty `.op` (`dotcards.c`, `ft_cktcoms`): a circuit with no non-ground nodes -- e.g. a deck whose every component is commented out - produces an `op` plot with no data vectors, so `pl_dvecs` is NULL; the former `assert(pl_dvecs != NULL)` aborted (SIGABRT; ngspice's autotools build defines no NDEBUD so asserts are live) and with `-DNDEBUG` the next line dereferenced NULL. FIX: guard `if(plot_cur && plot_cur->pl_dvecs)` and skip the empty `op` printout (the solver had handled the empty matrix correctly; this was purely output formatting). (5) KLU pole-zero (`cktpzset.c`): a `bsearch` miss for the drive pointer was reported with a `printf` and then dereferenced anyway (`job->PZdrive_pptr = matched->CSC_Complex`); latent (SMPmakeElt creates the element just above, so it is always in the bind table and the miss cannot occur today) but the guard was objectively broken. FIX: add the missing `else/NULL` path, matching the correct idiom in `cktsetup.c`. (6) `.include` recursion (`inpcom.c`, `inp_read`): `inp_read` tracked `call_depth` but never limited it, so a file that `.includes` itself -- directly, via an `A->B->A` cycle, or via a `.lib` section that `.includes` its own file -- recursed until the C stack overflowed (SIGSEGV); `.lib` files are de-duplicated by `find_lib` and read once, so the crash is always via `.include`. FIX: cap at `INP_MAX_INCLUDE_DEPTH=50`; a cycle now reports an error (a 49-level chain still reads fine). (7) `.func` recursion (`inpcom.c`, `inp_expand_macro_in_str`): the expander re-expands a substituted function body with no depth guard, so `.func f(x)={f(x)}` (or a mutual `f<->g`) expanded forever and overflowed the stack (each frame also holds a `params[FCN_PARAMS]` array ~8KB, so it fails near ~1000 deep). FIX: a `macro_depth` counter capped at `INP_MAX_MACRO_DEPTH=100`, reset on every error path. The netlist-recursion caps mirror the include/expression-

depth hardening OpenVAF-r received in Enhancement-148. No functional change to correct decks -- the fixes turn crashes on malformed or degenerate input into graceful errors. Verify (examples/crashfix, 17 checks x both solvers): every repro now exits gracefully (no signal -- crashes are detected from the child's SIGSEGV/SIGABRT exit code, which the pre-fix binary returned) while every valid form still works (iplot -w 3 v(1); altermod nm vto=0.7; meas tran vmax MAX v(1) -> 1.0; a normal .op; KLU .pz; a legitimate nested .include; .func sq(x)={x*x} -> sq(4)=16). 17 checks. Regression 172/172 (new example folder).

E-215 *src/main.c, src/osdi/osdiload.c, src/include/ngspice/osdiitf.h, examples/plusargs_examples/**

command-line plusargs (+name[=value]) served to Verilog-A models. Lets ngspice -b deck.cir +corner=ff pick a corner or toggle a feature flag at run time without editing the deck -- the simulator half of E-215 (the compiler half is in the openvaf report). main.c treats a +-prefixed command-line argument as a plusarg rather than an input file (both the batch and interactive file loops) and calls the new ngspice_register_plusarg (osdiitf.h). get_simparams (osdiload.c) exposes each plusarg on the OsdSimParas channel a compiled model already reads for \$simparam\$str (E-25), as namespaced simparams: numeric \$test\$plusargs\$name=1 (present), \$valset\$plusargs\$name=1 (given as name=value), \$valnum\$plusargs\$name (value via strtod), and string \$value\$plusargs\$name (value text). The base sim_params arrays are spliced onto the extended arrays only the FIRST time get_simparams sees any plusarg present, so a run with NO plusargs allocates nothing and is byte-for-byte unchanged; the extended arrays are built once (plusargs are constant for the run) and only the base numeric values + analysis_name are refreshed per call. No OSDI ABI change (the string/number channel already existed). Verify (examples/plusargs, 12 checks both solvers): a conductance set from +boost/+gain=25/+scale=2.5/+corner=ff|ss plus unmatched/unrelated/no-value edge cases. Regression 175/175 (new example folder).

E-216 *src/frontend/com_optimize.c, examples/pareto_examples/**

NSGA-II multi-objective / Pareto optimization. Adds a -method nsga2 to the optimize command that trades TWO OR MORE competing objectives and returns a Pareto FRONT of non-dominated designs, rather than the single optimum the scalar methods return (E-130 Nelder-Mead, E-194 PSO, E-195 DE, E-196 SA) -- the multi-objective generalization of E-206 design centering. Objectives are given as repeated -minimize / -maximize ; the existing optimize infrastructure is reused wholesale (the -param/-dparam/-mparam knobs, the -analysis-driven expression objectives via opt_eval_expr, the normalized [0,1] search space and the opt_rand RNG). New: a vector-valued objective evaluator opt_eval_objs (maximized objectives negated to a common minimization convention) and nsga2() implementing standard NSGA-II (Deb 2002) -- fast non-dominated sorting (Pareto rank via domination counts), crowding distance (per-front normalized neighbour gaps, boundaries at infinity), binary-tournament selection by the crowded-comparison operator, real-coded SBX crossover (eta=15) + polynomial mutation (eta=20, rate 1/n), and elitist parent+offspring (2N->N) survivor selection so a non-dominated design is never lost. The command prints the final front (each design's objectives + parameters, sorted by the first objective) and publishes each objective column as a permanent vector pareto1/pareto2/... so the front plots directly (plot pareto2 vs pareto1). nsga2 needs >=2 objectives and a single -analysis stage; population defaults to 20+4*np (override -swarmsize), generations -maxiter, -seed for reproducibility. The scalar methods and their single-objective/least-squares/centering paths are unchanged (a lone -minimize still seeds the scalar objective for nm/psa/de/sa). Verify (examples/pareto): the Schaffer-1 benchmark -- minimize $f_1=px^2$ and $f_2=(px-2)^2$ over px in $[-1,3]$, whose Pareto set is exactly px in $[0,2]$ tracing $f_2=(\sqrt{f_1}-2)^2$, with the knob range wider than the front so NSGA-II must discover $[0,2]$. The returned front is a reasonable size, every point is genuinely non-dominated, it spans the trade-off (px reaches ~ 0 and ~ 2), and the published vectors lie on the analytic curve to machine precision ($8.9e-16$). 6 checks. Regression 176/176 (new example folder).

E-217 *src/frontend/com_pyplot.c, src/frontend/plotting/pyplot.c, src/frontend/plotting/pyplot.h, src/frontend/plotting/plotit.c, examples/pyplohist_examples/**

pyplot -hist: histogram plots. `pyplot -hist <sig> ...` renders each listed signal's VALUE distribution as a matplotlib histogram (`plt.hist`) instead of a trace vs time/frequency -- for the amplitude spread of a transient, a Monte-Carlo measurement, or a noise sample. Unlike -eye (E-208, a distinct analysis) -hist is a render MODE, so it is small and reuses the whole pyplot pipeline: `com_pyplot` detects the -hist marker anywhere in the arg list, unlinks it, and dispatches the remaining [name] signal-list to `plotit` with a distinct device string `pyplohist`; `plotit` resolves the signal expressions exactly as for a normal plot (`v(out)`, `db(...)`, vector arithmetic, node names) and passes the resolved vectors to `ft_pyplot` with a hist flag; `ft_pyplot` writes the same .data table and emits `ax.hist(value_column, bins, alpha, label)` per signal instead of `ax.plot(x,y)`. All existing pyplot facilities carry over unchanged (set `pyplot_terminal=png/svg/pdf` headless render, `pyplot_subplots` panels, style sheets, `figsize`, `backend`); two histogram-specific options are added -- `pyplot_hist_bins` (bin count, default matplotlib 'auto') and `pyplot_hist_density` (normalize to a density). Correctness details: the x-axis is the signal value and the y-axis the count (labels set accordingly); histogram panels do NOT share an x-axis (`sharex=False` -- different signals have unrelated value ranges, vs a line plot's shared time/freq axis); the FULL value length is histogrammed -- the data-table loop normally walks the shared scale vector, but a histogram of a raw let vector (e.g. a Monte-Carlo result) can have a scale whose length differs from the values', so for hist it walks the longest VALUE vector and pads shorter ones with NaN (the generated script filters via `~np.isnan`), fixing a silent truncation to the scale length; overlaid histograms get `alpha=0.6` transparency and a legend while a single one is drawn opaque. No change to the ordinary pyplot trace path or to pyplot -eye. Verify (examples/pyplohist): two signals with closed-form distributions -- `ramp=i/(N-1)` uniform on $[0,1]$ (flat histogram), `sine=sin(2pi*i/100)` arcsine on $[-1,1]$ (U-shaped, a sinusoid dwells near its peaks so the edges tower) -- checked analytically from the generated .data (uniform bins within 15% of the mean; sine edge bins » middle), plus the -hist path taken (`plt.hist` not `plt.plot`, non-shared x), the full 20000-sample length not truncated, and a valid non-trivial PNG. 6 checks. Regression 177/177 (new example folder).

E-218 *src/frontend/com_pyplot.c, src/frontend/plotting/pyplot.c, src/frontend/plotting/pyplot.h, src/frontend/plotting/plotit.c, examples/pyplotcontour_examples/**

pyplot -contour: 2-D contour maps for 2-D parameter sweeps. `pyplot -contour <z> <x> <y>` renders a filled matplotlib contour map of a quantity z over the (x,y) plane -- the natural view of a 2-D parameter sweep (a device current over a (Vgs,Vds) grid, a gain over an (R,C) grid, a noise figure over a (freq,bias) grid). Like -hist (E-217) and unlike -eye (E-208, a distinct analysis), -contour is a render MODE dispatched over the ordinary signal list: `com_pyplot` detects the -contour marker anywhere in the arg list, unlinks it, and dispatches the remaining [name] z x y to `plotit` with a distinct device string `pyplotcontour`; `plotit` resolves the three expressions exactly as for a normal plot (`v(out)`, `db(...)`, vector arithmetic, node names) into a three-vector list (z first, then x, then y); the new `ft_pyplot_contour` writes an (x,y,z) data table and a matplotlib script that TRIANGULATES the (x,y) points (`ax.tricontourf`). Because the plane is triangulated rather than reshaped into a rectangular mesh, gridded OR scattered sweep data plots identically -- no grid-dimension metadata (nx,ny) is needed, and a sweep with a ragged tail still works; x/y/z come from any 2-D sweep leaving three equal-length vectors (nested .dc, the sweep command family E-146/E-190, a .step grid, or .control vectors). All existing pyplot facilities carry over (set `pyplot_terminal=png/svg/pdf` headless render, style sheets, `figsize`, `backend`); three contour-specific options are added -- `pyplot_contour_levels` (level count, default matplotlib-auto), `pyplot_contour_lines` (overlay labelled black contour lines on the filled map), `pyplot_contour_cmap` (colormap, default viridis). The colorbar is labelled with z's name and the axes with x's and y's. Robustness: -contour needs exactly three vectors, else it is rejected with a clear message (pyplot -contour needs exactly three vectors) rather than a confused plot; a degenerate 1-D (collinear) sweep cannot be triangulated, so the generated script catches the qhull/matplotlib failure and exits with a helpful line (a contour

needs a genuine 2-D sweep -- the points must not be collinear) instead of a bare traceback. No change to the ordinary pyplot trace path, pyplot -hist, or pyplot -eye. Also removed a stale unused-variable warning (ft_pyplot's scale) left by E-217. Verify (examples/pyplotcontour): a closed-form surface $z=x^2+y^2$ over x,y in $[-2,2]$ (a paraboloid whose contours are concentric circles, $z=0$ at the centre and $z=8$ at the corners) -- checked analytically from the generated .data: the -contour path is taken (tricontourf not plot/hist, a colorbar labelled z, axes x/y), the data table has three columns (x,y,z) for all 1681 rows, the column mapping is correct (z reconstructs x^2+y^2 to 9e-16), the sweep is genuinely 2-D (x and y each span the full range, z runs from ~0 at the centre to ~8 at a corner), and a valid PNG is rendered. A second demo (bridge_dc_demo.cir) exercises the feature on genuine simulation output: a diode-OR bridge whose output $V(c)$ follows whichever of its two inputs is higher, swept with a nested .dc (v1 drives node a, v2 drives node b, so $V(a)/V(b)$ are the two swept values at each point); pyplot -contour v(c) v(a) v(b) maps the output over the ($V(a),V(b)$) plane -- a max-like corner surface. It checks the nested .dc yields the flattened 51x51 = 2601-row 3-column grid, the axes are the two real swept sources (each spanning $[-1,1]$), and the output is ~0 when both inputs are low and rises when either input is high (the diode-OR signature, which confirms the columns map onto a real circuit result). 19 checks. Regression 178/178 (new example folder).

E-221 *src/frontend/inpcom.c, examples/busnodes_examples/**

array/bus node ranges in the netlist. ngspice has no bus/array node syntax: a node is an opaque string, and a range like $a[0:1]$ is read as a single literal node name (so $R1\ a[0:1]\ r=2k$ gives $R1$ only one node token and misparses $r=2k$). E-221 adds a netlist preprocessing pass that expands a range token $base[lo:hi]$ into the scalar node sequence $base[lo]\ base[lo\pm1]\ \dots\ base[hi]$, so buses can be written compactly: $R1\ a[0:1]\ r=2k \rightarrow R1\ a[0]\ a[1]\ r=2k$; $X1\ n[0:3]\ mysub \rightarrow X1\ n[0]\ n[1]\ n[2]\ n[3]\ mysub$; $.subckt\ mysub\ p[3:0] \rightarrow .subckt\ mysub\ p[3]\ p[2]\ p[1]\ p[0]$. The range supplies node tokens POSITIONALLY -- the 'two terminals' reading: $R1\ a[0:1]$ is a single resistor from $a[0]$ to $a[1]$, not an array of resistors; multi-terminal instances, subcircuit calls and .subckt port lists expand the same way. Semantics: $base[lo:hi] \rightarrow base[lo]\ base[lo\pm1]\ \dots\ base[hi]$, DESCENDING when $lo>hi$ (Verilog convention $d[3:0] \rightarrow d[3]\ d[2]\ d[1]\ d[0]$); $a[0:0]$ is a single element; the scalar names use the same bracket form so a bus $a[0:1]$ and an explicit $a[0]$ denote the SAME node (the two forms interoperate); only a WHOLE whitespace-delimited token that is exactly $base[int:int]$ is expanded, so already-scalar names ($a[0]$), non-integer ranges ($a[x:y]$), malformed ones ($a[1:2:3]$), device values, model names and XSPICE %vd[...] port groups are left untouched; .controlendc blocks are skipped and only element lines and .subckt lines are processed (.model/.param/other dot cards ignored); a range wider than 8192 elements is left literal (the device parser then reports it) so a typo like $a[0:1000000000]$ cannot exhaust memory. Implementation: one pass `inp_expand_buses` in `frontend/inpcom.c`, run in `inp_readall` right after whitespace normalization and BEFORE subcircuit expansion and device parsing -- so every downstream consumer (numparam, .subckt expansion, the device parser, OSDI) sees the already-scalar node list and needs no changes; continuation lines are stitched earlier in `inp_read`, so each logical line is processed whole; the token test parses `base` (a plain node name) then `[:]` filling the rest of the token and emits the element list with DSTRING, a non-match copying the token verbatim. Verify (examples/busnodes): 6 checks, both solvers, all from node voltages/branch currents (solver-independent) -- $R1\ a[0:1]\ r=2k$ connects $a[0]-a[1]$ so $I(R1)=(1-3)/2k=-1mA$; the scalar $R2\ a[0]$ addresses the same node as the bus element ($I=+0.25mA$, only correct if $a[0]$ is one shared node); a subcircuit with a descending port bus $p[1:0]$, called with $X1\ n[0:1]$, connects positionally ($I(Rs)=-5mA$); and non-integer ($b[x:y]$), malformed ($c[1:2:3]$) and scalar tokens are left literal while a .control let $w=a[0:1]$ is not rewritten. One file; no device/solver/OSDI change; a deck with no `base[lo:hi]` token is untouched (a fast path skips any line without `'[`). Regression 181/181 (new example folder).

E-222 *src/frontend/inpcom.c, src/frontend/subckt.c, src/misc/string.c, examples/parserfuzz_examples/**

ngspice netlist-parser hardening (fuzzing): seven crashes/hangs -> clean errors. Fuzzing the netlist parser -- mutating real decks (byte flips, truncation, line duplication, delimiter/bracket/

keyword/directive/number injection, and the new E-221 bus-range tokens) and running each under ngspice -b -- found seven ways to CRASH (SIGSEGV/SIGABRT, real memory-safety bugs since ngspice is C) or HANG on malformed input; a 12000-iteration campaign dropped the crash+hang rate from 26 to 1 (96%, all hangs eliminated). ROOT CAUSES, all fixed: (1) misc/string.c gettok_instance() returns an empty token WITHOUT advancing *s when it sits on '(' or ')', and frontend/inpcom.c get_number_terminals()'s OSDI ('n') case was the only multi-token case with no iteration cap -- a line starting with 'n' containing '(' (e.g. 'nan.func f(x)=...', nan lexing as a number) spun forever tmalloc-ing each pass; gettok_instance() now always advances (consumes the bracket as a one-char token) and the 'n' loop gained the same cap as its siblings. (2) inp_expand_macro_in_str(): a truncated '.model m' (no model type) ran nexttok off the end, leaving the scan pointer NULL, then strchr(NULL); NULL guards added on the .model skip and the scan loop. (3) frontend/subckt.c doit(): MAXNEST bounds subcircuit DEPTH, but a subckt that instantiates itself with branching factor ≥ 2 blows up to 2^{MAXNEST} instances (an effective hang) before the depth cap trips -- a total-instantiation cap now catches recursive subcircuits. (4) doit(): a bare 'X' invocation (nothing after the refdes) makes the find-last-token walk run off the FRONT of the line buffer -- an out-of-bounds read ending in strcmp(NULL); such a line is now skipped. (5) inp_modify_exp(): two copies into a fixed buf[512] with no bound -- an unterminated 'v' (no closing ')') and a long identifier token (a run of '|', which is an accepted char) overran the STACK buffer (stack smashing); both loops are now bounded by the buffer size. (6) doit(): a malformed '.subckt' can register a NULL name; the invoked-name match eq(su_name,...)=strcmp() then deref NULL; guarded. (7) translate(): gettok_noparens() returns NULL at the end of a malformed controlled-source line, then strcmp(NULL,'POLY'); guarded. Method: the fuzzer classifies OK/clean-error/CRASH/HANG over self-contained seed decks exercising the parser surface (devices, .subckt, .param, .model, .control, expressions, continuations, buses); each crash was triaged to its faulting instruction with lldb (conditional strchr/strcmp-with-NULL breakpoints), sample for the hangs, and Guard Malloc (libgmalloc) to pin the stack overrun, then read at the source; fixes applied one at a time, verified against the recorded crash corpus, and a fresh 12000-iteration run confirmed convergence. No valid deck changes behaviour. Verify (examples/parserfuzz): 10 checks -- a minimal deck per root cause (nan.func, .model in .control, a self-recursive subckt, a bare X line, an unterminated v, a 4000-'|' identifier, 4000 '{' in a subckt E-source) now yields a clean/bounded outcome (no signal/abort, no hang), and three valid decks (subckt hierarchy with an E-source and a parameter, a v(x)*v(x) B-source expression, a POLY source in a subckt) compute V to the analytic value. The recorded 27-input crash corpus is all clean. RESIDUAL: one fuzz input in 12000 still crashes in XSPICE (xspice/mif/mifgetmod.c MIFgetMod) -- an 'a'-device whose model name matches a non-code-model (e.g. a diode .model) is processed as a code model, reading device->modelParms as the wrong type; a code-model type-confusion in the XSPICE subsystem, out of scope for this netlist-parser pass and flagged as a follow-up. Three files (frontend/inpcom.c, frontend/subckt.c, misc/string.c); no device/solver/OSDI change. Regression 181/181 (new example folder).

E-223 *src/xspice/mif/mifgetmod.c, examples/xspicemodel_examples/**

XSPICE a-device model-type validation (MIFgetMod): fixes the one residual crash left by the E-222 netlist-parser fuzzing pass. An a device is an XSPICE code-model instance; the last token on its card is a model name, which MIF_INP2A (xspice/mif/mif_inp2.c) hands to MIFgetMod (xspice/mif/mifgetmod.c). MIFgetMod looks the model up in the pass-1 table and, if it has not been instantiated yet, processes its .model card AS a code model -- it casts INPmodfast to MIFmodel, reads the device's XSPICE DEVpublic fields (param, num_param), and matches the .model parameters through the MIF parameter path -- with NOTHING checking that the resolved model is actually a code model. If an a-device's model name happens to match a NON-code-model .model (a diode, a BJT, a resistor model), MIFgetMod walked all of that machinery over a structure that is not a MIFmodel and over DEVpublic fields that a built-in SPICE device leaves unset, reading unrelated memory as the wrong type -> SIGSEGV. The crash bit hardest when the non-code-model's parameter keywords COLLIDE with the a-device's: a diode .model has is/n, so the parameter loop found a keyword match and dereferenced the type-confused setModelParm path. The fuzz-

found trigger (the E-222 residual) was exactly this, reached through an E-221 bus-expanded a-device inside a subcircuit (a[0:0] D1 a b dm with .model dm d(is=1e-14 n=1)). FIX: a code model is exactly a device that carries a code-model evaluation function -- cmpp emits .cm_func = <fn> for every code model (xspice/cmpp/writ_ifs.c) and it is the pointer XSPICE calls to evaluate the model (mifload.c/evtload.c), while every built-in SPICE device sets .cm_func = NULL. MIFgetMod now validates DEVICES[INPmodType] -> DEVpublic.cm_func != NULL (and INPmodType < DEVmaxnum) BEFORE the MIFmodel cast and the parameter loop, returning a clean 'model X is not a code model; an a device requires an XSPICE code-model .model' error otherwise. The guard sits before the cast, so it catches the model whether or not an ordinary device of the same name already instantiated it. A NULL guard on the inner strcmp was rejected as whack-a-mole (the crash merely shifts to the next type-confused access); the model TYPE is the thing to validate. Legitimate code models are unaffected (cm_func != NULL). Scope: XSPICE only, one file (xspice/mif/mifgetmod.c); no netlist-parser, device, solver, or OSDI change. Verify (examples/xspicemodel): 8 checks, both solvers -- six pathological decks (a-device -> diode with and without colliding params, BJT, resistor model, the bus-expanded-in-a-subckt fuzz repro, and an undefined model) now yield a clean bounded error (no signal/abort, no hang), and a legitimate adc_bridge code-model a-device still binds and simulates (analog divider node v(a)=0.5). Regression 182/182 (new example folder).

E-224 *src/frontend/parse.c, src/frontend/evaluate.c, examples/arraynodeprint_examples/**

Reference array/bus node voltages in **print/plot/let**. Enhancement-221 added array/bus node ranges to the netlist (R1 a[0:1] -> R1 a[0] a[1]), naming the nodes with LITERAL brackets so the node voltage vector is stored as a[0]. But the frontend expression parser uses [...] as its vector-INDEX operator, so in print/plot/let the token a[0] was parsed as 'element 0 of a vector named a'. There is no vector a, so print v(a[0]) and print a[0] silently failed (with a 'vector a is not available' warning) even though the node existed and solved correctly (it shows up in the .op node table and display). FIX: a literal-node fallback -- when the base name of an index expression is an UNRESOLVED name (a zero-length placeholder dvec) and the index is a constant and a node/vector named base[index] exists, use that node instead of indexing. It only fires when the base does not resolve, so ordinary vector indexing (realvec[3], where the base IS a real vector with non-zero length) is unaffected. Three frontend spots, because a[0] reaches the machinery three ways: (1) frontend/parse.c checkvalid() -- the pre-evaluation validity check rejected the whole term because base a is a zero-length placeholder; for an INDX node over an unresolved base it now accepts the term when the literal node base[index] exists. (2) frontend/evaluate.c op_ind() -- evaluates the bare a[0] form; it now resolves the literal node name BEFORE evaluating the bare base, so no spurious 'no such vector a' is emitted. (3) frontend/evaluate.c apply_func() -- the v() node-access special case needed arg->pn_value, which an INDX op node lacks (it errored 'bad v() syntax'); it now reconstructs the literal node name from the INDX argument and resolves it. RESULT: print a[0], print v(a[0]), array nodes inside expressions (v(a[0])-v(a[1])), and plot/let all work; the recovered vector is a normal waveform (indexable, aggregatable) in transient. Branch current through the bus resistor stays the device parameter @r1[i] (ngspice has no current-between-two-nodes syntax; i() takes a source name). Scope: frontend only, two files (parse.c, evaluate.c); no device/solver/OSDI change. Verify (examples/arraynodeprint): 8 checks, both solvers -- print a[0]/v(a[0])/v(a[1]), the expression v(a[0])-v(a[1]), and @r1[i] read the correct DC values; ordinary indexing unitvec(4)[2] is unchanged; a genuinely-missing node v(a[9]) stays a clean miss (no value, no crash). Regression 183/183 (new example folder).

E-225 *src/frontend/evaluate.c, src/frontend/com_measure2.c, src/frontend/fourier.c, src/math/cmaths/cmath4.c, examples/cmdfuzz_examples/**

Command/expression-evaluator crash hardening (fuzzing). With the netlist parser hardened (E-212/E-222/E-223), the fuzzer was turned on a previously-unfuzzed surface: the interactive .control command interpreter and the vector-expression evaluator (print/let/plot/meas/fft/fourier

and the operators `?:`, indexing `[ ]`, ranges `[ : ]`, arithmetic). Mutating command lines/expressions over a running circuit and running each under ngspice `-b` found memory-safety crashes (SIGSEGV / SIGABRT, real since ngspice is C); 62 crashes in the first 2500 iterations, driven to 0 by FIVE fixes in three groups. GROUP A -- degenerate-input heap overflows in the math transforms (fft/deriv/fourier each crashed on a too-short (< 2 -point) or synthetic vector like `fft(1)`, `deriv(vecmin(v(1)))` (a plot-derived scalar), `fourier 1k deriv(vecmin(v(1)))`; the malloc abort fired on a LATER allocation, so they looked non-deterministic): (1) `maths/cmats/cmath4.c cx_fft()` -- the Green's FFT (rffts) derefs bit-reversal tables `BRLowArray[(M-1)/2]` that `fftInit()` only allocates for $M > 2$, so an input of ≤ 4 points read unallocated memory; and `time/xscale` were sized by the data length while the scale loops fill `pl_scale` $\rightarrow$ `v_length` entries and the scale vector uses `fpts` points, so a vector shorter than the plot scale overran both -- pad to $N \geq 8$, size buffers for the largest fill, reject `length < 2/scale-too-short`. (2) `maths/cmats/cmath4.c cx_deriv()` -- the polynomial-fit derivative reads `degree+1` points per fit; a `length-1` input ran the fit/edge loops over too few points and overran the heap -- for `length < 2` return a zero vector (derivative of a single point is undefined). (3) `frontend/fourier.c` -- a degenerate input gives a zero time span and overran the interpolation grid -- require ≥ 2 data points. GROUP B -- `frontend/evaluate.c ft_ternary()`: the `?:` operator evaluated its condition and selected branch with no NULL check, so when either failed to evaluate (`1?0[3]:9`, where `0[3]` indexing-a-scalar returns NULL) it hit `vec_copy(NULL)` / `cond->v_link2` and crashed -- the dominant crash; fixed with NULL guards (an audit confirmed `op_ind`, `op_range`, binary-op and `apply_func` already guard NULL). GROUP C -- `frontend/com_measure2.c`: measure error messages were formatted into a shared `char errbuf[100]` (passed by pointer to the `measure_parse_*` helpers); a failed measure writes its ENTIRE expression string as the vector name (`sprintf(errbuf, "no such vector as '%s'", meas->m_vec)`), so a measure expression longer than ~80 chars (trivially `meas tran m MAX (v(1)+v(1)+...)`) overran it $\rightarrow$ SIGABRT (content irrelevant, only length, which is why it resisted reduction) -- fixed with a file-scope `MEAS_ERRBUF_SIZE` and a bounded `snprintf` at every write (28 sites). Note: `dowhile 1 ... end` (true constant condition, no break) is an intentional infinite loop, flagged as a HANG but correct. Scope: `frontend/math` only, four files (`evaluate.c`, `com_measure2.c`, `fourier.c`, `cmath4.c`); no device/solver/OSDI change. Verify (`examples/cmdfuzz`): 13 checks -- a minimal repro per root cause now yields a clean bounded outcome, and regressions confirm `fft(v(1))/deriv(v(1))`, a valid `?:`, and ordinary indexing still work. Regression 184/184 (new example folder).

E-226 *src/frontend/rawfile.c, examples/rawfuzz_examples/**

Rawfile-load crash hardening (fuzzing). Continuing the fuzzing campaign onto another untrusted-input surface: the nutmeg rawfile reader (`raw_read` in `frontend/rawfile.c`, reached by the `load` command). A `.raw` file is external data (often written by another tool), so its parser must survive malformed input. Mutating a valid `.raw` (produced by `write`) -- tweaking the header counts, corrupting the `Flags: Variables:` lines, adding/removing value rows, truncating, flipping bytes -- and `load-ing` each found a NULL-dereference crash (SIGSEGV, `memmove` into `0x0`). ROOT CAUSE: `raw_read` resets `flags = VF_PERMANENT` at the start of each plot and sets `VF_REAL` / `VF_COMPLEX` only when it parses a `Flags:` line; each vector is then allocated with `dvec_alloc(NULL, SV_NOTYPE, flags, npoints, NULL)`, which allocates `v_realdata` OR `v_compdata` according to those bits. A rawfile with NO valid `Flags:` line -- the line missing entirely, or carrying only unknown flags (`Flags: xyz`) -- leaves `flags` with neither `VF_REAL` nor `VF_COMPLEX`, so `dvec_alloc` allocates NO data array at all, and the value-reading loop does `fread(&v->v_realdata[i], ..) / writes v->v_compdata[i]` into a NULL buffer $\rightarrow$ `memmove` into `0x0` $\rightarrow$ SIGSEGV. (A bad No. Points: / No. Variables: count -- huge, negative, non-numeric -- was already handled cleanly; only the missing TYPE was fatal.) FIX: when the header is complete and vector allocation begins (at the `Variables:` line), if `flags` carries neither `VF_REAL` nor `VF_COMPLEX`, default to real (the common case) with a warning instead of allocating a typeless vector; the `VF_PERMANENT` reset is per-plot so the guard runs for every plot in a multi-plot file, and a valid file is unaffected. Also fuzzed in the same pass: the interactive `.control` command

interpreter and shell layer (variable substitution `\$var`, `alter/altermod/alterparam`, analysis reruns, structured control `if/while/foreach/repeat/dowhile`) -- clean over 7500 iterations (already hardened by E-212/E-222/E-225). Scope: frontend only, one file (`rawfile.c`); no device/solver/OSDI change. Verify (`examples/rawfuzz`): 4 checks -- a valid `.raw` is written, then loaded with its Flags: line removed and with an unknown Flags: `xyz`, each now a clean bounded outcome (was SIGSEGV); a regression confirms a valid `.raw` still round-trips write -> load with the right length. Regression 185/185 (new example folder).

E-227 *src/frontend/snp2va.c, examples/snpfuzz_examples/**

Touchstone `.snp` parser crash hardening (fuzzing). Continuing the fuzzing campaign onto the Touchstone S-parameter parsers -- external network-data files (VNAs, EM solvers, other tools), so their parsers must survive malformed input. Two exist: `rdsnp` (the E-72 Touchstone-v1 reader) and `pre_snp` (the E-200/E-201 convert-to-Verilog-A path in `frontend/snp2va.c` that vector-fits the data and compiles it through `openvaf-r`). Mutating valid `.s1p/.s2p/.s3p` files -- corrupting the # option line, data values (NaN/Inf/overflow tokens, added/removed columns), duplicating/deleting rows, truncating, flipping bytes -- AND mismatching the port count encoded in the `.snp` filename extension, then running each through both readers: `rdsnp` was clean (4,500 iters); `pre_snp` hit a heap-corruption crash (SIGSEGV) 103 times, ALL on a filename with an absurd port count (e.g. `.s2147483647p`). ROOT CAUSE: `snp2va.c` infers the port count from the extension (`N = atoi(dot+2)`) with no upper bound; a huge `N` slips past the parse itself because the record stride `rec = 1 + 2*N*N` and the per-record inner-loop bound `N*N` use the same wrapped int (`N*N` overflows to a small/zero value, so the token layout stays self-consistent and no error is raised), but `N` is stored in `out->N` and used DOWNSTREAM to size the vector-fit allocations (`malloc(nf*N*N*sizeof(cplx))`, `malloc(N*N*sizeof(cplx))`) -- the overflowing/mismatched allocation and the writes that follow corrupt the heap (the program aborts later on an unrelated `malloc`, hence flaky). The brute-force fallback (used when the filename gives no usable count) already caps inference at 512 ports, so 512 is the parser's own sanity bound -- the filename path just wasn't held to it. FIX: hold the filename-derived count to the same limit -- `if (N > 512) N = 0`; falls back to inferring `N` from the data. A valid file is unaffected (its extension is far below the cap) and a genuinely mis-extended file now recovers `N` from the data layout instead of trusting an absurd filename. Scope: ngspice frontend only, one file (`frontend/snp2va.c`); no device/solver/OSDI change. Verify (`examples/snpfuzz`): 2 checks -- a valid `.s2p` copied to a `.s2147483647p` name loads via `pre_snp` without crashing (run repeatedly, was SIGSEGV), and a valid `.s2p` still converts through the full pipeline (`pre_snp -> .va -> openvaf-r -> .osdi`). Regression 186/186 (new example folder).

E-228 *src/osdi/osdiregistry.c, examples/osdifuzz_examples/**

OSDI `.osdi` loader descriptor-count hardening (fuzzing). A `.osdi` model is an external, compiled shared library that ngspice `dlopens` and reads descriptor metadata from (`load_object_file`, reached by the `pre_osdi` command). The loader strided/indexed the descriptor arrays by the counts the file declares -- `OSDI_NUM_DESCRIPTOR`s (outer loop stride by `*OSDI_DESCRIPTOR_SIZE`) and each descriptor's `num_params` (inner loop indexing `param_opvar[param_id]` and dereferencing its `name[]` pointers), `num_opvars/num_terminals/num_noise_src/...` (offset arithmetic) -- with NO bounds. A `.osdi` that still `dlopens` but declares a count larger than its real array (a truncated file, a byte-corrupted one, or one built by a compiler whose OSDI descriptor ABI differs -- the drift the loader's own comments warn about) read past the array -> SIGSEGV during `pre_osdi` (same unbounded-count class as E-227, here on a binary rather than a text surface). FIX: two generous ceilings in `osdiregistry.c` (`#define OSDI_MAX_DESCRIPTOR 4096u`, `#define OSDI_MAX_DESC_COUNT (1u<<20)`) far above any real compact model (the biggest ship a few dozen modules and ~1k params); after reading `OSDI_NUM_DESCRIPTOR`s, reject the file (clean diagnostic, `INVALID_OBJECT`) if it exceeds the first, before it sizes the registry or strides the array; inside the descriptor loop, reject if any of a descriptor's count fields (`num_nodes/num_terminals/num_params/num_instance_params/num_opvars/num_noise_src/num_jacobian_entries/num_collapsible/num_states`) exceeds the second, before those fields index/size anything

-- a single load-time choke point protecting every downstream consumer (setup, eval), since a rejected file never registers its devices. HONEST SCOPE: this targets the egregious/out-of-range count (the signature of a truncated/byte-corrupted/ABI-drifted object, where corruption reliably lands -- an ABI mismatch reads a pointer's bits as a count -> huge value); it is NOT a claim that the loader is proof against every malformed .osdi -- a count that lies within the plausible range is an internally inconsistent binary indistinguishable from a valid one, and a .osdi is native code whose mere loading executes it, so a hostile object is out of scope by construction. On macOS byte-corrupting a signed .osdi also breaks its code signature so dlopen rejects it before the loader runs; the residual exposure is chiefly Linux (no signing gate) and toolchain/ABI drift. Scope: ngspice only, one file (osdi/osdiregistry.c); no device/solver/compiler change. Verify (examples/osdifuzz): 3 checks -- two malformed .osdi built from tiny C stubs (one declaring a huge OSDI_NUM_DESCRIPTORs, one with a valid count but a huge per-descriptor num_params built from the real OsdiDescriptor struct in osdi.h) now load-reject cleanly (both a SIGSEGV rc=139 on the pre-fix binary), and a positive control confirms a real openvaf-r-built .osdi still loads and simulates (v(a)=1). Regression 187/187 (new example folder).

E-229 *src/frontend/com_dl.c, src/frontend/commands.c, src/spicelib/devices/dev.c, src/spicelib/devices/dev.h, examples/osdireload_examples/**

pre_osdi -f: reload a recompiled OSDI model without restarting. A Verilog-A developer's inner loop is edit .va -> recompile .osdi -> re-run, but interactively ngspice loaded each .osdi once: load_osdi (dev.c) is guarded by a path dedup, a dlopen-handle dedup (osdiregistry.c), and a device-name dedup (a type already in the global DEVices[] is ignored because the model-card lookup scans front-to-back and the first/stale entry wins -- see E-223); underneath, dlopen caches by path (a re-open returns the cached handle, never re-reading the recompiled file). So a recompiled model never took effect without restarting. FIX -- an opt-in -f/-force flag: com_osdi (com_dl.c) parses -f anywhere in the argument list and passes a force flag to load_osdi (path, force); on a forced reload of an already-loaded path it STAGES a byte-for-byte copy of the (recompiled) file under a fresh unique path and loads that (a new path is always re-read, sidestepping dlopen's cache; this avoids dlclose, which on macOS does not reliably unmap and where overwriting a still-mapped file faults the process), then removes the staged copy once it is mapped (the mapping keeps it alive on POSIX), and osdi_add_device(..., replace) swaps the registered device to the freshly loaded descriptor IN PLACE at its existing table index so no other device type's index shifts. The previous SPICEdev and the library mapping it points into are intentionally left resident: a circuit still built against the old model keeps a valid device, and freeing descriptor-owned memory would risk a double free (documented caveat: don't re-run a pre-existing circuit built against the prior model version after reloading it). Behaviour is unchanged without -f (plain re-load still skipped, now with a hint pointing at -f); the osdi command help gains the flag. Scope: frontend + device registry, four files (frontend/com_dl.c, frontend/commands.c help text, spicelib/devices/dev.c, spicelib/devices/dev.h); no solver/analysis/OSDI-ABI/compiler change. Verify (examples/osdireload): compiles a resistor model at 1k then 2k, drives ngspice in pipe mode to load the 1k version (i(v1)=-1mA), recompiles the same file to 2k mid-session, confirms a plain re-load is skipped (still 1k), then osdi -f reloads it and the operating point changes to the 2k result (i(v1)=-0.5mA) -- one process, no restart. Regression 188/188 (new example folder).

E-231 *src/frontend/com_gnuplot.c, src/frontend/plotting/gnuplot.c, examples/csv_examples/**

CSV output for **wrdata (set wr_csv + wrdata -csv)**. wrdata only ever wrote whitespace-separated columns (each vector prefixed by its own scale column); the knobs wr_singlescale/wr_vecnames/wr_onespace/numdgt tune spacing and precision but never the separator, so a genuine comma-separated export needed a post-processing pass. The only -csv in the tree was show -csv (device-parameter tables, not data). ADDED a first-class CSV mode. (1) set wr_csv -- a boolean read in ft_writesimple (plotting/gnuplot.c) alongside the existing wr_* vars; when set it takes a dedicated, self-contained branch that writes a single shared scale column once (implies wr_singlescale, so the existing equal-length sanity check applies), a header row of vector names (

implies `wr_vecnames`), and comma-separated values with `"%.*e"` (no sign-padding, no trailing separator) honouring `numdgt`; complex `.ac` vectors become two columns (real, imag). The branch writes and returns before the space-formatted writer, so DEFAULT `wrdata` output is byte-for-byte unchanged. (2) `wrdata -csv <file> <vec...> --` a per-call alias accepted in ANY argument position: `com_write_simple` (`com_gnuplot.c`) scans the wordlist for a `-csv` token, enables `wr_csv` for just that write, and restores the prior global state afterward (no leak). Because `plotit` COPIES its wordlist (never mutates/frees the caller's), the `-csv` node is spliced out for the call and relinked afterward, leaving the caller's list intact to free; the head case is detected as `csvnode == wl` (the incoming head's `wl_prev` is not necessarily null), and no-vector forms fall through the `if (wl)` guard writing nothing, node still relinked. Scope: `ngspice` frontend only, two files (`frontend/com_gnuplot.c`, `frontend/plotting/gnuplot.c`); no solver/analysis/device/OSDI/compiler change; default output preserved exactly. Verify (examples/csv_examples, 9 checks): default output unchanged (no commas); set `wr_csv` writes a header + comma-separated values with the right numbers; `wrdata -csv` in first/mid/last position is byte-identical to the option form; CSV numbers equal the default-format numbers bit-for-bit; the flag does not leak into the next plain `wrdata`; `.ac` emits real/imag columns under a frequency scale; `.tran` emits a time scale with uniform csv-parseable rows. Regression 190/190 (new example folder).

E-232 *src/math/KLU/klusmp.c, examples/solverfix_examples/**

KLU solver-glue correctness hardening (source audit). A deep read of the active SMP dispatch layer -- `klusmp.c`, the ONLY sparse-matrix-package layer compiled when KLU is enabled (`sparse/Makefile.am` builds `spsmp.c` only if `!KLU_WANTED`; `klusmp.c` routes BOTH solvers via its `CKTkluMODE` branches) -- found four latent defects, all fixed, all behaviour-preserving for any circuit that actually solves (prior functional KLU bugs E-113/114/115/171/172 were already fixed and are confirmed still in place). (A) `SMPluFac` and `SMPsolve` dereferenced `KLUmatrixCommon->status` BEFORE their `== NULL` guard; the complex twin `SMPclufac` checks `NULL` first with an early return -- reordered both to match (`SMPsolve`, being void, now reads `status` only inside the non-`NULL` else). Common lives for the matrix lifetime so it is never actually `NULL` here; the fix aligns the paths and satisfies a static analyzer. (B, the substantive one) the complex solves `SMPcSolve/SMPcaSolve` copied the RHS with a plain identity map (`RHS[i+1]`), while the real `SMPsolve` routes RHS through the node-collapse permutation `KLUmatrixNodeCollapsingNewToOld`; if a structural-zero column ever collapsed a node (`map != identity`) the AC/noise/pz RHS would be silently mis-ordered relative to the DC/tran solve. Dormant -- the map is the identity for any circuit that factors (an empty MNA column is structurally singular and does not solve), which is why KLU AC/noise/pz have always matched Sparse -- but the two paths were inconsistent; both complex solves now apply the identical gather/solve/scatter so a future change to collapse cannot quietly break complex analyses. (C) `SMPcZeroCol` indexed `KLUmatrixAp[Col-1]` with no ground guard (`Ap[-1]` if `Col==0`); added the `Col >= 1` guard its sibling `SMPfindElt` already has (pole-zero passes indices `>= 1`, so it never fired). (D) `SMPmultiply`'s real branch ended with a dead `iSolution = iRHS`; (reassigns a by-value local; a real matrix has no imaginary product) -- removed. Scope: `ngspice` only, one file (`maths/KLU/klusmp.c`); no analysis/device/OSDI/compiler change and no behavioural change for any solvable circuit. Verify (examples/solverfix, 4 checks): on an ASYMMETRIC net (a VCVS makes the MNA matrix non-symmetric, so the noise ADJOINT solve genuinely differs from the forward solve) KLU still agrees with SPARSE 1.3 to the bit on AC, on the noise spectrum (exercising the most-changed `SMPcaSolve`), and on pole-zero roots (`SMPcaSolve + SMPcZeroCol + SMPcAddCol`), plus a clean DC-op + transient run under KLU. Regression 191/191 (new example folder).

E-233 *src/math/KLU/klusmp.c*

KLU glue deeper audit: finish the null-check reorder + correct the collapse map. Follow-up to E-232 from a deeper read of the same file; two more latent defects, both behaviour-preserving (one path never triggers, the other is confirmed-dead code). FIRST two scarier candidates were investigated and CLEARED empirically, not just by inspection: (i) a suspected per-analysis memory leak

-- .pz/.sens reuse ckt->CKTmatrix and re-call SMPconvertCOOtoCSC + SMPpreOrder without freeing the prior arrays/Symbolic, which LOOKS like a leak -- was REFUTED by instrumenting the lifecycle over op->pz->sens: every convert/preorder sees a FRESH matrix (Ap==NULL, Symbolic==NULL) and SMPnewMatrix/SMPdestroy are balanced (5 new / 4 destroy / 0 overwrite-without-destroy); ngspice tears down and rebuilds the matrix per analysis, so nothing leaks (no fix made -- no bug). (ii) node-collapse is DEAD code: instrumenting the collapse branch and sweeping the whole 191-example regression (every OSDI model, pz, sens, noise) plus pathological topologies fired it 0 times, because OSDI collapses nodes at allocation time (osdisetup.c: a collapsed node gets no matrix column) so the COO never has a gap and Gmin fills every diagonal. FIXES: (3) E-232 fixed the dereference-before-NULl-check in SMPluFac/SMPsolve but the SAME pattern survived in SMPcReorder and SMPreorder (they read KLUmatrixCommon->status before the ==NULL guard) -- reordered both to check NULL first with an early return, matching SMPcLUfac, and corrected two mislabeled diagnostics (the Common-NULl and Symbolic-NULl branches each printed a stray 'KLUnumeric object is NULl'). Common is never actually NULl, so behaviour is unchanged. (2) in SMPconvertCOOtoCSC the structural-zero-column compaction recorded NewTo0ld[reduced] = MatrixC00[i].col, but with TWO OR MORE gaps the RHS is a partially-reduced index from an earlier pass, not the true original, so the map would resolve to the wrong node and the real SMPsolve gather/scatter would mis-address the RHS -- fixed by chaining through the existing map (NewTo0ld[reduced] = NewTo0ld[MatrixC00[i].col], identity on the first pass so single-gap behaviour is unchanged); defensive since the path is confirmed dead. Scope: ngspice only, one file (maths/KLU/klusmp.c); no analysis/device/OSDI/compiler change and no behavioural change for any circuit. Verify: behaviour-preserving, so no new example -- the existing examples/solverfix_examples (KLU vs SPARSE bit-agreement on AC/noise/pz over an asymmetric net) plus the full 191/191 regression remain unchanged; fix 2's dead path cannot be exercised by construction (correctness is by the chaining argument). Regression 191/191.

E-234 *src/frontend/com_loadpull.c (new), src/frontend/com_loadpull.h (new), src/frontend/commands.c, src/frontend/com_commands.h, src/frontend/Makefile.am, examples/loadpull_examples/**

loadpull: power-amplifier load-/source-pull analysis. Load-pull (sweep the load impedance a PA output sees over the Smith chart and contour Pout/gain/PAE/efficiency to find the optimum load) is a PA-design staple ngspice lacked, and open-source load-pull is scarce. NEW loadpull command rides the existing .tran engine + alter (as optimize/sweep do) + pyplot -contour (E-218). loadpull -load <R> <L> <C> -out <node> -drive <Vsrc> -f <f0> [-supply <Vsrc>] sweeps Gamma over ‘

E-235 *src/frontend/com_stb.c, examples/stbfix_examples/**

stb: fix a latent use-after-free in the probe lookup. The defensive follow-up flagged in E-234. com_stb resolved its voltage probe with INPretrieve(&name, symtab), which replaces the pointer with the interned symbol-table string -- the same memory the voltage source's own name field points at -- and does NOT free the old copy; the following tfree(name) therefore double-freed the source's live name (and the symbol-table entry). It never bit in practice because stb runs once with no re-setup, but it is real memory corruption and the identical defect that bit loadpull in E-234 (there the per-point re-setups re-created the drive's branch node from that freed, allocator-reused memory). Fixed by dropping INPretrieve (top-level device names need no subckt translation -- findInstance does its own name match) and lowercasing a private copy instead (ngspice stores instance names lowercased), so only our own copy is freed; as a bonus the probe lookup becomes CASE-INSENSITIVE -- stb Vprobe Iprobe used to fail "no such probe source 'Vprobe'" and now resolves. The now-unused complex helper stbsub was removed to keep the build warning-free. Scope: ngspice frontend only, one file (com_stb.c); no solver/analysis/device/OSDI/compiler change; loop-gain results unchanged. Verify (examples/stbfix, 3 checks): a mixed-case probe name resolves and reports a loop gain (failed before); the loop gain is still correct (a gain-1e5 buffer loop -> ~100 dB DC loop gain, measured 99.91 dB); 60 repeated stb runs stay stable. Regression 193/193 (new example folder).

E-236 *src/frontend/com_measure2.c, src/frontend/com_measure2.h, src/frontend/measure.c, src/tclspice.c, examples/measovf_examples/**

.meas: fix a stack-buffer overflow on long measurement names. Found in a memory-safety deep dive; reproduces on the stock binary. `get_measure2()` formats each measurement result line with `sprintf(out_line, "%-20s= %.*e ...", mName, ...)` into the caller's fixed stack buffer `char out_line[1000]` (`measure.c`). `mName` is the measurement NAME token, taken verbatim from the `.meas <analysis> <name> <func> ... card` via `cp_unquote` -- an unbounded user string with no length check. A `.meas` name longer than ~1000 chars overruns `out_line`; on macOS the stack canary fires and the process aborts (SIGABRT, exit 134), elsewhere it is plain stack corruption. All ten result-formatting sites share the flaw. E-225 had already hardened the sibling `errbuf[100]` in this same file to `snprintf` (after a measure-syntax fuzz campaign), but the fuzzing never exercised a long measurement NAME, so `out_line` was missed. Fix: thread the destination buffer size through `get_measure2(..., size_t max_out_line, ...)` and convert all ten `sprintf(out_line, ...)` to `snprintf(out_line, max_out_line, ...)`. The three callers pass the real size -- `measure.c` batch `.meas` passes `sizeof out_line`, the autostop check and the `tclspice.c` interactive path pass `NULL/0` (all ten sites are already guarded by `if (out_line)`, so they never reach `snprintf`). An over-long name is safely truncated to 999 chars; the measurement value itself is computed and stored exactly as before. Scope: ngspice frontend only; no solver/analysis/device/compiler change; measurement results unchanged. Verify (`examples/measovf`, 2 checks): a `.meas` with a ~4000-char name no longer crashes (exit 0, was 134); normal measurements still correct (MAX 1.0, AVG 0.6, 0.1->0.9 rise time 0.8 ns). Regression 194/194 (new example folder).

E-237 *src/frontend/dotcards.c, src/frontend/vectors.c, examples/nameovf_examples/**

.print/.plot/.four: fix stack-buffer overflows on long vector/node names. Sibling of E-236, same deep dive. The SPICE2-compat rewrites for output tokens, and the vector-name helper they feed, each copied an unbounded user name into a fixed `B_SIZE_SP[512]` stack buffer: `fixem()` (`dotcards.c`) rewrites `v(a,b)->v(a)-v(b)` (+ `vm/vp/vi/vr/vdb` variants) via `sprintf(buf, "v(%s)-v(%s)", a, b)` (19 writes); `gettoks()` (`dotcards.c`) rewrites `i(x)->x#branch` via `sprintf(buf[513], "%s#branch", x)`; `vec_basename()` (`vectors.c`) does `strcpy(buf, v->v_name)`, reached by `.print/fft/spec/linearize`. A `.print/.plot/.four` output token whose node/branch name(s) exceed the buffer overran the stack -- macOS aborts (SIGABRT/SIGTRAP, shell exit 133), elsewhere plain corruption. Fixing `fixem` alone was NOT enough: with valid long node names the rewrite succeeds and the long `v(a)-v(b)` name then flows into `vec_basename` and crashes there, so all three needed hardening. Fix: size each scratch buffer to its input -- `fixem` allocates `strlen(string)+32` and every write is a bounded `snprintf(buf, bufsz, ...)`; `gettoks` builds the string with `fprintf`; `vec_basename` replaces the fixed buffer + `strcpy` with `copy()` of the source (allocates to fit), preserving branch semantics. No truncation, so a valid long differential still computes the correct value. The `vectors.c` recompile also surfaced 3 pre-existing `-Wconversion` warnings (`v->v_type`, enum `simulation_types`, passed to `dvec_alloc's` int param), silenced with `explicit (int)` casts (build stays warning-free). Scope: ngspice frontend only; no solver/analysis/device/compiler change; output values unchanged. Verify (`examples/nameovf`, 4 checks): long `v(a,b)` on nonexistent nodes no longer crashes; a VALID long differential `v(A,B)` with `v(A)=2,v(B)=1` runs and prints `v(A)-v(B)=1.0` exactly (no truncation); long `i(x)` via `.four` (`gettoks`) no longer crashes; short `v(1,2)` still rewrites to `v(1)-v(2)=2.0`. Regression 195/195 (new example folder).

E-238 *src/frontend/dotcards.c, examples/malftoken_examples/**

gettoks: fix a NULL-deref crash on a malformed differential token. Third find of the same deep dive (after E-236/E-237), from fuzzing degenerate output tokens; reproduces on the stock binary. `gettoks()` parses the output tokens of `.save/.print/.plot/.four` (via `ft_savedotargs`) and `.measure` vars (via `com_measure2`). For a differential `v(a,b)` it finds the comma c and close paren

`r(r = strchr(t, ','), c = strchr(t, ','))` and splits the second operand at `r`: `if (c != r) { *r = '\0'; ...}`. A MALFORMED token like `v(1, has a comma but NO ')'`, so `r` is NULL while `c` is not -- `c != r` is then true and `*r = '\0'` dereferences NULL -> SIGSEGV (EXC_BAD_ACCESS at 0x0; lldb: `gettoks <- ft_savedotargs`). A one-character typo (missing paren) in a `.print/.save/.plot/.four/.measure` token crashes the simulator. Fix: require `r` non-NULL before splitting -- `if (r && c != r)`. When there is a comma but no ')', the split is skipped and the token degrades to a harmless parse; every well-formed case is unchanged (`v(a,b)` both non-NULL `c!=r` -> split; `v(a) c==r` -> skip; `i(x)` sets `c=r=NULL` -> skip). Scope: ngspice frontend only, one guard in `dotcards.c`; no solver/analysis/device/compiler change; well-formed parsing unchanged. Verify (examples/malftoken, 3 checks): `.print tran v(1,` no longer crashes; the malformed shape across `.save/.plot/.four/.meas + i()/vdb()/vm()` no longer crashes; well-formed `v(1,2)` still rewrites to `v(1)-v(2)=1.5`. A broad malformed-token sweep (prefixes×bodies×cards, hundreds) is crash-free after the fix. Regression 196/196 (new example folder).

E-239 *src/spicelib/parser/inpptree.c, examples/funcarity_examples/**

Expression parser: fix a NULL-deref on a one-argument **min/max/pow/pwr**. Fifth find of the same deep dive (after E-235–E-238), from fuzzing the behavioral-source expression parser; reproduces on the stock binary. `PTeval()` (`ifeval.c`) evaluates the two-argument functions `pow/pwr/min/max` by reading their operands as `tree->left->left` and `tree->left->right` -- it assumes the argument is a PT_COMMA pair. The function-node builder `PT_mkfnnode()` (`inpptree.c`) looks the name up but NEVER validates argument count, so a one-argument call like `min(1)` builds a node whose `left` is a scalar PT_CONSTANT (its `->left` is NULL); at load time `PTeval` does `tree->left->left -> NULL` deref -> SIGSEGV (`PTeval <- IFeval <- ASRCload <- CKTload`). Reachable from any B-source, E/G source, or `.param` expression -- a one-line typo crashes the simulator during setup. Crash is specific to these 4 functions and the scalar (non-comma) one-arg form: `min()` is caught by the existing `if (!arg)` guard, `min(1,2,3)` yields a PT_COMMA (no deref), and single-arg forms of other functions (`hypot(1)`, `atan2(1)`) fall through the single-arg path and error cleanly. Fix: reject the wrong arity at parse time in `PT_mkfnnode` -- `if ((POW`

E-240 *src/xspice/icm/analog/s_xfer/cfunc.mod, bin//codemodels/analog.cm (rebuilt), examples/sxfer_examples/*

XSPICE **s_xfer**: fix an out-of-bounds crash on a static-gain transfer function. Sixth find of the same deep dive (after E-235–E-239), from fuzzing XSPICE a-device instantiation; reproduces on the stock binary. First find in the XSPICE code-model library rather than the core simulator. The `s_xfer` analog code model (a Laplace transfer function $H(s)=\text{num}/\text{den}$ in controller-canonical form) allocates its integrator state array to `den_size` elements, then in the DC/transient branch unconditionally read `pout_pin = *(integrator[1])` (the $d(\text{out})/d(\text{in})$ partial) -- assuming a first- or higher-order denominator. A 0-order denominator (`den_size == 1`, i.e. a STATIC-GAIN transfer function like `s_xfer(num_coeff=[k] den_coeff=[1]) = constant k`) allocates integrator with a single element, so `integrator[1]` is one past the end -> read garbage -> SIGSEGV at load (`cm_s_xfer <- MIFload <- CKTload`). Depends only on `den_size==1`, NOT the coefficient values: `den_coeff=[1]` (valid unity denominator), `[5]`, `[0]` all crash; `[1 1]/[1 2 1]` fine -- so it bit ordinary static-gain input, not just degenerate decks. AC branch already safe (loops bounded by `den_size/num_size`). Fix: guard the read -- `if (den_size > 1) pout_pin = *(integrator[1]); else pout_pin = (num_size>0) ? *gain * *(num_coefficient[0]) : 0.0` (for `den_size==1`, `out = gainnum[0]/in`, so the partial is `gainnum[0]`); *the output value was already computed correctly. Zero denominator (H=inf) now gives clean non-convergence instead of a crash. Fix is in a .cm code model, so analog.cm was regenerated via cmpp and redeployed under bin//codemodels/*; the ngspice binary is unchanged. Scope: XSPICE analog code-model library only (`cfunc.mod` + rebuilt `analog.cm`); no core simulator/solver/analysis/device/compiler change; dynamic-filter results unchanged. Verify (examples/sxfer, 4 checks): static-gain `s_xfer(num=[3] den=[1])` no crash & exact ($H=3 \rightarrow 12\text{ V}$); another static gain $H=0.4 \rightarrow 1.6\text{ V}$; zero denominator no crash; dynamic $H=1/(s+1)$ still correct (

E-241 *src/frontend/com_fft.c, examples/fftnorm_examples/**

fft/spec: fix the amplitude normalization for non-power-of-2 records. First numerical-correctness (wrong-number, not crash) find of a correctness-audit campaign that checked ngspice analyses against analytic + numpy oracles. `com_fft.c` has two paths: FFTW3 (exact length-point transform, `scale = length/2`, correct) and a Green's radix-2 FFT (no FFTW3) that zero-pads the `length` input samples up to the next power of two `N`. The non-FFTW path normalized the single-sided amplitude by the PADDED size (`scale = N/2`) instead of the true sample count (`length/2`), so every FFT whose sample count is not a power of two reported amplitudes too small by `length/N` -- up to 2x. A `.tran` essentially never yields exactly 2^k points, so this bit the common case silently. The DC bin is the clean discriminator (always exactly bin 0, no window/scalloping ambiguity): a signal with DC offset `D` read `D*length/N`, e.g. a 1025-sample record (padded to 2048) reported a DC of 2.0 as 1.0009. `spec` carried the same error in its power normalization (`intres = N*N` instead of `length*length`; the FFTW path already used `length*length`). Fix: `scale = length/2` and `intres = length*length` in the non-FFTW path, matching FFTW. The frequency axis was already padding-aware (`freq[i] = i/span * length/N`), so only the magnitude scale needed the change; power-of-2 records (`length == N`) and FFTW-linked builds are unchanged. Validated against numpy as an independent oracle: after the fix ngspice's `fft` matches `numpy.fft.rfft` of the same zero-padded record (normalized by `length`) bin-by-bin to $\sim 1e-9$. Scope: ngspice frontend only, two normalization constants in `com_fft.c`; `fft/spec` amplitudes now change for non-power-of-2 records (they become correct). Verify (examples/fftnorm, numpy-free, 4 checks): DC offset 2.0 reads 2.0 on a non-power-of-2 record; same on a maximally-padded record (1025->2048, where the bug halved it to ~ 1.0); DC reading independent of sample count across four record lengths; `spec` PSD tone peak length-independent. Regression 199/199 (new example folder).

E-242 *src/spicelib/devices/nport/* (new device dir), src/frontend/snp2va.c, src/frontend/snp2va.h, src/frontend/com_presnp.c, src/spicelib/parser/inp2n.c, src/spicelib/devices/dev.c, configure.ac, src/spicelib/devices/Makefile.am, src/Makefile.am, examples/nport_native_examples/, bin/ngspice (rebuilt)*

Native C n-port device + **pre_snp -native**: a pole/residue Y-parameter block stamped straight into the matrix, no OpenVAF compile. `pre_snp` (E-200/201/205) converts Touchstone -> Verilog-A -> `openvaf-r` -> `.osdi`; the fit is cheap but the compile is the bottleneck and walls at ~ 24 -32 ports ($O(N*N_p)$ laplace_nd sections, $O(N^2)$ terms for full-rank coupling). This adds a built-in `nport` device (new `src/spicelib/devices/nport/`) that evaluates the same model $Y_{ij}(s) = d_{ij} + s*e_{ij} + \sum_k res_{ijk}/(s-p_k)$ directly. It stamps as a multi-terminal admittance (four-corner conductance per (i,j); no per-port branch unknowns -> N equations not $2N$): DC stamps $Y(0)$, AC the complex $Y(jw)$ into the (real,imag) slots, and transient a trapezoidal companion -- `d` a conductance, `s*e` a capacitor, each `res/(s-p)` a first-order state $dx/dt = p*x + u$ whose shared pole states `x_jk` are advanced once per load and parked in `CKTstate0` (the stamp phase recovers the history $B = x_{n+1} - a*u_j$, so no per-instance scratch). Complex poles come as conjugate pairs so the summed current is real. It rides the generic `N` dispatcher (`inp2n.c` broadened from OSDI-only to also accept the `nport` model): `N1 p1..pN ref mymodel / .model mymodel nport(file="m.nport"), ref` an explicit reference terminal (tie to 0 for a ground-referenced Touchstone block; the extra terminal makes floating/differential blocks expressible). KLU supported via `nportbindCSC.c` (real/complex/complex->real bindings; a ground row/col keeps its Sparse trash pointer, like the RLC devices). `snp2va.c` was refactored so the shared parse+vector-fit lives in `snp_fit()`, feeding two emitters: `pre_snp -osdi` (default, unchanged VA->OSDI route) and new `pre_snp -native`, which writes the compact `.nport` for this device -- same fit, no compiler, scaling to hundreds of ports. Verify (examples/nport_native, 6 checks x 2 solvers, all vs closed-form oracles, no `openvaf-r` needed): RC one-port e-term (AC $1/Y$ + transient exponential); RLC one-port complex conjugate pole pair through resonance; Π two-port off-diagonal d/e cross-coupling vs analytic MNA; `pre_snp -native` fits a known 2-port `.s2p` and reproduces its

Y; a 20-port 6-pole full-rank model vs an analytic linear solve. Cross-checks (dev notes): native AC == OSDI backend to 1e-11 on 137mhz_bpf.s2p; a 100-port block solves an AC point in ~90 ms (4.85e-10 vs analytic); KLU == Sparse bit-for-bit. Scope: ngspice only, additive; default -osdi path unchanged so existing decks are unaffected; no solver/analysis/compiler change. Regression 200/200 (new device dir + example folder).

E-243 *src/frontend/snp2va.c, examples/presnp_examples/verify_presnp.py, bin/*/ngspice (rebuilt)*

pre_snp -osdi: emit an explicit **ref** terminal so the instance line is identical for both **-osdi** and **-native**. E-242's native nport device carries an explicit reference terminal (N1 p1..pN ref model), but the original pre_snp -osdi emitted a Verilog-A module with only the N signal ports referenced to implicit global ground (N1 p1..pN model) -- so the two backends were not drop-in interchangeable. snp2va.c now emits module m(p1..pN, ref) and takes every branch relative to ref: I(pi,ref) <+ ... V(pj,ref) ... (and laplace_nd(V(pj,ref), ...), ddt(V(pj,ref))). ref is a PLAIN electrical port -- NOT a ground-declared node: that keeps it connectable (tie to 0 for a ground-referenced block, or any node for floating/differential), and it appears in every I(pi,ref) stamp so it is well-posed (declaring ground ref would make it an internal 0V node -- no longer a terminal -- and risks the E-116 all-zero-row/structurally-singular-KLU failure a ground node in no Jacobian entry caused). The instance line is now identical for both backends. Compatibility: the -osdi module goes from N to N+1 terminals, so existing -osdi decks must add the ref node (tie to 0 to reproduce the old behavior); the presnp example decks were updated. Verify: the identical line N1 p1 p2 0 model through both backends matches to 1e-11 on 137mhz_bpf.s2p (ref=0 reproduces the prior result exactly); examples/presnp (9 checks: 2-/3-/4-/8-port pre_snp cases, AC+tran+ordering) passes with the ref terminal. Scope: ngspice only, the VA emitter in snp2va.c; -native / .nport device / vector fit / .osdi compile path all unchanged. Regression 200/200.

E-244 *src/spicelib/devices/nport/nportsetup.c, src/frontend/com_pyplot.c, examples/crashfix2_examples/, bin//ngspice (rebuilt)*

Crash-hardening round 2: two user-triggerable crashes in recently-added code (both reproduce on the shipped binary). Found by argument-fuzzing the newer devices/commands. (1) nport device (E-242): NPORTsetup never checked that the instance line connected all N+1 nodes the .model's port count claims. A .nport with more ports than the instance wires (N1 a 0 mod for a 2-port), or nports beyond NPORT_MAXTERMS, made setup stamp an UNBOUND (-1) terminal -> the generic N dispatcher leaves unconnected terminals as -1, and passing a negative row/col trips the sparse builder assert Row>=0 && Col>=0 (spbuild.c:267 -> SIGABRT) / out-of-bounds. Fix (nportsetup.c): reject a .nport whose port count leaves no room for the reference (N+1 > NPORT_MAXTERMS) and verify every node [0..N] >= 0 before stamping -> clean E_BADPARG with a diagnostic naming the shortfall. (2) pyplot (E-217/E-218): the -hist/-contour render-mode markers were stripped by UNLINKING AND FREEING a node of the command's OWN argument wordlist -- but that list is owned/freed by the command loop. When the marker was the FIRST word, com_pyplot freed the list HEAD, which the command loop then freed again -> use-after-free/double-free (SIGSEGV) on pyplot -hist v(out) / pyplot -contour (pyplot v(out) -hist, marker not first, was fine -- so the E-217/E-218 examples, which never put it first, always passed.) Fix (com_pyplot.c): detect the marker by scanning (no mutation) and build a filtered COPY without it, freed by us at the end; the caller's list is untouched. Bonus: -hist as first arg now renders the histogram it silently degraded to a line plot before. Verify (examples/crashfix2, 9 checks x2 solvers): under-bound + over-max nport decks exit gracefully with the diagnostic; a correctly-bound 2-port still simulates; pyplot -hist/-contour first-arg and -hist-alone no longer crash; -hist-first now emits plt.hist(); trailing-marker form still works. Scope: ngspice only, device input validation + command-arg ownership; no numerical change, valid usage unaffected. Regression 201/201.

E-245 *src/frontend/com_measure2.c, src/frontend/spiceif.c, examples/crashfix3_examples/, bin//ngspice (rebuilt)*

Crash-hardening round 3: two NULL-deref crashes in CORE ngspice command parsers (stock code, both reproduce on the shipped binary). Found by argument-fuzzing meas and altermod. (1) meas: `measure_parse_stdParams` (`com_measure2.c`) splits each trailing param token on '=' with `strtok`; a LONE '=' token (a stray one, e.g. `meas tran m find v(x) when v(x)=0.5 = y`) makes `strtok` return NULL for the name, which was then passed to `strcasecmp()` -> NULL deref SIGSEGV. The `pValue==NULL` branch was guarded but `pName` was dereferenced first. Only the interactive meas command hits it (the `.meas` dot-card pre-tokenizes differently); `key=val/=val/val=` are all fine, only a standalone '=' produces the NULL. Fix: treat a NULL `pName` as a clean syntax error before `strcasecmp`. (2) altermod: `altermod nm c` where the 2nd token is a bare device-type letter (c/e/i/...) or a digit makes `com_altermod` treat it as another MODEL to alter with no `param=value`, so `paramlookup` (`spiceif.c`) is called with `param==NULL`; its model-parameter loop passed that straight to `eq()/strcmp()` (the instance-parameter loop just above already guarded `!param`) -> NULL deref SIGSEGV. Input-dependent: only tokens resolving to a real device type crash (c/e/i/0/1/2), others are dropped earlier. Fix: guard the model loop against a NULL param (and defensively a NULL keyword), matching the instance loop. Verify (examples/crashfix3, 12 checks x2 solvers): the stray '=' meas forms and altermod nm <c

E-246 *src/xspice/icm/analog/pwl/cfunc.mod, src/xspice/icm/analog/pwlts/cfunc.mod, bin//codemodels/analog.cm* (rebuilt), *examples/pwlfix_examples/*

XSPICE **pwl/pwlts**: fix an out-of-bounds heap read on `x_array/y_array` length mismatch. Second find in the XSPICE code-model library (after E-240's `s_xfer`), from fuzzing code-model parameters; confirmed with AddressSanitizer. `pwl` and `pwlts` take two INDEPENDENT vector parameters `x_array` (breakpoints) and `y_array` (ordinates); the XSPICE framework allocates each to its OWN length. Both models size their working table from `x_array` only (`size = PARAM_SIZE(x_array)+2`) then copy BOTH arrays with that single index (`y[i]=PARAM(y_array[i-1])`), so a `y_array` SHORTER than `x_array` reads past the end of the `y_array` parameter -> heap-buffer-overflow READ (ASan: `cm_pwl cfunc.c:321 / cm_pwlts cfunc.c:202`), pulling adjacent/uninitialised heap into the interpolation table (undefined behaviour; can crash outright once the mismatch runs past the mapped heap). Reachable from any deck with such an a-device (`.model m pwl(x_array=[0 .25 .5 .75 1 1.5 2 2.5] y_array=[0 1])`). The sibling models already guard this -- `onshot` checks `cntl_size != pw_size`, `multi_input_pwl` checks `PARAM_SIZE(x) != PARAM_SIZE(y)` -- but `pwl/pwlts`, the oldest PWL models, predate the check. A sweep of the other analog models with paired arrays found no further cases (sine/square/triangle copy into their own `cntl_size`-sized buffers, so a short array zero-pads rather than overrunning; `summer/mult` are validated by the framework against the connection size). FIX: add the same equal-length guard -- `if (PARAM_SIZE(x_array) != PARAM_SIZE(y_array)) {cm_message_send(size_error); return;}` -- before the alloc/copy in both models, aborting setup cleanly with `x_array` and `y_array` must have the same length instead of reading out of bounds. Fix is in a `.cm` code model, so `analog.cm` was regenerated via `cmpp` and redeployed under `bin/*/codemodels/`; the ngspice binary is unchanged. Scope: XSPICE analog code-model library only; no core simulator/solver/analysis/device change; valid equal-length `pwl/pwlts` usage unaffected. Verify (examples/pwlfix, 5 checks x2 solvers): a valid `pwl` interpolates exactly (in 0.5 -> 1.0 V); `pwl` with `x` longer than `y`, and the other way, both report the size error without crashing; a valid `pwlts` still runs; a mismatched `pwlts` reports the size error; self-skips if `codemodels` unavailable. The OOB was reproduced and shown fixed under an AddressSanitizer build of the code models. Regression 203/203 (new example folder).

E-247 *src/xspice/icm/table/table2D/cfunc.mod, src/xspice/icm/table/table3D/cfunc.mod, bin//codemodels/table.cm* (rebuilt), *examples/tablefix_examples/*

XSPICE **table2d/table3d**: fix out-of-bounds read + interpolation UB on degenerate or too-small tables. Third area of the XSPICE code-model deep dive (after E-240 `s_xfer`, E-246 `pwl/pwlts`), found by fuzzing table dimensions; confirmed with ASan/UBSan. Both models read a lookup table from a file whose first lines declare the axis point counts (`ix, iy [, iz]`); the loader (`init_local_data`)

never validated them, so a small table broke two ways. (1) OUT-OF-RANGE RAMP (heap OOB read): for an input past the table edge the model ramps the derivative to zero using `xcol[1]-xcol[0]` (low side) or `xcol[ix-1]-xcol[ix-2]` (high side); a single-point axis (`ix==1`) has no `xcol[1]/xcol[-1]` -> heap-buffer-overflow READ (ASan: `cm_table2D cfunc.c:298`); `table3d` has the same for all three axes. (2) ENO INTERPOLATION (OOB/UB): the Madagascar ENO of order `p` reads a $2*(p-1)$ -point stencil per axis (order 2 needs 2 pts, 3 needs 4, 4 needs 6, 5 needs 8); the order param DEFAULTS to 3 (needs ≥ 4 pts/axis) but the loader only clamped it UP to a minimum of 2, never DOWN to what the table supports -> the default order on any table with a <4 -point axis (a 3×3 table, or a 2-plane z axis like the shipped `test-3d-1.table` with `iz=2`) ran off the stencil -> UB/OOB in `eno2.c/eno3.c`. FIX (both models, right after the `order>=2` floor): compute `mindim = min(ix,iy[,iz])`; reject `mindim<2` with a clean error (a 1-point axis can be neither ramped nor interpolated); and clamp `interporder` to `mindim/2+1` (the inverse of the $2*(p-1)$ stencil rule) so a small table interpolates at a reduced but VALID order instead of reading out of bounds. Tables large enough for the requested order are byte-identical (every shipped example has `mindim/2+1 >= order`; the smallest are 7×7). Fix is in `.cm` code models, so `table.cm` was regenerated via `cmpp` and redeployed under `bin/*/codemodels/`; the `ngspice` binary is unchanged. No regression coverage impact: the regression `table_model` example uses OSDI Verilog-A table models, not these XSPICE code models. Scope: XSPICE analog code-model library only; no core simulator/solver/analysis change; tables large enough for the requested order unaffected. Verify (examples/`tablefix`, 6 checks $\times 2$ solvers): a valid 8×8 order-3 `table2d` interpolates exactly (`out=x+10y` at $(2.5,3.0)$ -> 32.5); a single-x-point `table2d` and a degenerate `table3d` (`iz=1`) are rejected with a clean error; small 3×3 / $3 \times 3 \times 3$ order-3 tables run (order clamped) without crashing. The OOB read and the ENO UB were reproduced and shown fixed across a full (dimension $\times$ order) grid under ASan/UBSan builds. Regression 204/204 (new example folder).

E-248 `src/spicelib/devices/cpl/cplsetup.c`, `bin//ngspice` (rebuilt), `examples/cplfix_examples/`

CPL coupled-line device: fix two out-of-bounds accesses (conductor count + R/L/C/G matrix sizes). First find in a CORE `ngspice` device (the XSPICE dive E-240/246/247 was the code-model library); found by fuzzing the `p`-device conductor count and matrix sizes, confirmed with ASan/UBSan; reproduces on the shipped binary. A CPL line has `noL` conductors (the instance dimension, set from the node count on the `p` card); each symmetric R/L/C/G matrix is its upper triangle, $noL*(noL+1)/2$ values. `CPLsetup/ReadCpL` (`spicelib/devices/cpl/cplsetup.c`) validated NEITHER count. (1) UNDER-SPECIFIED MATRIX (heap OOB read): `ReadCpL` fills the matrices with `f = CPLmodPtr(here)->Rm[counter]` (and `Lm/Cm/Gm`) where `counter` runs $0..noL*(noL+1)/2-1$, but `Rm/Lm/Cm/Gm` are allocated to the count the user actually gave (`Rm_counter`, ...); fewer values than the triangle needs -> read past the end (ASan: heap-buffer-overflow READ at `cplsetup.c:474`). (2) TOO MANY CONDUCTORS (fixed-array overflow): `ReadCpL` uses `RLINE *lines[MAX_CP_TX_LINES]` and `CPLLine.in_node[MAX_CP_TX_LINES]` with `MAX_CP_TX_LINES==8` (`include/ngspice/swec.h`), indexed by `noL`; a `p` card with >8 conductors writes past them (UBSan: index 8 out of bounds for '`NODE *[8]`' at `cplsetup.c:430`). Both reachable from a valid-syntax netlist; on the release binary the reads pull adjacent heap -> heap-layout-dependent garbage (here a downstream fatal-error exit, but could equally be wrong results or a crash). FIX: validate both up front in `CPLsetup`, right after `noL = here->dimension` and before any allocation -- reject `noL<1`

E-249 `src/spicelib/devices/urc/urcsetup.c`, `src/spicelib/devices/ltra/ltraset.c`, `bin//ngspice` (rebuilt), `examples/tlinefix_examples/`

URC + LTRA transmission-line input validation: reject a degenerate/unbounded lump count and negative line parameters. Two CORE transmission-line devices accepted out-of-range parameters that gave a silently wrong result or a resource-exhaustion hang instead of a clean error (same class as E-248 CPL). (1) URC (`urcsetup.c`): the uniform-RC line is expanded at setup into a ladder of `n` (the `n=` lumps) resistor+capacitor(/diode) sections -- one node group per lump -- and `n` is also an exponent of `pow(k,n)`; the user's `n` was NEVER bounds-checked. `n<=0` ran the loop zero times ->

no lumps -> output node left silently unconnected ($v(out)=0$, a wrong answer, no diagnostic); a large n (typo $n=100000000$, or $2e9$ near `INT_MAX`) exhausted memory / hung building the ladder (even ~ 20000 lumps takes tens of seconds; the source even carried a may want to limit lumps to ≤ 100 comment). Fix: require $1 \leq n \leq \text{URC_MAX_LUMPS}$ (1000; auto-computed count is $\sim 3..30$), clean `E_BADPARM` otherwise, validated after the auto-lump default so it covers both given and computed counts. (2) LTRA (`ltraset.c`): the lossy line picks its RLGC special-case with $!= 0$ tests, so a NEGATIVE L or C passed and fell through to $\sqrt{L/C}/\sqrt{L*C}$ in `LTRAtemp` -> NaN characteristic impedance/delay and a degenerate all-zero run (a zero/too-few-param line was already the clean "at least two ... must be ... nonzero" error). Fix: reject any negative $R/L/G/C$ up front (per-unit-length params must be physical). Scope: core ngspice, two device setups; binary rebuilt; no solver/analysis/numerical change; a validly-specified URC/LTRA line is unaffected. Verify (examples/`tlinefix`, 8 checks x2 solvers): a valid 5-lump URC divider and a valid LTRA lossless line still simulate; URC $n=0/-3/100000000$ each rejected cleanly (the huge case INSTANTLY, no hang); URC $n=1000$ (max) still runs; LTRA negative C rejected cleanly while $C=0$ keeps the pre-existing nonzero error. Regression 206/206 (new example folder).

E-250 `src/xspice/icm/digital/d_lut/cfunc.mod`, `src/xspice/icm/digital/d_genlut/cfunc.mod`, `bin/codemodels/digital.cm` (rebuilt), `examples/dlutfix_examples/`

XSPICE `d_lut/d_genlut`: fix an undefined-behaviour $1 \ll n$ shift on the input-port count. Found with UBSan while sweeping the XSPICE code-model library (E-240/246/247 dive). Both digital lookup-table models size their table as $2^{(\#inputs)}$ via a left shift: `d_lut size=PORT_SIZE(in); tablelen=1<<size (cfunc.c:143-144)`, `d_genlut isize=PORT_SIZE(in); entrylen=(1<<isize) (cfunc.c:154/177)`. The input-port count was NEVER bounded: a left shift of a 32-bit int by ≥ 32 is UNDEFINED BEHAVIOUR (UBSan: "shift exponent 32 is too large for 32-bit type 'int'" at `cfunc.c:146/:177`), and a count in the high twenties asks for a multi-gigabyte table alloc. Reachable from a valid-syntax netlist -- an a-device with $\sim 30+$ digital inputs fanned into the LUT (e.g. all tied to one pulled-up net). On arm64 the release UB is benign (shift amount taken mod 32 -> $1 \ll 32 = 1$, no crash) so the table size is silently WRONG rather than loud -- which is exactly why it's worth fixing. Fix: cap the input-port count at `D_LUT_MAX_INPUTS / D_GENLUT_MAX_INPUTS` (24; a real LUT has only a handful of inputs, and the `table_values` string is 2^n chars) BEFORE the shift, `cm_message_send` clean error + return otherwise. Fix is in `.cm` code models, so `digital.cm` was regenerated via `cmpp` and redeployed under `bin/*/codemodels/`; the ngspice binary is unchanged. Scope: XSPICE digital code-model library only; no core simulator/solver/analysis change; a normal LUT is unaffected. Verify (examples/`dlutfix`, 4 checks x2 solvers): valid 2-input `d_lut` and `d_genlut` still simulate; a 32-input `d_lut` and `d_genlut` are each rejected with a clean "out of range" error, no crash. The $1 \ll 32$ UB was reproduced under a UBSan build and shown fixed. Regression 207/207 (new example folder).

E-251 `examples/hb_examples/verify_hb.py` (verification only -- no ngspice source change)

Harmonic-balance tight nonlinear correctness verification. A rigorous tightening of the HB (E-134/135) correctness cross-check. The existing checks compare the HB spectrum against ngspice's own transient+fourier steady state, but at $K=8$ harmonics and a 1.5-3% tolerance. This proves -- and enforces in the regression -- that HB converges to the EXACT periodic steady state, not merely within a few percent. AUDIT: HB and tran+fourier are independent code paths, so their agreement is a real oracle once the transient reference is accurate. Using a purely RESISTIVE rectifier ($c_{jo}=0/tt=0$ -> steady state reached instantly, so the transient's only error is its $O(dt^2)$ timestep, which is Richardson-extrapolated to $dt \rightarrow 0$) and sweeping $K=8..48$, the HB harmonics converge MONOTONICALLY to the $dt \rightarrow 0$ transient; at $K=48$ the match is $\sim 1e-7$ (h_1/h_2 exact, $h_3 -1.9e-7$, $h_4 -3.1e-6$, $h_5 +8.9e-7$). The 13% apparent 5th-harmonic 'error' at $K=8$ is entirely HB truncation (aliasing of the discarded harmonics), NOT a modelling error -- it vanishes as K grows. TIGHTENED CHECK (`verify_hb.py` [10]): on the resistive rectifier, HB at $K=24$ matches a fine-dt (period/2000) transient fourier to $< 0.5\%$ for DC..4th (6x tighter than the loose checks, observed worst-case $\sim 2e-5$), and raising K from 8 to 24 strictly shrinks the 5th-harmonic mismatch (encoding 'converges

as K grows' so a future regression that broke HB's harmonic content would be caught). Scope: verification only (verify_hb.py); no ngspice binary / HB-engine / analysis change; documents that HB is EXACT, not merely within tolerance. Regression 207/207.

E-252 *src/xspice/icm/analog/xfer/cfunc.mod, src/xspice/icm/analog/file_source/cfunc.mod, bin/codemodels/analog.cm (rebuilt), examples/filefix_examples/*

XSPICE xfer/file_source: fix two heap OUT-OF-BOUNDS WRITES in the file parsers. Found while sweeping the file-parser code models (d_state/d_source/xfer/file_source); confirmed with ASan. Unlike the earlier code-model finds (E-246/247/250, reads or bounded errors) these are OOB WRITES -- silent heap corruption. (1) xfer (read_file): reads a transfer-function file (Touchstone-style # option line + data), sscanf's up to 9 values/line and a state machine stores EVERY value (freq/real/imag triples), so a line with more than one record stores >3; but the alloc check reserved only 3 (if(i+3>size), ALLOC=1024) -> a multi-record line writes past the buffer at the 1024-double boundary (ASan: heap-buffer-overflow WRITE, cm_xfer cfunc.c:232). Fix: reserve the sscanf max of 9 (if(i+9>size); the loop appends at most count<=9/line since offset>=1 && span>=offset+2 are enforced). (2) file_source: stores one record/line = a timepoint + size channels = stepsize(=size+1) doubles, but reserved only size (vecallocated - size), one short; at the realloc boundary (count lands exactly at vecallocated-size once vecallocated grows out of stepsize alignment) the last channel writes one double past the end (ASan: heap-buffer-overflow WRITE, cm_filesources cfunc.c:326). Fix: reserve a full record (-stepsize). Both are heap OOB WRITES reachable from a valid-syntax netlist + crafted data file; release build corrupts adjacent heap silently rather than always crashing (UB either way). Siblings checked: d_source validates per-line token count vs declared width; d_state's fixed line buffer is fgets-bounded -- no analogous overrun. Fix is in .cm code models, so analog.cm was regenerated via cmp and redeployed under bin/*/codemodels/; the ngspice binary is unchanged. Scope: XSPICE analog code-model library only; no core/solver/analysis change; a well-formed data file is unaffected. Verify (examples/filefix, 4 checks x2 solvers): valid transfer-function + file_source files simulate; an xfer multi-record file and a boundary-crossing file_source file run without overrunning. Both reproduced under ASan (cm_xfer/cm_filesources) and shown fixed. Regression 208/208 (new example folder).

E-253 *src/frontend/com_rfstab.c, src/frontend/com_rfstab.h, src/frontend/commands.c, src/frontend/com_commands.h, src/frontend/Makefile.am, bin/ngspice (rebuilt), examples/rfstab_examples/*

rfstab: two-port stability & gain report -- the first RF design-aid on top of the S-parameter machinery. ngspice can compute S-parameters (.sp / n-port / Touchstone) but not turn them into the stability/gain figures of merit an RF designer works with. After a .sp analysis publishes S_{1_1}..S_{2_2} vs frequency, the new rfstab command post-processes them into the standard linear-two-port metrics per frequency: determinant D=S₁₁S₂₂-S₁₂S₂₁; Rollett K=(1-

E-254 *src/frontend/plotting/pyplot.c, src/frontend/plotting/pyplot.h, src/frontend/plotting/plotit.c, src/frontend/com_pyplot.c, bin/ngspice (rebuilt), examples/pyplotsmith_examples/*

pyplot -smith: Smith-chart plotting mode -- the second RF design-aid (after rfstab, E-253). A Smith chart is the working surface for two-port RF design (reflection coefficients, matching trajectories, stability/gain circles), but ngspice's modern matplotlib output (pyplot) had no Smith mode -- only the legacy in-terminal plotter did. pyplot [name] -smith <complex vectors> draws each listed complex vector (S_{1_1}, S_{2_2}, a load reflection Gamma, a stability/gain-circle trace) as a curve in the reflection-coefficient plane (Re->x, Im->y) over the standard Smith grid: the unit circle

E-255 *src/spicelib/analysis/cktdisto.c, bin/ngspice (rebuilt), examples/distoexact_examples/*

.disto proven machine-exact vs Harmonic Balance / QPSS-HB + behavioral-source (B) distortion warning. RF/distortion correctness frontier, in the E-251 HB-proof mold. (1) PROOF: .disto (the 1990 Volterra small-signal distortion analysis) reports the pure 2nd/3rd-order kernels scaled by

the DISTOF1/DISTOF2 magnitudes; HB (hb) and two-tone QPSS-HB (qpss) are INDEPENDENT large-signal engines including all orders, whose k-th harmonic / mixing amplitudes converge onto the Volterra result with $\sim A^2$ leakage. So HB(A)->.disto as $A \rightarrow 0$ -- the HB-proof structure applied to distortion; because HB and .disto share ngspice's identical device model, model-constant ambiguity (VT, DC-op) cancels, so the agreement measures the .disto engine itself. Measured (diode + series R, both solvers): single-tone HD2 vs HB 2f tightens rel err $4.0e-5 \rightarrow 2.6e-7$ as $A:1e-3 \rightarrow 1e-4$ (HB->.disto), HD3 $\sim 4e-6$; two-tone IM3 (2f1-f2) vs QPSS-HB (2,-1) $\sim 1e-5$; .disto output scales EXACTLY as A^2/A^3 (DISTOF1/2 applied exactly, kernels amplitude-independent). The residual floors are the reference engine's own readout/

E-256 *src/maths/ni/niiter.c, src/spicelib/analysis/cktop.c, src/include/ngspice/cktdefs.h, src/spicelib/parser/ptfuncs.c, bin//ngspice (rebuilt), examples/bsrcconv_examples/*

Fix SILENT spurious DC operating point for singular-derivative behavioral sources. A behavioral (B) source whose small-signal conductance is INFINITE at the $v=0$ initial guess -- $B I = \sqrt{v(n)}$ ($dI/dv = 0.5/\sqrt{v} \rightarrow \infty$), $I = 0.1/v(n)$, $I = \ln(v(n))$, $v^{0.5}$ -- pins the node at $v \sim 0$ so Newton takes a vanishing step; the convergence test (iterate-to-iterate VOLTAGE change only) sees no change and declares "converged" at a point that grossly VIOLATES KCL (0.42A imbalance on a 0.6V/1ohm divider). The false convergence also PRE-EMPTED the gmin/source-stepping fallbacks (they never ran). Silent + data-dependent ($i = 1msqrt(v)$ converges fine; a .nodeset finds the true root). The B-source AC derivative itself is correct (matches analytic $dI/dv \sim 1e-10$ across the function library) -- the defect is purely the DC convergence ACCEPTING a KCL-violating point. FIX (3 coordinated parts): (1) niiter.c false-convergence guard -- after the voltage test declares convergence, also check the KCL residual $F = Gx - b$ (the node-current imbalance, already computed for the E-111 line search); if the worst node merit $> 100x$ the current tolerance the point is spurious -> decline convergence so CKTop falls through to gmin/source stepping, which regularizes the singular node and finds the TRUE point. Separation is decisive: false convergence sits at merit $\sim 1/reltol \sim 1000$, every legitimate circuit < 1 (measured $1e-7..0.35$). (2) First-attempt isolation -- a new one-bit CKTdcFirstTry flag (cktdefs.h) set by CKTop (cktop.c) ONLY around the initial plain-Newton attempt, so the guard never fires inside a convergence-aid sub-solve (gmin/source stepping, pseudo-transient/optran, convhelp); an earlier unconditioned version broke convhelp/ptcont/corenum/tempphys by firing inside their homotopies. (3) ptfuncs.c pwr(0,negative) guard -> HUGE instead of raw $+\infty$ (matches the existing /0 and sqrt(neg) guards) so a singular derivative stays finite rather than NaN-poisoning the Jacobian. Result-neutral for every circuit that already converged. Verify (examples/bsrcconv, both solvers): [1] $I = \sqrt{v}$ reaches analytic $v_0 = 0.178045$ with KCL satisfied ($i_{B1} = i_{R1}$), not the spurious $v \sim 0$; [2] $I = 0.1/v$ reaches upper-branch $v_0 = 0.887298$; [3] result-neutral: $v^2/v^3/\exp/\tanh$ unchanged. Full regression 212/212 incl. the convergence-aid suites convhelp/ptcont/corenum/tempphys. Scope: DC Newton loop + a one-bit CKT flag + a parser math guard; binary rebuilt; confined to the first operating-point attempt.

E-257 *src/maths/ni/niiter.c, bin//ngspice (rebuilt), examples/bsrcconv_examples/ (verify [4])*

Extend the E-256 false-convergence guard to the TRANSIENT operating point. E-256 fixed the silent spurious DC operating point for singular-derivative behavioral sources but gated the guard to MODEDCOP only; the transient operating point (MODETRANOP -- the bias a .tran starts from without uic) was left uncovered and still false-converged. So a .tran of a biased $B I = \sqrt{v(n)}$ started from the spurious $v \sim 0$ and showed a FAKE startup transient ($t=0$ $v(n) = 9.3e-17$ ramping $0 \rightarrow 0.178$ over $\sim 20ms$) when the biased circuit should sit at 0.178045 from $t=0$. E-256 gated to MODEDCOP conservatively because the pseudo-transient/optran aids run in transient modes -- but the guard already fires only when CKTdcFirstTry is set (the initial plain-Newton attempt), and optran runs as a FALLBACK with CKTdcFirstTry=0, so it is safe to broaden the mode condition to MODEDCOP

E-258 *src/maths/ni/niiter.c, src/spicelib/analysis/dctrcurv.c, bin//ngspice (rebuilt), examples/bsrcconv_examples/ (verify [5])*

Extend the false-convergence guard to the `.dc` sweep cold-start point. E-256/257 fixed the silent spurious operating point for singular-derivative behavioral sources for `.op` (MODEDCOP) and the transient `op` (MODETRANOP), but a `.dc` sweep still false-converged on its FIRST point: `.dc V1..` of a `sqrt(v)` source gave $v(n)=9.3e-17$ at the first sweep value (true root 0.178045) while every later point (warm-started from the previous solution, away from the $v=0$ singularity) was correct. Why the guard missed it: it fires only when `CKTdcFirstTry` is set, which happens only inside `CKTop` -- but in the default (non-HS) path the `.dc` sweep (`dcitrcurv.c`) solves each point with a `DIRECT NIiter` call, bypassing `CKTop`, and only falls back to `CKTop` if that solve FAILS; the false convergence "succeeds", so `CKTop` (and the guard) never ran. FIX (2 parts): (1) `niiter.c` -- the E-256/257 guard, which enumerated the `op` modes (MODEDCOP)

E-259 *examples/integaccuracy_examples/* (verification only -- no ngspice source change)*

Transient integration accuracy PROOF (oracle-based, both solvers). A permanent regression guard for the transient integrator, in the E-251 (HB) / E-255 (`.disto`) mold: rather than checking one waveform, it proves the integrator MATH PROPERTIES vs closed-form analytics + Richardson self-convergence. The audit that produced it swept the whole transient path and found it CORRECT. [1] ORDER of accuracy on RC decay $\exp(-t/RC)$: global error $\sim dt^p$ with the theoretical p -- TRAP $p=1.94$, Gear2 $p=1.93$, Backward-Euler $p=0.94$ (order test freezes the LTE step controller `trtol=1e11` so steps are uniform and the convergence is purely truncation order). [2] ENERGY signature on an LC $\cos(\omega t)$ over 30 periods: TRAP amplitude ratio 0.9999 (marginally stable/energy-conserving, AND the trapezoidal-ringing BE-switch is NOT firing spuriously), Backward-Euler 0.057 (dissipative) -- the fingerprint of a correct trapezoidal vs BDF. [3] BREAKPOINT: a PULSE edge into RC matches the piecewise-analytic charge (max err $6e-7$, pre-edge pinned 0). [4] RLC damped sinusoid vs $\exp(-at)(\cos \omega_d t + (a/\omega_d) \sin \omega_d t)$ (max err $4.3e-6$, correct ω_d /envelope). [5] NONLINEAR charge: a diode+CJO rectifier converges under Richardson (TRAP $dt > dt/2$) at order $p=2.00$, so the nonlinear device charge integration is 2nd order too. Both solvers give identical results (integrator is solver-independent). Separately confirmed (needs `.options dynorder + set ngdebug`, not asserted): higher-order Gear engages -- `maxord=6` reaches order 6 and rejects $\sim 10x$ fewer steps than order 2 (14 vs 143 on the RLC), so the high-order BDF coefficients are functional not a no-op. Scope: verification only; no source/binary change. Regression 213/213 (new example folder).

E-260 *examples/integaccuracy_examples/* (verify [6]; verification only -- no source change)*

LTE step-controller accuracy PROOF on a stiff circuit. Extends E-259 (which verified the integration METHODS with a FIXED step, freezing the LTE controller to isolate truncation order) with the complementary property: the ADAPTIVE local-truncation-error step controller actually DELIVERS the accuracy requested via `reltol` -- on a stiff circuit, the hard case. Two independent decays $v_f = \exp(-t/1\mu s)$, $v_s = \exp(-t/1ms)$ (stiffness ratio 1000) integrated with one adaptive step (the controller must take tiny steps to resolve the fast decay then coarsen for the slow tail). Swept over `reltol` the delivered error vs the closed form shrinks MONOTONICALLY with no plateau: `reltol` $1e-3 \rightarrow 1e-9$ gives `err(fast)` $9.0e-3 \rightarrow 2.0e-6$ while internal steps grow $1032 \rightarrow 1884$. The error scales $\sim reltol^{0.6}$ -- the theoretical global-error rate for a 2nd-order method under LTE control (`reltol` is a LOCAL per-step tolerance; $step \sim reltol^{(1/3)} \Rightarrow global\ err \sim reltol^{(2/3)}$). Separately confirmed (not asserted): extreme stiffness ratio $1e6$ ($\tau = 1ns/1ms$) still tracks `reltol`, and a re-excited fast mode (1MHz square wave) is handled by refining $\sim 6x$ more steps at each edge. Verify (`examples/integaccuracy [6]`, both solvers): on the stiff circuit `err` at `reltol` $1e-3/1e-5/1e-7$ is strictly decreasing and improves $\geq 10x$ (measured $\sim 247x$). Completes the transient-engine proof (E-259 methods + E-260 step controller). Scope: verification only; no source/binary change. Regression 213/213.

E-266 *src/spicelib/analysis/cktsetup.c, src/spicelib/analysis/cktpzset.c, src/include/ngspice/cktdefs.h, examples/solverannounce_examples/* (binaries rebuilt by CI)*

Announce the direct linear solver once, not on every analysis. `CKTsetup()` (and `CKTpzSetup()`) printed Using SPARSE 1.3 as Direct Linear Solver (or the KLU line) uncondition-

ally at the top of setup, and CKTsetup runs once per analysis -- so the sweep command, which sets the knob and re-runs the `-analysis` command for every point (`alter/altermod` in place, or `alterparam+reset` which re-sources the deck), reprinted the line on EVERY point (a 5-point sweep, 5 times; a 100-point sweep, 100). Monte Carlo, optimize, and the pso/de/sa optimizers -- all of which re-run the analysis per point -- were equally noisy. FIX: an announce-on-change helper `CKTannounceSolver(int klu)` (defined in `cktsetup.c`, prototyped in `cktdefs.h`, called by both `cktsetup.c` and `cktpzset.c`) prints the line only when the active solver differs from the last announcement. The last-announced solver is tracked PROCESS-WIDE (a function-static), not per-circuit, because a `.param` sweep re-sources the deck and rebuilds the circuit each point -- a per-circuit flag would still repeat. Consequences: a multi-point sweep/Monte/optimize run announces the solver ONCE; a single analysis still announces once (unchanged); a genuine solver switch (`.option klu/.option sparse`) has a different mode and re-announces; a fresh ngspice process starts un-announced, so batch runs and the dual-solver test harness (which re-execs once per solver) still print it once each, and the HB (`verify_hb`) and benchmark suites that detect KLU by grepping for the line still see it. Result-neutral: the line is informational and part of no numerical comparison; full dual-solver regression unchanged. Verify (examples/solverannounce, 4 checks): a 5-point sweep announces SPARSE once (was 5x); a single op still announces once; a `sparse->klu` switch re-announces both; a KLU analysis still prints the KLU line. Scope: 3 source files (one small helper + two call-site swaps + one prototype); no analysis/result change.

E-267 *src/frontend/com_sweep.c, examples/sweepbus_examples/* (binaries rebuilt by CI)*

sweep records array/bus nodes under their natural names (**ph[0]**, not **ph_0_**). Sweeping a circuit whose outputs are array/bus nodes -- a node named `ph[0]` (E-221) -- stored the result vectors under MANGLED names: `ph[0]` became `ph_0_`, `ph[1]` `ph_1_`, so the sweep plot listed `ph_0_`, `ph_1_`, ... The sweep command (E-146/-189/-190) builds each result-vector name from the output plus, for a `-vs` family or overlay waveform, an appended `_<knob>_<value>` segment; that segment carries a float (`_rt_1.5k`, with `./-/e` illegal in a nutmeg vector name), so the name builders `sw_familyname/sw_pointname` mapped every non-alphanumeric char (except `_`) to `_` -- but over the WHOLE string, INCLUDING the user's base output name, destroying a bus node's brackets. FIX (`com_sweep.c`): (1) a helper `sw_append_sanitized(base, suffix)` sanitizes ONLY the appended `_<knob>_<value>` segment, leaving the base name byte-for-byte intact, so `ph[0]` stays `ph[0]` (and a `-vs` family curve is `ph[0]_rt_2k`); ordinary names are unchanged (a plain node has no special chars; an explicit `-output name=expr` still uses the clean given name). (2) A bare `-output` token that is a bus range `ph[0:3]` now expands to one output per index (`ph[0]` `ph[1]` `ph[2]` `ph[3]`), matching the netlist bus expansion (E-221), so `sweep rt .. -output ph[0:3]` records the four taps directly. Values are unaffected -- only the NAME the result is stored under changes (and range-token expansion); each output is still evaluated exactly as before. Verify (examples/sweepbus, 4 checks, a 4-tap bus divider): `-output ph[0:3]` records four vectors named `ph[0]..ph[3]` (range expanded, natural names); the mangled `ph_0_..` names are gone; the recorded values are the correct divider ratios; a plain `-output vo=v(out)` is unaffected. Full dual-solver regression unchanged (the existing sweep suites use the `-output name=expr` form, whose clean names are untouched). Scope: one source file; no analysis/result change.

E-268 *src/frontend/spiceif.c, src/frontend/device.c, src/frontend/com_sweep.c, src/include/ngspice/ftext.h, examples/sweepwild_examples/* (binaries rebuilt by CI)*

Wildcard model-parameter knob **@*[param]**. Multiple `.model` cards can share a model-parameter NAME (two Verilog-A models both with a `wavelength` parameter). Nothing let you sweep that shared parameter across all such models in ONE knob: `sweep @dev1[wavelength]` (`altermod`) targets one model, and the `.param` idiom (`.param wl, wavelength={wl}` in each card, `sweep wl`) works but its `.param` knob commits with `reset` -- RE-SOURCING THE ENTIRE DECK at every point (slow) and needing the cards to reference the param. NEW: `@*[param]` sets param on every loaded model that has it, IN PLACE (`altermod`, no `reset`), so `sweep @[wavelength]` ... co-varies all of them fast; it also works standalone (`altermod @[wavelength]=1.55u`).

IMPL: (1) `spiceif.c if_setparam_wildcard(ckt, param, val)` -- model params are a device-TYPE property, so one `paramlookup` per device type decides whether that type's models carry param; every model of a matching type (walking `ckt->CKThead[typecode]->GENnextModel`) is set with `doset` (the same path a concrete `altermod @model[param]` uses); returns the count set and runs `CKTtemp` to push into live instances when called mid-run; models/types without the param are skipped. (2) `device.c com_alter_common` -- a wildcard device name `@*` routes to `if_setparam_wildcard`; a concrete `@model[param]` or device is unchanged; if no loaded model has the param, one clear warning. (3) `com_sweep.c sw_kind` -- `@*[param]` classified as an in-place altermod knob (like `@model[param]`), so the sweep applies it without reset; prototype in `fteext.h`. Additive -- `@*` was not previously a valid model name, so no existing alter/altermod/sweep behaviour changes. Verify (examples/sweepwild, 4 checks, both solvers): two `wlmodel` cards ($R = \text{wavelength}1k$) + one unrelated model without wavelength; sweep `@*[wavelength]` co-varies BOTH `wlmodel` devices (equal currents tracking $1/(\text{wavelength}1k)$); the unrelated model is untouched; a concrete `@dev1[wavelength]` still targets only `dev1`; `@*[<absent>]` warns and changes nothing. Full dual-solver regression passes. Four source files.

E-269 *src/frontend/spiceif.c, src/frontend/device.c, src/frontend/com_sweep.c, src/include/ngspice/fteext.h, examples/sweepwild_examples/* (binaries rebuilt by CI)*

Instance wildcard `@#[param]` (alias `@*[[param]]`). E-268's `@*[param]` sets a MODEL parameter on every model that has it; this adds the INSTANCE counterpart -- set an instance parameter on EVERY device instance that has it, in place (alter/altermod, no re-source), so sweep `@#[scale]` ... co-varies an instance parameter across all instances. Two spellings: `@#[param]` (canonical; # marks instance) and `@*[[param]]` (alias -- the outer `@dev[param]` parse leaves param as [param, whose leading [is stripped). IMPL: (1) `spiceif.c if_setparam_wildcard_instance(ckt, param, val)` -- instance params are a device-TYPE property, so one `paramlookup(..., do_model=0, ...)` per type gates membership; every instance (walking each model's `GENinstances->GENnextInstance`) is set with `doset(dev=inst, mod=NULL)` -> `setInstanceParm` (the path a plain alter `@dev[param]` uses). (2) `device.c alter_set` -- the device-name token picks the wildcard: `@#*` (token `#*`) or `@*[[param]]` (token `* + param` starting `[`) -> instance wildcard; `@*` -> model wildcard (E-268); else ordinary `if_setparam`; no-match -> one warning. (3) `com_sweep.c sw_kind` -- `@#*/@*[[...]]` classified as in-place altermod-style knobs (no reset); prototype in `fteext.h`. `@*=models`, `@#*/@*[[...]]`=instances; concrete `@dev[param]` unchanged. Additive -- these were not previously valid names. Verify (examples/sweepwild, 7 checks, both solvers) on a `wlmodel` with a MODEL param `wavelength` and an INSTANCE param `scale` ($R = \text{wavelengthscale}1k$): model wildcard co-varies both model cards AND reaches .model cards inside SUBCIRCUITS (they flatten to per-instantiation model copies); instance wildcard `@#[scale]` and alias `@*[[scale]]` co-vary every instance; concrete `@dev1[wavelength]/@NX2[scale]` target just one; absent-param wildcards warn. Full dual-solver regression passes. Four source files.

E-270 *src/frontend/com_sweep.c, examples/sweepbounds_examples/* (binaries rebuilt by CI)*

sweep validates its numeric bounds (ASan/UBSan fuzz find). An ASan/UBSan instrumented build fuzzing the newest command parsers (sweep/alter/altermod/loadpull/rfstab/stb/pyplot) surfaced a bug in sweep's spec parser `sw_parse_spec`. sweep reads each bound of its `<start>` `<stop>` `<step>` (and `'lin`

E-271 *src/frontend/com_let.c, examples/letoob_examples/* (binaries rebuilt by CI)*

let no longer reads one byte before its buffer on an empty LHS (ASan/UBSan fuzz find). The same command-parser fuzz that produced E-270 flagged a one-byte out-of-bounds READ in `com_let`. `com_let` flattens everything after `let` into a heap string `p`, splits the RHS off at the first `=`, then NUL-terminates the destination vector name at the first `[`. It next trims trailing whitespace from the name: for (`q=p+strlen(p)-1`; `*q<=' '` && `p<=q`; `q--`) ; `+++q='\0'`; . When the LHS is a bare bracket -- `let [[ = ...`, where the leading `[` has just been turned into `'\0'`, leaving

p an EMPTY string -- or is entirely whitespace, `strlen(p)==0` so q starts at p-1; the loop test evaluates *q BEFORE the `p<=q` guard can short-circuit, dereferencing `p[-1]` -- a read one byte before the allocation (ASan: heap-buffer-overflow READ of size 1 at `com_let.c`, from `let [[ = @#*[r] v1 -o i(v1))`). The stray byte does not fault deterministically on the shipped build (almost always valid heap metadata), but it is a genuine OOB read on malformed input and could fault if p sat at the start of a page. FIX (`src/frontend/com_let.c`): test the bound FIRST -- for (`q=p+strlen(p)-1; p<=q && *q<=' '`; `q--`) -- so `&&` short-circuits and q is never read while it points below p. An empty or all-whitespace name then falls through unchanged to the existing "bad variable name" check, which rejects it cleanly. No valid `let` is affected (a one-line reordering of a loop condition). Verify (examples/letoob, 5 checks): the fuzz-found `let [[ = ..`, a single-bracket `let [ = 5`, and an all-whitespace LHS each error quickly with bad variable name and no crash/hang; a plain `let a = 5` and an indexed `let a[1] = 7` still assign correctly; the original input under an ASan build no longer reports the overflow. Full dual-solver regression passes. One source file.

E-272 *src/frontend/device.c, examples/alternull_examples/* (binaries rebuilt by CI)*

alter/sweep no longer deref a NULL parameter for an m-named device (ASan/UBSan fuzz find). The same command-parser fuzz crashed on `sweep mag(v(b)) 0 1k 5k 1k. com_alter_common` (`src/frontend/device.c`) supports altering a device's PRINCIPAL value with no named parameter -- `alter r1 = 2k` -- in which case the parsed param is NULL. Just before applying the value it runs a binned-MOSFET guard: `'if ((dev[0]=='m') && (eq(param,"w"))`

E-273 *src/math/cmaths/cmath2.c, examples/mathcast_examples/* (binaries rebuilt by CI)*

`cmaths` integer operators guard their (**int**) casts against out-of-range doubles (ASan/UBSan fuzz find). Continuing the expression-layer fuzz flagged `cmath2.c:765` on `1e30 % 5`; it is one of a cluster where a routine casts a double operand to int with a plain cast (UB when the value is non-finite or outside int range). (1) `cx_mod` (the % operator) computed `(int) floor(fabs(op))` per operand -- `1e30 % 5` is UB (UBSan) and on the SHIPPED build returned a garbage 2 rather than an error (a silently wrong result). (2) `cx_vector/cx_cvector/cx_unitvec` set the result length with `(int) fabs(arg)` -- `vector(1e30)/unitvec(1e30)` produced a saturated (INT_MAX-ish) length and then allocated and filled it: a multi-gigabyte `calloc` (macOS over-commit lets it succeed) plus a billion-iteration fill loop, i.e. an apparent HANG on the shipped build (`rc=142` under a watchdog), not merely a sanitizer report. FIX (`cmath2.c`): range-check before every (`int`) cast. `cx_mod` floors each operand into a double, then `rchecks` it lies in `[0-or-1, INT_MAX]` before the cast; a non-finite/too-large operand takes the existing argument out of range for mod path (`rcheck->EXITPOINT->NULL`). A shared helper `cx_vec len()` validates the length for the three vector builders (`fabs/cmagnitude` are `>= 0`, so one `<= INT_MAX` check also rejects NaN/inf); an out-of-range length prints `vector length ... is out of range` and returns NULL, which `apply_func` already handles (it checks if `(!data)` return NULL). Valid expressions unchanged -- `17 % 5 == 2`, `vector(4) == [0,1,2,3]`, `unitvec(3) == [1,1,1]`, and any length within int range behaves exactly as before. Pre-existing (`cmaths`, not touched by the recent command work). Verify (examples/mathcast, 5 checks): `1e30 % 5`, `vector(1e30)`, `unitvec(1e30)` each error cleanly and fast (no UB, no hang, no garbage); valid `17 % 5` and `vector(4)` still evaluate correctly. Full dual-solver regression passes. One source file.

E-274 *src/frontend/evaluate.c, examples/idxcast_examples/* (binaries rebuilt by CI)*

An out-of-range vector index no longer invokes undefined behaviour (ASan/UBSan fuzz find). A fuzz of the expression layer flagged `evaluate.c:779` on `v(a) [1e308]`. `op_ind` (vector indexing `v[expr]`) rounds the index with `(int) floor(value+0.5)`; casting a double that is non-finite or outside int range to int is UB. The code ALREADY clamps the resulting index to `[0, majsize-1]` with a warning -- the only defect was the cast, which runs before that clamp. FIX (`evaluate.c`): a helper `idx_floor()` clamps the value to int range (NaN->0) before the cast, applied to all three casts (the real index and the real/imag parts of a complex `[lo,hi]` range index); `v(a) [1e308]`

then resolves to the last element with the pre-existing warning, exactly as a large in-range index already did. Verify (examples/idxcast, 5 checks): `vx[1e308]`, `vx[1e30]`, a NaN index each resolve cleanly (no UB); valid `vx[0]` and range `vx[0:2]` still return the right elements. One source file; no change to any valid index.

E-275 *src/math/cmaths/cmath4.c, examples/iffreal_examples/* (binaries rebuilt by CI)*

ifft() on a real vector no longer reads past its buffer (ASan/UBSan fuzz find). `cx_ifft` unconditionally did `ngcomplex_t *indata = (ngcomplex_t *) data;`. For a VF_REAL input, `data` is a plain `double[length]` (length 8 bytes), so `reading indata[i]` for $i < \text{length}$ walks length COMPLEX elements (length 16 bytes) -- twice past the buffer (ASan heap-buffer-overflow READ: a 528-byte/33-element region read as 66 complex). `cx_fft` (Enhancement-225) already distinguishes real and complex input; `cx_ifft` did not. FIX (cmath4.c): for a VF_REAL input build a proper complex array (`imag=0`) sized `length`, use it, and free it before returning; a VF_COMPLEX input still points straight at `data`; a `length >= 2` guard (matching `cx_fft`) rejects a degenerate input. Verify (examples/iffreal, 3 checks): `ifft(vx)` on a real vector runs with no overflow; the `fft->ifft` round-trip keeps the length; complex `ifft` still works. One source file; no change to a valid complex `ifft`.

E-276 *src/math/cmaths/cmath2.c, examples/rndcast_examples/* (binaries rebuilt by CI)*

rnd() guards its (**int**) cast against an out-of-range operand (ASan/UBSan fuzz find). `cx_rnd` (the `rnd()` random-integer function) turned each operand into a modulus with `j = (int)floor(dd[i]); rand() % j`. Casting a double outside int range to int is UB, so `rnd(1e30)` (or `inf`/NaN) tripped UBSan (cmath2.c:162). This is the same class as Enhancement-273, which hardened `cx_mod` and the `cx_vector` builders but not `cx_rnd` (at the time `rnd(1e30)` did not reproduce; the expression fuzz found the path). FIX (cmath2.c): a helper `cx_rnd_i()` clamps the value to int range (mapping NaN to 0) before the cast, applied to all three casts (the real operand and the real/imag parts of a complex operand); the result is unchanged for any in-range operand. Verify (examples/rndcast, 3 checks): `rnd(1e30)` and `rnd(inf)` resolve cleanly (no UB); a valid `rnd(5)` still yields an integer in `[0,5)`. One source file; completes the E-273 (int)-cast hardening for `cx_rnd`.

E-277 *src/math/cmaths/cmath4.c, examples/derivcx_examples/* (binaries rebuilt by CI)*

deriv() of a complex vector: fix a heap overflow AND a wrong result (ASan/UBSan fuzz find). The fuzz flagged cmath4.c:314 on `deriv((0.5, time))` (complex input). `cx_deriv` fits a degree-d polynomial over a sliding window; its COMPLEX branch diverged from its (correct) REAL branch in three places, each an out-of-bounds access on the last block AND a numerical error: (1) it read the data window `c_indata[j+i+base]` while fitting against the scale window `scale + i - degree + base` -- offset by degree, so it fit mismatched (x,y) pairs and read degree points past the end (heap-buffer-overflow READ); the real branch reads `indata + i - degree + base`. (2) The real-part output loop used `j <= i + degree/2` where the imag-part loop and the whole real branch use `j <= i - degree/2`, overrunning `scale[]/c_outdata[]`. (3) The tail indexed `scale[j+base]/c_outdata[j+base]`; the real branch's tail uses `j` (its FIXME notes `j+base` crashed), so a grouped (`base > 0`) complex derivative overran both arrays. FIX (cmath4.c, complex branch only): align to the real branch -- read `c_indata[j + i - degree + base]`, bound the real-part output loop with `i - degree/2`, index the tail with `j`. Every overflow is gone AND the complex derivative is now numerically correct. Verify (examples/derivcx, 3 checks): `deriv((0.5, time))` runs with no overflow; `deriv(t + 2t*i)` now returns (1, 2i) -- the correct derivative, mid-vector; a real `deriv(2t)` still returns 2; a focused `deriv` stress over complex/real/grouped inputs is clean under ASan. One source file; real derivatives unchanged, complex derivatives now memory-safe and correct.

E-278 *src/math/cmaths/cmath4.c, examples/scaleguard_examples/* (binaries rebuilt by CI)*

Transform functions `integ/deriv/ifft` guard a length/scale mismatch (ASan/UBSan fuzz find). Continuing the expression-layer fuzz, `integ/deriv/ifft` overran the heap on a vector whose length differs from its plot scale (cmath4.c:500 on `integ(unitvec(200))`, :1065 on `ifft(vector(5))`, :407 on `deriv(unitvec(200))`). `cx_integ/cx_deriv/cx_ifft` index both the data (by length)

and its plot scale; a SYNTHETIC vector (vector(n)/unitvec(n)/an expression result) carries the current plot's scale, whose length need not equal n. Two directions: (1) data LONGER than the scale (integ/deriv of unitvec(200) on a 66-pt plot) -- the loops read `pl_scale->v_realdata[i]` for `i<length`, past the end of the scale; (2) data much SHORTER than the scale (ifft(vector(5)) on a 66-pt plot) -- the Green's inverse-FFT sizes `datax` from `N=pow2(length)` but the output loop writes `tpts(=scale length)` points from `datax[2*i]`, so `N<tpts` overran `datax`. Only `cx_fft` had been guarded (Enhancement-225); the siblings had not. FIX (cmath4.c): `cx_integ/cx_deriv` reject a data vector LONGER than its scale (the direction that reads OOB) with a clean error -- a SHORTER data vector, as produced by `fft` (freq-domain, where group-delay `deriv(vp(3))` is legitimate), stays valid; `cx_ifft` grows its transform size `N` (and `M`) to cover the output length `tpts` before allocating `datax`, a no-op for a well-formed transform where `pow2(length)>=tpts` already (the `fft->ifft` roundtrip is unchanged). Verify (examples/scaleguard, 5 checks): `integ(unitvec(200))`, `deriv(unitvec(200))`, `ifft(vector(5))` resolve cleanly (no overflow); the `fft->ifft` roundtrip still returns a full-length vector; valid `integ(vx)/deriv(vx)` still work; a transform stress over mixed lengths/types is clean under ASan. One source file; completes for `integ/deriv/ifft` the length/scale guarding E-225 gave `fft`.

E-279 *src/frontend/com_let.c, src/frontend/options.c, src/frontend/com_measure2.c, src/mathscmaths/cmath4.c, examples/castguard_examples/* (binaries rebuilt by CI)*

The remaining unguarded (**int**) casts and the last unguarded scale-dependent transform (systematic audit, not fuzz). After the fuzz rounds a systematic AUDIT -- `grep every (int) floor/fabs(...)` of a runtime double, plus every `cmaths` routine taking a `struct plot *` and whether it guards its length against the scale -- found the tail of two already-fixed classes at sites the fuzzer never reached. (1) Unguarded `(int)floor(x+0.5)` of a user-supplied double (UB outside `int` range, the E-273/-274/-276 class): `com_let.c` (an index expression in an indexed `let`), `options.c` (set `numdgt / rawfileprec / measureprec` -- set `numdgt=1e30` tripped UBSan at `options.c:359`, set `rawfileprec=1e30` at :340), `com_measure2.c` (meas ... rise/fall/cross=<n> at :1580:1584:1588). (2) An unguarded scale-dependent transform (the E-278 class): `cx_mtimeavg` walks the scale data `dsc[j]` for `j < length-1`, so a vector longer than its plot scale -- `mtimeavg(unitvec(200))` on a shorter plot -- read past the scale (ASan heap-buffer-overflow @ `cmath4.c:1188`). FIX: a small clamping helper per file (`let_idx_int`, `opt_int`, `meas_int`) converts to `int` only after clamping to `int` range (NaN -> 0), so an out-of-range option/count saturates instead of invoking UB; values already in range are unchanged. `cx_mtimeavg` gets the same length-vs-scale guard `cx_integ/cx_deriv` received in E-278. Verify (examples/castguard, 7 checks): set `numdgt=1e30`, set `rawfileprec=1e30`, set `numdgt=-1e30`, meas rise=1e308, `mtimeavg(unitvec(200))` all clean where they previously tripped UBSan/ASan; a valid set `numdgt=6` and `mtimeavg(vx)` still work. Four source files, guard-only.

E-280 *src/frontend/com_let.c, examples/letidxoob_examples/* (binaries rebuilt by CI)*

An out-of-range `let v[i] = x` wrote past the end of the vector (heap-buffer-overflow WRITE). Hardening the index cast (E-279) exposed a far more serious defect underneath. `get_index_values` (`com_let.c`) parses either one index (`v[i]`) or a range (`v[lo:hi]`). It validates `low > high` and `high >= n_elem_this_dim` -- but those checks lived INSIDE THE RANGE BRANCH ONLY. The single-index path set `high = low` and returned completely unchecked, so `let vx[100] = 1` on a 66-element vector walked straight into the byte-offset arithmetic and performed a heap-buffer-overflow WRITE (ASan: WRITE of size 8) -- memory corruption from an ordinary typo, no exotic input required; a very large index additionally overflowed the `index * n_byte_elem` product in `int` (`com_let.c:771`). The range form `vx[0:999]` was correctly rejected all along. FIX: move both checks out of the range branch so they validate a single index too; an out-of-range assignment now reports `index/high range (N) exceeds the maximum value (M)` and changes nothing. Reads are unaffected -- `op_ind` clamps an out-of-range READ index with a warning (E-274) -- and every valid assignment, including the last element and the range form, behaves exactly as before. Verify (examples/letidxoob, 6 checks): `let w[10]=99` on a 10-element vector, `let w[999]=99`,

let `w[1e308]=99` each rejected cleanly where the first two corrupted the heap; the last valid index (let `w[9]=42`) and a mid-vector assignment still assign; the range form still rejected. One source file, moving two existing checks.

E-281 *src/math/cmath/cmath4.c, examples/derivgroup_examples/* (binaries rebuilt by CI)*

deriv() over-read on a partial block (grouping != length) (ASan fuzz find). `cx_deriv` walks its input in blocks: for (`base = 0; base < length; base += grouping`) for (`i = degree; i < grouping; i += 1`) ... fitting a window around `i + base`. `grouping` is the vector's `v_dims[0]`; for an ordinary vector that equals `v_length`, so there is a single block and the window always fits. But a vector whose declared dimension differs from its length -- as produced by a binary op on operands of UNEQUAL length, e.g. `min(v(b), ac.v(b))` where a 66-point real and a 5-point complex vector yield length 66 with `dims[0] = 5` -- leaves a PARTIAL LAST BLOCK: `base` climbs to `length - 1` while the window still spans `base + grouping - 1`, reading past the end of the input (ASan heap-buffer-overflow READ @ `cmath4.c:326`, the same line E-277 had already realigned). FIX (`cmath4.c`): bound the inner loop with `i + base < length` in BOTH the real and complex branches, so the fit window can never reach past the input; a no-op whenever `grouping == length` (every ordinary vector), and a partial trailing block too short to hold a full window is simply skipped. Verify (`examples/derivgroup`, 4 checks): `deriv(min(v(b), ac.v(b)))` and `deriv(max(v(b), ac.v(b)))` clean where they previously over-read; an ordinary real `deriv(2t)` still returns 2; the complex derivative from E-277 is still correct (`deriv(t + 2t*i) = 1 + 2i`). One source file, a loop bound in each branch.

E-282 *src/frontend/plotting/agraf.c, examples/plotlabel_examples/* (binaries rebuilt by CI)*

asciiplot read past its axis-label buffer on a 3-digit exponent (audit lead). Following up the plotting (`int`)`floor(...)` casts noted during the E-279 audit turned up a different, real defect one layer over. `ft_agraf` allocates its two axis-label lines as `maxy + margin + FUDGE + 1` bytes (`FUDGE=7`) and NUL-terminates them at the last byte, budgeting the exponent width with `sprintf(buf, "%1.1e", 0.0); shift = strlen(buf) - 7;`. Formatting 0.0 ALWAYS yields a 2-digit exponent, so `shift` is 0 and the whole layout silently assumes two digits -- but real data can need three: denormal or very large values produce labels like `1.00e-320` (9 chars vs 8). The last label's `memcpy(&line2[i + margin - ((j<0)?2:1) - shift], buf, strlen(buf))` then runs one byte too far and overwrites `line2`'s terminating `'\0'`, so the following `out_printf("%s\n%s\n", line2, line1)` walks off the end of the heap allocation -- ASan reports heap-buffer-overflow READ of size 82 on an 81-byte region inside `vsnprintf`. On the shipped build it does not necessarily fault; whether it prints heap garbage or crashes depends on what follows the buffer. FIX (`agraf.c`): remember the allocation bound in `line_end` (`maxy` is REASSIGNED further down as `maxy = spacing*nsp`, so the bound cannot be recomputed at the label loop), clamp the label `memcpy` so it can never reach the terminator slot (handling a negative offset too), and re-assert `line2[line_end] = '\0'` after the loop. Verify (`examples/plotlabel`, 5 checks): `asciiplot` of denormal `1e-320`, `1e300`, `1e-300` data and of both extremes together are clean where they previously over-read; an ordinary plot still renders its legend, axis rule and points; rendering confirmed BYTE-IDENTICAL to the pre-fix binary. One source file; no change to any rendered plot.

E-283 *src/frontend/plotting/agraf.c, src/frontend/points.c, src/frontend/display.c, examples/plotcoord_examples/* (binaries rebuilt by CI)*

Plot coordinate math no longer casts a non-finite double to **int** (extreme-data fuzz). The same output/formatting fuzz that produced E-282 found three more sites where a plot coordinate goes through `mylog10()` -- which returns +/-inf for zero, denormal or overflowed values -- or through a division by a range that can be zero, and is then cast to `int` (undefined behaviour). All are reached by ORDINARY plot commands on pathological data. (1) `agraf.c`: the decade `mag = (int)floor(mylog10(...))` -- data that overflows (`-1e308 * 6 -> -inf`, so `ylims[0]` is `+inf`) or is zero/denormal makes the argument non-finite; the limits `lmt/hmt = (int)floor(ylims[0]`

/ `tenpowmag`) are equally exposed since `tenpowmag = pow(10, mag)` can itself be 0 or inf. (2) `points.c` `ft_findpoint()`: `(int)((mylog10(pt) - tl) / (th - tl)) * (maxp - minp) + minp` is 0/0 for a degenerate range (`lims[0]==lims[1]`, or log endpoints sharing a decade). (3) `display.c`: the four screen-coordinate casts (log and linear, x and y). FIX: per-site clamping chosen so the result stays MEANINGFUL, not merely defined -- `agraf_decade()` bounds the exponent by `DBL_MAX_10_EXP` (the widest decade a double can represent) and maps NaN to 0, `agraf_int()` clamps the `lmt/hmt` ratios into int range; `ft_findpoint` computes the fraction in double, maps NaN (degenerate range) to 0 and clamps to [0,1] so the result always lands in `[minp, maxp]`; `disp_clamp_coord()` clamps each screen coordinate into the viewport. Verify (examples/plotcoord, 5 checks): `asciiplot` of overflowing (-1e308), denormal and all-zero data, and a min/max over denormal data, are clean where they previously invoked UB; an ordinary `asciiplot v(b)` still renders. Rendering additionally diffed against the pre-fix binary and confirmed BYTE-IDENTICAL for ordinary data. Three source files, guard-only.

E-284 *src/frontend/spiceif.c, src/frontend/device.c, src/frontend/com_sweep.c, src/include/ngspice/ftext.h, examples/wildparam_examples/* (binaries rebuilt by CI)*

Wildcard parameter diagnostics, and the **sweep** knob label (reported from real use). A Verilog-A parameter `L_um` could not be swept as `sweep @*[L_um]`, and ngspice's message named '`L_um`' -- which reads like the mixed-case name had been mangled by ngspice's case-insensitive netlist tokens. The case theory is a RED HERRING: the OSDI parameter lookup IS case-insensitive (`@*[L_UM]` resolves `L_um` correctly, verified). The real cause is a LEVEL mismatch that the diagnostics `hid: @*[param]` (and `@#[param]`) is the INSTANCE wildcard, while a plain parameter `real L_um` in Verilog-A is a MODEL parameter -- an instance parameter needs `(* type = "instance" *)`. So the instance wildcard correctly matched nothing and reported no loaded instance has parameter '`L_um`': lower-cased, with no hint that the parameter exists at the MODEL level, and nothing pointing at the fix. Separately, `sweep` labelled every knob it applies in place as (model param) -- including the instance wildcards -- because `sw_kind()` returns `SW_MODEL` as a DISPATCH flag ("apply via altermod, no re-source") and the banner reused that flag as a description. FIX: (a) `spiceif.c` gains a probe `if_hasparam_wildcard(ckt, param, do_model)` reporting whether any loaded model (or instance) carries a settable parameter of that name -- it only probes, nothing is changed; (b) `device.c` uses it so a wildcard that matched nothing checks the OTHER level and names the form that would work, in both directions (... but a loaded model does -- use the model wildcard '`@*[L_um]`' (an instance parameter needs `(* type = "instance" *)` in the Verilog-A), and the mirror suggesting `@#[...]`), while a parameter that genuinely does not exist still gets the plain message with no misleading suggestion; (c) `com_sweep.c` gains `sw_knobdesc()`, which classifies the knob from the NAME TOKEN, so the banner reads (instance param, wildcard) for `@#[...]/@*[...]` and (model param, wildcard) for `@*[...]`. Verify (examples/wildparam, 8 checks) on a model with a mixed-case MODEL parameter `WaveLength` and a mixed-case INSTANCE parameter `L_um`: both cross-level hints appear and name the right form; a truly absent parameter gets no bogus hint; `sweep` labels instance and model wildcards correctly; the matching wildcard still sweeps to the right values; and ALL-CAPS `@*[L_UM]` still resolves the mixed-case `L_um`, confirming the lookup was never case-sensitive. Four files; diagnostics and one banner string, no change to which parameters any wildcard actually sets.

E-285 *src/frontend/plotting/plotit.c, src/frontend/plotting/agraf.c, src/frontend/plotting/gnuplot.c, src/frontend/com_measure2.c, examples/veclenmix_examples/* (binaries rebuilt by CI)*

Output paths indexed one vector by another's length (extreme-data fuzz). The remaining findings from the extreme-data output fuzz all proved to be ONE class in four places: a vector's OWN length need not equal its plot SCALE's length -- any synthetic vector carries the current plot's scale, so let `y = vector(8)` on a 66-point transient plot has 8 points and a 66-point scale -- and a COMPLEX vector holds its data in `v_compdata`, leaving `v_realdata` NULL. (1) `plotit.c`: the transient re-sampling passed `v->v_realdata` together with `v->v_scale->v_length` to `ft_interpolate()`,

which indexes the DATA by that length, reading ~58 elements past a shorter vector; for a complex vector it passed NULL outright -- a HARD SEGV on the shipped build (`asciiplot sqrt(-1*vector(10))` returns `rc=139`). (2) `agraf.c`: the bracketing indices lower/upper are computed against the X scale and bounded by `xscale->v_length`, but were then used to index each plotted VECTOR. (3) `gnuplot.c` (`wrdata`): the "no more data" guard tested only `i >= scale->v_length` while the branch it protects indexes `v->v_realdata[i]`. (4) `com_measure2.c`: the plain tran/dc branch read `d->v_realdata[i] / dScale->v_realdata[i]` with no NULL check, though the ac/sp branches beside it already had one -- measuring a complex vector dereferenced NULL (present twice, in `measure_at()` and `measure_deriv_at()`). FIX: clamp each index to the vector it actually addresses -- `plotit.c` passes the length the data and scale SHARE and skips the transient resampling entirely when any vector is not real (`all_vecs_real()`), since that block replaces the shared scale and must convert all vectors or none (complex vectors keep plotting through the normal path); `agraf.c` clamps lower/upper into `[0, v->v_length-1]` per vector; `gnuplot.c`'s guard also requires `i < v->v_length`; `com_measure2.c` takes the real part when `v_realdata` is NULL, exactly as its ac/sp branches do. Verify (examples/`veclenmix`, 7 checks): a vector shorter than its scale, a vector longer than its scale, a complex vector (previously `rc=139`), `wrdata` of a short vector, and a measure over a complex vector are all clean and still render/measure; an ordinary `asciiplot v(b)` and `meas tran ... max v(b)` are unaffected. Ordinary plot output AND `wrdata` output diffed against the pre-fix binary and confirmed BYTE-IDENTICAL; all eight decks from the output fuzz are clean. Four source files, guard/clamp only.

(E-25's `simpam` exposure lives in the OSDI callback table populated at load time; its diff rides inside the `osdiload.c/callbacks` changes.)

Build-warning cleanup (post-E-127): a full rebuild under the repo's `-Wconversion` flags surfaced latent warnings. `configure.ac` had `-s` (a strip flag) in the compile CFLAGS, so clang warned "argument unused during compilation: '-s'" on every source file (~1526 warnings) — removed (binaries are stripped separately by the fold and CI). `-Wconversion` also flagged a real latent bug in `maths/ni/nipzmeth.c` (`b = 0.0` was an assignment where `b == 0.0` was intended) — fixed — and benign `double→bool` conversions in `maths/cmats/cm4th4.c` — made explicit. Full-build warnings dropped 1903 → 394; the remainder are `-Wconversion` warnings in third-party SuiteSparse (KLU) and CMC device models (HiSIM), which are left to their upstream sources.

Every entry above is guarded by a verify script under [examples/](#) and was regression-checked against the full example arsenal, the compiler's integration suite, and the VA_TEST industry-model corpus at the time it landed. | [E-296](#) | `src/frontend/plotting/pyplot.c` | pyplot appearance controls. Seven additive set variables: `pyplot_grid` (on/off/x/y, overriding the grid-follows-axis default), `pyplot_legend` (off, or a matplotlib location -- underscore form since set keeps only the first word), `pyplot_markers` (a cycling marker `o s ^ D v * P X` ON the line, distinct from `pointstyle=markers` which drops the line), `pyplot_axhline/pyplot_axvline` (SI-aware reference lines via `ft_numparse`), `pyplot_dpi`, `pyplot_transparent`. With none set the emitted script is byte-identical to before. Verify (examples/`pyplotmore`): each control appears in the script AND the script executes; the default path is unchanged. One file. | | [E-297](#) | `src/frontend/plotting/pyplot.c`, `plotit.c`, `com_pyplot.c` | pyplot **-fft** magnitude spectrum. `ft_pyplot`'s `bool hist -> int mode` (LINE/HIST/FFT). `-fft <sig>` reuses the time-domain (t,y) table; the emitted Python RESAMPLES onto a uniform grid (`np.interp`) before `np.fft.rfft`, because transient data is adaptively sampled and a raw FFT over non-uniform samples is wrong. One-sided, scaled by `2/sum(window)` so a pure tone reads back its amplitude. Options `pyplot_fft_window` (`hann/hamming/blackman/rect`), `pyplot_fft_db`, `pyplot_fft_points`, `pyplot_fft_logf` (`log-f` in Python, dropping the DC bin -- NOT the command's `xlog`, which would validate the `t=0` TIME scale and abort). Verify (examples/`pyplotmore`): a `2.0@1kHz + 0.5@3kHz` tone reads 2.0 and 0.5 (amplitude oracle, `rel<2%`). Three files. | | [E-298](#) | `src/frontend/plotting/pyplot.c`, `plotit.c`, `com_pyplot.c` | pyplot complex-aware AC views **-bode/-nyquist/-polar**. An ordinary `pyplot v(out)` on AC data silently keeps only the REAL part (0.5 not 0.7071 at -3 dB). New `ft_pyplot_ac()` emits a `<vec-index>` `<freq>` `<re>` `<im>` table -- the imaginary part is CARRIED -- grouped by the first column so multiple/

ragged vectors stay separate; `-bode = 20log10|H| dB / unwrapped phase deg vs log-f stacked`, `-nyquist = imag vs real (equal aspect)`, `-polar = |H| at angle(H) on a polar projection`. Reuses the shared file/launch scaffolding and `pyplot_*` settings; auto-names `bode/bode-2/...` via the shared counter. Verify (examples/pyplotmore): RC low-pass at `fc` gives `-3.010 dB` and `-45.00 deg` (exact first-order values, from the preserved complex data). Three files. | | [E-299](#) | `src/frontend/plotting/pyplot.c` | `pyplot` overlay robustness + hover cursor + export note. Cross-run overlay via `ngspice's plotname.vector` refs (`pyplot tran1.v(out) tran2.v(out)`) already worked, but the data table was sized by the FIRST vector's scale, so a longer second run was truncated; it now walks the LONGEST scale across the plotted vectors (each still uses its own scale for x, shorter ones pad with NaN which matplotlib skips; a single run is unchanged since all share one scale). `set pyplot_cursor` adds a hover crosshair via matplotlib's built-in `Cursor` widget (no extra package), interactive-only (not emitted for a hardcopy). Pan/zoom/save (matplotlib toolbar) and the `<name>.data` export were already provided. Verify (examples/pyplotmore): a finer second run keeps every sample; cursor present in a window script, absent from a hardcopy. One file. | | [E-300](#) | `src/frontend/plotting/pyplot.c` | `pyplot` **`pyplot_mplcursors`** cursor backend. On top of E-299's `pyplot_cursor` (built-in `matplotlib.widgets.Cursor` crosshair), `set pyplot_mplcursors` selects the `mplcursors` package -- data cursors that snap to a trace and show its (x,y) on hover -- AND turns the cursor on by itself. Unset, the built-in crosshair remains the default. The emitted script imports `mplcursors` inside a `try` and FALLS BACK to the built-in `Cursor` on `ImportError`, so a deck written where `mplcursors` exists still runs where it does not. Window-only (not emitted for a hardcopy), like `pyplot_cursor`; `mplcursors` must be importable by the interpreter named by `pyplot_python`. Verify (examples/pyplotmore): the `mplcursors` branch + fallback are emitted on `pyplot_mplcursors` alone, `pyplot_cursor` alone still emits only the built-in `Cursor`, neither appears in a hardcopy; the `mplcursors` branch was executed under an interpreter that has the package (calls `mplcursors.cursor(hover=True)`) and the fallback under one that does not. One file. | | [E-301](#) | `src/frontend/plotting/pyplot.c` | `pyplot` **`pyplot_cursor`** is the single cursor master switch. E-299 (built-in `Cursor`) and E-300 (`mplcursors`) had drifted: `pyplot_mplcursors` turned the cursor ON by itself, so there was no one "off" knob. The enable condition went from `pyplot_cursor OR pyplot_mplcursors` to `pyplot_cursor` alone -- off by default, the ONLY thing that enables any cursor; `pyplot_mplcursors` now only SELECTS the backend when the cursor is on and does nothing on its own. Predictable, and matching every other optional `pyplot` behaviour. Verify (examples/pyplotmore): the full gating truth table -- default off, cursor-alone -> built-in, `mplcursors-ALONE` -> off, both -> `mplcursors` (with fallback), off in a hardcopy. One condition, one file. | | [E-302](#) | `src/frontend/com_measure2.c` | **`.meas avg`** did not clip its window to `[from,to]`. Found by a closed-form oracle (a same-binary comparison is structurally blind to a uniformly present error). `measure_minMaxAvg()` trapezoid-summed only the samples INSIDE the window and divided by their span, without interpolating either boundary -- so the first and last partial intervals were dropped. `rms/integ (measure_rms_integral)` already clip exactly, so `ngspice's` own two answers for one window disagreed: `avg=1.27114` vs `integ/(to-from)=1.27324`, closed form `1.2732395`; and the echoed `to=` was the first sample OUTSIDE the window (`2.50280e-04` for a requested `2.5e-4`). `O(dt/window)` error: `1.6e-3 -> 4e-7` here. Fix interpolates both ends and reports `sprev` as the window end, all guarded to `AT_AVG` so `min/max/min_at/max_at` keep whole-sample semantics. The `dc` branch is deliberately EXCLUDED -- a `dc` sweep may descend and enters its window at the high end, where a clip extrapolates (an early version of the fix turned a descending-sweep `avg` of `0.270250` into `-3.79e+11`); every `dc` measurement is byte-identical. Known remaining: **`.meas dc avg`** still truncates (`0.270250` vs `0.2708333`), and on a descending sweep `integ` returns `0.0` with `from=nan`. Verify (examples/measwindow): 28 checks vs closed-form integrals, windows landing BETWEEN samples; fails 14/28 on the pre-fix binary. | | [E-303](#) | `src/frontend/com_measure2.c` | **`.meas dc avg`** window truncation, both sweep directions. Completes E-302, which fixed `tran/ac/sp` but excluded `dc` because a sweep may DESCEND (`dc v1 2 0 -0.001`) and enters its window at the HIGH end, where an unguarded clip extrapolates outside the bracketing samples. **`.meas dc avg`** averaged over `[first sweep point inside, last inside]` instead of `[from,to]`: `0.270250` against a closed-form mean-of-`x^2` of `0.2708333`, echoing `to=7.49000e-01` for a requested `0.75`, while **`.meas dc integ`** over the same window was already correct. Fixed direction-agnostically in `measure_minMaxAvg()`: the clip works from the ACTUAL crossing between the previous raw sample and the current one -- open the window

at the boundary crossed on the way in, close it on the way out -- so it fires at from->to ascending, to->from descending, and never on a sweep that does not cross. Guarded to AT_AVG, so min/max/min_at/max_at keep whole-sample semantics. The echoed window reports the upper end of the range actually covered, so a descending sweep no longer prints the nonsensical from= 0.25 to= 0.25. STILL OPEN (different function, measure_rms_integral): on a descending sweep integ/rms return 0.0 with from= nan -- the loop meets the first sample already above to and interpolates with index i-1 == -1, an OUT-OF-BOUNDS read, then breaks with an empty array. Verify (examples/measwindow): 38 checks, the same window run in BOTH sweep directions vs closed form; 32/38 on the pre-303 binary. | [E-304](#) | src/frontend/com_measure2.c | **.meas dc integ/rms** heap-buffer-overflow on a descending sweep. Completes E-303. On dc v1 2 0 -0.001 these returned 0.0 with from= nan (rms printed nothing). measure_rms_integral() builds its clipped window assuming the scale ASCENDS -- it clips at from going in and to going out, and the widths x[i+1]-x[i] must be positive for the Simpson/trapezoid sums. A descending sweep starts at the top, so the FIRST sample is already above to: the end-clip fired at i==0 and interpolated against index i-1 == -1, reading v_realdata[-1]. AddressSanitizer: heap-buffer-overflow ... measure_interpolate com_measure2.c:195 / measure_rms_integral com_measure2.c:1415. It then broke with a one-element array, so while (i < xy_size-1) ran ZERO times (sums 0) and first was never set (m_measured_at unwritten = the nan). Fix: walk the samples in order of INCREASING SCALE rather than index order -- the identity for ascending data, so tran/ac/ascending-dc are byte-identical -- and make the has-a-previous-sample guard the traversal position n > 0 instead of i > 0, correct in both directions. integ 0.0 -> 0.135416 (closed form 0.135416667); rms -> 0.307458 (0.307459347). Also removed two dead reads in the toVal < 0.0 tail that indexed d->v_realdata[] with the INTEGRATION loop counter (past the end of the data) into variables nothing reads again, and took the window end from xScale[v_length-1] -- the smallest scale value when the sweep descends, not the last traversed. Verify (examples/measwindow): 44 checks, both sweep directions for avg/integ/rms vs closed form; 40/44 on the pre-304 binary; and the OOB is PROVEN gone -- ASan reports it before the fix and is clean after. | [E-305](#) | src/math/misc/randnumb.c, src/include/ngspice/randnumb.h, src/frontend/com_sweep.c, src/frontend/commands.c | **wcd** -- worst-case distance / MPFP high-sigma. The named remainder of the statistical suite, and the complement E-150's own write-up says it does not implement. highsigma uses direction-free scaled-sigma IS; wcd works in STANDARDISED NORMAL space and asks a geometric question -- which point of the failure region is most probable? With the margin as g(u)>0 for pass, that is the boundary point closest to the origin; its distance beta is the worst-case distance and, the density being spherical, P_fail = Phi(-beta) -- the sigma number a designer quotes. Hasofer-Lind/Rackwitz-Fiessler iteration with forward-difference gradients: cost BOUNDED at a few iterations of (1+ndim) sims, versus the 1e6-1e9 samples plain MC needs merely to SEE a 4.5-6 sigma event. -is N refines FORM (exact for a linear margin, approximate when it curves) with mean-shift IS centred on the MPFP, carrying the exact ratio phi(z)/phi(z-u*): 986 failures seen in 2000 shifted samples for a 3.4e-6 event. Implemented as two modes behind the EXISTING single Gaussian-draw funnel mc_sample_gauss(), so it composes with LHS/SSS rather than duplicating them: MC_MODE_WCD makes draws deterministic (dimension d returns a chosen u[d], so the deck is a plain function g(u)) and MC_MODE_SHIFT does the shifted sampling. Dimensionality is DISCOVERED (a deck's Gaussian-draw count is unknown until it has been evaluated once) via a running maximum cleared per evaluation -- a single margin evaluation triggers several deck-copy passes and a later pass that draws nothing would otherwise wipe the count. Verify (examples/wcd): 19 checks vs the ANALYTIC GAUSSIAN TAIL, never a previous build -- FORM is exact for a linear margin so beta is exact to machine precision at 3/4/5/6 sigma and P matches Phi(-beta) to ~1e-8; a 2-D case puts the MPFP on the symmetric point (4/sqrt2, 4/sqrt2), which is what shows the search finds the right DIRECTION and not a 1-D degenerate answer; a nominal-already-failing case gives signed beta=-3. The four existing statistical suites pass unchanged. | [E-306](#) | src/math/cmats/cmath4.c | The E-241 twin: **fft()** expression function normalised by the padded N. E-241 fixed an amplitude normalisation that divided by the ZERO-PADDED transform size instead of the input length, in the **fft** COMMAND (frontend/com_fft.c). The identical mistake survived in math/cmats/cmath4.c **cx_fft** -- the vector-expression function reached by **let F = fft(v)**, a separate implementation of the same computation. Found by continuing the oracle campaign that produced E-241, and located

with E-241's own discriminator: bin 0 is unambiguous (no window/scalloping) and $X[0] = D\text{length}$ for a DC offset D , so dividing by N reads back $D\text{length}/N$. On one signal (4001 samples padded to 4096, DC 2.0) the COMMAND read 2.000000 and the EXPRESSION read $1.953613 = 2.04001/4096$. NOT a convention question: `cx_fft` holds TWO complete implementations (complex-input and real-input) and in BOTH the FFTW branch already used the input length $((\text{double})\text{length})/2.0, \text{pts}$ while Green's radix-2 used the padded size $((\text{double})N)/2, (\text{double}) N$ -- the correct version sitting a few lines from the wrong one, an internal contradiction like `avg` vs `integ` in E-302. HAVE_LIBFFTW3 is undefined in this build so Green's is live. Fixed both Green branches to the input length. INDEPENDENT confirmation: nothing here touches `ifft`, yet `ifft(fft(x))` went $2.3\text{e-}02 \rightarrow 1.1\text{e-}16$ (the pair only inverts when the forward normalisation is right), so `cx_ifft` was audited and deliberately left alone. Full audit of every Green-kernel caller: `com_fft.c` `fft x2` and `spec/PSD x2` correct (E-241); `cx_ifft` correct; `trannoise/1-fcode.c` cannot pad (`n_pts` grown to 2^{n_exp} by construction); `fft/ifft` are the only transform functions in the expression table so there is no spec twin. Verify (examples/fftexpr): 6 checks vs closed form (DC bin, round trip padded and unpadded, complex-input bin 0 = $|\text{mean}(H)|$); 3/6 on the pre-fix binary. || E-311 | `src/frontend/measure.c` | **param/expr** measurements now work in a **.control** block. Found oracle-checking the less-common **.meas** modes. The **param/expr** measure types are handled only by `do_measure()`'s second pass (via `nupa_eval`); the interactive **.control** `meas` command (`com_meas`) skips `do_measure` and calls `get_measure2()` directly, which has no **param/expr** case -- so `.meas tran pp param='a1-a2'` (dot-card) worked but `meas tran pp param='a1-a2'` (**.control**) failed with 'no such function as ...', down to a bare `param=a1`. Fix: in `com_meas` the prior results (`a1,a2`) are already single-valued ngspice VECTORS, so a **param/expr** measurement is `let <name> = (<expr>)` via the ordinary vector evaluator -- detected BEFORE `com_meas`'s single-valued-vector substitution loop (which would mangle the expression), reassembled from the tokens after `param=/expr=`, delimiter single quotes stripped, and re-lexed with `cp_lexer` so a spaced expression tokenises as the shell would. Works: `param=a1-a2` (unquoted), `param='a1-a2'` (quoted no-space), `param={sqrt(a1*a1 + a2*a2)}` (braces, spaces OK), `expr=`. Normal `meas` types (`max/min/avg/when/...`) route through `get_measure2` unchanged (byte-identical vs shipped). KNOWN LIMITATION left in place: `param='sqrt(a1*a1 + a2*a2)'` (single quotes + internal spaces) is mangled by the **.control** shell's own quote pre-expansion UPSTREAM of `meas` -- use braces; the dot-card handles it (no shell tokenizer). Verify (examples/measparam): 8 checks vs closed-form triangle values; 7/8 fail on the pre-fix binary. || E-312 | `src/xspice/cm/cm.c`, `src/xspice/icm/analog/s_xfer/cfunc.mod` | XSPICE integrating code models are second-order accurate. Found oracle-checking the `s_xfer` Laplace transfer function against a closed-form transient: its error fell only as $O(h)$, not the $O(h^2)$ every native ngspice storage element delivers. Two independent causes. (a) SHARED across all integrating code models (`s_xfer`, `int`, `d_dt`, ...): `cm_static_integrate()`'s TRAPEZOIDAL order-2 arm was a self-flagged stand-in -- `cur = -0.5*ag[0]*intgr[1]` with a compensating `geq *= 0.5`, carrying its own `/* WARNING - This code needs to be redone */` -- which works out algebraically to backward Euler ($y(n)=y(n-1)+h u(n)$), a first-order method. Fixed to the true rule $y(n)=y(n-1)+(h/2)(u(n)+u(n-1))$; the "one previous value" the WARNING said was missing is `u(n-1)`, stored in the SPARE state `double cm_analog_alloc` already reserves per integrator (`doubles_needed = bytes/sizeof(double)+1`), rotated through the `CKTstates[]` history for free -- the standard SPICE companion-model idiom of keeping a state and its rate in adjacent slots. `geq` is now `ag[0]` like the order-1 and Gear arms; a guarded legacy fallback preserves old numerics only where no spare exists (no real code model). (b) SPECIFIC to `s_xfer`: its controller-canonical feedback read the fed-back integrator states from the PREVIOUS timestep (`cm_analog_get_ptr(i,1)`), making the loop explicit/lagged one step, which caps $H(s)$ at $O(h)$ on its own even after (a). Now reads current-iteration states (offset 0) so the loop is solved implicitly within each Newton step (the DC/init pass already used offset 0). Both TRAP and Gear now converge $O(h^2)$. Convergence verified vs an exact $1/(1+\tau s)$ sine oracle (error ratio ~ 2 per h -halving before, ~ 4 after) and, for stability of the now-implicit feedback, vs SciPy `signal.lsim` across five transfer functions -- 1st-order, a $\zeta=0.05$ near-oscillatory resonator, a 3rd-order Butterworth, a stiff pair with poles three decades apart, and one with a numerator zero -- all converge and match to $<2.2\text{e-}4$. Verify (examples/sxferorder): 6 checks under both solvers, including a self-contained convergence-ORDER test (ratios ~ 4 , fitted $p \sim 2.1$) that fails on the pre-fix binary with no reference build required. 246-example regression green under both solvers. || E-315 | `src/spicelib/analysis/tfanal.c`, `src/spicelib/analysis/cktic.c`, `src/spicelib/analysis/distoan.c` |

Three analysis crashes on adversarial-but-valid input -> clean errors (command/netlist fuzz, E-222..228 / E-270..285 family). [6] `.tf` on a SINGULAR circuit -- a dangling inductor (l1 2 3 1, nodes 2,3 floating) makes the operating point singular; `tfanal.c` IGNORED CKTop's return, so the matrix was never factored and the following `SMPsolve()` asserted `IS_VALID(Matrix) && IS_FACTORED(Matrix)` (`spsolve.c:137, SIGABRT`). Fix: propagate the CKTop error before solving. [7] a second `.pz` over a URC device -- CKTic zeroes the RHS vectors with a loop the compiler vectorises into `memset(CKTrhs,...)`, but here `ckt->CKTrhs` is NULL, so it wrote through a NULL pointer (SIGSEGV in `platform_memset`). Fix: CKTic returns cleanly when the RHS vectors/matrix are unallocated. [8] `.disto` with NO distortion sources (a plain resistor) -- `distoan.c`'s output section left `OUTpBeginPlot`'s result unchecked, so on failure `acPlot` stayed NULL and `OUTattributes(acPlot,...)` dereferenced it (SIGSEGV at `OUTattributes+268, addr 0x28`); the first plot checked its result but the five output-section plots did not. Fix: capture and check `OUTpBeginPlot` at each output plot, bail on NULL `acPlot`. Legitimate `.tf/.pz/.disto` unaffected -- guards fire only on the failure paths (valid `.tf` still gives 0.5 on a 1k:1k divider). Verify (examples/ngcrashanalysis): 4 checks. | | [E-316](#) | `src/frontend/com_measure2.c` | `.meas AVG` ended one timestep short of `to`. Found continuing to oracle-check `.meas` against its own contract (E-302/303/304 family): AVG is the same quantity as `INTEG/(to-from)` over the identical window. `measure_minMaxAvg()`'s final window-clip (added by E-302, which made AVG integrate a trapezoid over `[from,to]`) guarded the WHOLE accumulation with `!AlmostEqualUlp(svalue, meas->m_to, 100)`, so when the first out-of-window sample fell within 100 ULPs of `to` the entire final trapezoid `[sprev,to]` -- a full timestep -- was skipped. AVG's window then ended one timestep short of `to`: for a $2.5e-7$ timestep the AVG window echoed `to=6.45990e-04` while `INTEG` (correct) echoed `to=6.46240e-04`, exactly $2.5e-7$ apart, so `avg != integ/(to-from)` by $\sim 1.6\%$. `measure_rms_integral()` (`INTEG/RMS`) does NOT have this bug -- it always adds the final point, interpolating to `to` only when the sample overshoots, else using the `~to` sample as-is. Fix: move the `!AlmostEqualUlp` guard so it gates only the INTERPOLATION, not the accumulation, matching `INTEG` -- the final trapezoid is now always added, and AVG covers the full `[from,to]` window (`AVG==INTEG/(to-from)`, rel 1.6e-6). MIN/MAX keep whole-sample semantics (branch is `AT_AVG-only`). Verify (examples/measavgwin): `AVG==INTEG/(to-from) + AVG` window reaches `to`, both fail on the pre-fix binary; the E-302/303/304 `measwindow` suite (44 checks) and E-311 `measparam` still pass. | | [E-318](#) | `src/spicelib/devices/vsrc/vsrcload.c` | SFFM/AM voltage sources dropped the DC offset before the delay. Found in the correctness campaign, oracle-checking transient waveform sources. `vsrcload.c` evaluated SFFM and AM with `if (time <= 0) value = 0;` (after subtracting TD), returning 0 instead of the quiescent value at `time<=0` -- so an SFFM/AM source's OPERATING POINT (common `TD=0`) was 0 not VO, and over any pre-delay window it read 0, injecting a spurious startup transient (an `SFFM(2,0.001,...)` `~= 2V` source into an RC started at 0V and charged to 0.785V, vs a DC-2 source flat at 2V). Three oracles show VO is correct: the SIN case in the SAME function holds `VO+VAsin(phase)` at `time<=0`; ngspice's OWN current-source SFFM (`isrcload.c`) has NO such zeroing and evaluates the formula directly (the two SFFM implementations in one binary disagreed); SIN/PULSE/EXP/PWL all preserve their offset. Fix: hold the waveform's `time=0` value -- SFFM value = `VO + VAsin(phase + MDIsin(phasem))`; AM value = `VO + (VMO + VMAsin(phasem))sin(phase)`; (both reduce to VO for zero phases). No example/regression circuit used SFFM/AM, so nothing else changes; the current-source path was already correct and is untouched. Verify (examples/sffmoffset): SFFM->1.5, AM->2.0 over a delayed window (SIN control->1.5), all failing on the pre-fix binary. | | [E-319](#) | `src/frontend/com_qpss.c` | Transient-form QPSS leaked the fundamental into every mixing bin. Found in the correctness campaign via the RF steady-state oracle (a QPSS steady state must equal the HB-form and a long transient + DFT). The transient-form `qpss <expr> <f1> <f2> [periods] [maxorder]` (as opposed to ... `hb K1 K2`) runs a two-tone transient then computes each 2-D harmonic `k1f1+k2f2` by a *TRAPEZOIDAL integral* of `v(t)exp(-j2pi f t)` over the "last period" `[tt[i0], tt[end]]` of the RAW transient grid. That window is not exactly the beat period `T=1/gcd(f1,f2)` (`tt[i0] >= wstart` but not `==`, short by up to one `tstep`), is non-uniform, and has non-periodic endpoints, so the large DC/fundamental leaks `~tstep/T` into EVERY mixing bin. On a LINEAR two-tone RC (every product with `|k1|+|k2|>=2` is analytically exactly 0) every bin read `~5.8e-4|dominant line|` (`~45 dB`), scaling LINEARLY with the DC line (the leakage signature); confirmed against the HB-form (`~1e-16`) and a plain `.tran + uniform DFT` (`~1e-10`). Fix: resample the last beat period onto a UNIFORM grid over exactly `[wend-T, wend)` (linear interpolation) and Fourier-project with the rectangular rule -- for

commensurate tones every harmonic $k1f1+k2f2=mfb$ completes an integer m cycles in T , so a uniform DFT over exactly T is exact. The spurious floor drops ~ 4 decades to $\sim 6.6e-8$ (~ 122 dB). Real products are unchanged: the `qpss_examples` strong IM3 ($3.75e-4$) still matches the built-in to 4 digits, and a single-tone diode's (2,0) matches the HB-form to 5. Honest limit: the residual floor is grid-limited (identical across 4/16/64 settling periods), so the very weakest products (below ~ 120 dB) remain floor-limited -- for those the exact HB-form is the tool; but the old -45 dB floor masked far more. The ... `hb` form was already exact and is untouched. Verify (`examples/qpssleak`): linear-circuit mixing products $< 1e-5$ of the fundamental (pre-fix $\sim 7e-3$); `qpss_examples` (12 checks) still pass. | | [E-320](#) | `src/frontend/com_sweep.c`, `src/frontend/numparam/{spicenum.c,xpressn.c,numparam.h,numpaif.h}` | Sweeping a `.param` no longer forces a full circuit reset at every point. A `.param` has no live binding into the built circuit -- `numparam` folds each `{expr}` on a device line to a fixed numeric literal at parse time -- so the only way to change one is `alterparam` (rewrite deck text) + `reset` (re-parse + `subckt-expand` + `CKTsetup` + `matrix REORDER`), per point, even though topology never changes. E-320 adds a fast path to the sweep command: when every swept `.param` feeds ONLY addressable top-level device/model VALUES, `com_sweep` per point overrides the swept symbol(s) in the retained `numparam` dico (`nupa_add_param`), refreshes the derived-param closure (new `nupa_recompute_params` replays the dependent `.param` lines), re-evaluates each captured device value against the dico (new `nupa_eval_expr` -> the `numparam` formula() evaluator), and pushes it into the live instance with a resolved direct set (`ft_sim->setInstanceParm = CKTparam -> DEVparam`), refreshing each touched device type once via `DEVtemperature` -- NO reset, so the matrix ordering/factorization pattern is preserved. The template is the `temper` machinery (a temperature-dependent device value re-evaluated + alter'd in place every .dc step); the difference is that a parse tree cannot re-read an arbitrary `.param` cell (only time/temper/hertz bind to live ckt storage), so re-eval goes through `numparam` not `IFEval`. Two optimizations: each `@dev[param]` target is resolved to (instance, param-id) ONCE at arm time (no per-point lex or device-list walk), and identical value expressions are grouped so each unique expression is evaluated once per point. Conservative classifier: arms only when a scan of the ORIGINAL (pre-expansion) deck finds every swept-param occurrence to be a top-level device/model value slot (positional principal, or `key={expr}` named); DISARMS to the exact reset path on any `subckt-body` use (expression not retained past flattening), structural use (param in a node position, an instance/model/subckt name, a subckt call arg, `.if.elseif`, `.temp`, an analysis card, `.option`, `.ic`, `.nodeset`, `.global`), or a derived `.param` (a non-swept param defined from a swept one). Because a disarm reproduces today's behaviour exactly, a sweep can only get faster, never change result. Measured on a 2500-stage ladder, 120-point sweep: param feeding ONE device in a large circuit $2.91s \rightarrow 0.26s$ (11.1x), param feeding ALL 5000 devices $3.42s \rightarrow 1.33s$ (2.6x); fast matches reset to $\sim 7e-9$ (`numparam` value-string formatting only), `subckt` fallback bit-identical. Scope: the sweep command (instance values fast, model values via textual `altermod` fallback); `subckt-internal` params and the `optimize/Monte-Carlo` `.param` reset sites stay on the reset path. Verify (`examples/paramfastsweep`): top-level param arms + matches closed-form $R2/(rval+R2)$; `subckt` param falls back yet identical. | | [E-321](#) | `src/frontend/com_sweep.c`, `src/frontend/numparam/{spicenum.c,numpaif.h}` | `.param` fast-sweep, subcircuit tier -- `subckt-INTERNAL` device values now take the fast path. E-320 was limited to top-level values: a swept param feeding a device inside a subckt disarmed to reset, because `numparam` freezes each `{expr}` to a literal during flattening. The expression is recoverable: a flattened instance card (`r.x1.r1 in out 1e3`) carries `card->linenum` back to its `subckt-DEFINITION` body line (`R1 a b {rval}`), a top-level template whose original text `numparam` retains in `dicoS->dynrefptr[linenum]` (two instances of a subckt share the `linenum`). So capture now walks the FLATTENED deck (`ci_deck`) and, per device card, recovers the value expression via the new `nupa_get_dynref(card->linenum)` while the card's own first token gives the full hierarchical instance name -- top-level and subckt-internal captured by one mechanism, no name reconstruction. (Empirically the key is `card->linenum`, NOT `card->linenum_orig`, which uses a different counter -- `inpcmc.c` `line_number` vs `line_number_inc` -- and is off by one.) SAFE across subckt scope, since a subckt can shadow/derive-from/pass-through a swept global: (1) a Pass-1 scan of the original deck INCLUDING subckt bodies disarms on a swept param in a subckt header default, a subckt-call (X) argument, a subckt-body `.param` that shadows (swept LHS inside a subckt) or derives from (swept RHS) the param, or any structural slot (node/name, `.if`, `.temp`, analysis card, `.option`, `.ic`, `.nodeset`); (2) an

arm-time SELF-CHECK re-evaluates each captured expression against the GLOBAL dico at nominal params and requires it to reproduce the value numparam already baked into the flattened card (read straight off the card, no ask-API), so any shadow/pass-through that slips past (1) disarms the whole path; (3) a disarm falls back to the exact reset path. Validated by a 10-deck battery -- 5 that must arm (top-level, direct subckt use, multi-instance, nested subckt, scaled expr) and 5 that must disarm (local .param shadow, formal-param shadow, formal pass-through, subckt-local derived, top-level derived) -- where EVERY deck's fast output equals reset bit-for-bit and every arm/disarm decision is as expected; the subckt fast path matches closed-form $R2/(rval+R2)$ to $9e-10$ and reset to $1.9e-9$. Speedup follows fan-out: a param feeding one device inside a subckt in a large fixed circuit $2.92s \rightarrow 0.24s$ (12.0x); a param feeding every device (400 subckt instances x 8 R) $2.50s \rightarrow 1.64s$ (1.5x). Scope: the sweep command (subckt-internal instance values fast, model values via altermod fallback); optimize/Monte-Carlo .param reset sites stay on the reset path. Verify (examples/paramfastsweep): top-level arm + subckt-internal arm + local-shadow fallback, all vs closed form (4 checks). || [E-322](#) | src/frontend/com_optimize.c, src/frontend/com_sweep.{c,h} | .param fast-sweep, optimizer tier -- the **optimize** command shares the fast-path engine. Every -dparam (symbolic .param) objective evaluation used to do alterparam + reset (re-parse + rebuild the whole circuit), and an optimization runs tens to hundreds of evaluations. E-322 exports the E-320/321 capture-classify-selfcheck-push engine (sw_fp_build / sw_fp_apply / sw_fp_free, de-static'd + declared in com_sweep.h) and wires it into the optimizer: opt_fp_arm() collects the OPT_DECKPARAM knob names and calls sw_fp_build ONCE before the search; opt_eval (Nelder-Mead / Levenberg-Marquardt / PSO / DE / SA) and opt_eval_objs (NSGA-II) push the candidate's deck-param values in place with sw_fp_apply instead of alterparam+reset; sw_fp_free at cleanup. The conservative classifier and arm-time self-check are unchanged, so a deck-param feeding a subckt shadow / structural slot / derived param disarms and the optimizer runs exactly as before. Two opt-outs: (1) -center design-centering (E-206) keeps reset -- its inner Monte-Carlo RE-SAMPLES process variation on each reset, a re-draw not a deterministic value push; (2) a device-count guard keeps the reset path on SMALL circuits -- the in-place apply has a fixed per-eval cost (numparam re-eval + dico ops) roughly independent of circuit size while a reset's cost grows with the deck, crossing at ~80 device instances, and because the in-place values differ from reset in the last few digits (numparam value-string formatting) an extremely tight -tol on a tiny circuit could otherwise send the two paths to different iteration counts; so opt_fp_arm counts flattened device instances and engages only at ≥ 80 . Correctness: across NM/PSO/DE the fast-path optimum is IDENTICAL to reset (122-device ladder fit: $r_{top}=361.914$ under NM and DE, 382.872 under PSO), and NSGA-II arms and returns the same Pareto front; all 41 existing optimize verify checks still pass. Speedup: a 4000-device fixed ladder, single -dparam knob, Nelder-Mead to convergence $0.94s \rightarrow 0.17s$ (5.6x); grows with circuit size, guarded to the sub-30ms reset on tiny circuits. Scope: optimize -dparam knobs, all methods except -center (-param/-mparam were already in-place). Verify (examples/optimize): a large-circuit -dparam optimization arms the fast path and still converges (2 new checks). || [E-323](#) | src/frontend/com_optimize.c, src/osdi/osdiinit.c, src/include/ngspice/osdiitf.h | .param fast-sweep, OSDI-aware optimizer guard. E-322 gated the optimizer's fast path behind a device-count guard (arm at ≥ 80 device instances) because on a small circuit a reset re-parses faster than the fast path's fixed per-eval overhead. That threshold was calibrated on RESISTORS and is wrong for OSDI (compiled Verilog-A) devices: a resistor reset just re-parses an element line, but an OSDI reset RE-INITIALIZES each compiled model instance (re-runs its setup/temperature callbacks), ~30x costlier per device. Measured (fast path on vs off, same binary, OSDI-resistor ladder): the fast path wins at 3 OSDI instances (2.1x), scaling to 3.2x at 161 -- where resistors needed ~80 to break even. So the E-322 count guard would keep a SMALL OSDI -dparam optimization (common in device model extraction/fitting) on the slower reset path, forgoing 1.5-3x (a 300-instance OSDI fit measured 3.0x fast vs reset, same optimum). Fix: the crossover is about reset COST not device COUNT, so opt_fp_arm now walks the built circuit's instances and WEIGHTS each by device kind -- primitive=1, OSDI=OPT_FP_OSDI_WEIGHT=30 (measured ~3 OSDI ~ the reset cost of ~80 primitives) -- and compares the weighted total to the same 80 primitive-equivalent threshold. A handful of OSDI devices arms the fast path; a small all-resistor circuit still keeps the cheap reset. OSDI types are identified by a stable marker: every OSDI SPICEdev is built by osdi_create_spicedev and shares the OSDIparam instance-parameter setter (no built-in device uses it), exposed via a one-line osdi_devtype_is_osdi(type) helper

in osdiinit.c; the optimizer call is `#ifdef OSDI`-guarded so a build without OSDI is unaffected (every instance weighs 1). Correctness is unchanged from E-322 -- the guard only decides WHETHER to arm, not what the fast path computes -- so a small OSDI -dparam optimization now arms and converges to the SAME optimum as the reset path. Verify (examples/optimize): a small OSDI -dparam optimization (compiled optres.va) arms the fast path and converges (2 new checks); all 43 optimize checks pass. | | [E-338](#) | ngspice-46/src/frontend/inpcom.c | A full 64-bit bus range HUNG ngspice and grew without bound -- 3.6 GB after 4 s, 7.6 GB after 9 s, from ONE netlist line, in the released binary. R1 a[-9223372036854775808:9223372036854775807] r=2k. Found by fuzzing the array/bus node feature. ROOT CAUSE -- THE GUARD COULD NOT SEE THE REAL WIDTH: E-221 expands `base[lo:hi]` into scalar node names behind a deliberate `BUS_MAX_WIDTH` (8192) ceiling, computed as `long width = (hi >= lo) ? (hi - lo + 1) : (lo - hi + 1)`. `strtol` saturates at `LONG_MIN/LONG_MAX`, so the full span makes `hi - lo + 1` OVERFLOW a signed long -- undefined behaviour, which in practice wrapped to a small value, so the guard saw a tiny width and let it through. The loop then stepped from `LONG_MIN` toward `LONG_MAX`, about $1.8e19$ iterations, appending to a string each time. The guard was not missing: it was computing its input with the very overflow it existed to prevent. ONLY THE FULL SPAN TRIGGERS IT -- the one-sided cases `a[-9223372036854775808:0]` and `a[0:9223372036854775807]` have width $\sim 9.2e18$, which fits and is correctly rejected. That is exactly why a corpus using only 32-BIT extremes (`a[-2147483648:2147483647]`, width 4.29e9) missed it: that width is representable, so the guard worked. FIX: compute the span in UNSIGNED arithmetic (exact for every pair of longs) and compare the SPAN rather than the width, so the `+ 1` cannot overflow either; `strtol` setting `ERANGE` now also rejects an endpoint too large to represent rather than letting it be silently clamped to `LONG_MIN/LONG_MAX`, values that then look like an ordinary acceptable range. A netlist is untrusted input, so this is a resource-exhaustion defect reachable from a single line with no diagnostic and no bound. FUZZ: 432 decks, 0 crashes/hangs/runaway -- 54 range tokens (32- and 64-bit numeric extremes, malformed brackets, missing parts, nesting, control characters, XSPICE `%vd[...]` groups, a 4000-character base name, 40-deep bracket nesting) crossed with 8 placements (device nodes, `subckt` call, `.subckt` port list, value position, model name, `print v()`, bare `print`, `let`). CORRECTNESS: 13/13 numeric checks -- ascending AND DESCENDING ranges map ports in the right order, verified through an ASYMMETRIC subcircuit (1k on one port, 10k on the other) so a reversed mapping changes the measured current instead of being invisible; `.subckt bus4 d[3:0]` expands descending and maps positionally across four distinct currents; R1 a[0:1] 2k is ONE resistor (series $2k+4k = 0.5$ mA); a bus element and an explicit scalar are the SAME node; `v(a[0])` and bare `a[0]` resolve correctly (E-224) while ordinary vector indexing still works; non-integer/malformed/already-scalar tokens are untouched; the guard boundary expands at 8192 and stays literal at 8193. Verify (examples/busoverflow): 5 checks. | | [E-339](#) | ngspice-46/src/frontend/parse.c | `v()` with THREE OR MORE node names SIGSEGV'd ngspice -- found by fuzzing the `pyplot` command family, but the defect is in the SHARED EXPRESSION PARSER. `print v(in,out,0)`, `let q = v(in,out,0)` and `pyplot v(in,out,0)` all died; `plot` SURVIVED, which is why an obvious spot-check passes and this was easy to miss. ROOT CAUSE -- TWO PATHS WITH DIFFERENT OWNERSHIP: `v()` takes at most two node names (`v(a)` or the differential `v(a,b)`), and a third was never rejected. `PP_mkfnnode`'s comma branch recurses, `PP_mkfnnode(MINUS, PP_mkfnnode(func, left), PP_mkfnnode(func, right))` then `free_pnode(arg)` -- that branch CONSUMES its argument, while the normal path at the end of the function BORROWS it (`p->pn_left = arg; pn_use++`). `free_pnode` recurses into children guarded by `pn_use`, so: with TWO names the children are plain nodes, each recursive call takes the BORROW path and bumps `pn_use` to 2, and the outer `free` merely decrements -- safe; with THREE OR MORE one child is ITSELF a comma node, so the recursive call takes the CONSUME path and frees it, and the outer `free_pnode(arg)` then walks into that already-freed child -- a DOUBLE FREE. Node existence is irrelevant: all-present `v(in,out,0)`, all-missing `v(a,b,c)` and mixed `v(in,out,zz)` crash alike -- it is purely the arity. FIX: reject the invalid arity where it is detected (Error: `v()` takes at most two node names.) rather than relying on the inconsistent ownership. Repairing the mismatch instead would mean changing `PP_mkfnnode`'s contract for every caller in order to make a syntactically invalid expression work; the arity is not legal in the first place, so rejecting is both smaller and more honest. The ownership mismatch is recorded as the underlying hazard. FUZZ: 122 `pyplot` invocations, 0 crashes/hangs/unexpected

writes after the fix -- the eight mode flags (-eye -hist -contour -smith -fft -bode -nyquist -polar) crossed against missing/extra/repeated/conflicting arguments, hostile signal lists (unclosed `v()`, empty `v()`, 8000-character names, control characters, 200 signals) and hostile output base names (3000 characters, spaces, quotes, shell metacharacters, unicode, leading -). NOT A DEFECT, flagged and dismissed: `pyplot ../name` and `pyplot /abs/path` write outside the working directory -- that is the documented `pyplot [file] plotargs` contract and matches `wrdata/write`, so the fuzz policy was wrong rather than the command. Verify (examples/vfuncarity): 5 checks. | | [E-341](#) | `ngspice-46/src/frontend/com_sweep.c` | **sweep -analysis reset** and **-analysis remcirc** SIGSEGV'd -- found by fuzzing `sweep/` optimize for RECURSION and STATE. `sweep` runs its `-analysis` argument as a command at every point, dispatched by name through `sw_run_cmd`, and nothing stopped a user naming `reset` or `remcirc` -- which free and rebuild (or remove) the very circuit the sweep is iterating over. The loop carried on with its resolved knob bindings, the old CKTcircuit and the old plot. *WHY THE FIX REJECTS UP FRONT RATHER THAN RECOVERING: the obvious repair -- notice the circuit changed and break out -- was tried FIRST AND MADE THINGS WORSE. The sweep's post-loop plot finalisation touches the same freed state, so breaking out merely moved the crash, AND it turned a previously WORKING case (-analysis 'optimize ...', which resets internally but re-establishes its own state) into a NEW crash. That attempt was reverted. Rejecting a destructive analysis before the loop starts cannot regress anything, and nothing legitimate is lost: a sweep re-sources the deck itself when a knob needs it, so a user reset in the per-point analysis has no purpose. optimize is deliberately NOT rejected -- it resets internally, recovers, and works today. FUZZ (E-270 already covered numeric bounds, so this EXTENDED THE STRATEGY rather than repeating seeds): ROUND 1, argument surface -- 684 invocations, CLEAN -- 20 knob kinds (instance param, @modell[param], .param, the E-268/269 wildcards @[p]/@[p]/@[p], malformed @-forms) x 29 sweep specifications (start/stop/step, lin|dec|oct, list, plus missing/reordered/non-numeric) x 31 option shapes, plus 51 optimize invocations across all seven methods (nm lm pso de sa nsga nsga2), all three knob kinds and every objective form at degenerate values. ROUND 2, recursion and state -- WHERE THE BUGS WERE -- 23 cases: sweep inside sweep (same knob, three deep), optimize driving a sweep and vice versa, repeated sweeps into one plot, -overlay accumulation, -warm with and without a prior run, fast-path and reset-path knobs alternating, and the analysis mutating the circuit under the loop. ALSO FOUND, CHARACTERISED, NOT FIXED: a sweep near SW_MAXPTS looks like a hang because point cost grows with point count -- 171 us/point at 1000, 761 at 5000, 2994 at 20000 (20x the points for 300x the time, roughly QUADRATIC). Profiling puts it in DCOp -> OUTpBeginPlot -> cp_getvar -> cp_usrvars -> cp_enqvar -> dup_string: cp_getvar materialises the ENTIRE user-variable list (a duplicated string per entry) and that list grows with the plots a long sweep accumulates, while OUTpBeginPlot calls it once per analysis for printinfo and interp. Making the construction LAZY was tried and MEASURED TO GIVE NO IMPROVEMENT (those variables are normally unset, so the search falls through to the materialised list anyway) and was reverted rather than shipped unmeasured; a real fix means making cp_usrvars() cheap or caching it with proper invalidation -- a general ngspice change outside a crash fix. Recorded with its measurements so it need not be rediscovered. Verify (examples/sweepanalysis): 5 checks. | | [E-342](#) | `ngspice-46/src/frontend/options.c`, `variable.c`, `variable.h` | Two ownership bugs in the synthetic user-variable list: a heap-use-after-free (SIGSEGV) and a double free (SIGABRT). cp_usrvars() synthesizes \$plots, \$scurplot, \$scurplottitle, \$scurplotname and \$scurplotdate and returns a list its CALLERS free. That only works if the list owns every node. Two places assumed it did. [A] cp_enqvar() does NOT always allocate -- it clears its tbfreed out-parameter and returns a BORROWED pointer into plot_cur->pl_env when that live list already defines the requested name. cp_usrvars() declared tbfreed, passed it to all five calls and NEVER READ IT, so it relinked the borrowed node (tv->va_next = v, orphaning the live list's tail) and the caller then freed a node somebody else still owned. Not hypothetical: a rawfile Option: line writes pl_env, so a loaded rawfile containing Option: plots = 1 was enough -- SIGSEGV for all five names, ASan confirming allocate/free/use across three different functions (raw_read -> cp_setparse, cp_getvar -> free_struct_variable, cp_usrvars -> cp_enqvar). Fixed by honouring the flag and copying a borrowed variable; the five near-identical blocks became one loop so the rule cannot be applied to four of five sites. New var_copy() is written as the deliberate DUAL of free_struct_variable() -- deep, because that function recurses into va_vlist and a shallow copy would only move the bug. [B] found while fuzzing the fix for*

[A] and INDEPENDENT of it, no file involved: `cp_remvar()` freed the variable it looked up even when nothing had unlinked it. Only the US_OK arm unlinks; `plots` returns US_READONLY and the `curplot` names return US_DONTRECORD, leaving the node inside `uv1` to be freed a second time by the trailing `free_struct_variable(uv1)`. `unset plots` aborted in `malloc` on a plain deck. Fixed by tracking whether the node is genuinely ours to free -- which also closes the same latent corruption for variables found in `variables`, `pl_env` and `ci_vars`. The `ci_vars` borrowed path is NOT reachable for these names today (`plot_cur` is always set, so all five return earlier) -- measured, not assumed -- but the fix keys on the ownership flag rather than the source list, so it covers that path anyway. Fuzz: 640 combinations (5 names x 8 Option: value shapes x 16 follow-up sequences), each run twice, plain and under ASan, all clean. No leak from the copy: 20000 iterations through `cp_usrvars()`, peak RSS 7.1 MB vs 7.2 MB control (LeakSanitizer unavailable on this host, so measured rather than asserted). Behaviour preserved and DIFFED against the old binary, not eyeballed. Found by tracing the E-341 performance finding through `cp_enqvar()` and asking what happens when `tbfreed` comes back 0. Verify (examples/usrvarown): 6 checks. || [E-343](#) | `ngspice-46/src/frontend/options.c`, `variable.c`, `include/ngspice/cpextern.h` | The $O(N^2)$ sweep cost E-341 documented but did not fix: FIXED, 26.6x at 16000 points (39.55 s -> 1.49 s). `cp_getvar(name,...)` looks up ONE name, but its first statement built ALL FIVE synthetic user variables (`$plots`, `$curplot`, `$curplottitle`, `$curplotname`, `$curplotdate`), searched the result and freed it again. Synthesizing `$plots` walks the live plot list and copies a string per plot, so every call cost $O(\text{number of plots in the session})$ regardless of what was asked for; `beginPlot()` does two such lookups per analysis ('`printinfo`', '`interp`') and a sweep creates a plot per point, so point `k` paid $O(k)$. WHY THE FIRST ATTEMPT FAILED (the whole lesson): E-341 tried making the construction LAZY -- build the list only after the variables search misses -- and measured 1.0x. `printinfo` and `interp` are normally UNSET, so that search always misses and falls through to needing the list anyway; laziness only helps the lookups that were never the problem. The gate has to be on WHICH NAME IS REQUESTED, not on when the list is built. Neither hot lookup is one of the five, so the fix builds NOTHING at all. `cp_usrvars()` is kept for `cp_vprint()`, which genuinely prints all five. Safe because both gated callers already search `pl_env` and `ci_vars` separately, which is the only other thing `cp_enqvar()` could return; ownership still follows E-342. Measured: 7.3x / 13.0x / 18.9x / 24.1x / 26.6x at 1k / 2k / 4k / 8k / 16k points -- the speedup GROWS with `N`, which is what removing a quadratic looks like. The regression suite itself, same 274 examples, went 748 s -> 479 s, because every analysis in every example was paying two of these lookups. NOT LINEAR YET, and the remainder is characterised rather than hidden: profiling the NEW binary at 64000 points puts 89% of what is left in `plot_alloc()`, a different function, which scans the whole plot list with a case-insensitive compare to pick a unique name. Two fixes were considered and REJECTED HERE: an exact-membership index (correct, but `plot_list` is mutated in ~9 places and a missed insertion hands out a DUPLICATE plot name), and a cheaper superset index (contained, but CHANGES BEHAVIOUR -- today plot names are reused after `destroy all`, and a superset would skip past them). Both are changes to plot lifetime, not to this lookup path. Semantics diffed byte-for-byte against the old binary, including a deck with `printinfo` and `interp` explicitly SET across `op/tran/ac`. E-342's two vectors re-checked. ASan clean. Verify (examples/sweepscale): 5 checks. || [E-344](#) | `ngspice-46/src/frontend/com_sweep.c` | **.model** params now take the fast sweep's DIRECT SET -- the one tier E-320 left on the textual path. Model params now cost exactly what instance params cost (they were 2.1x-2.8x slower). The E-320..323 fast **.param** sweep resolves each swept device value to a slot ONCE and pushes it with `ft_sim->setInstanceParm`, skipping the per-point reset -- but `sw_fp_resolve()` bailed on `if (!ckt || b->mod)`, so every model bind ran `altermod <model> <param> = <value>` through `sw_run_cmd` on EVERY point: `snprintf` + `cp_lexer` + `com_alter_common` argument re-parse + `finddev()` resolving the model BY NAME + a string walk of the model parameter table, all to reach the one `setModelParm` that mattered. None of that can change between points. Fix: `sw_fp_resolve()` resolves model binds to (`GENmodel`, `type`, `param-id`) at arm time and `sw_fp_apply()` pushes them with `ft_sim->setModelParm`. The parameter-table search instance binds already used was factored into `sw_fp_find_parm()` and SHARED, so both tiers apply identical rules (IF_SET, IF_REAL, keyword match, or the positional principal -- which only instances have, a **.model** line has no principal slot). Anything unresolvable (model not built, or a non-real param such as an integer `level`) keeps `rok=0` and its textual command: the fallback is the correctness backstop, NOT dead code, and an example check exercises it deliberately. The banner now reports fallbacks -- ..., `N` via `alter/altermod` -- appended ONLY when something fell back, so

ordinary output is unchanged and a slower textual push no longer hides behind the same banner as a direct one. THIS IS A SPEED CHANGE, NOT A CORRECTNESS FIX, and that shaped the verification: BEFORE writing any code the textual path was checked against reset on a semiconductor resistor and on a compiled OSDI model and matched EXACTLY, so 'identical to reset' became the whole acceptance criterion. Measured (same binary): 200 models x 200 points 0.095 s -> 0.034 s (2.8x); 400 R models x 400 pts 0.36 -> 0.12 s (3.1x, and fast-vs-reset 2.4x -> 7.5x); 60 OSDI models x 400 pts 0.07 -> 0.03 s (2.2x, fast-vs-reset 1.7x -> 4.0x). The real result is the model/instance cost ratio landing on exactly 1.00x -- no residual per-point resolution left. Correctness: a 12-case battery run twice (fast and reset) and compared -- resistor rsh/tc1, capacitor cj, diode is, MOSFET vto, BJT bf, OSDI model r, two models sharing one param, model+instance mixed, derived-.param (disarms as before), non-real integer param (falls back) -- ALL exact; plus subckt-internal models (arm + exact) and a subckt that locally shadows the swept name (correctly disarms). optimize shares the engine (E-322) so -mparam inherits the direct set; optimize_examples 43/43 unchanged. Verify (examples/modelparamset): 7 checks, closed form to 1e-10 (deck raises numdgt so the comparison is not capped by the default 7-digit print). Monte Carlo remains the only tier not on this path -- it needs a re-DRAW of process variation, a different mechanism. || [E-345](#) | ngspice-46/src/frontend/vectors.c, com_fft.c, linear.c, postcoms.c, spec.c, mw_coms.c | The residual O(N) E-343 profiled but rejected fixing: FIXED. The sweep is now LINEAR -- 87x at 64000 points (54.97 s -> 0.632 s). plot_alloc() and plot_add() pick a plot name by counting a shared, monotone plot_num up until <abbrev><plot_num> is not the typename of any plot in plot_list. Plots are PREPENDED, so the colliding probe hits at the head in O(1); it is the SUCCESSFUL probe -- proving a name ABSENT -- that walks the whole list, with a tolower per character. One full walk per plot created, and a sweep creates a plot per point. Only the MEMBERSHIP TEST changed: a hash index of the typenames currently in plot_list answers it in O(1). The search still starts at the same shared plot_num and still counts up by one, which matters more than it looks -- plot_num is shared ACROSS abbreviations and only advances on a collision, so the sequence is op1 tran1 op2 ac2 op3 not op1 op2 op3; and a number freed by destroy all is REUSED, which is exactly why E-343 rejected a 'remember every name ever issued' superset cache. Both copies of the loop (plot_alloc and plot_add carried the same one) now share plot_unique_typename(). E-343's OTHER objection -- an exact index is unsafe because plot_list is mutated in ~9 places and a missed insertion hands out a DUPLICATE plot name -- was answered by REMOVING the nine places rather than tracking them: plot_new() is now the SINGLE insertion point (six callers open-coded the same two lines: com_fft x2, linear x2, postcoms, spec), killplot() calls the new plot_forget(), com_removecirc() (which rewrites plot_list wholesale) calls plot_forget_all(), and the index builds LAZILY from plot_list so the static const plot needs no registration. PROVEN, not asserted: plot_name_taken() carries a PLOTNAME_SELFHECK block that answers the same question the OLD way and abort()s on disagreement; the full suite was run with that build -- 276/276 OK, ZERO disagreements. The block stays in the source, unset, so it can be re-run against future changes. A REAL BUG was hit on the way and is worth recording: converting the open-coded prepends with a REGEX produced a DOUBLE INSERTION, because three sites ALREADY called plot_new() with the redundant two lines wrapped AROUND it, so the edit turned the trailing line into a second plot_new(). The second call ran new->pl_next = plot_list when plot_list was already new -- a SELF-LOOP -- and vec_gc spun on the circular list forever. Found by a stress case (fft) that hung, located by SAMPLING the hung process, not by guessing. A mechanical bulk edit needs each site read: the pattern being replaced was not the pattern actually there. Names verified IDENTICAL to the previous binary across 15 stress cases (mixed analyses, destroy all, destroy first/middle/last, setplot new, fft, linearize x2, spec, write/load x2, load-then-destroy, twelve-then-destroy, remcirc, sweep-then-destroy, nested destroy). Measured: per-point cost pinned at 10 us and the doubling ratio converged on exactly 2.0x -- the first time this sweep has been linear. With E-343, a 16000-point sweep went 39.55 s -> 0.163 s, about 240x. Verify (examples/plotname): 7 checks. || [E-346](#) | ngspice-46/src/frontend/com_sweep.c | A BUG FIRST, then the last tier: the fast .param path was silently FREEZING random draws that the reset path re-drew -- and fixing that IS the Monte Carlo tier. The fast sweep only re-evaluates brace expressions that MENTION the swept name. But numparam INLINES a .param's expression into the device line during preprocessing, so .param rv = agauss(...) + R2 a b {rv} arrives as r2 a b {(agauss(...))} -- carrying the random call but NOT the swept name, so it was skipped

and never touched again. (Same discovery: `nupa_recompute_params`, added in E-320 to refresh the derived-param closure, replays ZERO lines on such a deck -- there is no `.param` line left.) Measured over 5 points: reset gave $rv = 1097.7/1015.2/952.5/988.7/969.2$, fast gave 1161.1 FIVE TIMES. E-320's guarantee that results never change, only speed, had not held for any deck with a random `.param` since it shipped. Fix: capture brace expressions that draw from the RNG (`agauss/gauss/unif/aunif/limit/mvnorm`) even with no swept name present. Once that exists MONTE CARLO NEEDS NOTHING MORE -- a sample is a re-draw plus an in-place push, and `montecarlo` arms the same machinery with NO swept knob (`sw_fp_build(NULL,0)`). Bit-identical to reset required THREE things, each found by a test that failed first: (1) DECK ORDER -- random binds sort ahead of the expression-grouped ones by capture index, since only they consume the RNG, so the stream matches however the deterministic binds are grouped; (2) NO CACHING -- a random bind is never served from the by-expression-text cache, because two devices with identical text must draw INDEPENDENTLY (verified: two `{agauss(1000,300,3)}` resistors give 1113.1 and 1099.5, not equal); (3) the SAMPLE BOUNDARY -- a deck re-copy signals one MC sample via `nupa_signal(NUPADECKCOPY) -> mc_sample_advance()`, which rewinds the per-sample dimension counter and steps the Latin-Hypercube sampler; skipping the re-source skipped that signal, which is exactly why plain draws matched reset (240/400 both) while `-lhs` did NOT (239 vs 258) and `-lhs -warm` did not (252 vs 258). Raising the boundary in the apply path made all four agree. The E-321 arm-time self-check is SKIPPED for random binds -- re-evaluating one draws a NEW value so it could never reproduce the baked one, and the draw would perturb the very stream that must match; their safety rests on the structural disarms, which were extended so a random gets the same scrutiny a swept name gets (a random reaching `.temp`, a `subckt` call or any structural slot disarms). Verified against an INDEPENDENT ORACLE, not a golden number: the same experiment handwritten as a reset loop in the control language -- `montecarlo 124/200, oracle 124`. Battery of 12 cases run on BOTH paths with discriminating yields (60.3/57.7/27.7/84.0/89.7/90.8%): plain, `-lhs`, `-warm`, `gauss`, `aunif`, `limit`, two randoms, same-expression-twice, random model param, two specs, derived random (disarms), structural random (disarms) -- all identical. Speed 1.8x/2.3x/2.4x at 20/200/1000 devices -- smaller than the sweep tiers because an MC sample re-draws and re-pushes EVERY random value; the saving is the re-source, not the pushes. CHANGES EXISTING RESULTS: a sweep over a deck with a random `.param` now re-draws it per point instead of freezing it -- which is what makes the fast path agree with reset again. Regression 278/278. Verify (`examples/mcfastpath`): 7 checks. || [E-348](#) | `ngspice-46/src/spicelib/parser/inp2dot.c`, `src/spicelib/analysis/dcpss.c` | `.pss` segfaulted on a short argument list -- and, separately, on a COMPLETE one whose **harmonics** was 0. Found by a new sweep shape: take a KNOWN-GOOD invocation of each command and run every PROPER PREFIX of its argument list. Existing campaigns could not see this -- `cmdfuzz` mutates argument VALUES (every value it supplies is present) and the parser fuzzers feed malformed FILES; a command that reads its fifth argument after checking only that a first exists is invisible to both. Across 38 commands `pss` was the ONLY offender: 0 args hung, 1-4 args SIGSEGV; `qpss` (8 args) and `optimize` (12) truncate cleanly. The stripped binary gave only the shape -- `ldr d0, [x19]` with `x19 == 0`, then `fdiv`, then a store -- and the UBSan build named it: `dcpss.c:4990, load of null pointer of type 'double', i.e. Mag[0] = Phase[0] / (double)ndata`. THREE failures stacked. (1) `dot_pss()` reads seven required values with no end-of-line check, and `INPgetValue()` cannot report 'nothing left to read' -- it returns 0 -- so a short card reached the analysis with points/harmonics silently 0. (2) `DCpss()` never validated: `CKTharms` is the LENGTH of every array the DFT writes into, and `TMALLOC(double, 0)` is `tmalloc(0)`, which returns NULL. (3) `DFT()` writes index 0 OUTSIDE any loop -- `Mag[0]`, `Phase[0]`, `nMag[0]`, `nPhase[0]`, `Freq[0]` -- so its `numFreq` bound never protected them. That third point is why a parser guard ALONE would have been insufficient: `pss 1meg 1u out 1024 0 50 5u` is well-formed at full arity and crashed by the same route, a second entrance that never passes the truncation check. Fixed at all three layers: a `PSS_NEED_ARG` macro naming the missing argument (the idiom `dot_dc()` already uses a few hundred lines earlier in the same file, which `dot_pss` simply never adopted); `DCpss()` rejecting `fguess <= 0`, `points < 1` and `harms < 1` before anything is sized from them; and `DFT()` guarding its own unconditional index-0 stores. WHAT WAS DELIBERATELY NOT TIGHTENED, because it was measured rather than assumed: every parameter was tested at 0 and negative, and `sc_iter` and `steady_coeff` are harmless at both (`rc=0`), so they are still accepted -- narrowing the accepted input on no evidence could break a working deck.

NO VALID ANSWER MOVED: 22 correct invocations (op/dc/ac/tran/tf/pz/sens/noise/disto/four/pss plain and uic/pac/pnoise/hb/sp) were run on the shipped and rebuilt binaries and every printed number compared EXACTLY -- all identical, the one apparent difference being Total analysis time, i.e. the clock. Truncation sweep 0 crashes from 5. Verify (examples/pssargs): 7 checks. || [E-349](#) | ngspice-46/src/spicelib/parser/inp2dot.c | One transposed letter in a node name killed the process -- **op** then **tf v(ouy) v1** and the session is gone, with every loaded circuit, plot and let-defined vector. Seven commands did it: tf, pz, noise, sens, pss, pxf, psp. Found by a STATEFUL CROSS-COMMAND fuzz -- random sequences of state-mutating commands followed by a canary circuit whose answer is known exactly (1 V across a 1k/3k divider, v(b) must be 0.75). The harness was first aimed at the pre-E-342 binary, where it found 38 aborts in 200 sequences, so a clean run means the detector works rather than that it is asleep. ROOT CAUSE: INPtermInsert() is CREATE-OR-FIND. That is right for a device card, where the device DEFINES its nodes (R1 in out 1k is what brings out into existence), and wrong for an analysis card, which only REFERS to one. A mistyped name was quietly created as a new unconnected node. The milder consequence is arguably the worse bug -- before setup nothing crashes and the invented node reads back as a perfectly plausible v(bogus) = 0.000000e+00, no error, no warning. The fatal one: from the .control section CKTsetup() has already snapshotted the node list into prev_CKTlastNode and sized the matrix, so the extra node makes the tail check ending CKTunsetup() fail -- and that check calls controlled_exit(EXIT_FAILURE). The check is CORRECT and was firing on a real inconsistency; the defect is upstream of it. This never bit in a plain netlist purely because of ORDERING: a .tf card in the deck is parsed BEFORE CKTsetup() runs, so the invented node is inside the snapshot and the tail still matches -- it is reachable only interactively, which is exactly where a typo is most likely and losing the session hurts most. Fix: analysis cards resolve through INPtermSearch(), the lookup-only half of the pair that already existed alongside INPtermInsert() and was simply never used here; an unknown name is reported (no such node: ouy) instead of created. DECK ORDER STILL WORKS, and this is the constraint that shaped the fix rather than an afterthought -- a .tf card may legally sit ahead of the devices that define its nodes, so creation remains permitted while the circuit is NOT yet set up, and is refused only once CKTisSetup is true, i.e. deck parsing is over. Applied at 14 call sites across dot_noise, dot_tf, dot_sens, dot_pss, dot_pac, dot_psp, dot_pnoise, dot_pxf and dot_hb; dot_pz reached the same insert INDIRECTLY through INPgetValue(IF_NODE) and its four node arguments are now read explicitly so they take the same path. Verified: 0 of 14 commands kill the process, from 7; the same reference as a deck card still parses; a .tf card placed BEFORE its devices still returns transfer_function 0.75, output_impedance 750, input_impedance 4000; 22 valid invocations compared digit-for-digit against the shipped binary, all identical; and the stateful fuzz that found it went 58/60 to 60/60. Verify (examples/nodetypo): 6 checks. || [E-350](#) | ngspice-46/src/frontend/com_sweep.c | A sweep never put its .param back, and after two sweeps reset could not either. .param rload = 3k, sweep it 1k..5k, sweep it again 2k..10k, then reset + op gave 0.909 -- rload still 10k while the deck said 3k. One sweep was fine; the second broke it. ROOT CAUSE: the two implementations disagreed about the state they LEAVE BEHIND, which is exactly what E-320 guarantees they never do ("a sweep can only get faster, never change its result") -- it held for the numbers, not for the aftermath. The reset path drives each point with alterparam, which edits the parameter PERMANENTLY, so a later reset re-sources a deck that now carries the last swept value; the fast path never touches the deck, so reset did restore. Same command, same netlist, different circuit afterwards, decided only by which path happened to arm. WHY THE SECOND SWEEP: arming self-checks each captured expression against the value numparam baked into the flattened card at nominal (a guard against a subckt-local shadow), but the first sweep left the dico holding its last swept value while the flattened card still carried the nominal one -- so the check could not pass for an honest reason and DISARMED. The second sweep of a parameter therefore fell silently back to the reset path: no error, no banner, the fast .param path armed line simply stopped appearing -- and the reset path is the one that edits the deck permanently, which is why the damage needed exactly two sweeps to surface. Both symptoms, one root cause: the sweep never restored the parameter. FIX: capture each swept .param before the first point and restore it at cleanup -- deck text via sw_set_deck (what reset re-sources) AND the numparam dico via nupa_add_param + nupa_recompute_params (what the NEXT sweep reads and what arming self-checks against). BOTH HALVES ARE NEEDED and the first attempt shipped only the deck half: with the dico

still dirty the second sweep read the first sweep's last value as its own nominal and dutifully restored THAT, giving 0.833 instead of 0.909 -- better, and still wrong. The capture is all-or-nothing (if any swept name cannot be read back nothing is restored) rather than putting some parameters back and not others. Restoring is the direction that makes the two paths agree; making the fast path permanent instead would have satisfied the letter of the E-320 invariant while leaving reset unable to undo a sweep and every repeat sweep disarmed, and it matches how ngspice already treats a swept source in .dc, which does not leave it parked at the last step. DELIBERATELY UNCHANGED: the live circuit still holds the last swept point, so a bare op afterwards answers with that value -- true of both paths before this change and unchanged by it; only the deck and dico return to nominal, so reset means something again. Verified: reset gives 0.75 after 1/2/3/mixed-spec sweeps (was 0.909 from the second on); 3 of 3 sweeps arm the fast path (was 1 of 3); derived params and subckt-internal values restore exactly; and 12 sweep batteries compared digit-for-digit against the shipped binary -- plain/list/dec, repeated sweeps, derived, subckt-internal, model params, Monte Carlo draws twice over to catch RNG-stream drift, and a two-knob family, 5-11 numbers per case, none vacuous -- ALL IDENTICAL, so nothing a sweep computes moved. Regression 282/282. The example is a proven trigger, not decoration: on the pre-fix binary 4 of its 5 checks fail, and the one that passes is precisely the invariant that must hold on both. Verify (examples/sweepstore): 5 checks. | | [E-351](#) | ngspice-46/src/osdi/osdisetup.c, ngspice-46/src/osdi/osdidefs.h | **sens** killed the process on any OSDI model carrying an INTERNAL NODE -- which is every production compact model. BSIM, HICUM, PSP and EKV all allocate internal nodes for their terminal resistances, so .sens was effectively unusable with real Verilog-A models, and it failed by taking the session down: "Internal Error: node allocation in DEVsetup() during sensitivity analysis, this will cause serious troubles !, please report this issue !" then exit(1). FOUND BY a campaign built on the observation that the previous OSDI sweep (83/83 clean) was LOPSIDED -- 40 dc and 22 ac checks against only 2 sens, 2 disto and 4 pz. Those thin three are exactly where an OSDI device is most likely to differ from a built-in, because they do not merely solve the circuit, they consume the DERIVATIVES the model returns. Three axes were run (OSDI vs equivalent built-in on both solvers; Sparse vs KLU on the same OSDI deck; repeat-invariance): 181/184, and all three failures were this one bug, on the single model in the set with an internal node. ROOT CAUSE: cktsens.c re-invokes every model's DEVsetup() to stamp the perturbation matrix and requires that doing so allocate NO nodes -- it snapshots CKTlastNode around the call and calls controlled_exit(EXIT_FAILURE) if it moved. The requirement is reasonable and every built-in device meets it by guarding allocation on "not already allocated": the inductor does if (here->INDbrEq == 0) { CKTmkCur(...); here->INDbrEq = tmp->number; } and INDunsetup() zeroes INDbrEq again. OSDI had NO such guard -- its setup called CKTmkVolt/CKTmkCur for each internal node unconditionally, so the second call, the one sens makes on a circuit that is still set up, allocated a fresh set and tripped the check. The check was CORRECT and firing on a real inconsistency; the defect is upstream of it. FIX: the same idiom the built-ins use. Each instance records in OsdIExtraInstData the node numbers its internal nodes were given, and a later setup reuses them instead of allocating. OSDIunsetup() deletes those nodes so it also clears the record -- the OSDI counterpart of INDunsetup() zeroing INDbrEq -- because otherwise a genuine re-setup after teardown would hand out numbers for nodes that no longer exist. Reuse is conditional on the node COUNT matching, since collapsing depends on parameters that alter can move; if it changes, the record is dropped and nodes are allocated afresh. NOT MERELY NON-FATAL -- CORRECT, checked two independent ways because "it stopped crashing" proves nothing: an internal-node model (r=3000 + rs=10) and a plain one (r=3010) are the SAME circuit (operating points agree to 14 digits), so their sensitivity to the shared built-in R1 must be identical -- and is, at -1.8718770344518e-04 from both decks, every printed digit, against a closed form $-RL/(R1+RL)^2 = -1.87187890622571e-04$ (the 1e-6 residue is the sensitivity method's own perturbation step, not the fix). The model's own parameter agrees too: n1_r 6.21886087563079e-05 vs closed form 6.21886679809205e-05. Verified: campaign 184/184 (from 181/184); sens completes on both solvers; sens twice in one session gives 41 bit-identical values; a reset -> sens -> reset cycle returns the operating point to 7.50623441396507e-01, guarding that the node record never outlives the nodes; all 7 other analyses on the same model unaffected. Regression 283/283. The example is a proven trigger: on the pre-fix binary 5 of its 7 checks fail, and the two that pass are precisely the ones that should -- the model WITHOUT an internal node, and the analyses

other than sens. Verify (examples/osdisens): 7 checks. || [E-352](#) | ngspice-46/src/osdi/osdidisto.c (new), ngspice-46/src/osdi/osdi.h, osdiinit.c, osdiregistry.c, osdiext.h, ngspice-46/src/include/ngspice/osdiitf.h, ngspice-46/src/spicelib/analysis/cktdisto.c | SUPERSEDED BY E-359 (the capability stands and is still verified by the same oracles; the compiler-emitted-tensor IMPLEMENTATION below was replaced by numerical differentiation of the analytic jacobian in ngspice, removing the compile-time regression, the 30MB objects, the runtime penalty and the ABI bump). **.disto** now includes a Verilog-A model's nonlinearities; previously it printed a warning and left them out. Distortion is a VOLTERRA analysis: it needs each device's Taylor expansion of $I(v)$ to THIRD order, not the operating-point linearisation the Jacobian provides. Built-in devices hand-code those coefficients (diodset.c computes id_{x2}/id_{x3}), which is why only FOUR of ~58 built-ins implement DEVdisto at all -- diode, BJT, BSIM1, MES -- and the OSDI ABI stopped at first derivatives, so cktdisto had nothing to ask an OSDI device for. KEY ENABLER, verified before anything was built: OpenVAF's autodiff is ALREADY arbitrary-order (mir_autodiff's DerivativeIntern chains through previous_order, with a comment anticipating 8th order), and nested ddx(ddx(ddx(...))) reproduces closed form to every printed digit. So the compiler work was not implementing higher-order AD but ASKING for it: build_taylor_tensors re-runs autodiff over its own first-order results for the second, and again for the third. NO 3-VARIABLE CEILING: ngspice's framework caps at three controlling variables (Dderivs holds derivatives "w.r.t 3 variables"; the DFx helpers take up to 27 scalar arguments with no fourth-variable form, DF_n2F12 taking a struct because the list outgrew the calling convention). That cap belongs to those hard-coded helpers, not to the mathematics -- the contribution is a symmetric tensor contraction -- so osdidisto.c contracts directly over however many inputs a device has and never calls D1x/DFx. Proven with an ideal mixer $I(out)=kV(a,ref)V(b,ref)$ whose ONLY nonlinearity is the mixed partial (both diagonal 2nd derivatives identically zero, so a dropped cross term gives exactly zero): matches closed form $0.5kA^2R$ EXACTLY. TWO SILENT-ZERO BUGS found on the way, both indistinguishable from a linear model: (1) the tensors were OPTIMISED AWAY -- ensure_optbarriers protects residuals/noise/jacobian but the Taylor values feed nothing else in the MIR, so the optimiser deleted them and every coefficient arrived as zero (they also take the mfactor scaling, since m parallel devices inject m times the distortion current); (2) NOTHING STORED THEM -- eval's jacobian/residual stores are gated on CALC* flags that .disto never sets. Plus one wrong-by-a-constant bug only a reference implementation could catch: D1nF12 returns $0.5S2vF12(\dots)$ and S2vF12 ALREADY sums both index orderings, so without that 0.5 the mixed-tone products came out exactly 2x -- and because IM3 consumes the f1-f2 kernel its error was a non-obvious 2.246x. KNOWN LIMITATION AT THE TIME, SINCE LIFTED BY E-353: a model whose contribution goes through \$limit emitted NO tensors (the residual depends on the limited value, not the raw voltage read; build_jacobian handled that via intern.lim_state and the tensor pass did not). Limiting is standard in production diode/BJT/MOS models, so this was a real restriction; E-353 folds the limited values into the derivative chain and removes it. What remains unreachable is a nonlinearity in a GROUND-REFERENCED probe (the tensors are indexed by model input and a bare $V(a)$ is not one, having no hi/lo pair). Registering DEVdisto removed cktdisto.c's blanket warning, which would have turned such a model into a SILENT ZERO -- worse than before -- so OSDIdisto warns for itself. VERIFIED against four oracles, two involving no simulator at all: HD2/HD3 vs a closed-form polynomial (0.0 and $1.2e-16$); the A^2/A^3 scaling laws (4.0000, 8.0000); f1+f2, f1-f2 and IM3 vs the built-in diode ($1.87e-06$, the bound the 0.24 ppm \$vt difference justifies); IM3 vs transient+FFT in a DIFFERENT DOMAIN, converging 5.7% -> 1.4% -> 0.21% as the drive halves, exactly the A^2 signature of the 5th-order content a 3rd-order truncation excludes; and the multi-variable cross term (0.0). TWO HARNESS TRAPS recorded because both would have produced a confident wrong answer: the first IM3 transient was measuring SPECTRAL LEAKAGE from the fundamentals rather than IM3 (exposed by the A^3 law giving 2.93 instead of 8, not by the value looking wrong); and a first parameter choice made HD3 IDENTICALLY ZERO (the two contributions cancel exactly when $c3 = 2c2^2/Y_{tot}$, and $2(1e-4)^2/2e-3$ is precisely the $1e-5$ chosen) -- a zero that would have passed a naive check for the wrong reason, the same trap as the old campaign's "HD2 ~ 0 on a LINEAR network". The suite now refuses to score a both-zero comparison. Regression 284/284. The example is a proven trigger: on the pre-fix binaries 5 of its 6 checks fail, and the one that passes is the one that should. Verify (examples/osdidisto): 6 checks. || [E-353](#) | ngspice-46/src/osdi/osdidisto.c, examples/limitdisto_examples/, examples/osdidisto_examples/ | SUPERSEDED BY E-359 (\$limit models still work -- the

suite below passes unchanged -- but the chain-fold no longer exists; differencing the real jacobian makes the problem unable to arise). **.disto** now works for Verilog-A models that use \$limit -- i.e. every production compact model. E-352 gave OSDI devices distortion analysis, but only for models reading their controlling voltages DIRECTLY; a model passing one through \$limit contributed NOTHING, and limiting is how every production diode/BJT/MOS model converges, so the feature did not reach the models people actually run. ROOT CAUSE: \$limit hands the model a LIMITED copy of a branch voltage and the residual is written in terms of that copy, so differentiating the residual by the RAW voltage read yields zero -- nothing in the expression depends on it. The model is not linear, but every derivative the tensor pass asked for was. This was never a problem for AC or transient: build_jacobian has always folded the limited values back in via intern.lim_state; E-352's build_taylor_tensors simply did not perform the same fold. FIX: each model input now maps to a CHAIN of autodiff unknowns rather than a single one -- new taylor_unknown_chain returns the limited values from intern.lim_state.raw (each with the sign it contributes) followed by the input's own unknown. The 2nd-order pass sums over every PAIR drawn from the two chains and the 3rd-order pass over every TRIPLE, each term carrying the XOR of its entries' signs; an entry is negated when the model limits a REVERSED branch, \$limit(V(ref,a),...). VERIFICATION uses an exact expectation with no appeal to another simulator: \$limit is a convergence aid, so at convergence the limited value equals the actual one and a model written with it is THE SAME DEVICE as the same model written without it -- every distortion product must agree. Both spellings are generated from ONE body string, which is what guarantees they differ only by \$limit and cannot drift apart. Five shapes cover the chain combinatorics (where this can go wrong -- a model limiting a single branch never puts more than one entry on either side of a pair): two independently limited diagonals; a cross term with BOTH inputs limited (2x2 pairs); a cross term with ONE input limited (asymmetric chains, 2x1); a THIRD-order cross term whose chain has 3 entries, so the triple loop genuinely runs; and a reversed-branch limit, the only spelling producing a NEGATED chain entry. All 13 live comparisons agree, worst 7.1e-10 -- convergence noise, since the two spellings settle 8.3e-11 apart (limiting changes the Newton path, not the solution) and d2/dv2 of exp(v/vt) amplifies that by 1/vt^2. THREE WAYS THIS COULD HAVE PASSED WHILE WRONG, each caught and closed because each produces a confident green: (1) BOTH-ZERO comparisons -- 'no distortion' is precisely the pre-fix behaviour, so scoring 0 == 0 as agreement would certify the bug; the suite scores a both-zero product as a FAILURE except where the mathematics requires zero (a bilinear term kv1v2 has no third derivative, so its IM3 is correctly zero and is asserted as such). (2) COMPARING MAGNITUDES -- a dropped sign on a negated chain entry flips the result and leaves |.| untouched, so the reversed-branch shape, the ONLY one testing the sign logic, would have scored its own target bug as a pass; products are compared as COMPLEX values and the shape is pinned against the forward one (+1.2488e-01 vs -1.2488e-01, exact negatives). (3) DEGENERATE DECKS -- three shapes initially read zero everywhere for reasons unrelated to \$limit: the probed node was pinned by an ideal voltage source so no injected distortion current could move it, and each nonlinearity saw only ONE tone and so had nothing to intermodulate; fixed with series resistors and both tones on both drives. A FOURTH trap was ruled out rather than fixed: several shapes agree at exactly 0.00e+00, which would be the signature of OpenVAF having elided the \$limit altogether (identical code compared against itself); dumping the chains under OPENVAF_DAE_DEBUG settles it -- every plain build reports len 1 on every input, every \$limit build reports 2 or 3. The rest of the E-352 suite is unchanged: closed-form HD2/HD3 still 0.0 and 1.2e-16, two-tone vs the built-in diode still 1.9e-06, the closed-form mixer still exact. The OSDI ABI is UNCHANGED -- this is a compiler fix, so 0.8 descriptors from before and after are interchangeable. NGSPICE SIDE: no functional change -- osdidisto.c already consumes whatever tensors a model publishes, which is why the fix is compiler-only. What did change is the no-tensors warning: it named \$limit as the case that falls here, which is now false, so both the message and the comment above it name the remaining unreachable case instead (a nonlinearity in a GROUND-REFERENCED probe, not a model input because it has no hi/lo pair). E-352's example check [6] asserted that warning on a \$limit model and so became a REGRESSION the moment this landed -- it now uses a ground-referenced model, which still legitimately trips the guard (3 jacobian entries, 2 nodes, no model inputs), keeping the never-silently-zero guarantee under test. Verify (examples/limitdisto): 7 checks; osdidisto suite back to 6/6. | | [E-359](#) | ngspice-46/src/osdi/osdidistonum.c (new),

osdidistonium.h (new), osdidisto.c, osdi.h, osdiregistry.c, Makefile.am, src/include/ngspice/osdiitf.h, examples/osdidisto_examples/* | **.disto** for Verilog-A rebuilt on NUMERICAL differentiation of the analytic jacobian; the compiler-side tensor machinery is DELETED. E-352 obtained the 2nd/3rd-order Taylor coefficients by having the compiler emit a symbolic closed form. The capability was right but the cost was out of all proportion to what .disto does: on ASMHEMT 707k intermediate derivatives -> 1.4M MIR instructions -> a 30.2MB .osdi and a 126s compile against a 3.6s baseline (35x; hisimhv 20x), plus -- because the tensors were computed inside eval() -- a 3.3-3.9x RUNTIME penalty on every DC sweep, transient, AC and noise run, and an OSDI 0.8->0.9 ABI bump that made every already-compiled model unusable without a rebuild. ROOT MISTAKE: .disto needs the coefficients at ONE point, once per instance, once per analysis; E-352 computed a closed form valid at EVERY point in order to evaluate it once. WHAT REPLACES IT: the model already publishes its FIRST derivatives analytically -- that is the jacobian every analysis loads -- so the higher ones follow by differencing THAT at the operating point ($d2 R_r/dv_p dv_q = d/dv_q[J_{rp}]$; $d3 = d2/dv_q dv_s[J_{rp}]$). Differencing an already-analytic first derivative rather than the residual is what keeps it accurate. Two properties make it cheap: the tensors are evaluated at the DC OPERATING POINT so they are frequency-INDEPENDENT (built once at D_SETUP, reused by every mode and every frequency), and they are built in NODE coordinates where an instance has 5-20 nodes rather than the 52 'model inputs' ASMHEMT reports. Entries keep the exact (row, col...) shape the analytic path produced, so the whole verified Volterra contraction is reused UNCHANGED -- only the tensor source and the kernel differ. NODE COORDINATES ALSO LIFT A REAL LIMITATION: E-352 indexed by model input (a branch voltage with a hi/lo pair), so a nonlinearity in a GROUND-REFERENCED probe $V(a)$ had no pair, was never recorded, and the device reported 'contributes no distortion' and returned zero; it now works and is pinned to a closed form (-4.16666665e-02 vs exact -4.16666667e-02). E-353's \$limit chain-fold disappears for the same reason -- differencing the real jacobian sees what the simulator sees, so there is no chain to fold, and E-353's seven-shape suite (written specifically to break that logic) passes unchanged against an implementation containing none of it. ACCURACY, measured against EXACT CLOSED FORMS rather than against the old implementation: cubic 4.0e-09/5.4e-09, exponential diode 2e-10/~1e-10, internal-node 3.6e-12/1.2e-08, ground-referenced 4e-09, MEXTRAM vs the analytic implementation 5e-09/1e-07 -- all below the 1.9e-06 floor the built-in-diode oracle already sits at (a 0.24 ppm \$vt difference). THE STEP SIZE IS THE WHOLE GAME: truncation is $O(h^2)$ and roundoff $O(\epsilon/h)$ at 2nd order but $O(\epsilon/h^2)$ at 3rd, so one step cannot serve both (a single 1e-3V step put the result 8.5e-5 out); and the right yardstick is not the operating voltage but the CURVATURE scale ($v_t = 26mV$ for a diode, volts for a polynomial), so a voltage-relative step left a cubic biased at 0V 11% wrong. The 2nd-order pass measures $|dJ/dv|/|J| = 1/V0$ directly and the 3rd-order step is derived as $h = 1.2e-4V0 \sim V0\epsilon^{1/4}$. ONLY .disto IS APPROXIMATE: the jacobian is still exact symbolic autodiff and the compiler now emits nothing distortion-related, so a model's .osdi is what a pre-E-352 compiler would have produced -- op/dc/ac/tran/noise verified IDENTICAL to 15 digits against a build with the tensor pass disabled, and an op before and after a .disto agrees exactly (the probe restores the unperturbed evaluation as its last act). THREE BUGS THAT PRODUCED PLAUSIBLE WRONG NUMBERS rather than failures, all now documented in osdidistonium.c with the measurement table that settles them: one step for both orders (8.5e-5); a voltage-relative step (11% on a cubic at 0V); and indexing write_jacobian_array_resist as if parallel to jacobian_entries[] when it is a DENSE array of the RESISTIVE entries only (the two coincide only for a model with no charge storage, which is exactly why simple models agreed and MEXTRAM was out by 3-10x). A fourth belonged to the prototype and mattered most: COLLAPSED NODES -- $V(a,b) <+ 0$ shorts nodes together (MEXTRAM merges three onto one) and treating them as independent perturbs the shared node once per alias, a silent ~0.2% error that briefly looked like a defect in the SHIPPED analytic implementation. It was not: E-352/E-353 are correct on every model tested. NGSPICE SIDE: the new osdidistonium.c (~300 lines) is the whole implementation -- distinct-global-node discovery (collapse-aware), a +-h jacobian sweep for 2nd order and a mixed second difference over node pairs for 3rd, emitted as sparse (row, col...) entries. osdidisto.c keeps its Volterra modes verbatim and only changes where the entries come from (numcache, built at D_SETUP) and the kernel (node voltage rather than an input node PAIR). The has_taylor gate and the OSDI version floor are gone -- all this needs is eval, jacobian_entries and write_jacobian_array_resist, which every OSDI ≥ 0.4 object has, so models compiled before any of this work get distortion WITHOUT

being recompiled. Verify (examples/osdidisto): 6 checks, now including a ground-referenced nonlinearity pinned to closed form; examples/limitdisto 7 checks unchanged. Regression 285/285. || [E-360](#) | ngspice-46/src/osdi/osdidisto.c, examples/osdidisto_examples/ | A circuit with TWO OR MORE different Verilog-A models reported distortion for only the LAST one; every other model contributed ZERO, silently. Introduced by E-359 and shipped with it. CAUSE: DEVdisto is dispatched once per DEVICE TYPE -- cktdisto.c walks DEVices[] and calls each type's handler in turn -- and every distinct .osdi registers as its own device type, so a circuit using two Verilog-A models calls OSDIdisto TWICE for every mode, D_SETUP included. E-359 cached the numerical tensors in ONE GLOBAL array and cleared it at D_SETUP, so model B's setup destroyed model A's tensors; in the mode passes A's instance pointers then failed the consistency check and every instance was skipped. That check is the only reason this produced NO numbers rather than WRONG ones -- without it model A would have contracted model B's tensors against A's kernels and produced confident garbage. FIX: the cache is keyed by descriptor (OsdNumModelCache{descr, ent, n, cap}; numcache_for(descr,create) finds-or-adds, numcache_reset drops only that model's entries), so each model only ever clears its own; lookup is linear in the number of MODEL TYPES, a handful, not in instances. WHY THE TESTS MISSED IT: every distortion test -- the six E-352 checks, the seven E-353 \$limit shapes, every model in the corpus campaign -- used EXACTLY ONE .osdi. The bug needs two. Single-model coverage was thorough enough to hide a whole-feature failure. The new check [7] compares the cubic branch's value WITH a diode present against the same branch ALONE, so it fails on the silent zero rather than merely asserting non-zero; against the pre-fix binary it reports '0.00000000e+00 alongside a diode vs 6.24999998e-03 alone'. Also stress-tested: 50 instances (cache growth past its initial capacity of 32; first and last match the single-instance reference) and three model types with .disto run TWICE in one session (every branch correct on both runs, so the per-model reset is idempotent). Verify (examples/osdidisto): 7 checks. Regression 285/285. || [E-361](#) | ngspice-46/src/osdi/osdidistonum.c, ngspice-46/src/spicelib/analysis/distoan.c, examples/osdidisto_examples/* | Two sanitizer findings in .disto that every ordinary build tolerates silently. (1) HEAP-BUFFER-OVERFLOW, mine from E-359: numdisto_jac_at copied the solution vector with for (i = 0; i <= n; i++) where n = CKTmaxEqNum, but CKTrhsOld holds CKTmaxEqNum entries -- ngspice's own code copies it as CKTmaxEqNum * sizeof(double) -- so the last valid index is n-1 and this read one element past the end on EVERY .disto run with a Verilog-A device. ASan: 'READ of size 8 ... 0 bytes after 32-byte region', a 4-double region faulting at index 4, matching exactly. It is a READ, so nothing was corrupted and the computed values are unchanged by the fix, but it is undefined behaviour on the main path; the same off-by-one was in the node-range guard (gp > n where the valid range is gp < n). (2) (int)NaN IN THE SWEEP POINT COUNT, pre-existing ngspice in distoan.c and reproducible with NO OSDI device in the circuit at all: a linear sweep with start == stop leaves DfreqDelta at zero so the tolerance term is 0/0, and a decade/octave sweep with ZERO steps does exp(log(10)/0) = inf and then evaluates 0 * inf -- converting NaN to int is undefined, so a nonsense request quietly ran with whatever point count the hardware produced instead of being refused. FIX: zero-step decade/octave sweeps are rejected the way an unknown step type already is, and the linear division is guarded (where DfreqDelta is non-zero the term is floor(DfreqDeltareltol/DfreqDelta) = floor(reltol) = 0 for any sane reltol, so zero is the value consistent with every other case and normal sweeps are bit-identical). NOTE: this second one had been firing under the project's OWN test deck all along -- osdidisto check [4]'s mixer sweeps lin 1 1e6 1e6, precisely the start == stop trigger -- and nothing ever failed because an un-sanitized build tolerates it. Neither defect produced a wrong number: one is an out-of-bounds READ, the other an undefined conversion that happened to yield a usable value on this hardware. That is the class only a sanitizer finds, and the argument for running the suites under one after any change involving raw index arithmetic. Verify (examples/osdidisto): 8 checks, the new one asserting a zero-step sweep produces no output rather than inventing some (an ordinary build cannot observe the UB itself). Both suites 8/8 and 7/7 under ASan+UBSan with values unchanged. Regression 285/285. || [E-362](#) | ngspice-46/src/spicelib/analysis/{distoan,dctran,cktsens,dctrurv}.c, examples/sweepguard_examples/ (new) | Fuzzing analysis-card sweep parameters against an ASan+UBSan build: SEVEN pre-existing defects, all reachable from ordinary input, three of the counts involved being used as ALLOCATION SIZES and one reaching the allocator NEGATIVE. E-361 found one (int)NaN conversion in .disto's point count; that looked like a family rather than a one-off, so this drives every sweep-taking analysis (.disto/.ac/

.noise/.sens/.sp/.dc/.tran/.four/.fft) with degenerate and extreme parameters. FINDINGS: distoan.c DECADE and OCTAVE -- the count goes +-inf (a zero or denormal start frequency makes log() diverge) or exceeds INT_MAX, and the (int) cast is undefined, feeding DstorAlloc(..., NoOfPoints+2); distoan.c LINEAR -- DnumSteps+1 is plain INT arithmetic and overflows at INT_MAX BEFORE any double-valued guard can run; distoan.c LINEAR -- the final count could be negative, reaching the allocator as ASan's 'requested allocation size 0xfffffffffff0'; dctran.c -- CKTtimeListSize overflows and is used both by TMAPLOC and by osdiaccept.c to size a buffer; cktsens.c x2 -- the same cast pattern driving the sensitivity sweep; dctrcurv.c -- .dc has NO bound at all, advancing by a step and comparing against the stop value, so dc V1 0 1 1e-30 (a typo for 1e-3) runs forever with no diagnostic, and a step below the ULP of the start never advances the value at all. .tran already declines the equivalent request and a zero step is already refused, so .dc was the outlier. ONE FIX MADE THINGS WORSE FIRST: clamping CKTtimeListSize to a representable value regressed tran 1e6 1e30 1e6 from erroring to HANGING, because ngspice's existing rejection of absurd transients WAS this very overflow producing an allocation size that failed -- clamping made the allocation succeed and the run proceed forever. The accidental rejection had to become a deliberate one (E_PARMVAL); absurd transients now return rc=1 exactly as before and ordinary ones are bit-identical. TWO FINDINGS WERE NOT BUGS and were deliberately not 'fixed': an enormous but finite sweep is SLOW, not hung, and a timeout cannot tell them apart -- sens ac dec 1000000 over 300 decades measured 0.05/0.20/1.78/17.4s at 1/10/100/1000 steps (linear, so simply large), and ac oct 1000000 1e300 1e6 takes 12.8s identically on the shipped and fixed builds, exceeding the fuzzer's timeout only under ASan's ~5x slowdown. The scaling check that distinguishes them is documented in the harness. Final fuzz run 400 iterations: SANITIZER=0, CRASH=0, no real hangs. Verify (examples/sweepguard): 3 checks pinning eleven repros plus six ordinary sweeps that must be unaffected; a proven trigger -- against the pre-fix build it reports three HANGs. Its 'produces no output' check passes even pre-fix, because an ordinary build cannot observe an undefined conversion, only its consequences, which is why the sanitizer harness (fuzz_analysis.py) ships alongside it rather than being folded into the regression. Regression 285/285. | [E-364](#) | src/osdi/osditrnoise.c (new), src/osdi/{osdidefs.h,osdiload.c,osdisetup.c,osdi.h,Makefile.am} | TRANSIENT NOISE for OSDI devices: Verilog-A noise sources are now injected into .tran. .noise linearises about an operating point, so it cannot show jitter, noise-induced switching or oscillator phase noise; ngspice has had trnoise for independent sources for years, but OSDI devices were the only silent components in an otherwise noisy circuit. NO ABI OR COMPILER CHANGE was needed -- the compiler already emits a full parametric description that NOTHING read: noise_source_type[], load_noise_params() (per-source power and exponent, i.e. $S(f)=\text{power}/f^{\text{exp}}$ at the CURRENT bias), and each source's node pair and correlation name. load_noise_params and noise_source_type were dead ABI, partly because the NOISE_TYPE_* names lived only in the compiler-side header and had never been mirrored into ngspice's osdi.h (now fixed). ACTIVATION IS AUTOMATIC with no option to set: it turns on when the circuit already has a trnoise source and adopts that source's noise timestep, so all generators share one grid; it is deliberately NOT keyed on 'the device declares noise sources', which every real compact model does and would mean always-on. AMPLITUDE LAW, derived not fitted: f_alpha is Kasdin's fractional-noise method, $H(z)=(1-z^{-1})^{-(a/2)}$ so $|H|^2 \rightarrow (2\pi f)^{-a}$, and discrete white noise of deviation Q on a grid ts has one-sided density $2Q^2ts$, giving $Q = \sqrt{|\text{power}|(2\pi f)^a/(2ts)}$ -- with $a=0$ collapsing to the white case so both kinds share one line. The first attempt used the white law for flicker: wrong by $\sqrt{1/(2\pi f)} = 399\times$ at $ts=1\mu s$, and measurement against .noise gave $397\times/396\times$, which is what confirmed the derivation instead of inviting a fudge factor. A FIXED noise grid is essential -- scaling by the ADAPTIVE timestep makes the injected power track the LTE controller and the spectrum an artefact of the integrator. power is re-read every timepoint (shot noise follows the current), correlated same-name sources (LRM 4.6.4) share one stream, and the signed power from E-42 is carried into the amplitude for coherent addition. Injection is RHS-only (an independent current source, no Jacobian entry) and is stamped in the SERIAL post-eval loop. Verified: thermal 1.7% vs exact analytic, shot 0.6-4.5% across a 45x current range (which proves the bias is re-read), flicker 0.9939 +/- 0.0091 vs .noise with fitted slope -0.993. noise_table is deliberately NOT injected (a tabulated spectrum needs frequency shaping, not a scalar amplitude); it warns once and stays fully accounted for in .noise. | [E-365](#) | src/include/ngspice/cktdefs.h, src/spicelib/analysis/{cktpzset.c,cktsetup.c}, src/

frontend/com_hb.c | **pz** followed by **hb** returned a SILENTLY WRONG harmonic-balance result (DC term 0.2500 instead of the correct 0.2439, 2.5% out) and, underneath it, read freed memory. CKTpzSetup opens with NIdestroy(ckt); NIinit(ckt); -- freeing ckt->CKTmatrix and building a DIFFERENT one -- while leaving CKTisSetup asserted, so every device's cached matrix-element pointer (bound by CKTsetup into the old matrix) dangles. com_hb then guarded its setup with if (ckt->CKTmatrix == NULL || SMPmatSize(...) <= 0), which asks 'is there a matrix?' -- and after a pz there is a perfectly good non-empty one -- so CKTsetup was skipped and CKTload read the stale pointers (ASan heap-use-after-free in VSRCload, via HBAanalyze/com_hb; freed by spDestroy<-NIdestroy<-CKTpzSetup<-PZan). FOUND BY A NEW KIND OF CAMPAIGN: every prior ngspice fuzzing round varied the INPUT (netlist, model card, command, expression, rawfile, .snp, OSDI loader, analysis-card parameters), and each input runs in a fresh process, so none could reach a bug living in what one analysis leaves behind for the next. This one held the netlist fixed and fuzzed the ORDER of analyses and state mutators within one session -- 500 iterations, 4 sanitizer reports, all the same cause. Same shape as E-360 (a second Verilog-A model silencing the first in .disto). SCOPE measured, not assumed: pz then op/dc/ac/tran/noise/disto/tf/sp/pss is clean, and hb after any other analysis is clean; portnum matters only because it lets hb reach the load. The bug is NOT confined to a failing pz -- the success path corrupts identically. FIX: a new CKTbindStale flag records that an analysis replaced the matrix while CKTisSetup stayed asserted; CKTpzSetup sets it, CKTsetup clears it (a successful setup is what makes the bindings valid), and com_hb honours it with a BALANCED CKTunsetup()/CKTsetup() pair -- a bare CKTsetup() returns E_NOCHANGE since CKTisSetup is still 1, and calling it without the unsetup would re-run DEVsetup on already-setup devices and double-allocate their internal nodes. Circuits that never run pz are untouched. The example suite needs NO sanitizer: the consequence is a number, so it pins that hb after pz equals hb alone (fixed 4/4, before 2/4 with max deviation 4.55e-02). | | [E-366](#) | src/frontend/com_qpss.c, src/include/ngspice/klu-binding.h | Two more sites of the E-365 stale-binding class, found by continuing the SAME sequence-fuzzing campaign against the FIXED build. Widening the pool (RF/steady-state commands, sweep/wcd, and SOLVER SWITCHING option klu/option sparse between analyses -- added deliberately, since KLU and SPARSE keep separate per-instance pointer sets) took the yield from 500 iterations/1 signature to 700/3. (1) com_qpss.c carried the identical guard com_hb.c had -- if (ckt->CKTmatrix == NULL || SMPmatSize(...) <= 0) -- which asks 'is there a matrix?' when the question is 'do the device bindings point into it'; after a pz there IS one, so CKTsetup was skipped and CKTload read freed memory. Same fix: honour CKTbindStale with a BALANCED CKTunsetup/CKTsetup. com_checkpoint.c also matches the grep but only sizes a buffer, so it is unaffected. (2) CREATE_KLU_BINDING_TABLE in klu-binding.h did if (matched == NULL) { printf("not found"); } here->ptr = matched->CSC; -- it REPORTED the NULL and then dereferenced it, so every lookup miss was undefined behaviour and silent on an ordinary build (UBSan: member access within null pointer of type 'BindElement'). Reachable from option klu + pz + any AC-family analysis (ac/sp/stb), because pz rebuilds ckt->CKTmatrix and the device's COO pointer is no longer in the new bind table; SPARSE is unaffected since binding tables are a KLU construct. The companion CONVERT_KLU_BINDING_TABLE_TO_COMPLEX/_TO_REAL macros dereferenced the same unresolved binding and are now guarded, and a miss sets binding to NULL rather than leaving the freed value -- leaving it stale would turn a NULL deref into a use-after-free. The macro lives in a shared header, so this covers every device using it. STILL OPEN and documented: under KLU the pole-zero block at the end of VSRCbindCSCComplex reads through a binding that is NOT null but STALE (CREATE_KLU_BINDING_TABLE is skipped for that entry by its branch > 0 precondition), so the existing NULL test passes; that needs the bindings TORN DOWN when pz rebuilds the matrix, not another guard, because a guard cannot distinguish stale from live. A central rebinding in CKTdoJob was tried and REVERTED -- tracing showed CKTbindStale is already clear by dispatch time, so it fixed nothing, and a guard that fixes nothing is worse than none. Campaign findings 3 -> 1. | | [E-367](#) | src/frontend/typesdef.c, src/frontend/com_sweep.c | **sweep** produced plots nobody could name. (1) plot_alloc() names a plot from the plotabs[] table in typesdef.c and falls back to the literal "unknown" when there is no entry -- and EVERY plot type this project added was missing: sweep, sweepwave, hb, envelope, eye, loadpull, rfstab, stb. sweep therefore created unknown4/unknown7/unknown10. All eight registered. ORDER IS LOAD-BEARING: ft_plotabbrev returns the FIRST entry whose pattern is a substring of the plot name, so "sweepwave" must precede "sweep" or a waveform plot would be abbreviated "sweep" and

collide with the point plot (the same rule already explains why a plot named "spectrum" matches the earlier "sp" entry, never "spect"). The struct is {p_name, p_pattern} -- the SECOND field is matched and the FIRST is returned, which reads backwards from the table and is now documented in-source, because getting it wrong yields a table that compiles and matches nothing. (2) The summary printed 'sweep' as a LITERAL, so the one hint the command gave for returning to an earlier sweep -- setplot sweep -- always failed with "no such plot named sweep"; it now prints the real name (sweep3), in BOTH summary branches (single-knob and the -vs/-family curve family), and the stale header comment in com_sweep.c documenting the plot as named sweep is corrected. The numbers are not 1,2,3 because plot_unique_tynename draws from a counter SHARED by every plot type and advances it only far enough to avoid a collision -- a 3-point sweep burns op1/op2/op3 and lands on sweep3. That is pre-existing behaviour and is NOT changed; what changes is that you no longer have to guess. FOUND ALONGSIDE, LEFT OPEN: print alle is dead in every build configuration -- findvec_alle() sits behind #if defined(XSPICE) && defined(SIMULATOR) with a #ifndef SIMULATOR stub returning NULL beside it, and -DSIMULATOR is set only on ngspice_CPPFLAGS/libngspice_la_CPPFLAGS (main.c, ngspice.c, sharedspice.c) while every src/frontend source is compiled ONCE into libfte.la with AM_CPPFLAGS alone, which both the executable and the shared library link. Confirmed by inspection: the string "DigitalData" is absent from the binary. Adding -DSIMULATOR to src/frontend/Makefile.am was tried and REVERTED -- ngproc2mod/ngmultidec/ngmakeidx also link libfte.la but not the XSPICE libraries, so enabling the block breaks them at link time under --enable-oldapps. The guard is unsatisfiable BY DESIGN, and a real fix means moving the event-node lookup into a simulator-only object. Two latent items behind that dead code are recorded so they are not re-discovered: findvec_alle overwrites the pl_tynename plot_alloc just assigned with a hardcoded copy("dig1") (leaking it, leaving the registered completion keyword pointing at a name no plot has, and giving every digital plot the same name), and get_all_type has a misplaced paren -- tolower(word[0] != 'a') instead of tolower(word[0]) != 'a' -- defeating its own case-insensitivity, currently harmless only because callers lowercase first (verified: print ALLV and print allv behave identically). Verified by examples/sweepname_examples: 6/6 fixed vs 1/5 on the pre-fix binary (got ['unknown3'], no 'into plot' message). Regression 291/291. | | [E-368](#) | src/frontend/typesdef.c | The periodic small-signal analyses named their plots wrong; E-367's claim that the naming was fixed was too broad. E-367 found its eight missing plot types by grepping for STRING LITERALS passed to plot_alloc(), which is blind to the path every standard and RF analysis uses: outitf.c calls plot_alloc(run->type) with a RUNTIME string handed to beginPlot(analName) by each analysis (mostly ckt->CKTcurJob->JOBname). Checking properly found seven more, four of them COLLIDING with an unrelated analysis -- worse than "unknown", because the name looks right. ft_plotabbrev returns the FIRST entry whose pattern is a substring of the plot name, and each of these contains a more general pattern: PXF Analysis matched nothing at all -> unknown; PAC Analysis contains "ac" -> collided with ordinary AC; PSP Analysis contains "sp" -> collided with S-parameters; PNoise Analysis, qnoise and phasenoise all contain "noise" -> collided with ordinary noise AND with each other; Frequency Domain Periodic Steady State Analysis (QPSS) contains "periodic" -> collided with PSS. Each now has its own pattern placed BEFORE the general one its name contains. TWO ORDERING TRAPS: "qnoise" contains "pnoise", so the qnoise entry must precede the pnoise entry; and PSS/QPSS differ only by their leading words (Time Domain vs Frequency Domain Periodic), so the QPSS pattern must carry "frequency domain periodic" rather than the "periodic" they share. Verified by examples/periodicnames_examples: 9/9 fixed vs 4/9 pre-fix (pac->ac2, pxf->unknown2, pnoise->noise2, PSS/QPSS both pss). The FOUR REGRESSION CONTROLS -- ordinary noise, ac, tran, sp -- pass on BOTH binaries, which is what shows the change is targeted: those names are relied on elsewhere in the repo (setplot noise1, print noise1.noise_spectrum in noisecorr/noisejw/stdaudit/tempphys/physcheck) and none moved; no deck that runs pnoise selects a plot by name either. HARNESS TRAP RECORDED: the setplot listing has a blank line immediately after its header, so a non-greedy regex reaching to the first blank line captures NOTHING -- an early harness reported "no plot" for all 25 analyses and called the run clean. SCOPE STATED HONESTLY: envelope/eye/loadpull use literals and were registered by E-367, so they are correct by construction but no deck here reaches those paths; sens produced no plot under every invocation tried, consistent with the already-open sens breakage and not a naming problem. Regression 292/292. | | [E-369](#) | src/spicelib/devices/vsrc/vsrcbindCSC.c | Closes the E-365/366 stale-binding class. E-366 fixed two sites and left the third OPEN on purpose, because

it could not be closed with another guard: the binding was NOT NULL but STALE, and no NULL test distinguishes those. THE ASYMMETRY IS THE BUG: VSRCbindCSC assigns the pole-zero binding only INSIDE if (here->VSRCibrIbrPtr), and that pointer is allocated ONLY by a pole-zero analysis -- so on any later analysis the gate is false, the assignment never happens, and the binding keeps its previous value, pointing into the BindStruct that SMPdestroy() freed when the pz matrix went away. Both consumers (VSRCbindCSCComplex, VSRCbindCSCComplexToReal) then dereference it, because THEIR guard is VSRCbranch != 0 -- a property of the device -- rather than "was this binding re-established for THIS matrix". ASan on option klu ; pz ; ac: heap-use-after-free READ of size 8 in VSRCbindCSCComplex, freed by SMPdestroy, reallocated by SMPconvertCOOtoCSC. FIX: clear the binding BEFORE the gate, so a stale value can never survive a matrix rebuild -- one line, fixing both consumers at once since both NULL-test it; the duplicated (x != 0) && (x != 0) copy-paste slip is collapsed in the same pass. NO OTHER DEVICE NEEDS THIS: vsrc is the only one that hand-writes a pz binding block (grep "Pole-Zero Analysis" across every bindCSC.c returns one file); every other binding goes through the klu-binding.h macros, which are self-consistent -- CREATE and CONVERT are gated on the SAME (a>0 && b>0) condition, so a skipped create implies a skipped convert. HONEST LIMIT, stated in the doc and the example: the PRE-FIX binary PASSES every behavioural check -- the freed BindStruct entry still held the right CSC_Complex pointer, so the UAF read plausible values and the numbers came out correct, verified deliberately on a 40-node ladder with a tran between the pz and the ac to churn the heap. This is a MEMORY-SAFETY fix, not a wrong-answer fix; the behavioural checks are a regression guard, and the check that DISCRIMINATES runs under a sanitizer (the example runs it when NGSPICE_ASAN is set and SKIPS LOUDLY otherwise, so nobody mistakes a pass for proof). UB that happens to give the right answer today is still UB -- the same read against recycled memory is how E-365 presented, 2.5% wrong and silent. examples/klubind_examples 8/8. Regression 294/294. | | [E-370](#) | src/spicelib/devices/urc/urcinit.c | Every .pz re-expanded the URC subcircuit, creating nodes past the allocated RHS. FOUND BY RUNNING AN EXISTING FIXTURE UNDER A SANITIZER: a sweep of all 185 shipped decks with each solver forced produced exactly one real finding, in ngrashanalysis_examples/pz_urc.cir -- the E-315 fixture, passing ever since E-315 stopped it SIGSEGVing. Nothing had run it under a sanitizer; the crash was gone, the corruption underneath was not. ASan: heap-buffer-overflow READ of size 8 in RESload (CKTload->NIiter->CKTop->PZan), buffer allocated by NIreinit. RESload indexes ckt->CKTrhsOld by node number, so the read past the end means a resistor's node number exceeded the allocated RHS. CAUSE: urc declared .DEVsetup = URCsetup AND .DEVpzSetup = URCsetup, but URCsetup is not a matrix-entry allocator like its neighbours -- it is a SUBCIRCUIT EXPANDER (CKTmkVolt per lump, CKTtctElt per element) with NO idempotency guard, and CKTpzSetup calls DEVpzSetup for every device on EVERY pz job. Each .pz therefore expanded the URC again, creating fresh internal nodes AFTER NIinit had sized the RHS. That is exactly why RESsetup/CAPsetup are safe in the same slot and URCsetup is not: they only allocate matrix entries, which is idempotent. URC was the only expander wired up this way. Even a SINGLE .pz triggered it (CKTsetup expands once, CKTpzSetup expanded again); the overflow took a second pass to cross the allocation boundary. Visible symptom on an ordinary build: doAnalyses: device already exists, existing one being used + aborted run. FIX: .DEVpzSetup = NULL. The URC stamps nothing itself -- DEVload, DEVacLoad and DEVpzLoad are all NULL -- and the RES/CAP instances it creates are ordinary elements under their own device types, for which CKTpzSetup already calls RESsetup/CAPsetup; that is what actually re-binds the matrix. NOT A SOLVER BUG despite being found in a KLU/Sparse hunt: it reproduces identically under both. BEHAVIOURAL CHANGE, not hidden: on the E-315 fixture the fixed build gets FURTHER -- instead of aborting on a spurious duplicate device it runs the analysis and reports singular matrix: check node 3, which is CORRECT (that fixture's node 3 is genuinely floating); the duplicate expansion had been masking a real topology diagnostic behind a spurious one. verify_ngrashanalysis.py still 4/4. examples/urcpz_examples 10/10 fixed vs 5/10 pre-fix. Regression 294/294. | | [E-371](#) | src/frontend/vectors.c, src/frontend/outitf.c, src/frontend/spec.c, src/frontend/postcoms.c, src/frontend/com_fft.c, src/frontend/rawfile.c | Per-type plot numbering, and a date on every plot. Reported from user observation: a 500-point sweep produced sweep500, not sweep1, and the plot's Date field was empty. (1) NUMBERING. plot_unique_typename() counted up a SINGLE counter shared by every plot type. A sweep runs one analysis per point and KEEPS its plot (deliberate -- overlay and

waveform inspection rely on it), so a 500-point sweep created op1..op500, each collision pushing the shared counter, and by the time the sweep's own plot was named the counter stood at 500. The number looked like the point count; it was really "how many plots exist". Each abbreviation now carries its own counter, so the first sweep is sweep1 whatever else ran. A tran/ac/tran/ac/noise/sweep/sweep deck goes from sweep12 op12..op3 noise3 to sweep2 op10..op1 noise2; the two agree except in the pathological case where one type's plots number an unrelated type. WHAT HAD TO BE PRESERVED: E-345 exists because this path was QUADRATIC over a sweep (89% of a 64000-point sweep in plot_alloc->cieq->tolower), fixed by starting the search from the shared monotone counter plus a hash membership index. A per-type counter restarting at 1 each time would have reintroduced exactly that quadratic, so the counter is REMEMBERED per type (nghash on the lowercased abbreviation) and the search resumes from it -- E-345's timing check still passes at 22->21 us/point, ratio 0.93 over a 4x longer sweep. destroy HAS TO WALK IT BACK DOWN: destroy all must restart numbering at tran1, which examples/lifecycle_examples (E-81) pins -- and that is what CAUGHT the first version of this change. destroy unlinks plots one at a time through plot_forget() rather than plot_forget_all(), so plot_forget() now lowers that type's counter to the freed number. That also makes a single destroy op2 recycle the number (the shared counter went on to op4), which is the behaviour destroy all always had, so it makes the two consistent rather than adding a rule. (2) DATES. Only the analysis path set one: outtif.c did pl->pl_date = copy(datestring()) after plot_alloc, so every plot created DIRECTLY by a command -- sweep, hb, envelope, eye, loadpull, stb, rfstab, qpac -- left pl_date NULL and print rendered it as (null). The stamp now happens in plot_alloc(), the single point where every plot is created, covering all eight command paths and any future caller. The five sites that set their own date became redundant and were removed, except rawfile.c, where a loaded rawfile's OWN date must win -- that one frees the stamp before replacing it, so centralising does not leak. VERIFIED by updating and extending examples/plotname_examples -- the E-345 example whose expectations this deliberately alters, and the file that documents this behaviour: 9/9 fixed vs 4/9 on a true pre-fix build (all six files reverted). The tran-has-a-date row PASSES on both, as the control showing the analysis path always had one and only the command paths changed. Regression 294/294. | | [E-372](#) | src/frontend/variable.c | **unset plots** reported a bogus "Internal Error". Found by a PLOT-LIFECYCLE sequence fuzzer -- random analyses interleaved with plot-management commands (destroy, setplot, unset plots, write/load, fft, linearize, remcirc) under ASan. That area was chosen deliberately: it is where E-342 (borrowed-pointer UAF via unset plots) and E-345 came from, and E-371 had just added a new allocation and new pointer arithmetic to it. Fired in 18/120 cases and minimised to ONE command on the SHIPPED binary: unset plots -> "Error: plots is read-only." followed by "cp_remvar: Internal Error: var 112". TWO DEFECTS in that one line. (1) THE MESSAGE PRINTED AN ASCII CODE, NOT A NAME: the signature is cp_remvar(char *varname) and the format was "...var %d", *varname, which dereferences the string to its FIRST CHARACTER -- 112 is 'p' for plots, and unset curplot reported 99, which is 'c'. The diagnostic never named the variable it complained about (verified against the ordinals). (2) IT WAS NOT AN INTERNAL ERROR: the branch is guarded by if (*p), and *p non-NULL only means the variable WAS FOUND in one of the lists walked just above -- the normal state for plots and curplot -- so it fired on 100% of ordinary unsets of them. An internal error valid input reaches every time carries no signal. FIX: drop both spurious prints. For US_READONLY the "%s is read-only" message is already the complete answer and nothing is unlinked or freed there (correct for a read-only variable), so removing the noise changes no behaviour; for US_DONTRECORD the case's own comment says "Do nothing..." and its siblings curplotname/curplottitle/curplotdate were always silent, so curplot now matches them. The other Internal Error in the same function ("US val %d", i) is CORRECT and left alone -- i really is an int and that default branch is unreachable from valid input. CHECKED AS A CLASS: a grep for a %d conversion fed a dereferenced char across the frontend returns no other occurrence. VERIFIED by examples/unsetvar_examples (no sanitizer needed): 9/9 fixed vs 5/9 pre-fix. The FOUR CONTROLS -- curplotname, curplottitle, curplotdate and an ordinary set/unset round-trip -- pass on BOTH binaries, which shows the change is confined to the two affected cases and did not break unset itself. Re-running the fuzzer against the fixed build: 400 iterations, 0 findings, 382/400 cases holding live plots at the end (a real clean result, not a harness that stopped working); harness committed under examples/unsetvar_examples/fuzz/ and excluded from the regression sweep.

PROCESS NOTES RECORDED: the investigation began by checking whether E-371 had leaked (it added a `datestring()` allocation per plot) -- it had not, `killplot()` frees `pl_date` and peak RSS stayed flat 7152->7168 kB across 202->6060 plots created and destroyed. That leak check was FIRST attempted with `ASAN_OPTIONS=detect_leaks=1` and came back clean MEANINGLESSLY, because macOS/arm64 answers "detect_leaks is not supported on this platform" -- a sanitizer that silently does not run is indistinguishable from a clean result unless you check. Regression 295/295. || [E-373](#) | `src/frontend/rawfile.c` | A rawfile write/load round trip lost the x-axis column and renamed the sweep axis. FRESH AXIS: E-226 fuzzed rawfile LOADING with malformed input looking for crashes; this asks instead whether a rawfile written by ngspice and then loaded by ngspice still holds the same data. Round-trip identity is a perfect oracle (no reference implementation needed) and catches silent fidelity loss a crash fuzzer cannot see; it also lands on freshly-changed code, since E-371 had just touched `rawfile.c`. Two independent defects, NEITHER of which corrupted a value. (1) `print` LOST THE X-AXIS COLUMN for every loaded plot -- before `write`: Index `v-sweep v(mid)` became after `load`: Index `v(mid)`, so nothing indicated which x-value each row belonged to. `print` prepends the scale column only when `pl_ndims` is non-zero (`postcoms.c`); `outtif.c` initialises it to 0 and sets it to 1 when analysis data arrives, but `pl_ndims` appeared NOWHERE in `rawfile.c`, so every loaded plot carried 0 and failed the gate. PROVEN NOT INFERRED: before writing the fix, a one-line probe setting `pl_ndims=1` on load was inserted, shown to restore the column exactly, and reverted. THE op CASE IS THE CONTROL THAT MATTERS: `pl_ndims=1` could have made `print` INVENT a column for a plot with no real scale (an op plot's variable 0 is just data, and the reader assigns `pl_scale` to variable 0 unconditionally) -- it does not, because `print`'s inner `pl_scale && !vec_eq(bv, pl_scale)` test still suppresses it, and op prints the same header before and after; checked explicitly, since that is why the fix is safe. (2) THE .dc SWEEP AXIS WAS RENAMED `v-sweep -> v(v-sweep)` in the written file. The `v(...)` form means "the voltage AT node X" -- right for a node probe, wrong for a sweep AXIS. A .dc plot's scale is the synthetic, voltage-typed vector `v-sweep`, which has no `v(` prefix, so the writer wrapped it. It did NOT compound (verified over three successive write/load passes: the prefix test leaves `v(v-sweep)` alone), so the damage was a one-time rename. Fix: add `|| v == pl->pl_scale` to the `write-as-is` condition. Checked how wide that else-branch's reach was: node voltages are already named `v(mid)` internally and take the first branch, and let vectors are `SV_NOTYPE`, so the plot's scale was the ONLY thing it was ever reached by; time and frequency escape it by not being voltage-typed. SCOPED OUT AND RECORDED, NOT SILENTLY IGNORED: ASCII rawfiles are 1 ULP lossy. The writer emits `%.*e` with `prec = DEFPREC`, i.e. 16 significant digits, while reproducing an IEEE double exactly needs 17. Measured with `numdgt=16`: 33 of 59 rows differ, ALWAYS in the scale column and ALWAYS at the 17th digit (4.0000000000000004e-11 reads back as 3.999999999999998e-11, one ULP); data values identical, worst relative deviation 4.33e-16 over 118 values. The in-source comment on `raw_prec` even claims "default 15 (max)", which is wrong. Raising the default precision changes the on-disk format for every ASCII rawfile ngspice writes -- broader than these two defects -- so it is documented and NOT made. The example's `ascii` value check therefore compares to 1e-15 while the binary one demands bit-exactness. VERIFIED by `examples/rawtrip_examples`: 19/19 fixed vs 6/19 pre-fix. Controls pass on BOTH binaries (the op rows and both value comparisons), which is what shows the values were never the problem. Regression 296/296. || [E-374](#) | `src/frontend/trannoise/wallace.c`, `src/math/misc/randnumb.c`, `src/main.c`, `src/sharedspice.c` | **setseed** did not seed TRANSIENT NOISE. Found by a correctness campaign over the command set: two identical decks with `setseed 42` produced DIFFERENT noise waveforms, while `setseed 42` then `rnd(100)` returned exactly 50.0 both times -- so the command worked, just not for the generator transient noise draws from. CAUSE: `#define WaGauss` selects the Wallace normal generator, and `initw()` opened with `srand((unsigned int) getpid()); // srand(17); TausSeed();`. `initw()` runs at STARTUP and fills its two pools from that `getpid()`-derived stream, and `GaussWa` draws from those pools; a later `setseed` reset `srand` and the `Tausworthe` state but nothing refilled pools that already existed. FIX, in three parts (moving the seeding alone is not enough): (1) `initw()` no longer calls `srand()`, so it seeds from whatever state the caller established and can never clobber a user seed; (2) the two startup call sites do the `srand(getpid())` themselves, so an UNSEEDED run stays random per process; (3) `com_sseed()` calls `destroy_wallace()`; `initw()`; so a new seed actually rebuilds the pools. TRAP RECORDED: the first version of the `com_sseed` change was guarded `#if defined(WaGauss) && defined(SIMULATOR)`, and `SIMULATOR` is NOT defined for library sources like `randnumb.c` -- exactly the situation E-367 documented after finding

print alle dead for the same reason. The whole fix compiled out silently and setseed still did not reproduce. Guard is now on WaGauss alone. That is twice this macro has disabled working code; treat `#ifdef SIMULATOR` outside `main.c/ngspice.c/sharedspice.c` as a bug until proven otherwise. VERIFIED by `examples/setseed_examples`: 5/5 fixed vs 4/5 pre-fix. Three properties, the third mattering as much as the first (a fix that pinned one constant sequence would satisfy the first and be useless): same seed -> byte-identical waveform; different seed -> different waveform; NO setseed -> still varies run to run. Two controls pass on BOTH binaries: `rnd()` was never broken, and the seeded stream is still real noise (3.5 mV peak-to-peak over 160 points), not a frozen constant. CONSEQUENCE: this falsified a claim in `docs/internals/ngspice_internals/ngspice_transient_noise_analysis.md` that setseed fixes the stream -- untrue when written, true now; the note was corrected and its seed-averaged statistics stand, having been means over genuinely independent runs. Regression 297/297. || [E-377](#) | `src/osdi/osdi.h`, `examples/simparamdiag_examples/` | OSDI diagnostics were unreadable and carried no severity. Found by the correctness campaign over all 94 `\$`-prefixed system functions, when a wrong `\$simparam\$str("analysis")` reported OSDI(debug) n1: unknown `\$simparam_stranalysisOSDI(debug) n1: unknown \$simparam_str...` FOUR defects in one line. (1) NO SEPARATOR: `concat("unknown \$simparam_str", name)` joins with nothing, so the function name and its argument run together and the reader cannot see where one ends -- the numeric `\$simparam` path had the same bug; now unknown `\$simparam\$str "analysis"`, and reported with the `\$simparam\$str` spelling the user actually writes rather than the internal `\$simparam_str`. (2) NO NEWLINE: `osdi_log` writes `fprintf(dst, "%s", msg)` and no message carried a `\n`, so consecutive reports concatenated into one line -- which also HID the repetition, the old output looking like two lines where the fixed one shows 373 (the same 373 reports were always there). (3) WRONG SEVERITY FOR EVERY MESSAGE IN THE OSDI LAYER -- the one with reach beyond `\$simparam`: `ngspice's osdi.h` had `#define LOG_LVL_MASK 8`, but the level occupies the low THREE bits (DEBUG 0 .. FATAL 5) so the mask must be 7; 8 selects bit 3, which no level ever sets, so `lvl & LOG_LVL_MASK` was 0 = LOG_LVL_DEBUG for EVERY level. `\$display`, `\$info`, `\$warning`, `\$error` and every fatal diagnostic alike were labelled "OSDI(debug)" and written to STDOUT -- severity invisible to the user and to any log scraping, and `\$error/\$warning` never reaching stderr. OpenVAF's own copy of the header (`osdi/header/osdi_0_4.h`) has always said 7, so the two sides of the ABI disagreed on how to decode the level. Measured: before, all four severities print "OSDI(debug)" on stdout; after, OSDI n1:/OSDI(info) n1: on stdout and OSDI(warn) n1:/OSDI(err) n1: on stderr. (4) A LEAK: the message was malloc'd, passed to `osdi_log` and never freed -- free was not even declared in the runtime's `N0_STD` extern block -- so a failing operating point leaked 373 allocations, not one; both call sites now go through one helper that frees. THE REPETITION IS FIXED BY E-378: making the messages visible exposed that there were 373 of them, left open here as "a convergence-path change" -- an understatement. CKTop read the fatal's `E_PANIC` return as "did not converge" and walked the `gmin/source-stepping` ladder, re-evaluating the device each pass, then blamed `timestep too small`. E-378 aborts the operating point on a fatal, so the report now appears once. Regression 302/302. Verify (`examples/simparamdiag`): 12 checks, a proven trigger -- pre-fix 1/12, and that single pass is a negative check that succeeds for the wrong reason on the old binary. || [E-378](#) | `src/spicelib/analysis/cktop.c`, `examples/opfatal_examples/*` | A Verilog-A `\$fatal` now ABORTS the operating point instead of driving the convergence ladder. CKTop reads any non-zero return from `NIiter` as "did not converge" and answers it by working through `gmin` stepping, source stepping, pseudo-transient continuation and `optran`. A Verilog-A fatal arrives through that same channel as `E_PANIC`, so the operating-point solver could not tell "this model refuses to evaluate" from "this circuit is hard to solve" -- and responded to a fatal by trying harder. Every aid re-evaluates every device, so the model re-raised the same fatal on each pass: measured at 373 messages for one device, 746 for two, 1119 for three -- exactly 373N, one per device evaluation, 373 being how many evaluations the whole ladder performs before giving up. THE DAMAGING PART IS NOT THE VOLUME: the run then ended with Error: Transient op failed, timestep too small, naming CONVERGENCE, so a model with a typo'd `\$simparam` name told the user their circuit would not converge while the real cause sat several hundred lines above. THE SAME GUARD ALREADY EXISTED IN THE OTHER PATH: E-55 added it to the transient time-stepping loop in `dctran.c`, with a comment that the error was otherwise "swallowed by the retry logic"; the operating-point path never got it (guards before this change: `dctran.c` 2, `cktop.c` 0).

A `.tran` is affected too, since a transient computes its operating point first -- which is why the transient case still emitted 44 messages despite E-55's guard: it never reached the loop where that guard lives. THE FIX tests for `E_PANIC` after the plain Newton solve AND after each aid (a model may only fatal at a bias some later aid reaches), then reports the real cause. The test is EXACT rather than heuristic: `E_PANIC` is 1 and the non-convergence code `E_ITERLIM` is `E_PRIVATE+3 = 103`, distinct values, so it cannot mistake a stalled Newton solve for a fatal. Result: \$fatal in an .op 380 messages/"timestep too small" -> 1/"aborting"; unknown \$simparam 373 -> 1; \$fatal through .tran 44 -> 1. Regression 303/303. Verify (examples/opfatal): 8 checks, a proven trigger -- pre-fix 2/8, and the two that pass pre-fix are the ACCEPT checks (an ordinary circuit still solves; a circuit forced onto the aids with `.options noopiter` still converges through gmin stepping), which are there because a guard that aborted too eagerly would break every circuit that legitimately needs an aid. | | [E-380](#) | `src/spicelib/analysis/dctrcurv.c`, `examples/dcstate_examples/` | A `.dc` sweep inherited integration coefficients from a preceding analysis -- a 45% SILENT wrong answer. `dioload.c` gates its charge branch on `MODEDCTRANCURVE` | `MODETRAN` | `MODEAC` | `MODEINITSMSIG`, and `MODEDCTRANCURVE` is the DC SWEEP -- so a charge-storing device does take that path during a `.dc`, ending in `NIintegrate()`, which returns `geq = CKTag[0] * cap`. In a fresh session `CKTag[]` has never been computed, so it is zero, `geq` is zero, and charge contributes nothing to a DC sweep, which is correct. But `CKTag[]` is plain circuit state and `dctrcurv.c` never initialised it, so after any analysis that drives the transient machinery -- `pss` (a shooting method, so many transient cycles), `tran`, `envelope` -- it still held THAT analysis' coefficients, where `ag[0] ~ 1/delta` is large, and the sweep added a spurious `geq = ag[0]*cap` to every charge-storing device. Fix: zero `CKTag[]` at the head of the sweep, restoring exactly the fresh-session state. MEASURED on a 1k/1k divider with a diode across it where `v(mid) = V1/3` is exact: dc alone 0.166666667 / 0.333332016, after `pss` 0.093917448 / 0.185725364. The diode's own operating point came back SELF-INCONSISTENT (gd 2.51e-3 against `id/(nVt)` 1.71e-2, an `id` 3.4e7x too large for its own `vd`), which is what made it unmistakably a defect rather than initial-condition sensitivity. Across preceding analyses: `op/ac/sp/hb` unaffected at 0%, `tran` 1.7% -> 0%, `envelope` 8.8% -> 0%, `pss` 45.0% -> 0%. FOUND BY cross-analysis STATE fuzzing with a NUMERIC oracle -- for every ordered pair, `result(B after A)` must equal `result(B alone)`, so B is its own reference and no reference implementation is needed. Rounds 1-2 of that campaign (E-365/E-366) used ASan and could only see memory damage; a 45% wrong answer leaves no sanitizer trace. THE ORACLE NEEDED TWO CORRECTIONS, both worth recording: the first pass reported 150/196 "mismatches", ALL spurious -- ngspice announces the solver once per session (E-266) so that line appears for the first analysis and not the second, and console output from A and B interleaves across `stdout/stderr` so an echoed marker does NOT partition the text; self-consistency caught neither, because both reference runs carried the SAME contamination and only the paired runs differed. Fixed by dumping each analysis' plot to a FILE, which cannot be interleaved. The second pass gave 30 mismatches, of which 28 were `rfstab/qpss` not creating their own plot so the probe captured the PREVIOUS analysis' data. TWO DEAD ENDS, recorded so they are not re-tried: (1) `CKTmode` leaking `MODEINITSMSIG` out of `PSS` -- real, `PSS` sets it ~20 times and never restores it, but not the channel since `dctrcurv.c` fully reassigns `CKTmode`; (2) the state-ring rotation pulling stale `CKTstates[]` into `CKTstate0` -- this was IMPLEMENTED AND INSTRUMENTED to prove it ran, and the answer was UNCHANGED; the stale value is the coefficient, not the stored charge. A third guess, that `dcpsc.c` leaves the matrix complex (5 `spSetComplex` calls, 0 `spSetReal`), was ruled out because `SMPLuFac` and `SMPPreorder` both call `spSetReal` first -- that asymmetry is still worth a look but is not this bug. Regression 304/304. Verify (examples/dcstate): 11 checks, a proven trigger -- pre-fix 7/11, where the 4 failures are exactly the defect checks and ALL SEVEN accept checks pass on BOTH binaries. That accept half is the point of the file: this touches the DC sweep's convergence path, so a fix that zeroed too much would break continuation. It covers a 21-point nonlinear sweep that genuinely relies on point-to-point continuation, a single-point sweep, a repeated sweep in one session, a nested two-source sweep, a circuit with 100x the junction capacitance, and a plain `.op`. | | [E-381](#) | `src/frontend/com_stb.c`, `examples/stbrestore_examples/` | `stb` handed its probe sources back ZEROED, so a following `.ac` returned all zeros. `stb` measures loop gain by injecting through two EXISTING sources -- it drives one with `ac = 1` while holding the other at `ac = 0`, then swaps them -- and when finished "restored the probes to quiescent" by setting BOTH to zero. Zero is only quiescent if that is where the probe started: a source carrying a real AC drive had that value destroyed, and any following `.ac` came back with EVERY node

exactly 0.00000000e+00, with no warning and no error (@v1[acmag] 1.0 -> 0.0; vm(mid) 0.3333333333 -> 0.0000000000). FIX: read each probe's acmag before the first injection and write those exact values back, instead of forcing zero. Only the magnitude is saved, because the injection writes `ac = N` which sets acmag and leaves acphase untouched -- verified rather than assumed, and the example asserts it so the assumption cannot rot. SCOPE is narrower than it first looks: in the documented usage the probes are dedicated injection sources sitting at `ac 0`, where restoring zero happens to be right. It becomes wrong when an already-driven source is named as the probe, which is legal and is what the fuzzer did (stb V1 V2 with V1 the deck's own `ac 1` source), and nothing indicates the drive has been destroyed. FOUND BY cross-analysis STATE fuzzing with a NUMERIC oracle -- `result(B after A) == result(B alone)` -- the same instrument as E-380; re-running the 196-pair campaign against the E-380 binary left four mismatches and this was the only genuine one. The other three were --> hb deviations of $1e-17..1e-23$ on near-zero harmonics whose RELATIVE deviation reached 30%, which looks alarming and means nothing -- the classic no-absolute-floor trap, the same one that produced a false $2.12e-07$ "mismatch" in the metamorphic campaign; with `atol=1e-12` all three are within tolerance. WHAT THIS CLOSES OUT: the chase began from an asymmetry in `dcps.c` -- five `spSetComplex` calls and zero `spSetReal` -- which turned out NOT to be a defect. Matrix mode is managed per factorization by the SMP layer (SMPcLUfac/SMPcReorder set complex, SMPluFac/SMPreorder set real), so there is nothing to restore; `dcps.c` sets it manually because it assembles `G + jC` by hand (`spSetComplex` -> `CKTload` for the real halves -> `CKTacLoad` for the imaginary halves -> `factor`), which is the intended pattern and the reason it is the only file in the tree calling `spSetComplex` directly. Regression 305/305. Verify (examples/stbrestore): 7 checks, a proven trigger -- pre-fix 4/7. The four that pass on BOTH binaries are the accept half (a probe that started at `ac=0` still ends at `ac=0`; `acphase` survives; `stb` still reports a loop gain; `stb` run twice reports the same one), and `stb`'s own answer is BIT-IDENTICAL across the fix (`|T[0]| = 9999995.004562`, `phase -0.00100110002572`) with its existing suite examples/stb_examples passing 5/5 unchanged. | | [E-382](#) | `src/frontend/com_loadpull.c`, examples/lprestore_examples/* | **loadpull** left the user's tuner at its last swept point. It sweeps the R, L and C of the user's own matching network across the Smith chart, and when finished left them wherever the last grid point put them -- the old code said so itself, `/* restore something sane on the load (last set values are fine) */`, a comment with no code under it. The last set values are not a result, merely where the loop stopped: `RL 50` -> 84.83 ohm, `LL 1e-15` -> 1.34e-8 H. Any following analysis then ran against the wrong network -- on the test deck an `.ac` moved from 0.4789 to 0.6765, a 41% error, silently. THE SAME CLASS AS E-381 (`stb` handing its probe sources back ZEROED): both commands borrow parts of the user's circuit, drive them, then guess at what "putting them back" means -- one guessed zero, the other guessed "wherever we stopped". sweep already had it right (E-350 captures each swept parameter's nominal value and restores it at cleanup) and this follows that precedent. Two neighbours were checked and are NOT defects: `optimize` leaves the best point because that is its deliverable, and `aging` writes back the accumulated dose because that is the whole command. A TRAP WORTH RECORDING, and the reason this row is precise: a `@name[param]` string built in C must be LOWER-CASED and fails SILENTLY if it is not. Two wrong explanations preceded this one, so what follows is what was MEASURED. (1) THE PARSER REJECTS UPPERCASE -- A/B at a single call site, same context, same instant: `raw=<@RL[resistance]> len=0` versus `lower=<@rl[resistance]> len=1`; `ft_getpnames_from_string("@RL[...]", TRUE)` returns NULL. (2) IT FAILS INSIDE PPparse, BEFORE `checkvalid` -- instrumenting both branches showed "PPparse FAILED", and that path is return NULL with NO diagnostic, so `if_getparam` (which would have printed "no such device or model name") is never reached; `checkvalid` does warn on a zero-length vector and was confirmed not to fire. (3) TYPING THE SAME TEXT WORKS because the command never sees uppercase -- instrumenting `com_print`'s wordlist showed it already holding `@rl[resistance]` when the user typed `@RL[resistance]`, so the fold happens BEFORE command dispatch. That is why `print/alter/show/sweep` all accept uppercase (none passes it on) while a string a command builds internally from its own argument words bypasses the fold. (4) NOT ESTABLISHED: where that pre-dispatch fold lives -- not `com_print`, not `ft_getpnames_quotes`, not a command-table flag, and not a blanket fold of `.control` lines, since unquoted echo UNQUOTED MixedCaseWORD preserves case. Left stated rather than guessed at. The first attempt blamed placement instead (reading before `loadpull`'s priming transient), a red herring settled

only by instrumenting the actual code path. Regression 306/306. Verify (examples/lprestore): 7 checks, a proven trigger -- pre-fix 3/7. The four failures are the defect (R and L left moved, the 41% .ac shift, and source-pull mode leaving its own tuner at 70.69 ohm); the three passing on BOTH binaries are the accept half -- loadpull still reports a peak Pout (3.9576 dBm, identical across the fix), running it twice still reports the same optimum, and CL was already unchanged because the last grid point happened not to move it. examples/loadpull_examples passes 4/4 unchanged. || [E-383](#) | src/frontend/typesdef.c, examples/plotorder_examples/, examples/sweepname_examples/verify_sweepname.py, examples/plotname_examples/verify_plotname.py | Four plot-type entries in **plotabs[]** that could never be reached, so four analyses' plots were named after a DIFFERENT analysis. ft_plotabbrev() returns the FIRST entry whose pattern is a SUBSTRING of the plot name, and E-367/E-368 both wrote that ordering rule into the source -- then E-367 broke it with its own entry: envelope went in at the BOTTOM of the table, below { "op", "op" }, and "envel-OP-e" contains "op", so the entry was dead and every envelope plot came out op1. setplot envelope1 found nothing, and the plot took a number out of the operating-point sequence. Auditing every plot name in the tree that reaches plot_alloc() -- the literals, the names qp_emit_plot() passes, and the descriptive analName strings arriving via plot_alloc(run->type) -- found three more of the same shape: qpac->pac1 (collides with .pac), qpxf->pxf1 (collides with .pxf), and spectrum->sp1 (collides with .sp; BOTH spec and fft set pl_name to that literal, so both were affected). A COLLIDING name is worse than an unnamed one because it looks right: .pac plus qpac in one session gave pac1/pac2 with nothing to tell them apart. qpnoise was already correct, since E-368 had placed it ahead of pnoise/noise. The spectrum fix is a deliberate change to two stock ngspice commands' plot names (spec and fft) -- { "spect", "spect" } had been sitting in the table unreachable for exactly that purpose, and E-367's comment had recorded the result as an unfixable quirk. WHERE it goes is the constraint: the noise analysis names its plot "Noise SPECTral Density Curves ...", so one row too high would have renamed every ordinary noise plot and broken setplot noise1 across this repo; it sits below { "noise", "noise" }. E-345's example asserts the fft name directly and its expectation was updated WITH this change rather than around it -- the full regression is what surfaced that, which is why it is run before the fix is called contained. plot_alloc("digi") in findvec_all() looks like the same defect and is deliberately NOT given an entry: the next line overwrites pl_typename with a hardcoded "dig1", so an entry would be dead on arrival, and the path proved unreachable from every invocation tried (print alle, plot alle, event nodes confirmed via eprint) -- adding an untestable entry is the defect being removed. Since two of the four defects were introduced BY the enhancements fixing this table, the ordering invariant is now asserted against the table itself rather than left to deck coverage. loadpull leaving 17 tran plots behind was checked and is NOT a defect -- sweep keeps 334 op plots for the same documented reason. Verified by 25 checks (25/25 fixed, 12/25 pre-fix; the 11 accept checks pin op/tran/ac/noise/sp/.pac/.pxf/qpnoise and E-367's sweep/eye/hb on BOTH binaries). || [E-384](#) | src/spicelib/devices/vsrc/vsrcpar.c, src/spicelib/devices/vsrc/vsrcset.c, src/spicelib/devices/vsrc/vsrcrtemp.c, src/spicelib/analysis/span.c, src/osdi/osdiregistry.c, src/include/ngspice/devdefs.h, src/spicelib/devices/dio/dio.c, src/spicelib/devices/mes/mes.c, examples/sensstate_examples/ | A transient after **sens** came back with EVERY NODE EXACTLY ZERO -- including the node the source drives -- plus five more defects around **sens** and the RF ports. THE BIG ONE IS NOT IN sens: VSRCPARAM's pwr and freq cases set here->VSRCFUNCTIONTYPE = PORT UNCONDITIONALLY, and EVERY voltage source carries those parameters, not just ports. sens perturbs every settable real parameter of every device, so it wrote pwr/freq on ordinary sources, flipped them to PORT (power 0), restored the VALUE afterwards and never the function type -- the deck's SIN was gone for the rest of the session. alter @v1[pwr]=0 did the same damage in one line with no sens involved. Blast radius measured before the fix: tran and envelope ZEROED, hb/pss failing with |F|=nan while blaming the user's circuit, op/dc ~60x less accurate (6.7e-8 vs 1.1e-9 against a tight-tolerance reference), ac/tf/pz shifted 1e-7..2e-6; only reset cleared it. Fixed by claiming the source as PORT only when it carries no explicit waveform (!VSRCFUNCGiven). THE OTHER FIVE: (1) sens ended the PROCESS on any deck with a portnum source -- vsrcset.c allocates the port's res node with no already-allocated guard, unlike the branch node ten lines above, and cktsens.c calls DEVsetup a second time and controlled_exit()s when a node appears; a 3-line deck was enough. (3) sp with no ports called controlled_exit(EXIT_BAD) on what is an ordinary deck typo; it now reports, names the fix and returns E_NOTFOUND. (4) sp with z0 <=

0 SILENTLY demoted the port and ran with what was left -- a 2-port with $z_0=0$ on port 2 emitted a plot holding ONLY $S_{1,1}$ (no $S_{1,2}/S_{2,1}/S_{2,2}$) and an $S_{1,1}$ of 0.9512 where 0.9089 is correct; z_0 is a divisor ($Y_0=1/z_0$, $k_i=0.5/\sqrt{z_0}$) and a non-positive value is now a parameter error. (5) OSDI: `osdiregistry.c` tested `dt/temp` with case-SENSITIVE `strcmp` in one branch and `strcasecmp` in the next, so a model declaring `DT` had ngspice's `dt/dtemp` aliases built over its own parameter and `dtemp=50` was accepted with no diagnostic then silently ignored -- the identical mistake E-335 fixed one line above for `m`. (6) Two parameter-table inconsistencies that ngspice's OWN `check_ifparm` reports and nothing in this repo ever ran: `dio`'s `tref` aliases `tnom` but was `IOPUR`, dropping `IF_NONSENSE` (the flag that keeps a parameter OUT of sensitivity analysis), so two spellings of one parameter disagreed about being sensitivity-able -- a new `IOPXUR` macro spells it correctly; and `mes`'s `m` was flagged `IF_REDUNDANT` while carrying its own id. An independent checker over all 972 device files found these two and no others. WHY THIS SURVIVED A CAMPAIGN AIMED AT IT: the sequence fuzzer has `sens` in its pool, but its netlist carries `portnum`, so every `sens` case died instantly on defect (1) and `sens` was never exercised; and its oracle is crash-only, which a transient full of zeros does not trip. TWO DEAD ENDS recorded so they are not re-tried: `CKTnumStates` (really is clobbered by `cktsens`, was IMPLEMENTED as the fix, symptom did not move, reverted rather than shipped untestable) and `CKTmode/CKTstates` leaking (both `#ifdef notdef` dead code; `dctran` reassigns `CKTmode` at entry). Instrumenting `VSRCload` settled it: `fctype=2` (SIN) before `sens`, `fctype=10` (PORT) after. Verified by 17 checks (17/17 fixed, 7/17 pre-fix; the 6 accept checks pin bit-identical S-parameters, a real port source's `pwr/freq`, `sens`'s own analytic numbers, a plain transient and the OSDI model's own `DT` on BOTH binaries). || E-385 | `src/spicelib/devices/vccs/vccsset.c`, `src/spicelib/devices/cccs/cccsset.c`, `src/spicelib/devices/vsrc/vsrcpar.c`, `src/spicelib/devices/vsrc/vsrctemp.c`, `src/frontend/com_sweep.c`, `examples/staterestore_examples/*` | Three more commands that left the user's circuit changed behind them -- found by a STATE-RESTORATION AUDIT that goes after the CLASS rather than instances (E-380 .dc coefficients, E-381 stb probes, E-382 loadpull tuner, E-384 `sens/PORT` were the first four). Oracle: for every (instance, parameter) pair, the value after a command must equal the value before. (A) `sens . . . ac` KILLED VCCS AND CCCS SOURCES -- `@g1[gain] 1e-3 -> 0` and a following `.ac` returned 0.0 where the answer is 1.0. Not a bug in `sens`: `VCCSparam` folds the multiplier into the coefficient when `gain` is written (`if (VCCSmGiven) VCCScoeff *= VCCSmValue`) and `VCCSmValue` was never defaulted to 1 -- `res` does exactly that in `ressetup.c`. `sens` perturbs every settable real parameter, so it wrote `m` (setting `mGiven`), read it back as 0, wrote that 0 back as the restore, and the next `gain` write multiplied by zero. Explains the scope exactly: VCCS/CCCS have a settable `m`, VCVS/CCVS do not, and only the first two were hit; and why one frequency point was harmless while three were fatal. (B) `sweep` NEVER PUT AN `alter/altermod` KNOB BACK -- `@r1[resistance] 2000 -> 2199`, following op 3.8% wrong. E-350 covered only the `.param` path. Fix follows E-350's all-or-nothing rule; needs its own reader because `sw_eval_expr` returns 0.0 on failure, indistinguishable from a knob that is legitimately zero. (C) AND E-350's OWN PATH RESTORED ONLY HALF OF WHAT IT MOVED -- it put the `numparam` table back, but on the E-320 FAST path the point loop pushes values straight into the live circuit, so devices kept the last point's values: `.param r1=3k + sweep r1 lin 3 1k 5k` left `@r2[resistance]=5000` and the next op 19% wrong. Nominals are now pushed back the same way each point was, before `sw_fp_free()` drops the binds. E-384's FIX COVERED ONLY HALF ITS CLASS: `if (!VSRCfuncTGiven)` left the DC-ONLY source unprotected (no waveform -> guard passes), so `sens` still turned it into a PORT and a following `tran` read 0.0 where the answer was 1.0. The decision moved to `VSRCtemp`, inside the block that already knows whether the instance is really a port; `VSRCparam`'s `pwr/freq` cases no longer touch the waveform selector at all. E-384's own example passes either way because its deck carries a SIN -- the audit is what caught it. HARNESS TRAPS, both load-bearing: the BEFORE snapshot must be BOUNDED by the AFTER marker or it swallows the AFTER block and (building a dict) later values overwrite earlier ones -- before == after, EVERY command clean; caught only by a POSITIVE CONTROL against a known-defective binary. And the control must be a UNION of benign analyses subtracted PER PAIR, not op vs op (same analysis reproduces the same operating point) and never by NAME (`p` is settable on some devices, an output on others). LIMITATION stated: 114/292 pairs are operating-point dependent and subtracted, so a corruption of one of those is masked. ALSO OBSERVED, NOT FIXED: `sens_cplx` reads uninitialised memory (denormal garbage varying

run to run). The campaign harness ships as examples/staterestore_examples/audit/audit_state.py (31 commands x 292 pairs), labelling each pair [INPUT]/[output] from the device tables. Verified by 20 checks (20/20 fixed, 9/20 pre-fix; the 8 accept checks pin an explicit $m=2$ still doubling the gain, sens's own analytic numbers, VCVS/CCVS, plain tran/ac, E-350's .param restore, and bit-identical S-parameters, on BOTH binaries; check 14 is an audit canary -- a deliberate alter the embedded audit must detect). | | [E-386](#) | src/spicelib/devices/{res,cap,ind,dio,bjt,vccs,vcvs,cccs,ccvs}/ask.c, src/frontend/spiceif.c, examples/senscplx_examples/ | The sensitivity queries returned the PREVIOUS query's value. print @r1[resistance] -> 1000, then print @r1[sens_cplx] -> 1000; print @r1[i] -> 5e-4, then print @c1[sens_cplx] -> 4.99999616e-04. Every *_QUEST_SENS_* case had the shape if (ckt->CKTsenInfo) { ...write... } return(OK); -- and CKTsenInfo is set only by the SENS2 analysis, which is not compiled in, so on any ordinary run the handler wrote NOTHING and still returned OK. The caller then read whatever was already in its IFvalue. FIRST SEEN AS UNINITIALISED MEMORY (denormal garbage 2.12736e-314 varying between runs) -- same defect in disguise: the frontend's IFvalue is a static reused by every query (spiceif.c, doask, which sets only pv.iValue), and sens_cplx is IF_COMPLEX, so a preceding REAL query leaves the imaginary half stale -- a double read of bytes that were never a double is exactly what a denormal like 2.1e-314 is. Interleaving queries makes it plain: the answer is the last thing you asked for. FIXED IN THE HANDLERS, not the caller, because two other callers pass an uninitialised STACK IFvalue: dctrcurv.c (which saves the result as a parameter's NOMINAL to restore after a .dc sweep) and cktsens.c's sens_getp (which feeds sgen the value it will write back). Both latent today -- they only ask for parameters whose handlers do write -- but a contract of "returns OK, may not have written" cannot be made safe downstream. All 60 cases initialise their output before the if: 10 device types (res, cap, ind, mutual, dio, bjt, vccs, vcvs, cccs, ccvs) x six queries (sens_dc/sens_real/sens_imag/sens_mag/sens_ph/sens_cplx). Zero is what the surrounding code already chose -- the MAG and PH cases explicitly set 0 when the response magnitude is zero. doask hardened as well (memset before use; pv.iValue set AFTER, since it is an INPUT). SCOPE MEASURED by querying every parameter of seven device types immediately after a sentinel: 44 returned the sentinel pre-fix, 2 after -- and those two are @r1[r]/@r1[ac], genuine aliases of resistance that really are 1000. Verified by 8 checks (8/8 fixed, 6/8 pre-fix). One check passes on BOTH binaries deliberately: querying the sensitivity parameters FIRST, with no preceding query, already returned 0 because the static starts zeroed -- that is what made the defect easy to miss, since a probe asking only the suspect parameter sees nothing wrong. The 4 accept checks pin ordinary parameters (keyed by INSTANCE -- gain exists on both a VCCS and a VCVS and a name-keyed check silently kept only the last), operating-point readbacks asserted SELF-CONSISTENTLY as $p = i^2 R$ rather than against a literal from a simpler deck, sens's own analytic numbers, and show all : all. | | [E-394](#) | src/frontend/inpcom.c, src/frontend/inp.c, src/osdi/osdiparam.c, src/osdi/osdisetup.c, src/osdi/osdiload.c, src/osdi/osdiinit.c, src/spicelib/devices/nport/{nport.c,nportparam.c,nportdefs.h}, examples/osdiplumb_examples/* | Six ngspice+OSDI plumbing defects; FIVE are the same shape -- it works for a BUILT-IN device and silently does not for an OSDI one. (1) THE SUBCIRCUIT MULTIPLIER NEVER REACHED AN OSDI DEVICE: X1 a 0 sub m=3 scales a built-in R/diode inside the subckt exactly while an OSDI device contributed 1x -- in DC, AC, transient, thermal noise, flicker noise, charge AND S-parameters -- and $\$m$ factor read inside the model was always the device's OWN m, never multiplied by any enclosing X at any nesting depth. The cause is ONE CHARACTER: inp_fix_subckt_multiplier appends $m=\{m\}$ to each device line inside a multiplied subckt but skips lines whose first letter names a device with no multiplier, and 'n' was in that skip list -- N is the OSDI dispatcher. PDKs wrap compact models in multiplied subckts, so this under-counted device area with NO diagnostic. N also hosts the native n-port, which has no multiplier and ERRORED on an unexpected $m=$, so nport now ACCEPTS m and reports a non-unity value rather than the multiplier being dropped in silence there too (applying a real multiplier would mean scaling a stateful convolution and its history terms -- no finding here concerns that device). (2) NESTED SUBCIRCUIT MULTIPLIERS DID NOT COMPOUND, for built-in devices as well -- only the OUTERMOST $m=$ survived (2 around 3 gave 2x; 2x3x5 gave 2x, not 30x) because an existing $m=$ was multiplied ONLY in HSPICE compatibility mode and otherwise a second $m=\{m\}$ was appended and won. A subckt that DECLARES m as a parameter is a separate case and was always right (X's m binds it and must not also multiply) -- pinned in the accept half. (3) INSTANCE temp=

WAS NOT CONVERTED CELSIUS->KELVIN: every built-in adds CONSTCtoK at parameter-set time (dioparam.c) and ngspice's own OSDI code acknowledges the convention where it hands tnom over as CKTnomTemp - CONSTCtoK, but the OSDI path stored the raw number and used it as the Kelvin device temperature -- temp=75 reached the model as \$temperature=75 and \$vt=6.5 mV instead of 30 mV, -2.5e+16 A on a VA diode where the correct answer is -4.85e-07 A; temp=0 made \$vt EXACTLY ZERO so limexp(V/\$vt) divided by zero and the op failed outright; temp=-40 gave a NEGATIVE absolute temperature and a negative thermal voltage; and nothing warned. temp also STACKED with dtemp (75+10 meaning 85) instead of overriding -- it now overrides and prints restemp.c's own "Instance temperature specified, dtemp ignored", with the same sensitivity-analysis silence. .temp/.option temp/dtemp/temp sweeps were all correct throughout; only this path was raw. (4) .option scale NEVER REACHED AN OSDI MODEL: each built-in applies it in its own parameter setter (cp_getvar("scale")) and nothing scales an OSDI instance parameter -- nor can it, the OSDI ABI carries no units -- so the Verilog-A channel \$simpnam("scale") is the answer, and real models ask for it (the EKV model in this project's own VA_TEST corpus has SIMPARSCALE \$simpnam("scale",1.0)) while ngspice's ten-entry simpnam table omitted it, leaving the model on 1.0 while a built-in MOSFET in the same netlist scaled l=2 -> 2e-6. shrink/imax/rthresh deliberately NOT added: ngspice has no such option, so the honest outcome is the model's own default. (5) .options savecurrents PRODUCED NOTHING FOR OSDI DEVICES -- @r1[i] appeared for a built-in resistor and the compact model beside it produced no current vector at all, @n1[i] did not exist, so the only way to see a terminal current was to EDIT THE MODEL. Every OSDI instance now answers to i_<terminal> and a two-terminal one also to the bare i that R/C/L use, taken from the device's own stamp into that node's KCL row (resistive residual + the integrated charge derivative in a transient) so it is EXACT, not a finite difference -- verified equal-and-opposite, KCL to 3e-16 on a 3-terminal device, and CdV/dt in a transient. SCOPE BOUNDARY STATED: savecurrents is a TEXTUAL pre-pass and a model's terminal NAMES are not knowable there, so >2-terminal devices are read per terminal by name (.save @n1[i_d]) rather than auto-saved. (6) \$simpnam\$str("analysis_name") CONTRADICTED analysis(): E-53 taught the ANALYSIS_ flags to consult the running job so an AC job's op phase reports ac, but the string channel stayed on the CKTmode-only derivation and OSDIfinalStep carried a THIRD derivation testing MODEAC without MODEINITSMSIG -- so inside ONE evaluation a plain op reported name=ac while analysis("ac") was false, and an AC job's op phase reported name=dc while analysis("ac") was true. The in-source comment claiming "the same convention as analysis()" had become FALSE. THE MISTAKE WORTH RECORDING: making the other two match OSDIload is the obvious repair and is WRONG -- examples/finalstep_examples pins that ac-qualified events stay SILENT in a plain op, and propagating OSDIload's derivation made final_ac fire there. OSDIload was the odd one out: it set CALC_REACT_JACOBIAN and ANALYSIS_AC together from MODEAC | MODEINITSMSIG, but those are DIFFERENT QUESTIONS -- the reactive Jacobian IS needed during MODEINITSMSIG (that pass computes small-signal capacitances after a DC solution) while ANALYSIS_AC is a NAME and a plain op is not an AC analysis. They are now separated: the name bit follows MODEAC plus E-53's job consultation in all three channels, OSDIfinalStep reverts to what it had, and a plain op reports exactly one phase (dc). A suite that pins behaviour you are about to "unify" is the cheapest possible review. VERIFIED: examples/osdiplumb 45/45 (14/45 against the shipped binary) -- 31 checks pin real defects, and every OSDI check carries a BUILT-IN CONTROL in the same netlist because "the OSDI number changed" is not evidence; regression 318/318. | | E-395 | src/spicelib/parser/inpdp.c, src/spicelib/parser/inpgmod.c | Setting an OSDI parameter and its aliasparam to different values was accepted in silence and one spelling lost. An OSDI parameter and each of its alias names are registered by osdiinit.c as SEPARATE IFparm entries that all carry the SAME .id, so N1 a 0 md w=1 width=4 writes ONE slot twice; setting the same name twice does likewise. Both are now reported, on the instance line (INPdevParse) and on the model card (create_model), by tracking the ids written while parsing that one line or card. SCOPED to OSDI devices -- aliasparam is a Verilog-A construct, so no built-in device's parsing changes -- and deliberately placed in the DECK parser, because alter legitimately re-sets a parameter after the deck is read and must stay silent; an instance line overriding a model-card default is likewise a different thing and is not reported. THE MODEL-CARD MESSAGE DELIBERATELY DOES NOT NAME A WINNER, because the two channels on a card disagree: a

MODEL parameter is written straight through so the LAST one on the card wins, while an INSTANCE-parameter default is pushed onto INPmodfast->defaults with wl_cons and therefore replayed in REVERSE, so the FIRST one wins -- stating a rule would be wrong half the time; the instance-line message, which has no such split, does say the last value is used. The rest of E-395 is compiler-side and is recorded in the openvaf report. Verify (examples/langguard_examples): 11 checks on this item alone -- 5 doubly-set forms reported, 6 legitimate forms silent AND giving the same number. || E-396 | src/osdi/osdiregistry.c, src/osdi/osdiinit.c | \ \$limit with an unresolvable limiting function SEGFAULTED the simulator with ZERO output. osdi_create_registry_entry resolves the model's OSDI_LIM_TABLE against a fixed set -- pnj lim (2 extra args), fet lim (1), limit log (1), limvds (0). On an unknown NAME or a mismatched ARITY it printf'd "ignoring..." and left func_ptr NULL; the compiled model then CALLED that pointer, so the next evaluation jumped to address zero. The warning never reached the user either: it went to STDOUT and the crash destroyed the buffer before it was flushed -- which is exactly why the symptom was a silent death rather than a warning followed by a crash. fetlim takes 1 extra argument here and 2 in the LRM (vto, vgst), so a model following the standard was one of the ways to crash. There is no safe continuation -- the \ \$limit call site is already compiled to an indirect call -- so an unresolvable entry is now a HARD LOAD FAILURE (INVALID_OBJECT) with a message naming the function, the arity given, the arity expected and the supported set, written to STDERR where a later abort cannot swallow it. A collision warning fired on almost the whole industry corpus. E-335's check compared KEYWORDS only, but a model that declares one of the loader's own names has two IFparm entries under that keyword BY DESIGN: osdi_create_registry_entry routes the built-in to the model's parameter (dtemp -> dt = param_id, m -> has_m, temp suppresses the loader entry), so both address the SAME id and nothing is unreachable. dtemp is a conventional CMC instance parameter, so 69 warnings fired across the 124-file corpus (PSP, MEXTRAM, VBIC, BSIM, HiSIM, L-UTSOI, EKV, MVSG, ASM-HEMT, JUNCAP200, r2/r3_cmc), none of them a real problem, and each worded "declared more than once differing only in case" when the spellings are IDENTICAL and there is no second declaration to find. osdi_warn_case_collisions now skips a pair whose .ids MATCH; the corpus emits 5, all genuine different-id clashes. A finding was WITHDRAWN here: this began as "a model parameter named m silently defeats E-394's subcircuit multiplier" because such a model gave 1x under X1 ... m=3 -- but the probe declared m and never USED it. A real CMC model declares m and scales by it, which is what has_m exists for, so the appended m={m} lands there and the multiplier is DELIVERED through it (verified 1x/3x/5x). The warning added for it was removed. The compiler-side half of this release is recorded in the openvaf report. Verify (examples/linguard_examples): the crash check asserts a POSITIVE return code and a message naming the supported set, because the defect's signature was rc = -11 with an EMPTY output stream -- a naive "did it fail?" test scores that as a pass. E-394's multiplier is re-pinned end to end (1x/3x/5x) THROUGH a model-declared m. Load sweep over the corpus: 107 loaded, 0 failures, 0 crashes, warnings 69 -> 5. || E-397 | src/osdi/osdiinit.c, src/osdi/osdiregistry.c, src/osdi/osdiparam.c, src/osdi/osdisetup.c, examples/instknobs_examples/* | temp, dtemp and dt could be WRITTEN and never READ on an OSDI device. osdiinit.c registered the loader's own three entries IF_REAL + IF_SET with no IF_ASK, while a model's own parameters get both -- nothing in the source defends the asymmetry. So print @n1[temp] answered "no such parameter" where every built-in reports one (@r1[temp] -> 27, @r1[dtemp] -> 0), show n1 listed none of the three, and -- worst -- a sweep over any of them ended with a spurious error AFTER completing correctly: the swept data was right and printed, then ngspice's end-of-run vector bookkeeping failed to resolve @n1[temp]. A diagnostic that fires on SUCCESS is worse than none; it cost this project a wrong answer (see below). It also made the three look as though they existed only when the Verilog-A declared them -- declaring one turns it into a MODEL parameter, which is askable -- when in fact ngspice supplies all three to every OSDI device by default. WHY A FLAG WAS NOT ENOUGH: the ids COLLIDED. osdiregistry.c gave the synthesized knobs num_params + num_opvars and +1, which is exactly the base osdiinit.c gives E-394's synthesized TERMINAL CURRENTS -- dt's id WAS terminal 0's id and temp's was terminal 1's, survivable only because the two groups were disjoint by DIRECTION (temperature knobs set-only, terminal currents ask-only). Adding IF_ASK on top would have made @n1[temp] return a terminal current: a wrong number where there had been an honest error. FIX: the synthesized ids move ABOVE the terminal range (+ num_terminals), making the three

id spaces disjoint; OSDIparam dispatches on param == entry->dt explicitly so the write side follows unchanged; OSDIask serves the two new ids, guarded on >= cur_base so a model that DECLARES dtemp/temperature (whose routed id is BELOW that base) still falls through to the ordinary readable-parameter path. CONVENTION MEASURED, NOT ASSUMED: a built-in resistor reports temp as the BASE temperature in DEGREES CELSIUS, following .temp/.option temp when the instance does not set it and NEVER including dtemp; dtemp is the offset, 0 when unset. An OSDI device now answers identically in all seven cases tested. ONE BEHAVIOUR CHANGE, not merely reporting: temp= overrides dtemp= (E-394 established that and prints the same message restemp.c does) -- but restemp.c also FORCES RESdtemp = 0, which was invisible while dtemp could not be read. Leaving the written value would now report an offset that has no effect, so it is cleared at both osdisetup.c sites; the device temperature is unchanged (temp=75 dtemp=10 is 348.15 K before and after). THE WRONG ANSWER THIS COST, RECORDED: asked whether sweep works with these knobs, this project answered no, on the strength of that trailing error -- the sweep had completed correctly and its five points were printed immediately above the message. The lesson is not "read more carefully" but that a diagnostic emitted after a successful operation will be read as a failure of it, by tools and people alike. Verify (examples/instknobs_examples): 127/127 fixed, 92/127 shipped -- 35 checks pin this. Every read-back case is checked twice, against the expected value AND against a built-in resistor in the same deck. The suite re-pins what the id move could have broken: E-394's terminal currents still resolve on a 3-terminal device and satisfy KCL to 1e-12 with @n1[temp] readable alongside; the 2-terminal bare i alias still resolves; a model-declared dtemp is still read from the model's own parameter; the physics is unmoved; and sweep over each of the four knobs yields the right point count with NO diagnostic. Corpus load sweep 107 loaded / 0 failures / 5 warnings -- identical to what E-396 left. Full regression 321/321. || [E-399](#) | ngspice-46/src/spicelib/parser/inpgval.c, ngspice-46/src/spicelib/analysis/cktop.c, ngspice-46/src/osdi/osdiload.c | Three ngspice-side defects from round 13: a value that depended on which door it came through, a diagnostic that stated the opposite of the truth, and three answerable \$simparam names. (1) AN INTEGER PARAMETER ROUNDED DIFFERENTLY FROM THE NETLIST THAN IN-MODEL: INPgetValue converted with floor(0.5 + x), rounding .5 toward +infinity, while Verilog-A's own conversion inside a model rounds half AWAY FROM ZERO per the LRM. .model mm dut(n=-2.5) therefore gave -2 where ii = -2.5 gave -3, and EVERY negative half-boundary disagreed (-0.5 -> 0 vs -1, -1.5, -2.5, -3.5) while positives always agreed. Now round(), exactly the LRM rule. This is a GENERIC parser path shared by every device; the two forms differ only at exact negative half-integers, where no built-in uses a meaningful value and the old answer was the surprising one, and the full 322/322 regression confirms nothing depended on it. (2) A PLAIN CONVERGENCE FAILURE WAS ANNOUNCED AS A MODEL ABORT: CKTop's ordinary 'every convergence aid failed' exit FELL THROUGH into the fatal: label, so a model containing no \$fatal at all was reported with 'a Verilog-A device raised \$fatal ... This is not a convergence failure -- see the OSDI(fatal) message above', with no OSDI(fatal) line anywhere above to read. Both claims false, and the text denied the true cause -- a singular matrix that had just defeated gmin stepping, source stepping and optran, each of which printed its own correct diagnostic immediately above. E-378 added that block for a genuine device-raised abort and it landed in the fall-through path of the normal failure route; the path returns converged now, leaving E-378's own goto fatal jumps as the only way in. Verified both directions. (3) THREE MORE STANDARD \$simparam NAMES -- iteration (STATnumIter), abstime (CKTtime), simulatorSubversion (0) -- the ones ngspice can answer truthfully. shrink, imax, imelt, rthresh stay out, keeping E-394's stated reasoning. What made the absence matter: an unknown name is not a soft miss, \simparam("iteration") with no default aborts the ENTIRE run. || [E-401](#) | src/include/ngspice/osdiitf.h, src/osdi/osdregistry.c, src/osdi/osdisetup.c | **V(a,b) <+ 0** between two module TERMINALS connected nothing at all -- a silent OPEN CIRCUIT where the model wrote a short. The LRM's own page-155 parares, whose purpose is to degenerate to a wire, gave i(v1) = 0.0 at r=1e-6 against -5.0e-4 for a real short, while r=1e-2 conducted correctly: crossing the model's own threshold flipped it from wire to open. This project's diode_lim.va test model read 1000 V of self-heating rise at its DEFAULT rth=0, where it ties its thermal terminal to ground. TWO CAUSES, NEITHER WRONG ALONE: collapse_nodes skips any collapse pair whose endpoints are simulator-allocated ("*Terminals can never be collapsed in ngspice because they are allocated by ngspice*")

instead of OSDI”), and the compiler’s DAE build emits no equation for a trivial potential contribution -- so the collapse was dropped and nothing replaced it. THE COMPILER-ONLY FIX IS WRONG AND THE REGRESSION PROVES IT: emitting the 0 V source and stopping there passes every targeted case and cargo test 210/0, then breaks FIVE regression examples, because a 0 V source between two nodes the NETLIST has already shorted is SINGULAR -- node collapsing is idempotent, an equation is not (HICUM with its thermal terminal grounded, the normal wiring: “Warning: singular matrix: check node n1#xf1”). Only the simulator knows the netlist, so the fix is two-sided: the compiler emits the source AND lists the branch in two new ADDITIVE exported symbols (OSDI_TERM_SHORT_COUNTS/_INFOS, the absdelay/last_crossing pattern -- no descriptor layout change, no ABI version bump), and ngspice drops the branch current when the terminals do not resolve to two distinct connected circuit nodes. Recorded only for branches whose current the model never reads, which is what makes dropping it safe. Terminal-to-INTERNAL collapse is untouched and still exact (BSIM4’s $V(s,si) <+ 0$), as is the op-dependent switch branch (the LRM page-114 relay). VERIFIED: regression 322/322 (the five failures gone), cargo test --workspace 210/0 with six snapshots regenerated, corpus 107 compiled by both, 0 rc differences, 20 byte differences -- the FIRST release in this run to change emitted bytes, all 20 thermal models (HICUM L0/L2, HiSIM-HV/SOI, BSIM-BULK/CMG/IMG, ASM-HEMT, MVSG, MOSVAR) that ground a thermal terminal, correlating exactly with the new metadata (changed models export TERM_SHORT symbols, the 87 unchanged ones export none). Structure changes, answers do not: HICUM L2 on a normally-wired circuit gives -1.86927e-02 on both binaries, self-heating off and on. || [E-402](#) | src/spicelib/parser/inp2n.c | An OSDI instance line with FEWER nodes than the model has terminals was accepted in silence, while every built-in rejects the same mistake (diode “could not find a valid modelname”, resistor “is not a valid resistor instance line, ignored!”, MOSFET “Simulation interrupted”). A 4-terminal device given 3 nodes still simulates and answers differently -- v(g) 1.333 fully connected against 2.000 mistyped, rc=0, empty diagnostic stream -- because the omitted pins DANGLE and every branch touching them goes dead. ROOT CAUSE: inp2n.c bounded the count on one side only (numnodes > *dev->terms errors; too few was never asked about), then bound the missing terminals to -1. REJECTING IT WOULD BE WRONG: that -1 is the same sentinel osdisetup.c reads as “deliberately unconnected”, i.e. the LRM optional-terminal feature 32 corpus models query via \port_connected; and a dangling node is no substitute, since it reports \port_connected == 1 where an omission reports 0. The lower bound therefore WARNS, naming each absent terminal from dev->termNames and spelling out that they are NOT grounded -- the natural assumption, and wrong by 67% on the measured node. Validated against BSIMSOI compiled with PORT_CONNECTED and instantiated with 4 of 7 terminals: the three it names (p, b, t) are exactly the optional ones its own logic is written for, and that case keeps working. Semantics unchanged; regression 322/322 with the warning firing 0 times across the suite, so every future firing is signal. || [E-403](#) | src/spicelib/devices/res/resnoise.c, src/spicelib/devices/bjt/bjtnoise.c, src/spicelib/devices/dio/dionoise.c, src/spicelib/devices/vbic/vbicnoise.c, src/spicelib/devices/hicum2/hicum2noise.c, openvaf/hir_ty/src/inference.rs | Writing temp= equal to the ambient inflated an instance’s thermal noise power by 9%. rd x 0 1k and rd x 0 1k temp=27 are the same circuit at the same temperature, yet read 4.069337e-09 and 4.248250e-09. Five device noise routines computed the delta handed to NevalSrcInstanceTemp (which evaluates $4k(CKTtemp + param2)G$, so param2 is a DIFFERENCE) as inst->Xtemp - ckt->CKTtemp + (model->Xtnom - CONSTCtoK) -- the third term adding the NOMINAL temperature in CELSIUS, 27 by default, to a difference. Instance temperatures are stored in Kelvin so the first two terms were the whole answer; the third is dropped in res, bjt, dio, vbic and hicum2 alike. HOW IT SURFACED: a bug hunt found sens silently changing every later noise in the session (2.878894e-09 alone against 3.005592e-09 after sens, sticky, no analysis clearing it). The route to the cause matters because the symptom pointed away from it -- the shift was EXACTLY constant at 1.08996 in power across resistor values and topologies (so a global scale, not a perturbation); onoise and inoise scaled equally (so the sources changed, not the transfer function); $1.089955 = 327.15/300.15 = +27$ K, and repeating at 100 C and -50 C gave +27.00 K EVERY time (a constant, not a scaling); and an OSDI device’s own white_noise was unaffected at ratio 1.00000 while the built-in resistor beside it shifted, which located the fault in the built-in device rather than in ckt->CKTtemp. sens restores parameter VALUES but leaves ...tempGiven set, flipping the noise routine into that branch; correcting the arithmetic fixes

both, since an instance at ambient then contributes zero either way. ALSO (compiler side): a type mismatch on `ac_stim` reported *"expected string literal, string literal or string literal"* -- `hir_ty` collected the requirement at the failing argument from every surviving CANDIDATE SIGNATURE and joined the list verbatim, and three of `AC_STIM`'s four signatures take a string there; the list is now deduplicated, leaving `\$limit`'s genuinely distinct *"string literal or function"* untouched. DELIBERATELY NOT SHIPPED: a save/restore for `CKTnumStates`, which `cktsens.c` overwrites with one device's state offset and never restores -- it was written, changed no measured result, and was reverted, because a fix must be no wider than its evidence. Regression 322/322, cargo test --workspace 210/0, corpus 107 compiled by both, 0 rc differences, 0 byte differences. || [E-408](#) | `src/frontend/inpcom.c`, `src/frontend/parse.c`, `src/frontend/vectors.c`, `src/frontend/device.c`, `src/spicelib/analysis/dctrcurv.c` | Three ways a bracketed name missed what it pointed at. (1) A LEADING ZERO split a node in two: `n[1]`, `n[01]` and `n[001]` were three distinct nodes -- `print all` listed all three -- while the vector lookup canonicalised the index first, so every probe returned the `n[1]` value (1.0 against the 2 V and 3 V actually driven). With one node it is worse: a netlist that built `m[02]` was unreachable by EVERY spelling including its own, since the lookup only ever asked for `m[2]`. E-221's own header settles it (the scalar names a bus produces use the same bracket form, so `a[0:1]` and `a[0]` denote one node), so the netlist now canonicalises the index, negatives included, as a byte-identical no-op when already canonical; a contradictory deck that used to answer 1.0 now correctly reports a singular matrix. (2) `@dev[param]` COULD NOT NAME ANY BRACKETED PARAMETER: `show` lists the bus terminal currents `i_a[0]..i_a[3]` (E-394) and array parameter elements `ap[0]..ap[2]` with correct values and the instance line can set them, but four separate splitters each cut the name at the INNER] -- *no such parameter i_a[0*. -- so `read` failed, `alter` failed SILENTLY (the current stayed put) and a dc sweep was a FATAL ERROR, with no workaround: the name the simulator prints was not one it accepts back. All four had to change because each serves a different command (`PPlex` lexes the token at all, `vec_get` serves `print/let`, `com_alter` serves `alter`, `DCTfindInstParam` serves E-62's sweep). E-269's wildcard alias `@*[param]` works BECAUSE of the old first-] split, so a name beginning with] deliberately keeps it -- all four wildcard spellings measured unchanged and `wildparam` still passes. (3) A [lo:hi] BUS RANGE expanded on instance lines, R/C node lists, subcircuit calls and port lists but NOT on the cards that name nodes rather than connect them: `.save/.print/.plot` produced nothing and `.ic/.nodeset` warned and ignored. Those were skipped by design -- `inp_expand_buses` admits lines carrying BARE node fields and an output card writes `v(a[0:3])` -- so a wrapper-aware form emits one copy of the whole token per index, carrying the wrapper and any =value; the spaced `.ic v(a[0:1]) = 0.25` absorbs its value into the token first. The four expanded `.print` columns check against the resistor dividers (0.5, 1/3, 0.2, 1/9), not merely against each other. DELIBERATELY UNCHANGED: a token naming two ranges has no unambiguous expansion and stays literal; and one unparseable probe voiding its whole `.print` card is general behaviour for any unresolvable probe, no longer reached by the bus case, and measured to stop at exactly one card. Compiler untouched. New example `busname_examples 30/30`; regression 325/325. || [E-409](#) | `src/frontend/com_sweep.c`, `src/frontend/spiceif.c`, `src/include/ngspice/fteext.h` | A wildcard **sweep** knob was never put back, and said so in a way that looked like a broken parameter name. `sweep @*[wavelength_nm] ...` ran correctly and then printed *Error: no such device or model name / PError: syntax error in line segment @[wavelength_nm] near wavelength_nm]** -- the stray] because the expression lexer's specials set contains *, so `@*[p]` cannot lex as ONE token and the parser reports the leftover text. THE MESSAGE WAS THE HARMLESS PART: `sw_read_knob()` is how E-385 captures the nominal it restores at the end of a sweep, it parses the knob name as an EXPRESSION, and that call can never succeed for a wildcard -- so the read reported failure and E-385's deliberate ALL-OR-NOTHING rule skipped restoring EVERYTHING. Measured: `@*[wavelength]` left two model cards that started at 2.0 and 9.0 both sitting at 3.0; `@#[scale]` and the E-269 alias `@*[[scale]]` the same; and with `-vs` it took a CONCRETE co-knob down with it (`@dev1[wavelength]` left at 3.0 although it restores perfectly alone), so any following analysis silently ran against the last swept point. ONE NUMBER COULD NOT FIX IT: a wildcard sets every matching model or instance to one value but the values it overwrites all differ, so undoing it needs one reading PER TARGET. New `if_saveparam_wildcard()/if_restoreparam_wildcard()` walk exactly the same targets in exactly the same order as the existing `if_setparam_wildcard{,_instance}` (same device-type loop, same model and instance

chains, same parmlookup(..., inout=set) predicate), reading through the parameter's askable twin and the same doask/doset pair a plain alter/altermod uses; saving stays all-or-nothing (a set-only or vector-valued parameter refuses the whole capture) and the replay is pinned to the circuit the readings came from, so a .param co-knob that re-resources the deck cannot have them replayed onto a different one. sw_read_knob() now recognises the three wildcard spellings and returns without parsing, which removes the diagnostic. WHY IT SURVIVED THE AUDIT THAT EXISTS FOR THIS CLASS: E-385's own oracle staterestore_examples/audit/audit_state.py sweeps @r1[resistance], a CONCRETE knob, and reports sweep clean on the DEFECTIVE binary too -- no wildcard form was ever exercised; the audit is unchanged and still reports the same four known offenders (hb, pz, sens, sens_ac) on both binaries. New example wildrestore_examples 24/24, and 14/24 on the pre-409 binary, so it detects the defect rather than describing it. Compiler untouched; regression 326/326. || [E-410](#) | src/frontend/spiceif.c, src/frontend/gens.c, src/frontend/com_sweep.c, src/spicelib/analysis/dctrcur.c, src/include/ngspice/fteext.h | A device inside a subcircuit could not be named the way the node beside it is. @r.x1.r1[resistance] worked; @x1.r1[resistance] -- the obvious spelling, and the one that matches the node x1.m -- named nothing. THE TYPE LETTER IS NOT AN ACCESSOR CONVENTION: r.x1.r1 is the device's real name, because ngspice flattens a subcircuit by rewriting its cards back into the deck and re-parsing them as ordinary element lines, and the parser takes the device type from the FIRST CHARACTER of the card (inppas2.c: c = *(current->line); switch (c)) -- the card x1.r1 a m 1k would be re-read as another SUBCIRCUIT CALL. Two details prove that is the reason rather than a plausible story: translate_inst_name() EXEMPTS x devices (if (tolower_c(*name) != 'x')) because a subcircuit instance already starts with the right letter, so x2 in x1 is plain x1.x2; and NODES, which have no type, keep plain hierarchical paths (x1.m, x1.x2.q) -- which is exactly the asymmetry. THE RECONSTRUCTION NEEDS NO SEARCH: the letter prepended is literally the leaf name's own first character (bxx_putc(buffer, *name)) and a device name must already begin with its type letter, so x1.r1 can only mean r.x1.r1; two device types cannot share a leaf name. New if_find_instance_hier() is consulted ONLY after the exact lookups fail, so every name that resolves today resolves to exactly what it does today. Following E-408's lesson that @dev[param] is resolved in several independent places, every consumer is routed through it: finddev() (print/let/alter/altermod), finddev_special(), DCTfindInstParam() for E-62's .dc @inst[param] sweep, and sw_fp_bind() for the sweep knob -- where the old behaviour was NOT EVEN AN ERROR: the knob silently failed to bind and the three swept points came out IDENTICAL. show needed its own answer, since its grammar takes the query's first character as the device-TYPE selector (type = *word++) and uses :/# rather than . as the subcircuit delimiter, so x1.r1 parsed as type x; dgen_hier_match() is consulted ALONGSIDE that grammar as a whole-word alternative and can only add a match (a first attempt widened the strcmp inside the decomposition instead and was measured as DEAD CODE, then reverted). BACKWARD COMPATIBILITY DIFFED, NOT ASSUMED: across 17 show forms 14 are byte-identical and the only 3 that changed are the new spellings; legacy :r1/#dmod/showmod/-v unchanged; an x leaf gets no phantom prefix; bogus names and bogus parameters still reported; hierarchical NODES untouched. E-49's hiername_examples is a different subject (hierarchical names INSIDE Verilog-A) and is unaffected. New example hierdev_examples 32/32, and 15/32 on the pre-410 binary. Compiler untouched; regression 327/327. || [E-411](#) | src/frontend/inpcom.c | A descending netlist bus range binds nodes to a device's terminals in REVERSE order, and was silent. nd1 d[3:0] 0 mm against a model declaring inout [3:0] a compiles clean, simulates clean and is wired backwards -- and it is the spelling that LOOKS consistent, since both sides say [3:0]. TWO CONVENTIONS DISAGREE: the COMPILER declares a bus port's terminals in ASCENDING bit order whatever direction is written (hir_def item_tree lowering sorts the endpoints and loops upward -- declare from lsb to msb (ascending), matching natural bit order; direction of the original [msb:lsb] only affects range checks), so [3:0] and [0:3] declarations give the SAME positional terminal list; while the NETLIST does honour direction, E-221 expanding d[3:0] to d[3] d[2] d[1] d[0]. The instance line's Nth node binds the model's Nth terminal, so a descending list reverses the mapping. MEASURED with one conductance per bit (a[k] -> (k+1) mS, so a node's current names the bit it landed on): wired d[0:3] the nodes read 1,2,3,4 mS, wired d[3:0] they read 4,3,2,1 -- and the [3:0] and [0:3] model rows are IDENTICAL, confirming the declaration contributes nothing. inp_expand_bus_token() now reports

msb-first expansion and `inp_expand_buses()` warns on element instance lines only. SCOPE CAME FROM SCANNING THE SUITE, not taste: a `.subckt` descending PORT list is a deliberate interface choice E-221 documents and `busnodes_examples` tests; `.save/.print/.ic` bind nothing; and a range too wide to expand never binds (E-338's guard), so warning there would be false. Verilog-A part-selects (`v[3:2]`, `.i(v[1:0])`) are compiler-side and untouched. THE OPT-OUT WAS CORRECTED AFTER TESTING: `set nobusdirwarn` does NOT work inside `.control`, because bus ranges expand while the netlist is read; it works from `.spiceinit`, verified both ways, and the message says which. New example `busdir_examples` 20/20 (13/20 on the pre-411 binary), pinning the binding itself as well as the diagnostic. Regression 327/327 with the warning firing 0 times across the suite, so every future firing is signal. Compiler untouched; no binding changed, only a diagnostic added. || [E-412](#) | `src/osdi/osdiload.c` | An OSDI operating-point variable read after `.ac/.noise` returned the SMALL-SIGNAL solution, and CHANGED WITH FREQUENCY. An opvar defined as nothing but `vbias = V(a,c)` read 0.4939733 after `op --` the true bias, from solving the circuit's KCL cubic by hand -- but 0.4823410 / 0.4819028 / 0.0473587 after `.ac` at 1 k / 10 k / 1 meg, a multi-point sweep leaving the LAST point's. An operating point cannot depend on frequency, which is what separates this from a tolerance argument: identical at `reitol` 1e-3, 1e-6 and 1e-10. CAUSE: E-53 fires `@(final_step)` by issuing one dedicated evaluation per instance when an analysis completes, at `CKTrhs0ld` -- which after a sweep holds the small-signal solution, not a bias. The model is genuinely RE-EVALUATED rather than mis-read, confirmed with opvars chosen to distinguish the two (`3V+1` and `V^2` matched the `ac` solution to nine digits while a constant opvar stayed 42). Built-ins are unaffected, having no such post-analysis evaluation -- the recurring *works for a built-in, silently not for OSDI* vein. NOT WRONG, established BEFORE fixing anything: the analyses themselves. A bias-dependent `white_noise(kf*V^2)` source integrated to `onoise_spectrum = 1.342573180366e-01`, matching the dc-bias prediction EXACTLY and not the `ac`-bias one (`7.21e-02`) -- so this is a READBACK defect, not a numerical one, and it is not sticky (any following `op/dc/tran` restores the correct value). THE EVALUATION CANNOT SIMPLY BE SKIPPED: it is the only thing that fires `@(final_step)`, and `@(final_step(\"ac\"))` plus the noise variant are supported and tested by `finalstep_examples`. `OSDIfinalStep` therefore snapshots the instance data around the call and restores it, for `MODEAC` and `MODEACNOISE` only -- event bodies still run and their `\\$strobe` output stands, while what the evaluation wrote is discarded, which is correct because those results are by design never loaded into the matrix or RHS. DC, DC-sweep and transient are deliberately NOT snapshotted, their final solution being a real operating point (measured unchanged). WHY IT SURVIVED: with a purely resistive device the dc and ac solutions coincide, so the natural small test cannot see it; the example is reactive on purpose. New example `opvarac_examples` 19/19 (12/19 on the pre-412 binary); `finalstep` and `simctrl` pass unchanged; regression 329/329. Compiler untouched. || [E-413](#) | `src/frontend/breakp2.c`, `src/osdi/osdiparam.c`, `src/include/ngspice/osdiitf.h` | `.options savecurrents` registered an EMPTY vector for a multi-terminal OSDI device. It produced `@nd1[i] : current, real, 0 long` -- present in the listing, empty in use, with no diagnostic -- while a built-in BJT in the same deck produced four filled waveforms (`@q1[ic]/[ie]/[ib]/[is]`). CAUSE: `inp_savecurrents()` is a TEXTUAL pre-pass over the deck, running long before any `.osdi` is loaded, so it cannot know a compact model's terminal names and emits the same bare `@dev[i]` that R/C/L use -- E-394's own comment says exactly that. E-394 defines the bare `i` alias only for TWO-terminal devices, where it is unambiguous; for anything wider the save named a parameter that does not exist. THE VALUES WERE NEVER WRONG, only uncapturable: read as scalars after `op` the terminal currents were exact (`1.000e-3 / 4.000e-3 / 1.200e-2 / -1.700e-2`, KCL sum 0) and an explicit `.save @nd1[i_d]` always worked -- what was missing was the WAVEFORM, which is what `savecurrents` exists to provide. Since the names cannot be known during a textual pre-pass, the expansion moves to where they can: `ft_getSaves()` runs at analysis start with the circuit set up, so a bare `@dev[i]` belonging to an OSDI instance that does not define it is expanded there into one entry per terminal, through a new `OSDIterminalNames()`. The two-terminal case is deliberately untouched (`@dev[i]` is real there, byte-identical to the pre-413 binary) and built-ins never enter the path. Also explains why `meas tran ... MAX @nd1[i_a]` had quietly returned a scalar snapshot instead of a maximum over time. New example `savecur_examples` 15/15 (7/15 on the pre-413 binary); regression 330/330. Compiler untouched. || [E-415](#) | `src/osdi/osdinoise.c`, `src/spicelib/analysis/cktsgen.c`, `examples/evtnoise_examples/` (new) | A

per-device noise total that was the wrong quantity, and one knob counted twice. (1) `nVar(i, j)` indexed `noise_vals` with a stride of `num_noise_src`, but each of the `NSTATVARS` state variables needs one slot per source PLUS one for the whole-device total accumulated at index `num_noise_src` -- which is exactly what `osdiregistry.c` allocates (`NSTATVARS * (num_noise_src + 1)`). With `n` sources `nVar(OUTNOIZ, n) = noise_vals[1*n+n]` and `nVar(INNOIZ, 0) = noise_vals[2*n+0]` are THE SAME ADDRESS, so `onoise_total_<dev>` came out bit-identical to `inoise_total_<dev>_<first source>`: an input-referred, single-source number reported as the device's output-referred total (2.24x out in the measured case), while every built-in device's total matched its own components exactly. The spectra and the grand totals were never affected -- `onoise_spectrum` matched two real 1k resistors to 9e-08 relative and the grand total already equalled the quadrature of the per-SOURCE entries -- so the regression pins the spectrum bit-identically against the pre-415 binary rather than against a derived value. (2) `m` is registered as an alias of `\$mfactor` with the SAME parameter id, so anything walking the parameter table saw the multiplier twice: `sens` listed both `<inst>_m` and `<inst>__mfactor` with identical non-zero sensitivities and summing the table double-counted it. `IF_REDUNDANT` -- the marking built-in devices give their alias keywords -- was tried and REJECTED by the regression: that flag also hides the keyword from listings, and `show n1` is required to list `m` (E-397 went 126/127). Marking the other spelling would invert the flag's meaning, since `spiceif.c` walks BACKWARDS from a redundant entry to find its principal. The de-duplication therefore happens where the repeat mattered: the sensitivity generator skips a parameter whose id an earlier keyword in the same table already claimed. `sens` lists it once, `show` still lists `m`, and `m` remains settable from an instance line, a `.model` card and `alter` (all three verified identical). | | E-419 | `src/math/ni/nicomcof.c`, `src/math/ni/niinteg.c`, `src/spicelib/analysis/dctran.c`, `src/spicelib/analysis/cktterr.c`, `src/spicelib/analysis/cktsopt.c`, `src/spicelib/analysis/cktdojob.c`, `src/spicelib/analysis/cktntask.c`, `src/include/ngspice/{cktdefs,tskdefs,optdefs}.h`, `examples/integmethod_examples/` (new) | Three new integration methods, all OPT-IN, plus the measurements that say when to use them. ngspice shipped trapezoidal (A-stable but NOT L-stable, so it RINGS at a sharp transition) and Gear 1-6; the only cures for the ringing were `xmu<0.5`, which costs an order, or `method=gear`, which costs accuracy everywhere. (1) **method=trbdf2** -- a one-step COMPOSITE: trapezoidal over $[t, t+gh]$ then BDF2 across $t, t+gh, t+h$. With $g = 2\text{-}\sqrt{2}$ (the root of g^2-4g+2) both sub-steps share the leading coefficient $2/(gh) == (2-g)/((1-g)h) == 3.4142/h$, so the Jacobian scaling does not change between stages. Order 2 and L-STABLE. Neither sub-step needed a new formula -- stage 1 is the existing trapezoidal order-2 rule at $\Delta t=gh$, stage 2 the Gear order-2 shape with unequal-step BDF2 coefficients -- and the two earlier charges stage 2 needs are supplied by ROTATING the state slots once mid-step (the rotation `dctran` already does on acceptance), the interior slot being spliced back out so the accepted history never contains the stage value. Self-starting, so it never needs the order-1 starter E-181 showed imposes an $O(h^2)$ floor. (2) **method=sdirk** -- Alexander's 3-stage order-3 L-stable SDIRK, restricted to STIFFLY ACCURATE tableaux ($a[s][j]==b[j]$, $c[s]==1$) so the last stage IS the answer; without that a step ends in a weighted COMBINATION of stage values and every value a device holds has to come out of a solve, not an assignment. Coefficients derived from gamma in code, since a mistyped Butcher digit does not crash -- it quietly costs an order. Fully implicit RK is out of scope: coupled stages need one $s_n \times s_n$ system. (3) **method=adams** -- Adams-Moulton at order `maxord`, weights derived PER STEP from the actual spacing (the textbook $(5,8,-1)/12$ are fixed-step and ngspice never takes fixed steps); normalising to $\tau=(t-t_{n+1})/h$ keeps the Vandermonde entries $O(1)$ where raw seconds would underflow. NOT stiffly stable above order 2. Error constants are derived, not borrowed: for an RK method $R(z)=1+\sum z^{k+1} b^T A^k e$, so $C = b^T A^p e - 1/(p+1)!$ -- a formula that reproduces the two constants ALREADY in `cktterr.c` exactly (1/12 trapezoidal, 1/2 backward Euler), which is what makes it trustworthy on new tableaux. TR-BDF2 0.04044 (trapezoidal's 1/12 overstated it 2.06x) and SDIRK 0.025897 (Gear-3's 3/22 overstated it 5.27x); the derived TR-BDF2 constant predicts a 2.06x per-step advantage and the fixed-step test independently MEASURES 2.02x. What measuring showed: per STEP the new methods win (SDIRK observed order 2.76 vs 1.80 for the second-order trio, $1.26\text{e-}07$ vs trap $1.66\text{e-}05$ at $h=2.5\text{e-}8$), but per NEWTON ITERATION -- the honest cost unit, since TR-BDF2 solves twice per step and SDIRK three times -- trapezoidal or Gear wins three of four circuits, and Adams-3 wins the oscillatory non-stiff case by $\sim 100\text{x}$. Recalibrating the constants barely moved that, which is the finding: the (cost,error) curve belongs to the method, so a wrong constant costs predictability of `trtol`, not efficiency. So TR-BDF2's

case is *L*-stability alone -- a 1ns edge arriving at a step pinned to 20tau leaves trapezoidal ringing 25 times and still 2.7e-02 off, while TR-BDF2 settles in 2 steps; trapezoidal's measured ratio is -0.8182, exactly the theoretical $(1-h/2\tau)/(1+h/2\tau) = -9/11$. None becomes the default. Two earlier claims corrected in the write-up: trap/gear do NOT hit an accuracy floor (that was a trtol sweep stopping at 0.5), and verify_syntaxhl.py was wrongly named as the orphan source. Explicit methods (forward Euler, Adams-Bashforth, explicit RK) are NOT included and cannot be: with ag[0]=0 a reactive element becomes a fixed current source and a series RC returns $v_{n+1}=v_n$ -- at Newton convergence the element's current IS state0[ccap] whatever ag[0] is, so charge, being the dependent quantity $q(v)$, cannot be prescribed by a conductance+current stamp. OSDI is verified explicitly, since it loads its Jacobian with CKTag[0] ALONE while TR-BDF2 stage 2 and Adams order>=3 need ag[1]/ag[2] -- it routes its reactive residual through NIintegrate, and all five methods agree to 0.0e+00 between SPARSE and KLU on an OSDI device. Verify (examples/integmethod_examples): 13 checks, both solvers; full regression 336/336. | | [E-418](#) | src/osdi/osdisetup.c, src/osdi/osdipzld.c, src/osdi/osdidefs.h, src/frontend/outitf.c, src/frontend/com_measure2.c, examples/saveguard_examples/ (new) | Four defects that share a shape: something was accepted, counted, or concluded, and never verified -- the fourth found by fixing the first, and older than it. (1) absdelay() and last_crossing() read 0.0 whenever their result was only OBSERVED rather than contributed -- the E-415 open item, now CLOSED. They are the only operators whose output row the compiler deliberately leaves empty (ngspice fills it from its history buffer via elements OSDIsetup allocates later), so the node appeared in no descriptor Jacobian entry and E-116's decoupled-node scan tied it to ground; eval() then read CKTrhs0ld[0]. A contribution -- even at weight 1e-30 -- is what put the node into a Jacobian entry, which is why only it rescued the value, and why transition() (same lowering) worked: its output feeds the slew residual. Confirmed rather than inferred: before the fix n1#implicit_equation_1 does not exist as a circuit node. Five lines, idempotent for contributed models, no OpenVAF change and no ABI change -- E-415 had looked in the compiler and reverted when the descriptor came out byte-identical, correctly. The rule is now written at the site: a node coupled only by ngspice-side stamping must be marked there. (2) .save never validated a device name. addSpecialDesc only interns the string and getSpecial's caller discards INPaName's E_NODEV/E_BADPARAM, so @nosuchdev[i], @r1[nosuchparam] and @*[i_p] each produced a registered vector that stayed 0 long, silently, while print/meas/wrdata all report the same name loudly. The check now calls INPaName -- the very routine the per-point read uses -- and a HIERARCHICAL spelling is rewritten rather than rejected, so E-410's @x1.n1[i_p] yields a full waveform under the user's own spelling. Unresolvable names are warned about and still added (dropping them would change the plot; a bracket-less @name is a statistic on another path). Wildcards are diagnosed, not expanded: every wildcard mechanism in the tree is setter-side and none yields instance NAMES. Also corrects a round-25 note -- a bad NODE name is silently dropped too, so .save validated nothing. (3) meas ... when invented a time: -inf, 1.15292e+05 seconds in a 3 us run, negative times, silently. The initialisation counted a crossing in the FIRST interval and the evaluation -- which deliberately starts one sample in -- applied that leftover count to a later, crossing-free interval, dividing by a difference that was exactly zero or ONE ULP (hence every bad value being a power of two times a power of ten). The first interval spans the operating point to the first timepoint, so it straddles thresholds for reasons unrelated to the waveform: an earlier attempt to evaluate it properly broke defaultttransition (expected 5.0e-07, got 1.0002e-10), which is what identified starting one sample in as deliberate rather than as the bug. So the count is no longer taken, every interpolation is gated on the interval actually BRACKETING the target (mirroring measure_at, which has always refused to interpolate outside a bracketing pair), and a .dc restart resets to sample 0 so the previous sweep's last point is never an endpoint. Clamping to [tstart, tstop] was rejected -- it would turn 1.15292e+05 into tstop, a plausible wrong answer in a delay measure. cross=1 improves outright: garbage becomes the genuine later crossing at 1.00025e-06. (4) pz filled the simulator-stamped row nowhere. Those rows are the simulator's to fill, so every load path has to fill them -- osdiload.c does for dc/tran and osdiacld.c for ac, but osdipzld.c did for neither: the row was identically zero, the matrix singular at EVERY trial s, every trial looked like a root, and pz blamed the netlist ("the input signal is shorted on the way to the output", naming neither cause nor device). This predates (1) for a model that CONTRIBUTES the value -- its row was already live -- and (1) would have widened it to every observed-only model. The AC stamp cannot be reused: it

is exact and bounded there ($|e^{-j\omega t_d}| = 1$), whereas for complex s the exponential overflows across pz 's own search range (up to $1e35$) and a transport delay has infinitely many roots, so no finite pole-zero set exists to find. `absdelay` is stamped as the zero-delay wire -- the linearization `absdelay_stamp_dc` already uses for the operating point -- and warned about once per instance (`OSDIpzLoad` runs hundreds of times per analysis); `last_crossing` is stamped exactly and NOT warned about, its small-signal sensitivity being exactly zero. Each case now reports the delay-free pole to the digit -- -1000000, -1000001, -1000002 for the three modules, each matching that module's own conductance at the output node. Verify (examples/saveguard_examples): 38 checks, 17/38 on the pre-418 binaries. Both solvers; full regression 335/335. || [E-417](#) | `src/osdi/osdisetup.c`, `src/osdi/osdiparam.c`, `src/osdi/osdiinit.c`, `src/osdi/osdiregistry.c`, `src/osdi/osdidefs.h`, `src/include/ngspice/osdiitf.h`, `src/spicelib/analysis/cktsens.c`, `src/frontend/breakp2.c`, examples/collapsestate_examples/ (new) | A node collapse is decided once in `setup_instance`, and `OSDItemp` re-decides it on every temperature update without being able to rebuild anything derived from it. Three consumers reported over the mismatch. (1) `sens` printed `ROUND OFF` as a derivative: with the collapse taken, four Jacobian entries map onto ONE matrix element, so the perturbed device's $+g/-g$ ($g = 1/\delta = 1e6$) cancel on that diagonal and what survives is $\text{eps} * E / (Y * \delta^2)$ -- $2.5289e-02$ where the true derivative is $2.2676e-04$ (112x; the hunt's original case was 14000x and sign-flipped), with a magnitude tracking $1/r_d$ and a `SIGN` that moves with unrelated parameters. It cannot be computed here -- `DEVsetup` runs only at the base value and re-running it would allocate nodes and trip `cktsens`'s own `CKTlastNode` guard -- so it is reported as 0 with a warning naming instance and parameter, leaving every other row of the table untouched. (2) A `.dc` temp sweep ran the whole range on ONE topology, in both directions: sweeping INTO a collapse leaves the terminal floating (exactly 0 A), sweeping OUT of one keeps the collapsed topology so the series element never appears (1 mA where the truth is 500 uA -- a plausible 2x error, the more dangerous of the two). `.temp` and `set temp` were always right because each re-does the setup; the sweep now warns once per instance, naming instance, model type and temperature. Rebuilding mid-sweep is not a fix but a re-architecture (nodes would be allocated and deleted, `SMPmakeEl`t called on an ordered and factored matrix -- a hard no under KLU -- and every other device's cached element pointers invalidated), and a hard ERROR would mean propagating five bare `CKTtemp(ckt)`; returns in `dctrcurv.c`, changing error handling for every device model in the tree. (3) `.options savecurrents` expanded `@dev[i_<term>]` at three or more terminals but bailed out at exactly two, where `@dev[i_p]` stayed a length-1 SCALAR and `meas/plot` silently used one point -- the shape E-413 existed to remove, surviving at a different arity. The expansion is now unconditional and the bare `@dev[i]` E-394 defines is KEPT beside it, with synthesized names de-duplicated against explicit `.save` entries. The detection has one trap that is the whole fix: `OpenVAF`'s collapse callback only ever STORES TRUE, so `OSDItemp` must CLEAR `collapsed[]` before re-deciding or a collapse that stopped happening is invisible -- precisely the direction the defect takes -- and must restore the snapshot before the error check, because the matrix implements the snapshot. E-416's own "deliberately not changed" note is corrected in place: its $\sim 2e-3$ terminal-vs-source gap was defect (3) measuring a scalar against a waveform (the plain R||C agrees to $1.31e-15$), and the gap that does exist on a multi-terminal model IS Newton convergence (HICUM/L2: $4.73e-04$ at `reltol` $1e-3$, $7.9e-15$ at $1e-6$). NOT fixed, and documented: `%m` prints the MODULE name because the compiler splices it into the format string `CONSTANT` at compile time (`hir_lower/src/fmt.rs`), so no simulator-side fix exists; the clean fix is a `handle->name` callback needing no ABI change, but the compiler initialises such slots to a NULL pointer, so a new `.osdi` on an older `ngspice` would call NULL (the E-396 `SIGSEGV` class) unless a fallback is emitted into the `.osdi` first. Verify (examples/collapsestate_examples): 22 checks, 13/22 on the pre-417 binaries -- nine flip. Both solvers; full regression 334/334. || [E-416](#) | `src/osdi/osdiparam.c`, `src/osdi/osdisetup.c`, `src/osdi/osdiregistry.c`, `src/osdi/osdiinit.c`, `src/osdi/osdidefs.h`, examples/collapsestate_examples/ (new) | A terminal current read exactly 0.0 whenever that terminal was COLLAPSED onto an internal node. `if (rd == 0) V(d,di) <+ 0; else I(d,di) <+ V(d,di)/rd;` is the standard compact-model idiom for an optional series resistance and `rd = 0` is the shipped default (BSIM `rdsmo`d=0, HICUM `re`=0), so the broken path is the COMMON one. The model writes its current into `di`'s residual and terminal `d`'s own slot stays zero, while `OSDIask` read only `descr->nodes[t].resist_residual_off` -- 0.0 for a terminal carrying the device's full 2.001 mA, through `@dev[i_<term>]`, `show` and `.options savecurrents`

alike. With `rd` AND `rs` both zero EVERY terminal read 0.0 and the reported currents summed to zero, so a Kirchhoff check passed on `0 = 0`. The solution was never wrong: `osdiLoad.c` stamps a collapse group into one matrix row, so the collapsed and uncollapsed forms give identical node voltages and source currents -- which is why nothing a user plots ever moved. `OSDIask` now sums the terminal's whole collapse group (a group holds at most one terminal, since `collapse_nodes` refuses to merge two simulator-allocated terminals), resistive residual plus the integrated reactive residual in a transient, advancing the state cursor past non-members exactly as `osdiLoad.c` does -- the reactive half is load-bearing, not a refinement: 1 nF at 1 V/us makes the terminal current ~1 mA of pure displacement current against a few uA of conduction. The grouping cannot be recovered after setup, because `write_node_mapping` replaces the local mapping with GLOBAL node numbers and those cannot express a group -- two terminals on one net share a global number without being collapsed, and ground additionally collects grounded terminals, nodes collapsed to ground, E-116's decoupled internal nodes and E-401's dropped term-short flow nodes (grouping by global index makes a grounded terminal read 1.1 mA where 1.0 mA is correct). It is therefore snapshotted the moment `collapse_nodes` returns, into one `uint32_t` per descriptor node trailing `OsdiExtraInstData`, holding `t + 1` so the calloced zero still means "setup has not run". An uncollapsed terminal owns only itself, so every uncollapsed number is bit-identical to before; an unconnected terminal (E-402) is unowned and reports 0.0. A differential over the 40 real compact models in `integration_tests/` shows how common the broken path was: 23 bit-identical, 14 differing in terminal-current lines only (all previously all-zero: HICUM/L2, PSP103/_nqs/103t, ASMHEMT, BSIM3, BSIMBULK, BSIMSOI, HiSIM2, HiSIM-HV n4/n5, DIODE, DIODE_CMC, BSIMCMG), and nothing else moved anywhere -- the models that were already right (BSIM4 `rdsmode=1`, MEXTRAM 505, VBIC 1.3, BSIM6, PSP102, JUNCAP200, EKV, MVSG) are bit-identical, as an uncollapsed terminal owning only itself requires. One cosmetic consequence: the sum accumulates into `0.0`, so a residual of exactly negative zero now prints `0` rather than `-0` in `print/show` (numerically identical; nothing in the suite depends on it). Verify (`examples/collapse_examples`): 31 checks over four `rd/rs` combinations, a three-node collapse chain, a ground-collapsed internal node beside a grounded terminal, a capacitive transient, `show`, `savecurrents` and E-397's instance knobs; 13/31 on the pre-416 binary. The grown instance block was checked rather than assumed: size-vs-offset recomputed against all 589 descriptors in the repo, and the suites re-run under guard `malloc` (faults on any access past the allocation) -- clean. Both solvers; full regression 333/333. |